DRIVEALERT: AN IOT-BASED DRIVER DROWSINESS MONITORING AND ALERT SYSTEM USING COMPUTER VISION

A Capstone Proposal
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DEDICATION

This study is dedicated to all drivers who continue to face the risks and challenges brought by drowsiness while driving. May this work serve as a meaningful step toward promoting safer driving practices through technology, real-time monitoring, and alert systems that can help address drowsiness-related driving risks.

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TABLE OF CONTENTS

ACKNOWLEDGEMENT i

DEDICATION ii

TABLE OF CONTENTS iii

LIST OF TABLES v

LIST OF FIGURES vi

CHAPTER I INTRODUCTION 1

Rationale of the Study 3

Objectives of the Study 6

Scope and Limitations 6

Significance of the Study 9

Flow of the Study 10

Definition of Terms 12

CHAPTER II REVIEW OF RELATED LITERATURE AND STUDIES 16

Theoretical Background 16

Related Literature 17

Related Studies 30

Comparative Matrix 32

CHAPTER III RESEARCH METHODOLOGY 38

Software Engineering Methodology 38

Planning-Initiation Phase 45

Business Model Canvas 45

Program Workflow 47

Validation Board (Stages 1 and 2) 50

Gantt Chart/PERT Chart 52

Functional Decomposition Diagram 57

Use Case Diagram 60

User Interface Design 63

Storyboard 106

Database Design 108

Entity Relationship Diagram 109

Data Dictionary 111

Network Design 133
Network Model 133
Network Topology 134
Development/Construction/Build Phase 135
Technology Stack Diagram 135
Software Specification 136
Hardware Specification 138
IoT Architecture 139
Program Specification 140
List of Modules 140
REFERENCES 142
APPENDICES 150
APPENDIX A – Transmittal Letter – Respondents 150
APPENDIX B – Research Instrument 151
APPENDIX C – Results & Findings – Driver 156
APPENDIX D – Results & Findings – Potential Trusted Contacts 163
APPENDIX E – Consultation Logs – Adviser 166
APPENDIX F – Consultation Logs – Censor 168
APPENDIX G – Censor's Certificate 170
APPENDIX H – Grammarly Results 171
TEAM COMPOSITION 172

LIST OF TABLES

Table No.	Table Name	Page No.
1	Comparative Matrix	33
2	Comparative Matrix Summary	36
3	Agile Sprint Plan	40
4	Business Model Canvas	45
5	Validation Board	50
6	Gantt Chart	53
7	User	112
8	ConnectionCode	113
9	TrustedContactConnection	113
10	DriverCalibration	115
11	ProvisionedDevice	116
12	DriverPreference	117
13	MonitoringLogPreference	118
14	MonitoringSession	118
15	Alert	119
16	AlertSign	120
17	StageSyncRecord	120
18	StageSyncRecordSign	122
19	MonitoringLog	122
20	MonitoringLogEvent	123
21	SharedStage3Record	125
22	Device	126
23	DeviceOwnership	127
24	DeviceDriverAccess	128
25	DeviceSubscriptionEvent	130
26	DeviceCredential	131
27	DeviceAccessGrant	132
28	DeviceIdentityMapping	133
29	Technology Stack	136
30	Software Specification	137
31	Hardware Specification	138
32	List Of Modules	140

LIST OF FIGURES

Figure No.	Figure Name	Page No.
1	Flow of the Study	11
2	Agile Methodology	38
3	Program Workflow (Drivers)	47
4	Program Workflow (Monitoring Session)	48
5	Program Workflow (Trusted Contact)	49
6.1	Functional Decomposition Diagram	58
6.2	Functional Decomposition Diagram cont'd	58
6.3	Functional Decomposition Diagram cont'd	59
7.1	Use Case Diagram	61
7.2	Use Case Diagram cont'd	62
8	Starter Screen (Driver and Trusted Contact)	64
9	Sign In Screen (Driver and Trusted Contact)	65
10	Create Account Screen (Driver and Trusted Contact)	66
11	Forgot Password Screen (Driver and Trusted Contact)	67
12	Setup and Device Screen (Driver)	68
13	Device Connection Screen (Driver)	69
14	Camera Alignment Screen (Driver)	70
15	Driver Calibration Screen (Driver)	71
16	Driver Home Screen (Driver)	72
17	Camera Preview Screen (Driver)	73
18	Before Monitoring Dialog (Driver)	74
19	Monitoring Activity Screen (Driver)	75
20	Alert History Screen (Driver)	76
21	Alert Detail Screen (Driver)	77
22	Trusted Contacts Screen (Driver)	78
23	Invite a Trusted Contact Screen (Driver)	79
24	Shared Alerts Screen (Trusted Contact)	80
25	Invite a Driver Screen (Trusted Contact)	81
26	Requests Screen (Trusted Contact)	82
27	Driver Settings Screen (Driver)	83
28	Subscriptions and Drivers Screen (Driver)	84
29	Invite an Additional Driver Screen (Driver)	85

30	Demo Subscription Activation Dialog (Driver)	86
31	Device Invitation Screen (Driver)	87
32	Active Additional Driver Access Screen (Driver)	88
33	Trusted Contact Settings Screen (Trusted Contact)	89
34	Account Screen (Driver and Trusted Contact)	90
35	History and Storage Screen (Driver and Trusted Contact)	91
36	Monitoring Logs Screen (Driver)	92
37	Session Log Screen (Driver)	93
38	Log Deletion Screen (Driver)	94
39	Driver Notifications Screen (Driver)	95
40	Trusted Contact Notifications Screen (Trusted Contact)	96
41	Warning Sound Screen (Driver)	97
42	About DriveAlert Screen (Driver and Trusted Contact)	98
43	Log Out Confirmation Dialog (Driver and Trusted Contact)	99
44	Sign In Screen (Web)	100
45	Create Account Screen (Web)	100
46	Dashboard Screen (Web)	101
47	Connected Drivers Screen (Web)	101
48	Shared Alert History Screen (Web)	102
49	Shared Alert Detail Screen (Web)	102
50	Requests Screen (Web)	103
51	Invite a Driver Screen (Web)	103
52	Account Screen (Web)	104
53	Settings Screen (Web)	104
54	History and Storage Screen (Web)	105
55	About DriveAlert Screen (Web)	105
56	Driver Storyboard	106
57	Trusted Contact Storyboard	107
58	Trusted Contact Storyboard (Web)	108
59	Entity Relationship Diagram	109
60	DriveAlert Network Model	134
61	DriveAlert Network Topology	135
62	DriveAlert IoT Hardware Architecture	139

CHAPTER I

INTRODUCTION

Safe driving depends not only on road conditions, traffic regulations, and vehicle performance but also on the driver's ability to remain alert while operating a vehicle. Driver drowsiness is a serious road-safety concern because reduced alertness can impair attention, reaction time, judgment, and decision-making while driving. In the Philippine setting, the Land Transportation Office, as reported by the Philippine Information Agency, emphasized that drivers should pull over and rest when they feel tired. Similarly, Fonseca and Ferreira identified driver drowsiness as a significant road-safety issue associated with impaired attentional control, slower reaction time, reduced decision-making ability, and an increased risk of crashes and severe injuries (Philippine Information Agency, 2025; Fonseca & Ferreira, 2025). When drivers continue operating a vehicle despite visible signs of drowsiness, their ability to maintain the sustained attention required for safe driving may become compromised.

In the Philippines, road traffic crashes remain a major public-safety issue. The World Health Organization Philippines and the Department of Transportation reported that road traffic deaths in the country increased by 39%, from 7,938 deaths in 2011 to 11,096 deaths in 2021. The same source reported that road traffic injuries are the leading cause of death among Filipinos aged 15 to 29 years old and cost approximately 2.6% of the country's gross domestic product. In response to this concern, the Philippine Road Safety Action Plan 2023–2028 was launched as the country's blueprint for road-safety initiatives, with the goal of reducing road traffic deaths by 35% by 2028 (World Health Organization Philippines & Department of Transportation, 2023). The Philippine Statistics Authority also reported that deaths due to land transport accidents reached 13,125 in 2023, accounting for 1.9% of all deaths in the country. Meanwhile, the Asian Transport Observatory reported that road crashes and serious injuries cost the Philippines approximately USD 11 billion in 2021, equivalent to about 3% of the country's gross domestic product (Philippine Statistics Authority, 2025; Asian Transport Observatory, 2025). Although Philippine statistics do not separately identify the number of crashes associated with driver drowsiness, Reyes (2025) reported that international estimates attribute up to 30% of road accidents to driver fatigue. The same report explained that sleepiness associated with drowsy driving slows reaction time, decreases awareness, impairs judgment, and increases the risk of crashes. These conditions demonstrate the need for preventive and technology-supported road-safety systems that can assist drivers before dangerous situations become more serious.

Human behavior and driver condition are also important considerations in road safety. Segun et al. (2024) identified human behavior, public awareness, infrastructure, technological

interventions, and enforcement as factors associated with road-crash incidents in Metro Manila. Road safety therefore involves not only road and vehicle conditions but also the driver's physical and mental condition. Because drowsiness may reduce alertness and responsiveness, technologies that monitor visible signs of drowsiness may help drivers recognize possible declines in attentiveness while driving.

Emerging technologies in intelligent transportation provide new methods for detecting possible driver drowsiness before it develops into a more serious safety risk. Computer vision and artificial intelligence have been widely explored in driver-drowsiness detection systems because they can analyze visible indicators such as eye closure, yawning, mouth movement, and changes in head posture. Albadawi et al. (2022) explained that advances in computing technology and artificial intelligence have contributed to the development of more capable driver-drowsiness detection systems. However, the practical implementation of these systems may still be affected by lighting conditions, device capability, detection reliability, and system accessibility. These limitations highlight the need for a driver-monitoring system designed to operate under practical in-vehicle conditions.

The background of this study is grounded in the integration of computer vision, artificial intelligence, the Internet of Things, and mobile-based driver monitoring. Camera-based systems can observe selected visible signs of drowsiness without requiring wearable sensors or direct vehicle integration. Lighting conditions remain an important consideration because camera-based facial measurements depend on usable image quality. DriveAlert uses an infrared-capable camera, and its performance under different available-light conditions will be evaluated using the final hardware configuration. Local processing is also important because warning systems should respond promptly even when internet access or mobile data is unavailable. Edge and localized processing can reduce latency and support privacy-sensitive Internet of Things applications, while ESP32 documentation supports Wi-Fi capability and local communication between the IoT device and the smartphone (Andriulo et al., 2024; Espressif Systems, 2025; Espressif Systems, 2026). When combined with Android-based processing and local offline Wi-Fi communication, these technologies can support a more accessible driver-drowsiness monitoring system.

This study proposes DriveAlert, an Internet of Things-based driver drowsiness monitoring and alert system that uses computer vision and artificial intelligence. The system is designed to monitor selected visible signs of drowsiness, specifically prolonged eye closure, yawning, and head nodding. When a selected behavior satisfies its established duration, repetition, or temporal-confirmation condition, DriveAlert activates a warning alert. Threshold-based alerting is appropriate because driver-drowsiness detection systems should not rely on a single uncertain video

frame and should instead evaluate whether a visible behavior is sustained or repeated before activating a warning (Albadawi et al., 2023; Essel et al., 2025). DriveAlert also stores distinct confirmed warning events for Driver review and applies staged intervention when qualifying warning activity recurs or persists. Warning Stage 1 provides the initial selected warning sound, and Warning Stage 2 retains that sound as an intermediate warning state. Warning Stage 3 combines the selected sound with a prerecorded spoken rest advisory and introduces Trusted Contact synchronization eligibility. These Warning Stages represent levels of system intervention and do not represent medical or physiological drowsiness severity.

Rationale of the Study

Driver drowsiness may not always be recognized immediately while a person is operating a vehicle. Although drivers may stop and rest once they become aware that they are sleepy, observable changes associated with reduced alertness may appear before they respond to their condition. Rajkumarsingh and Totah (2021) explained that drowsiness can affect judgment and decision-making, while Ayas et al. (2024) found that driver-state monitoring systems commonly combine detection with warning interventions. This creates a need for a supplementary system that can monitor selected visible signs of drowsiness and provide a timely warning without replacing the driver's responsibility to stop and rest.

Visible changes involving the eyes, mouth, and head are commonly examined in driver-drowsiness monitoring because they can be observed without attaching physiological sensors to the driver. Rajkumarsingh and Totah (2021) considered blinking duration, yawning frequency, and head position, while Albadawi et al. (2023) examined eye condition, mouth opening, and head pose as visual features associated with drowsiness. These findings are relevant to the selected indicators monitored by DriveAlert: prolonged eye closure, yawning, and head nodding. Monitoring these behaviors provides a non-contact method of identifying observable changes that may be associated with reduced alertness.

However, ordinary driver behavior may resemble a visible sign of drowsiness. Brief eye closure may represent normal blinking, mouth opening may occur while speaking, and downward head movement may result from briefly looking at another area inside the vehicle. Responding to every isolated movement may therefore produce unnecessary warnings. Albadawi et al. (2023) evaluated visual indicators across a temporal window, while Quiles-Cucarella et al. (2024) and Essel et al. (2025) applied time-based or consecutive-frame conditions before activating warnings. These approaches demonstrate the importance of confirming that a behavior is sustained or repeated rather than relying on a single uncertain observation.

Warning reliability is also important because detection has limited value when the resulting alert is unclear, easily ignored, or repeatedly activated by incorrect detections. Ayas et al. (2024) found that driver-drowsiness mitigation systems commonly use auditory warnings and may apply different levels of intervention. Weibull et al. (2024) also showed that false alarms can influence how drivers respond to later warnings. A drowsiness-warning system must therefore balance responsiveness and consistency so that the driver is alerted when the established detection conditions are satisfied without producing excessive interruptions during ordinary movements.

DriveAlert applies three Warning Stages to manage recurrent or persistent qualifying warning activity. The first distinct confirmed warning event occurring while no Warning Stage is active immediately activates Stage 1 and begins a 60-second evaluation period. If another distinct confirmed warning event occurs while Stage 1 is active, the system immediately progresses to Stage 2 and begins a new 60-second evaluation period. A subsequent distinct confirmed warning event during Stage 2 immediately activates Stage 3. When no new distinct event occurs, the system evaluates whether a qualifying behavior remains persistently active at the 60-second boundary. Persistent qualifying activity causes progression from Stage 1 to Stage 2 or from Stage 2 to Stage 3. If no qualifying behavior remains active, Stage 1 returns to the normal state, while Stage 2 de-escalates to Stage 1. Stage 3 is the maximum intervention level. Recurrent events during Stage 3 may reactivate its warning without restarting the current evaluation period. At the Stage 3 boundary, recurrent or persistent qualifying activity maintains Stage 3; otherwise, the system de-escalates to Stage 2. Stages 1 and 2 use the selected warning sound, while Stage 3 adds a spoken rest advisory and Trusted Contact synchronization eligibility. These stages describe system intervention rather than medical or physiological drowsiness severity.

An immediate warning addresses a detected condition at a particular moment, but the warning alone provides little information after it has ended. Without an organized record, the driver may be unable to review when an alert occurred, which visible sign was detected, or whether alerts appeared repeatedly during a monitoring session. Hidayat and Ibnugraha (2025) developed a system described as a driver-fatigue monitoring system that displayed detection results through a web-based dashboard, while Kashyap et al. (2025) stored driver-status information, detection timestamps, and historical records for later review. These studies show that recorded events can remain useful beyond the immediate warning by providing structured information for later review.

Stored drowsiness-alert events may also be summarized to make the recorded information easier to interpret. Retained alerts may be organized into alert counts, detected-sign summaries, the highest Warning Stage associated with their stored warning events, alert trends, and Stage 3 synchronization status. Monitoring-session summaries separately identify the highest intervention

stage reached during each session. These summaries allow users to review patterns in the available records without examining every alert individually. Their purpose is descriptive and should not be interpreted as medical analysis, accident prediction, confirmation of drowsiness, or proof that a driver is safe or unsafe to continue driving.

Drowsiness-alert records may also support secondary awareness through an approved Trusted Contact. DriveAlert stores all distinct confirmed warning events for Driver review but limits external synchronization according to its Warning Stage 3 rule. When DriveAlert first enters Warning Stage 3, one Stage 3 transition record becomes eligible for Trusted Contact synchronization. Individual confirmed warning events occurring during the active Stage 3 period remain in the Driver's alert history and do not each independently create synchronization records. When a 60-second Stage 3 evaluation period ends, DriveAlert creates one synchronization-eligible persistence update if at least one additional distinct confirmed warning event occurred during that period or a qualifying behavior remains persistently active at its boundary. Either condition maintains Stage 3 and begins another 60-second evaluation period. If neither condition is present, the system de-escalates to Stage 2 without creating a persistence update for the completed period.

The reviewed approaches provide visible-sign monitoring, warning delivery, event storage, dashboard viewing, physical alarm output, offline operation, and authorized record access in different combinations. Some systems concentrate mainly on detecting a visible sign and producing an immediate warning, while others emphasize historical records, dashboards, or remote monitoring. The need addressed by this study is therefore not the absence of driver-drowsiness technology but the opportunity to combine selected functions that support the driver during monitoring and preserve relevant information after a warning has been issued.

DriveAlert addresses these connected needs by monitoring prolonged eye closure, yawning, and head nodding and providing an audio warning when a selected behavior satisfies its established detection condition. It also stores distinct confirmed alert episodes locally, presents descriptive summaries of retained records for Driver review, and synchronizes only eligible Stage 3 transition records and Stage 3 persistence updates for access by an approved Trusted Contact when internet connectivity becomes available. Through this integration, DriveAlert connects visible-sign monitoring, warning delivery, event recording, record review, and authorized information sharing within one driver-focused system. It is intended as a supplementary driver-support tool and does not medically or physiologically confirm drowsiness, determine whether a driver is fit to continue driving, replace adequate rest, or control the vehicle.

Objectives of the Study

This study aims to develop DriveAlert, an Internet of Things-based system designed to assist drivers by monitoring selected visible signs of drowsiness, providing warning alerts, storing drowsiness-alert records, and synchronizing these records with a connected Trusted Contact when internet connectivity becomes available.

Specifically, this study pursues the following objectives:

1. To gather data on the occurrence of drowsiness and observable signs associated with it among drivers while driving.
2. To design and implement a computer-vision algorithm capable of detecting prolonged eye closure, yawning, and head nodding using driver-specific detection thresholds and temporal-confirmation conditions for triggering warning alerts.
3. To design and implement a local Wi-Fi communication architecture between the IoT device and the driver's smartphone that enables real-time drowsiness-sign monitoring without requiring internet or mobile data connectivity.
4. To implement an alarm system in which the IoT device receives wireless warning commands from the smartphone and produces predefined audio warning alerts through an onboard speaker.
5. To develop a dashboard that records and displays drowsiness-alert events and synchronizes drowsiness-alert records with a connected Trusted Contact when internet connectivity becomes available.

Scope and Limitations

This study focuses on the development and pilot evaluation of DriveAlert, an IoT-based driver drowsiness monitoring and alert system using computer vision. The system monitors prolonged eye closure, yawning, and head nodding through an IoT device equipped with an infrared-capable camera and an onboard speaker. Dedicated infrared LEDs and external infrared illuminators are excluded from the final prototype.

An Android mobile application processes the captured video locally using calibrated EAR and MAR thresholds, a fixed 15° downward head-pitch deviation relative to the Driver's calibrated neutral position, and sign-specific temporal-confirmation conditions. After a selected visible sign satisfies its complete confirmation condition, DriveAlert creates a confirmed warning event, applies the appropriate Warning Stage, requests the corresponding warning output, and stores the event locally for Driver review. Separate Stage 3 transition records and qualifying Stage 3 persistence updates become eligible for synchronization with an approved Trusted Contact when internet connectivity becomes available.

DriveAlert is intended for drivers of four-wheeled and larger vehicles. Motorcycles and other two-wheeled vehicles are excluded because of differences in driver position, camera placement, weather exposure, and operating conditions. The system will be evaluated only under controlled or simulated driving conditions and will not be tested for long-term use during actual road travel, heavy traffic, severe weather, uneven roads, or other unpredictable conditions.

DriveAlert is limited to monitoring three selected visible signs associated with drowsiness: prolonged eye closure, yawning, and head nodding. The system does not medically or physiologically confirm that a driver is drowsy. These behaviors may occur for reasons unrelated to drowsiness, while a drowsy driver may not always display them clearly. The system may therefore generate false alerts or miss relevant events. It does not detect sleep disorders, stress, illness, medication effects, alcohol influence, or other causes of reduced alertness, and it does not use physiological or wearable sensors.

Detection performance may be affected by camera placement, Driver position, head orientation, facial obstruction, eyeglasses, sunglasses, masks, glare, lighting conditions, sudden movement, reflections, and image quality. Although dedicated infrared illumination was included in the initial design, it was excluded from the final prototype due to limitations encountered during hardware development. The available infrared LED configuration did not provide sufficient facial illumination at the intended camera-to-Driver distance. Attempts to improve illumination by increasing the number of infrared LEDs also resulted in unstable operation of the breadboard power supply setup, including intermittent power loss and interruptions affecting connected components. These issues prevented the expanded LED configuration from operating reliably and limited further improvements to infrared illumination using the available hardware. These limitations apply specifically to the configuration tested in this study and do not establish that infrared illumination is unsuitable for driver monitoring in general. The final prototype therefore depends on available light, and its infrared-capable camera does not guarantee reliable assessment under extremely low-light or no-light conditions. DriveAlert pauses drowsiness-sign assessment when facial lighting is insufficient, even if a face remains detected. Assessment also pauses when reliable facial tracking is unavailable or fewer than two monitored signs remain available. Frames rejected because of insufficient lighting do not advance calibration's required valid measurement time. Assessment resumes when the applicable visibility, lighting-quality, and observation-freshness requirements are restored. These assessment pauses do not by themselves end the monitoring session. DriveAlert also provides operational guidance that is separate from drowsiness-stage warnings. During eligible active monitoring, sustained inactivity may produce a "No movement detected" status after ten seconds and a spoken movement advisory after twenty-five seconds. These timings are provisional

engineering settings. The advisory does not create a drowsiness-alert record, advance the Warning Stage, verify Driver identity, or establish that the camera input is from a live person.

DriveAlert uses guided facial-measurement calibration and spatial Driver-area selection rather than facial recognition or automatic Driver identification. Calibration records neutral facial measurements, closed-eye and open-mouth references, and the direction of downward head rotation. The face-processing pipeline may detect multiple face candidates, but drowsiness-sign assessment uses at most one selected Driver face. Selection depends on the configured Driver area and spatial continuity. Ambiguous, competing, or unreliable detections pause assessment. These safeguards do not verify identity or guarantee the exclusion of passengers in every scene. Camera movement, incorrect positioning, other visible faces, or changes in the Driver's position may affect tracking and detection. When reliable facial tracking is unavailable, assessment pauses and the Driver is prompted to restore the camera view.

The system does not automatically identify individual drivers. In shared vehicles, each driver must use the correct DriveAlert account, connect to the IoT device, and verify camera alignment before monitoring begins. Otherwise, drowsiness-alert records may be stored under the wrong account. Facial recognition and automatic driver identification are outside the scope of the study.

System performance may depend on the Android smartphone's processing power, memory, battery condition, and temperature. Devices with limited capability may experience slower video processing, delayed warnings, or reduced application stability. The pilot evaluation will use selected compatible Android devices; therefore, the results may not represent all Android devices and hardware configurations.

All distinct confirmed alert episodes are initially stored locally in the Driver's alert history, subject to user deletion and configured retention. Descriptive alert summaries reflect retained records. Removing a shared record from a Trusted Contact's personal history view does not delete the source Driver's cloud record. DriveAlert uses three Warning Stages with 60-second evaluation periods. A new distinct confirmed event immediately advances Stage 1 to Stage 2 or Stage 2 to Stage 3 and begins a new evaluation period. When no new distinct event occurs, persistent qualifying behavior at the evaluation boundary may cause the same progression. The absence of qualifying activity permits incremental de-escalation and eventual return to the normal state. Stages 1 and 2 use the Driver's selected warning sound, while Stage 3 adds a prerecorded spoken rest advisory and Trusted Contact synchronization eligibility. Each progression from Stage 2 to Stage 3 creates one transition record, including re-entry into Stage 3 after de-escalation. Each completed Stage 3 period creates at most one persistence update when an additional distinct event occurred

during the period or a qualifying behavior remains persistently active at its boundary. Individual warning events remain in the Driver's local history and do not independently create synchronization records. Sharing requires an approved Trusted Contact relationship and internet connectivity. Only eligible records from periods beginning on or after connection approval may be shared, and access depends on the relationship remaining approved. This feature supports secondary awareness and does not operate as an immediate emergency-notification service. Warning Stages do not represent medical or physiological drowsiness severity.

The study does not measure DriveAlert's effect on accident reduction, injury prevention, driver behavior change, or long-term driving habits. It does not cover commercial deployment, vehicle-manufacturer integration, vehicle-control systems, traffic enforcement, or automotive safety certification. DriveAlert cannot apply the brakes, steer the vehicle, reduce its speed, stop the engine, or determine whether the driver is fit to continue driving.

Local monitoring can operate without internet after the required account and device setup. Starting a monitoring session requires the appropriate Driver account, an active calibration, a configured and connected camera, and confirmed camera alignment. Account registration, cloud relationship management, remote record review, and external synchronization require internet connectivity.

The Android Driver interface supports local monitoring, alert-history review, and descriptive summaries. Trusted Contact interfaces provide access to eligible shared Stage 3 records rather than the Driver's complete local alert history or camera stream. Record synchronization and notification delivery are separate processes; successful synchronization does not establish that a Trusted Contact has received or viewed a notification.

DriveAlert is not intended to replace adequate rest, responsible driving practices, professional medical evaluation, or certified vehicle-safety systems. The study is limited to evaluating whether drowsiness-sign monitoring, warning activation, drowsiness-alert record management, dashboard functions, and Trusted Contact synchronization can operate together under controlled or simulated testing conditions.

Significance of the Study

The study brings benefits to various stakeholders involved in or affected by the development and implementation of DriveAlert.

Drivers. Drivers are the primary beneficiaries of DriveAlert. They may benefit from real-time monitoring of selected visible signs of drowsiness, predefined audio warning alerts with adjustable volume and an enforced minimum level, and a dashboard that records and summarizes drowsiness-alert events. Through the dashboard, Drivers can review alert counts, detected-sign

summaries, the highest Warning Stage associated with their stored warning events, alert trends, and Stage 3 synchronization status. Session summaries separately identify the highest intervention stage reached during an individual monitoring session.

Trusted Contacts. Trusted Contacts may benefit from accessing synchronization-eligible Stage 3 transition records and Stage 3 persistence updates when internet connectivity becomes available. These records provide secondary awareness when qualifying warning activity reaches or continues at DriveAlert's maximum intervention stage. The feature supports synchronized record review and does not function as live Driver monitoring or an emergency-response service.

Passengers. Passengers may indirectly benefit from DriveAlert because the system provides a warning when a selected visible sign of drowsiness satisfies its established detection condition. The warning may prompt the driver to reassess their condition and take appropriate action, such as stopping and resting.

Researchers. The researchers will gain practical experience in Internet of Things development, mobile application design, computer vision-based drowsiness-sign detection, local wireless communication, alert-system implementation, and system evaluation. The study will also strengthen their technical, analytical, problem-solving, documentation, and research skills.

Future Researchers. Future researchers may use this study as a reference for related projects involving driver drowsiness monitoring, computer vision, IoT-based safety systems, mobile processing, offline-capable warning systems, alert-record management, and driver-assistance technologies.

Flow of the Study

The procedure involves identifying prolonged eye closure, yawning, and head nodding as selected visible signs associated with drowsiness, processing the observed behaviors through the proposed system, and generating alerts and records when the established alert conditions are satisfied.

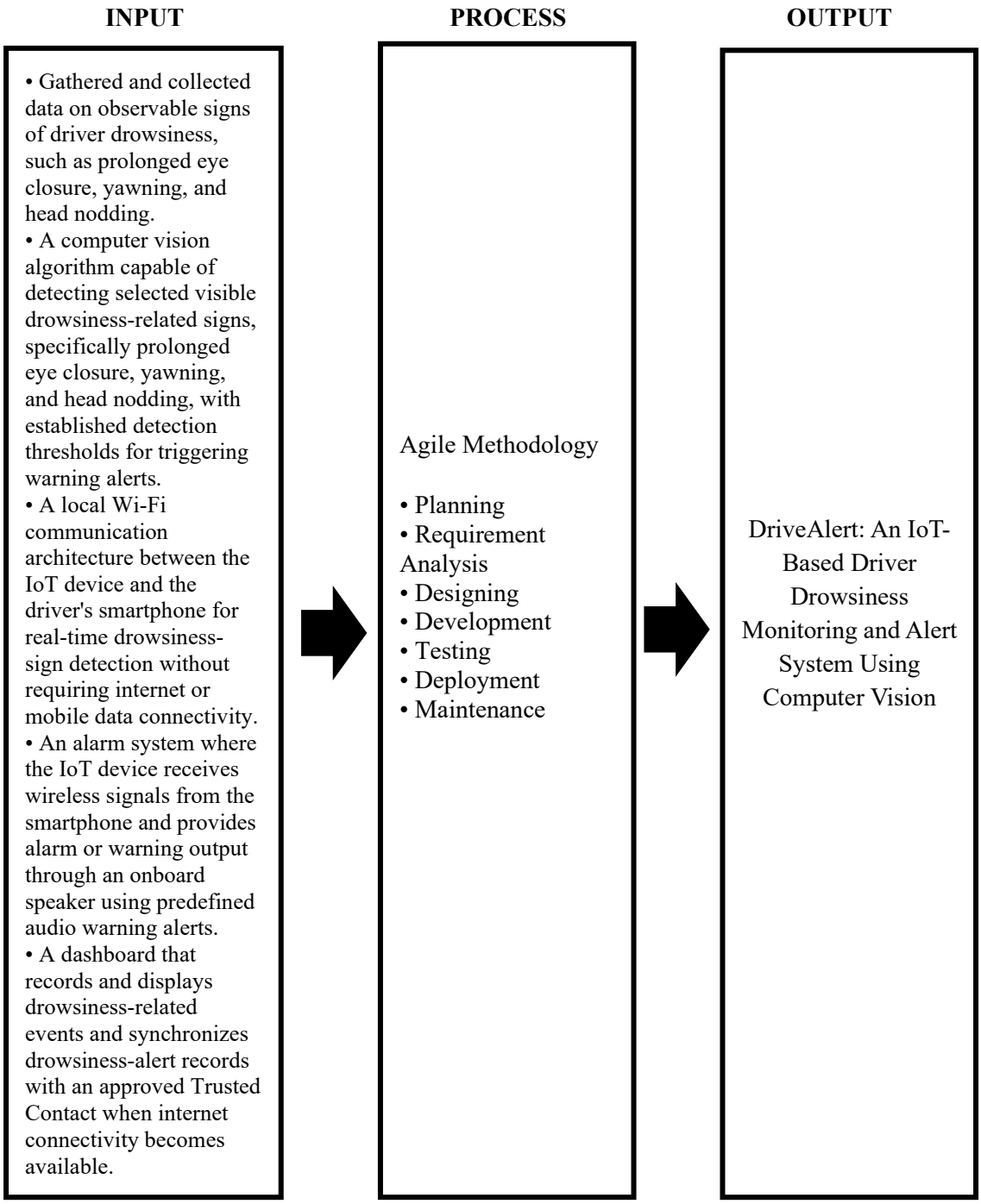


Figure 1: **Flow of the Study**

Figure 1 shows the flow of the "DriveAlert" study. The process begins with gathering and collecting data on observable signs of driver drowsiness, such as prolonged eye closure, yawning, and head nodding.

The next phase involves determining an algorithm that can monitor live video and detect selected visible signs of driver drowsiness, specifically prolonged eye closure, yawning, and head nodding. This phase also includes applying Driver-specific detection thresholds together with sign-specific temporal-confirmation conditions to determine when a confirmed warning event is created.

Following this, a direct and offline wireless connection will be designed between the IoT device and the smartphone of the driver. This connection allows the video to be processed instantly through the mobile application without depending on internet or mobile data.

To complete the alerting process, an alarm system will be implemented where the IoT device receives wireless signals from the smartphone and triggers an onboard speaker. The onboard speaker will play predefined audio warning alerts using an adjustable warning volume that cannot be muted or reduced below the enforced minimum level during active monitoring.

The study also includes the implementation of a mobile dashboard that allows the driver to review drowsiness alert history and basic drowsiness insights. The dashboard displays today's drowsiness alerts, weekly alert count, synchronization status, most frequently detected sign, highest Warning Stage reached, alert trend, and synchronization summary. A background synchronization system will also be developed to automatically synchronize eligible Warning Stage 3 transition records and Stage 3 persistence updates with an approved Trusted Contact when internet connectivity becomes available.

The study follows the Agile Methodology, progressing through iterative stages of planning, requirement analysis, system design, development, testing, deployment, and maintenance. This flexible approach supports continuous improvement and refinement based on feedback from users and testing environments.

Ultimately, the output of this study is "DriveAlert," an Internet of Things safety system designed to assist drivers through drowsiness-sign detection, warning alerts, drowsiness alert history recording, basic drowsiness insights, and Trusted Contact synchronization.

Definition of Terms

This section contains the specific meaning of the following terms concerning the context used to clarify the study.

Alert Trend. It refers to the basic insight showing whether the number of recorded drowsiness alerts is higher, lower, or unchanged compared with the previous period.

Artificial Intelligence. It refers to the system capability used by DriveAlert to assist in analyzing driver video and identifying possible drowsiness-related signs.

Computer Vision. It refers to the technology used by DriveAlert to process live video and examine visible movements of the driver's eyes, mouth, face, and head.

Driver. It refers to the primary user of DriveAlert who operates a four-wheeled or larger vehicle and uses the IoT device and mobile application to start monitoring sessions, receive drowsiness warnings, review drowsiness-alert records and summaries, and manage Trusted Contact access.

Drowsiness Alert. It refers to a confirmed warning event generated after one or more selected visible signs satisfy DriveAlert's complete Driver-specific measurement and sign-specific temporal-confirmation conditions

False Alert. It refers to a warning generated when ordinary or non-target driver behavior incorrectly satisfies the system's alert condition.

Fleet Manager. It refers to a person responsible for supervising a group of organizational or commercial vehicles and their drivers. In the current DriveAlert design, Fleet Manager is identified only as a potential customer segment and not as a separate system actor.

Highest Warning Stage. It refers to the highest Warning Stage within the context being summarized. A monitoring-session summary reports the highest intervention stage reached during that session, while an alert-history summary reports the highest stage associated with the stored warning events included in that summary. Neither value represents the Driver's medical or physiological level of drowsiness.

Internet of Things. It refers to the connected hardware and software components of DriveAlert that allow the IoT device and mobile application to exchange video data, connection information, and warning commands.

IoT Device. It refers to the DriveAlert hardware unit containing the infrared-capable camera, wireless communication components, and onboard speaker used during driver monitoring.

Local Offline Wi-Fi Connection. It refers to the direct local network between the DriveAlert IoT device and the driver's Android smartphone that does not require internet access or mobile data.

Local Processing. It refers to the processing of the transmitted driver video through the Android smartphone instead of sending the video to an internet-based cloud service for drowsiness-sign detection.

Missed Detection. It refers to a situation in which the system fails to identify a selected drowsiness-related sign that is visibly demonstrated by the driver.

Monitoring Session. It refers to the period beginning when the driver starts drowsiness monitoring and ending when monitoring is stopped. Drowsiness alerts detected during this period are associated with the same session.

Onboard Speaker. It refers to the physical speaker installed in the DriveAlert IoT device that produces the selected audio warning after receiving an alert command from the smartphone.

Prolonged Eye Closure. It refers to the driver's eyes remaining closed long enough to satisfy the established duration condition, distinguishing it from an ordinary brief blink.

Temporal Confirmation. It refers to the requirement that an observed behavior continue or repeat over a defined period or across consecutive video frames before the system treats the behavior as meeting its established alert threshold.

Threshold. It refers to the predefined duration, repetition, movement, or other condition that a detected drowsiness-related sign must satisfy before DriveAlert generates a warning.

Trusted Contact. It refers to a user connected to a Driver through an approved relationship who may review eligible shared Stage 3 transition records and Stage 3 persistence updates. The role does not provide access to the Driver's complete local alert history or full Driver-facing insights. Access depends on the relationship remaining approved. A Trusted Contact does not control monitoring or function as an emergency-response service.

Trusted Contact Synchronization. It refers to the automatic upload of eligible Stage 3 transition records and Stage 3 persistence updates for access by an approved Trusted Contact when internet connectivity becomes available. Sharing is limited to eligible records from periods beginning on or after connection approval and does not release the Driver's earlier complete alert history. Removing the approved relationship revokes the Trusted Contact's access.

Warning Stage Progression. It refers to the process through which DriveAlert changes its active intervention stage. A new distinct confirmed warning event immediately advances Stage 1 to Stage 2 or Stage 2 to Stage 3 and begins a new 60-second evaluation period. Persistent qualifying behavior may also cause escalation at an evaluation boundary. At Stage 3, recurrent or persistent qualifying activity maintains the maximum stage, while the absence of qualifying activity permits incremental de-escalation. Stage progression does not represent increasing medical or physiological drowsiness severity.

Warning Trigger. It refers to the system action that activates or reissues the warning associated with the applicable Warning Stage. A warning may be triggered by a new distinct confirmed warning event or by persistent qualifying behavior evaluated at a warning-stage boundary. Recurrent qualifying events during Stage 3 may also reactivate its warning without creating an additional stage.

Warning Stage. It refers to the current state of DriveAlert's warning intervention. Stage 1 provides the initial selected warning sound. Stage 2 retains the same sound as an intermediate warning state. Stage 3 represents the maximum intervention level, combines the selected sound

with a prerecorded spoken rest advisory, and introduces Trusted Contact synchronization eligibility. Each stage uses a 60-second evaluation period, although new distinct confirmed events may cause earlier progression. Warning Stages do not represent medical or physiological levels of drowsiness.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the related literature and studies gathered through the researchers' careful review and analysis. It provides the theoretical and technical foundation for the development of DriveAlert: An IoT-Based Driver Drowsiness Monitoring and Alert System Using Computer Vision. The review discusses concepts and studies related to driver fatigue and drowsiness, vigilance, human-centered warning systems, computer vision, Internet of Things, mobile-based detection, offline communication, alert mechanisms, drowsiness alert history recording, and Trusted Contact synchronization. This chapter also establishes how the proposed system relates to existing approaches and how DriveAlert contributes to the development of driver drowsiness monitoring and alert systems.

Theoretical Background

This study is anchored primarily in Vigilance Theory, which explains how sustained attention and alertness may decline during prolonged monitoring tasks. Mackworth (1948) introduced the concept of vigilance decrement through research showing that a person's ability to detect important signals may decrease over time. In driving, this theory is relevant because drivers must maintain attention for extended periods, particularly during nighttime travel, long-distance driving, and monotonous road conditions. McWilliams and Ward (2021) explained that low-demand and monotonous driving conditions may place drivers in a state of cognitive underload, contributing to vigilance decrement and reduced responsiveness to safety-critical events. For DriveAlert, Vigilance Theory provides a conceptual basis for understanding why maintaining alertness is important during prolonged or monotonous driving. The system itself is limited to monitoring prolonged eye closure, yawning, and head nodding as selected visible signs associated with drowsiness.

The study is further guided by Human Factors Engineering and Human-Centered Design principles. Human factors focus on designing systems according to human capabilities, limitations, behaviors, and possible errors. The Federal Highway Administration (n.d.) explained that human factors address the relationship between people and the systems they use, with consideration for human abilities and limitations. This is relevant to driver monitoring because warnings should be clear, timely, noticeable, and understandable without requiring complex interaction from the driver. Ayas et al. (2024) found that driver-drowsiness mitigation systems commonly use auditory warnings and may employ multistage interventions. These principles support DriveAlert's use of predefined audio warnings, warning escalation, clear monitoring status, and an interface designed

to minimize driver interaction during active monitoring while allowing drowsiness-alert records to be reviewed afterward.

Together, Vigilance Theory and human-centered design principles provide the theoretical and design foundation of DriveAlert. Vigilance Theory explains the importance of maintaining sustained attention during prolonged or monotonous driving, while Human Factors Engineering and Human-Centered Design guide how the system should communicate warnings and present information according to human capabilities and limitations. These foundations support DriveAlert's development as a driver-centered drowsiness monitoring and alert system that combines visible-sign monitoring, warning output, drowsiness-alert record review, basic drowsiness insights, and Trusted Contact synchronization.

Related Literature

Driver sleepiness, sleep deprivation, and prolonged working hours are associated with increased road-traffic risk. Saleem (2022) identified sleepiness and sleep deprivation as contributors to road traffic accidents, while the National Highway Traffic Safety Administration (n.d.) documented crashes and fatalities involving drowsy drivers. In the Philippine setting, Lu (2022) reported that prolonged working hours among bus and truck drivers may contribute to fatigue and sleepiness, thereby increasing road-accident risk. Segun et al. (2024) further identified human behavior and technology-supported interventions among the factors relevant to road safety. Together, these studies establish driver drowsiness as a road-safety concern and support the development of systems that monitor Driver behavior and provide timely warnings.

Driver fatigue and drowsiness are related but distinct concepts. Fatigue broadly refers to mental or physical exhaustion, whereas drowsiness describes a sleep-like reduction in alertness that may precede sleep onset. Although some literature uses fatigue as a broader term when discussing reduced alertness, DriveAlert is specifically limited to selected visible signs associated with drowsiness and does not claim to detect fatigue as a whole. Hassan et al. (2026) identified prolonged or slow eyelid closure, yawning, and head nodding as behavioral manifestations of drowsiness. Albadawi et al. (2022) and Jan et al. (2023) likewise identified eye closure, yawning, and head nodding among observable behaviors used in non-intrusive Driver-drowsiness detection. These findings support the selection of prolonged eye closure, yawning, and head nodding as the three visible signs monitored by DriveAlert.

Because these signs are externally observable, computer vision provides a non-invasive means of monitoring them without requiring physiological sensors. Albadawi et al. (2023) extracted Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and head-pose features from facial landmarks using MediaPipe Face Mesh. Essel et al. (2025) similarly examined eye closure and

mouth aperture, while Kim et al. (2023) used infrared video and facial landmarks to evaluate eye closure and head orientation. These studies demonstrate that facial landmarks can support a common visual method for examining eye-, mouth-, and head-related behavior. However, facial measurements may still be affected by lighting, camera placement, face orientation, eyeglasses, facial obstruction, and individual facial characteristics.

Eye Aspect Ratio represents the degree of eye openness using the relationship among selected eye landmarks and generally decreases as the upper and lower eyelids move closer together. Mouth Aspect Ratio similarly represents the degree of mouth opening and generally increases as the mouth opens wider (Albadawi et al., 2023). Head-pose estimation uses facial reference points to estimate changes in head orientation, while pitch represents upward or downward head rotation and may therefore support head-nodding analysis (Albadawi et al., 2023; Kim et al., 2023). Since ordinary blinking, speaking, and brief downward glances can temporarily produce similar measurements, a single threshold-crossing frame is insufficient to confirm a selected visible sign.

Previous studies provide numerical values that can serve as initial references for DriveAlert, but these values should not automatically be applied to every Driver. Albadawi et al. (2023) selected an EAR threshold of 0.20 after evaluating different candidate values and eye characteristics. Higher values increased the possibility of normally open but smaller eyes being classified as closed, while lower values reduced the number of detected closed-eye frames. The same study selected a MAR threshold of 0.50 after considering differences in mouth characteristics; values that were too high could miss wide-mouth events among subjects with smaller mouths, while values that were too low increased the likelihood of ordinary talking or laughing being classified as mouth opening. These findings support 0.20 for EAR and 0.50 for MAR as literature-supported reference values rather than universal operational thresholds.

Individual differences support the use of Driver-specific thresholds because fixed facial measurements may not generalize across users with different facial characteristics and normal head orientations. DriveAlert therefore uses four guided calibration phases, each requiring five seconds of valid measurements. During the neutral phase, the Driver faces forward with the eyes open to establish baseline EAR, MAR, and neutral head pitch. During the eyes-closed phase, closed-eye EAR is recorded as a calibration reference, although the operational EAR threshold remains 75% of the neutral open-eye baseline following Ersoy et al. (2026). During the mouth-open phase, the Driver keeps the mouth as open as possible, and the operational MAR threshold is set at 50% of the average open-mouth MAR following the personalized initialization approach of Quiles-Cucarella et al. (2024), expressed as MARThreshold = OpenMouthMAR × 0.50. During the head-

down phase, pitch measurements determine the direction of downward rotation relative to the neutral position. DriveAlert then applies a 15° downward deviation from the calibrated neutral head pitch as its operational head-pitch threshold. Xu et al. (2025) similarly used normal forward orientation as a reference, while Shang et al. (2023) reported drowsy head-pitch changes of approximately 12° to 20°, placing the selected 15° deviation within the reported range. These Driver-specific calibration values are stored and used during monitoring.

For prolonged eye closure, DriveAlert will use 2.0 seconds as its initial operational confirmation duration. Dong et al. (2026) analyzed distributions of normal and fatigue-related continuous eye-closure events and reported that separation between them became stable at approximately two seconds. Their system therefore used sustained eye closure exceeding two seconds as a fatigue-related criterion. Albadawi et al. (2023) also reported that ordinary blinking occurs on a substantially shorter timescale. DriveAlert consequently adopts 2.0 seconds only to determine whether an observed eye closure qualifies as prolonged eye closure and does not interpret the duration as medical confirmation that the Driver is drowsy. Prolonged eye closure will therefore be confirmed when the Driver's EAR remains below the Driver-specific EAR threshold continuously for at least 2.0 seconds. If the EAR reaches or rises above the threshold before the required duration is completed, the temporal counter resets and no prolonged-eye-closure event is confirmed.

Yawning requires a different duration because mouth opening associated with yawning is naturally a longer behavior. Dong et al. (2026) analyzed sustained mouth-opening duration and reported that separation between ordinary mouth movements and fatigue-related yawning became stable at approximately three seconds. Their system consequently used sustained mouth opening exceeding three seconds as its yawning criterion. Quiles-Cucarella et al. (2024) likewise used a personalized MAR threshold together with a three-second mouth-opening condition. DriveAlert will therefore confirm yawning when the Driver’s MAR remains above the Driver-specific MAR threshold continuously for at least 3.0 seconds. If the MAR reaches or falls below the threshold before the required duration is completed, the temporal counter resets and no yawning event is confirmed.

Head nodding requires a different confirmation approach because downward head movement is more ambiguous than sustained eye closure or mouth opening. A Driver may naturally look toward the dashboard, vehicle controls, or another location below the normal forward-facing position without being drowsy. Kholerdi et al. (2016) addressed this ambiguity by evaluating the persistence of downward or away-from-road head position rather than treating every downward movement as an immediate warning condition.

Kholerdi et al. (2016) considered a prolonged head position significant when the Driver's head remained dropped or turned away during at least 80% of the preceding three seconds. The authors explained that even when such a prolonged position was not caused by drowsiness, maintaining the head away from the normal forward-facing position for a substantial portion of the observation period could still warrant warning.

DriveAlert will therefore initially treat downward head rotation exceeding a fixed 15° deviation from the Driver's calibrated neutral head position as a candidate head-nodding event rather than immediately confirming the behavior. Calibration establishes the Driver's neutral head pitch and the direction of downward rotation; it does not determine a different angular deviation threshold for each Driver. The candidate will be confirmed when downward rotation exceeds this threshold in at least 80% of valid observations within a rolling three-second period. If this persistence condition is not satisfied, no head-nodding event is confirmed. The persistence condition is adapted from Kholerdi et al. (2016), while the fixed 15° deviation is a DriveAlert operational choice informed by the approximately 12° to 20° head-pitch changes reported by Shang et al. (2023).

The three selected signs are therefore not given identical confirmation procedures because they present different sources of false detection. Sustained EAR measurements help distinguish prolonged eye closure from ordinary blinking, while sustained MAR measurements help distinguish yawning from shorter mouth movements. Head movement is more ambiguous because normal Driver behavior can also involve downward pitch changes, making temporal persistence appropriate. DriveAlert consequently evaluates each selected sign according to its own characteristics rather than requiring prolonged eye closure, yawning, and head nodding to occur simultaneously.

During uninterrupted valid observation, DriveAlert groups additional confirmed signs into the current alert episode while a previously active episode sign remains currently confirmed. These additional signs do not independently create another alert episode or advance the Warning Stage. A continuously confirmed sign does not repeatedly create distinct events merely because its measurement remains beyond its threshold. Its detector must be re-armed before the same sign can produce another confirmation. Tracking loss, stale observations, or assessment interruptions may break episode continuity; the system does not assume that a physical behavior remained continuously observable during an unavailable interval.

The adopted detection thresholds and confirmation conditions will be evaluated during controlled testing using the actual DriveAlert IoT camera and Android processing environment. Testing will examine false alerts, missed detections, Driver-specific differences, landmark stability,

camera placement, performance under different available-light conditions, and processing performance. The purpose of this testing is to verify or refine literature-supported initial conditions under DriveAlert's intended implementation environment rather than assuming that numerical values obtained from another system will transfer perfectly.

Computer vision also supports local and non-invasive system implementation. In the Philippine setting, Macalisang et al. (2021) developed Drive-Awake using machine vision to examine eye closure associated with drowsy driving. Fonseca and Ferreira (2025) reviewed behavioral, physiological, and multimodal Driver-monitoring approaches and emphasized that system performance may vary according to datasets, camera conditions, and implementation environments. These considerations support evaluating DriveAlert under its actual deployment conditions.

Local mobile processing can reduce dependence on cloud infrastructure by processing data close to its source. Andriulo et al. (2024) discussed edge processing for connected devices, while Martinez-Alpiste et al. (2022) demonstrated the relevance of optimized real-time computer-vision processing for resource-constrained platforms. Li et al. (2024) and Tan and Cao (2024) further indicated that mobile inference may be affected by hardware capability, processing workload, battery condition, and temperature. These findings support DriveAlert's Android-based local processing while recognizing that smartphone capability may influence detection responsiveness.

Fourie and Myburgh (2022) demonstrated communication between ESP32-CAM devices and an Android smartphone through a local wireless network for real-time computer-vision functions. Their implementation transmitted camera frames through local Wi-Fi while the smartphone performed processing, demonstrating the technical feasibility of ESP32-based camera hardware communicating with Android without relying on cloud inference. Espressif Systems documentation likewise establishes ESP32 Wi-Fi capability, supporting DriveAlert's local IoT-to-mobile communication architecture.

Vehicle-mounted driver-monitoring systems must also consider road-safety regulations concerning device placement and driver visibility. In the Philippines, Republic Act No. 10913, or the Anti-Distracted Driving Act, regulates distracting uses of mobile communication and electronic computing devices while driving. Section 5(a) further establishes that the placement of hands-free mobile communication devices must not interfere with the driver's line of sight (Republic Act No. 10913, 2016). The Land Transportation Office (2021), through its Filipino Driver's Manual, Volume 2, illustrates safe device-placement areas and recommends positioning dashboard cameras behind the rearview mirror. Additionally, Joint Administrative Order No. 2014-01 addresses defective, improper, or unauthorized vehicle accessories and devices that are manifestly prejudicial to road

safety (Department of Transportation and Communications et al., 2014). These regulatory considerations provide a basis for DriveAlert's physical design and proposed installation approach. Accordingly, the IoT enclosure and mounting arrangement should be designed to minimize obstruction of the driver's forward visibility, avoid interference with vehicle operation, and maintain a suitable camera view of the Driver. The applicability of these considerations to the final prototype will be assessed during controlled installation testing.

Low-light monitoring is relevant because drowsiness may occur during nighttime, early-morning travel, or in dim vehicle environments. Facial measurements obtained through computer vision may be affected by lighting, camera placement, facial orientation, and image quality. DriveAlert uses an infrared-capable camera without dedicated infrared illumination. Its infrared capability does not itself provide light or guarantee usable facial measurements under extremely low-light or no-light conditions. DriveAlert therefore evaluates facial brightness and contrast before allowing calibration measurements or drowsiness-sign assessment. When lighting is insufficient, assessment pauses even if a face remains detected, and affected frames do not contribute to calibration’s required valid measurement time The system prompts the Driver to restore adequate visibility and resumes assessment when the applicable lighting-quality, recovery, and observation-freshness requirements are met. These checks are image-quality safeguards rather than measures of Driver drowsiness. Their suitability will be evaluated under different available-light conditions using the final hardware configuration.

Drowsiness-stage warning intervention begins only after a selected sign satisfies its applicable measurement threshold and temporal-confirmation condition. Separate operational guidance may be issued when facial visibility or lighting is inadequate or when the movement-advisory condition is satisfied. Such guidance does not itself constitute a confirmed drowsiness-sign event, create an additional alert-history episode, or advance the Warning Stage. Detection of a qualifying visible behavior and selection of the corresponding intervention are therefore treated as separate processes. Ayas et al. (2024), in a scoping review of Driver-state-monitoring intervention studies, found that multistage interventions were commonly used and that auditory warnings were among the most frequently employed warning modalities. The reviewed systems generally progressed from an initial caution toward stronger warnings, requested Driver actions, or additional interventions. These findings support progressive warning intervention rather than requiring every confirmed event to produce an identical response.

Schwarz et al. (2015) provide a direct basis for DriveAlert’s warning-stage transition architecture. Their drowsiness-mitigation system used a three-stage audio-visual warning in which the same drowsiness threshold was used to trigger each stage. Detection of drowsiness while no

mitigation was active initiated Stage 1. During Stage 1, the detection algorithm could either continue classifying the Driver as drowsy or return to an awake state. If the Driver returned to an awake classification, the mitigation was allowed to abate after a fixed delay. However, if another drowsy episode was detected before the current mitigation abated, or if the drowsy state persisted for 60 seconds, the system escalated to Stage 2. Schwarz et al. applied the same logic during Stage 2, allowing the mitigation to either abate to Stage 1 or escalate to Stage 3. Stage 3 represented the maximum intervention level; further escalation was not permitted, although the Stage 3 warning could be reissued when drowsiness persisted or recurred soon afterward. The system permitted only incremental escalation and abatement between adjacent stages.

Gaspar et al. (2017) later experimentally evaluated a related staged drowsiness-countermeasure architecture using 72 participants in a high-fidelity driving simulator. Their Driver-state algorithm issued awake or drowsy assessments every six seconds, and the staged interface consisted of three increasingly urgent warning levels. In the evaluated implementation, each countermeasure state expired after 60 seconds. If the Driver was classified as awake when the state expired, the system returned to the previous stage; if drowsiness continued, the system progressed to the next stage. When already at Stage 3, continued drowsiness caused the Stage 3 warning to be triggered again rather than creating an additional stage. Gaspar et al. found that the multistage condition reduced drowsy lane departures relative to the single-stage condition, particularly during the early-morning drives in which participants were expected to be drowsier. These findings provide experimental support for progressive multistage intervention while Schwarz et al. (2015) provides the more detailed basis for the transition mechanics adopted by DriveAlert.

DriveAlert will adapt the three-stage transition principles described by Schwarz et al. (2015) to its discrete computer-vision warning events. Because DriveAlert does not repeatedly classify the Driver simply as awake or drowsy, a new distinct confirmed warning event will represent a newly occurring qualifying episode for warning-stage progression. A distinct confirmed warning event is created only after prolonged eye closure, yawning, or head nodding satisfies its complete Driver-specific measurement and temporal-confirmation condition. A single continuous qualifying behavior does not repeatedly generate new events, and qualifying signs occurring during the same overlapping behavioral episode may remain associated with one alert record. These event-management rules prevent a continuous or overlapping episode from causing artificial stage escalation.

The first distinct confirmed warning event occurring while no Warning Stage is active will immediately activate Warning Stage 1 and begin a 60-second Stage 1 evaluation period. If another distinct confirmed warning event occurs before the Stage 1 period ends, DriveAlert will interpret

the new event as recurrence of qualifying warning activity and will immediately progress to Warning Stage 2. Progression to Stage 2 begins a new 60-second Stage 2 evaluation period. The system therefore does not require the complete Stage 1 period to expire when a new distinct qualifying episode has already occurred. This adapts Schwarz et al.'s (2015) rule in which another drowsy episode detected before the current mitigation abated could cause progression to the next stage.

If no additional distinct confirmed warning event occurs during the Stage 1 evaluation period, the system will evaluate the state of the qualifying behavior when the 60-second boundary is reached. If qualifying warning activity remains continuously active at the boundary, DriveAlert will progress to Warning Stage 2 and begin a new 60-second Stage 2 period. If the previously confirmed behavior has ended, no new distinct warning event has occurred, and no qualifying behavior remains active, the system will return to the normal state with no active Warning Stage. This persistence pathway adapts Schwarz et al.'s use of 60 seconds of persistent drowsiness as an alternative reason for escalation while retaining Gaspar et al.'s use of 60-second countermeasure-state boundaries for reevaluation.

Warning Stage 2 will follow the same transition principle. If another distinct confirmed warning event occurs while Stage 2 is active, DriveAlert will immediately progress to Warning Stage 3 and begin a new 60-second Stage 3 evaluation period. The new event must represent a genuinely distinct qualifying episode rather than the continuation of the event that previously caused Stage 2. If no new distinct warning event occurs during Stage 2 but qualifying warning activity remains continuously active when the 60-second Stage 2 period expires, the system will likewise progress to Stage 3. If no additional distinct event occurs and the qualifying behavior has ended by the Stage 2 boundary, the system will de-escalate to Warning Stage 1 and begin a new 60-second Stage 1 evaluation period. This preserves incremental progression and de-escalation between adjacent intervention stages.

Warning Stage 3 is the maximum DriveAlert intervention stage. Additional distinct confirmed warning events occurring while Stage 3 is active cannot produce a Stage 4. Instead, a new distinct event may reactivate the Stage 3 Driver-directed warning and will be recorded as recurrent qualifying warning activity during the current Stage 3 period. The 60-second Stage 3 evaluation period will not be converted into an additional intervention level. When the Stage 3 period expires, DriveAlert will evaluate whether recurrent qualifying warning activity occurred during the completed period or whether a qualifying behavior remains persistently active at the boundary. If either condition is present, DriveAlert will remain at Stage 3, begin another 60-second Stage 3 evaluation period, and may reissue the Stage 3 intervention. If neither recurrent nor

persistent qualifying warning activity is present, the system will de-escalate to Warning Stage 2. Continued absence of qualifying warning activity subsequently permits incremental de-escalation from Stage 2 to Stage 1 and from Stage 1 to the normal state. This maximum-stage behavior follows the principle described by Schwarz et al. (2015), in which Stage 3 could not escalate further but its warning could be reissued when drowsiness persisted or soon recurred.

The 60-second value is not used as a confirmation threshold for prolonged eye closure, yawning, or head nodding. Those visible signs retain their independent Driver-specific measurement and temporal-confirmation conditions. Instead, 60 seconds represents the warning-stage evaluation interval used to determine whether persistent warning activity requires escalation or whether the absence of renewed qualifying activity permits de-escalation. A new distinct confirmed warning event may therefore cause earlier progression from Stage 1 to Stage 2 or from Stage 2 to Stage 3 without waiting for the corresponding 60-second period to expire. Conversely, when no new distinct event occurs, the 60-second boundary provides a defined point for evaluating persistent activity or recovery. Schwarz et al. (2015) directly described progression when another drowsy episode occurred before mitigation abated or when the drowsy state persisted for 60 seconds, while Gaspar et al. (2017) experimentally evaluated three-stage warnings using 60-second countermeasure states.

Because Schwarz et al. (2015) and Gaspar et al. (2017) used a Driver-state algorithm that classified the Driver as awake or drowsy rather than DriveAlert's separate EAR-, MAR-, and head-pose-based events, DriveAlert's mapping of a new distinct confirmed warning event to a newly occurring qualifying episode is an adaptation rather than a direct replication of their detection algorithm. Similarly, DriveAlert interprets continued satisfaction of a selected sign's confirmed condition at a stage boundary as evidence of persistent qualifying warning activity rather than claiming that the Driver has been continuously and medically confirmed as drowsy. The suitability of this mapping and the 60-second evaluation interval will therefore be examined during controlled testing using the actual DriveAlert implementation.

For example, if a Driver first produces a confirmed prolonged-eye-closure event while no stage is active, DriveAlert immediately activates Stage 1. If the eye closure ends and the Driver later produces a distinct confirmed yawning event while Stage 1 remains active, the new event immediately progresses the system to Stage 2 and begins a new Stage 2 evaluation period. If a subsequent distinct head-nodding event occurs while Stage 2 is active, the system immediately progresses to Stage 3. In contrast, when additional signs become confirmed while an existing alert episode remains active under the episode-continuity rule, their signs are merged into that episode and do not independently advance the Warning Stage. Likewise, during uninterrupted valid

observation, keeping the eyes continuously closed after one prolonged-eye-closure confirmation does not repeatedly create new distinct alert episodes. Current-stage warning reminders may still be requested, while continued qualifying activity is separately evaluated at the applicable warning-stage boundary.

Warning Stage 1 will activate the Driver's selected predefined warning sound through the IoT device. This represents the initial Driver-directed intervention following a confirmed warning event. Auditory warnings are widely represented in the Driver-state-monitoring literature reviewed by Ayas et al. (2024), while Chandana and Sangeetha (2023) similarly used a sound warning at the first level of their three-level system.

Warning Stage 2 will retain the Driver's selected predefined warning sound without adding a spoken advisory. It serves as an intermediate warning state when qualifying warning activity recurs during Stage 1 or remains persistently active at the Stage 1 evaluation boundary. Although Stages 1 and 2 use the same audio output, they represent different positions within the warning-stage transition process. Ayas et al. (2024) support the broader use of staged interventions, while Chandana and Sangeetha (2023) provide precedent for introducing human-voice warnings after initial sound alerts. DriveAlert adapts these principles by reserving the spoken rest advisory for Warning Stage 3, where it is intended to distinguish the maximum intervention stage from the preceding sound-only warnings. The placement of speech at Stage 3 is a DriveAlert design decision rather than an arrangement directly prescribed by these sources.

Chandana and Sangeetha (2023) provide a more direct precedent for reserving external communication for a final intervention level. Their three-level system initially provided a sound warning, progressed to a human-voice warning following repeated warning activity, and used its third level as an alert-notification stage in which information including GPS data was transmitted to the vehicle owner or another concerned person. Their implementation therefore demonstrates an intervention structure progressing from an initial Driver-directed warning, to a stronger Driver-directed warning, and finally to external awareness.

Multi-feature Driver-monitoring approaches can reduce dependence on a single facial indicator when visual information is partially obstructed. Jia et al. (2021, 2022) demonstrated that monitoring could continue using the remaining observable facial features when the eyes or mouth were affected by sunglasses or face masks. DriveAlert applies this principle to supported mask and sunglasses detection and to facial measurements reported as unavailable by its processing pipeline. It does not claim to recognize every possible obstruction, such as all hands, phones, hair, scarves, or other objects. When one of the three monitored signs becomes unavailable while two remain reliable, DriveAlert continues in a degraded monitoring state and informs the Driver which

capability is unavailable. If any two signs become unavailable, DriveAlert pauses drowsiness-sign monitoring rather than relying on a single behavioral indicator and instructs the Driver to restore sufficient facial visibility before monitoring resumes.

Awasekar et al. (2018) provide additional precedent for placing external communication at a highest warning stage. Their Driver Fatigue Detection and Alert System used three fatigue stages based on its own eye-closure and yawning conditions. At Stage 3, the system combined its highest local intervention with communication to predefined contacts through an Android smartphone. DriveAlert does not adopt the fatigue-stage classifications used by Awasekar et al. because DriveAlert's Warning Stages represent levels of system intervention rather than physiological or medical levels of Driver drowsiness. Their work nevertheless provides additional precedent for associating external communication with a final or highest intervention stage.

Fitzharris et al. (2017) further demonstrated the value of supplementing Driver-directed warnings with external awareness in an operational Driver-monitoring environment. Their study examined a camera-based Driver-monitoring system that generated in-cab warnings and was subsequently combined with external monitoring and management feedback when fatigue events were identified. The external feedback condition produced a larger reduction in fatigue-event occurrence than the in-cab-warning condition alone under the study conditions. Although DriveAlert's Trusted Contact function differs from fleet-management intervention, the study supports the broader principle that external awareness can supplement immediate Driver-directed warnings when monitored warning activity persists.

Warning Stage 3 will combine the Driver's selected predefined warning sound with a fixed prerecorded spoken advisory recommending that the Driver stop in a safe place and take a rest. It will also introduce Trusted Contact synchronization eligibility. When DriveAlert progresses from Stage 2 to Stage 3, one Stage 3 transition record becomes eligible for synchronization with the Driver's approved Trusted Contact, subject to internet availability. The introduction of speech following earlier sound warnings adapts the progression described by Chandana and Sangeetha (2023), although DriveAlert places speech at Stage 3 rather than at their second intervention level. The exact advisory wording and its placement at Stage 3 are DriveAlert design decisions. Stage 3 represents the maximum system intervention level following recurrent or persistent qualifying warning activity; it does not establish a medically severe level of Driver drowsiness.

The three Warning Stages must therefore be interpreted as levels of system intervention rather than measurements of drowsiness severity. Warning Stage 1 represents the initial Driver-directed intervention following the first distinct confirmed warning event. Warning Stage 2 represents an intermediate warning state that retains the selected warning sound when another

distinct qualifying episode occurs while Stage 1 remains active or when qualifying warning activity persists through the Stage 1 evaluation boundary. Warning Stage 3 represents the maximum intervention level when qualifying activity again recurs during Stage 2 or persists through the Stage 2 evaluation boundary. DriveAlert does not infer that a Driver at Stage 3 has reached a medically established severity of drowsiness or is physiologically drowsier simply because the highest Warning Stage has been reached.

Continued qualifying warning activity while Stage 3 remains active also requires an external-awareness rule that does not generate a separate Trusted Contact synchronization record for every confirmed event. Schwarz et al. (2015) retained Stage 3 as the maximum mitigation stage and allowed the Stage 3 warning to be reissued when drowsiness persisted or recurred. Gaspar et al. (2017) similarly retained Stage 3 as the maximum countermeasure state and retriggered the Stage 3 intervention when continued drowsiness required another escalation. DriveAlert adapts this repeated maximum-stage principle to its Trusted Contact synchronization management.

When DriveAlert initially enters Stage 3, one Stage 3 transition record becomes eligible for Trusted Contact synchronization. Individual distinct confirmed warning events occurring during the subsequent 60-second Stage 3 evaluation period continue to be stored in the Driver's alert history and may reactivate the Stage 3 Driver-directed warning, but they do not each independently create a Trusted Contact synchronization record. Instead, they indicate that recurrent qualifying warning activity occurred during the active Stage 3 period.

When the 60-second Stage 3 period expires, DriveAlert evaluates both recurrent and persistent qualifying activity. If at least one additional distinct confirmed warning event occurred during the completed Stage 3 period, or if a qualifying behavior remains persistently active at the Stage 3 boundary, DriveAlert remains at Stage 3, begins another 60-second Stage 3 evaluation period, and creates one Stage 3 persistence update that becomes eligible for Trusted Contact synchronization. If neither condition is present, DriveAlert de-escalates to Stage 2 and no Stage 3 persistence update is created for that completed period.

If qualifying warning activity continues across several consecutive Stage 3 periods, at most one persistence update becomes eligible for synchronization for each completed Stage 3 period in which recurrent or persistent qualifying warning activity is established. Individual warning events remain available in the Driver's detailed alert history but do not independently produce external synchronization records. This separates detailed event recording from the frequency of Trusted Contact updates.

The exact rule of generating one Trusted Contact persistence update per completed qualifying Stage 3 period is not a notification frequency directly prescribed by Schwarz et al.

(2015), Gaspar et al. (2017), Chandana and Sangeetha (2023), Fitzharris et al. (2017), or the other reviewed studies. Schwarz et al. directly support maintaining Stage 3 as the maximum intervention level and reissuing its warning when drowsiness persists or recurs, while Gaspar et al. directly support repeated Stage 3 intervention following continued drowsiness at a 60-second countermeasure boundary. Chandana and Sangeetha provide precedent for external communication at the final intervention level, and Fitzharris et al. support supplementing Driver-directed warnings with external monitoring or feedback. DriveAlert combines these principles by aligning Trusted Contact persistence updates with its Stage 3 evaluation boundaries. The resulting persistence-update rule is therefore a literature-informed DriveAlert information-management adaptation and will be evaluated during controlled testing for usefulness, unnecessary repetition, and synchronization behavior.

For example, several distinct confirmed warning events occurring during one Stage 3 evaluation period do not generate an equal number of Trusted Contact synchronization records. The events remain individually available in the Driver's alert history, subject to the existing overlap and deduplication rules, while the completed Stage 3 period may produce one persistence update indicating that qualifying warning activity continued. This separates detailed event storage from external-awareness frequency.

Trusted Contact synchronization remains subject to Driver approval and internet availability. Only eligible records from periods beginning on or after approval of the Driver–Trusted Contact relationship may be shared. Approval does not release the Driver's earlier complete alert history, and continued access depends on the relationship remaining approved. If internet connectivity is unavailable when a Stage 3 transition or persistence update becomes synchronization-eligible, the corresponding record is retained in the synchronization queue until connectivity becomes available. Kashyap et al. (2025) demonstrated the use of Firebase and a web dashboard for remote review of Driver-status information, while Kumar et al. (2022) used SMS or email to provide drowsiness-related information to authorized persons or family members. These studies support the broader technical feasibility of making selected Driver information available to an authorized external person. DriveAlert limits this feature to secondary awareness and record review and does not operate as an emergency-response service, guarantee immediate assistance, determine whether the Driver is fit to continue driving, or permit the Trusted Contact to control the Driver's monitoring session.

The Driver's drowsiness-alert history and basic Drowsiness Insights feature are supported by literature on recorded-event visualization and dashboard-based review. Murphy et al. (2021) and Moens et al. (2021) demonstrated that dashboards can support review of recorded safety-related

events and trends. DriveAlert therefore allows Drivers to review confirmed warning events and basic descriptive summaries of stored records. These records and summaries are not intended as medical analysis, accident prediction, confirmation of drowsiness, or determination of the Driver's fitness to continue driving.

Overall, the reviewed literature supports the principal components of DriveAlert while showing that its three selected visible signs require different confirmation procedures. EAR, MAR, and head-pose measurements provide a basis for examining prolonged eye closure, yawning, and head nodding, while Driver calibration addresses individual differences in facial measurements and neutral head orientation. Temporal evidence supports sustained confirmation of prolonged eye closure and yawning, while temporal persistence provides a stronger basis for interpreting ambiguous downward head movement. Driver-monitoring literature further supports progressive multistage warning intervention. Schwarz et al. (2015) provide the primary basis for DriveAlert's three-stage transition mechanics by describing incremental escalation when another drowsy episode occurs before the current mitigation abates or when drowsiness persists for 60 seconds, together with incremental abatement and Stage 3 as the maximum intervention level. Gaspar et al. (2017) later experimentally evaluated a related three-stage architecture using 60-second countermeasure states and found that staged warnings reduced drowsy lane departures relative to single-stage warnings under the study conditions. Chandana and Sangeetha (2023) provide precedent for introducing human-voice warnings after initial sound alerts and external communication at a final intervention level. DriveAlert adapts these principles by retaining the selected warning sound during Stages 1 and 2 and introducing both a spoken rest advisory and Trusted Contact synchronization eligibility at Stage 3. Fitzharris et al. (2017) provide evidence that external awareness or feedback can supplement immediate Driver-directed warnings. DriveAlert adapts these principles to distinct confirmed computer-vision warning events and controlled Trusted Contact synchronization while treating its Warning Stages as levels of system intervention rather than medical levels of drowsiness. Mobile-processing and IoT literature further supports DriveAlert's local Android processing and local wireless communication, while its performance under different lighting conditions will be evaluated using the final camera configuration.

Related Studies

Deployed driver-drowsiness, fatigue, and alertness systems use mobile computer vision, wearable sensors, driver-condition assessment, and physical warning devices to help users recognize reduced alertness. Some systems monitor visible indicators such as eye closure and head movement, while others estimate fatigue using personal information, trip conditions, and reaction-

time testing. These systems demonstrate that mobile, wearable, and embedded technologies can provide immediate warnings, although they differ in detection methods, hardware requirements, offline capability, record management, and user support.

Drowsy Driving Alert is a deployed mobile application that uses artificial intelligence and the smartphone's front camera to monitor the driver's face and eyes. The application distinguishes normal blinking from extended eye closure and provides audio and visual alerts when drowsiness is detected. It can operate alongside Google Maps and Apple Maps, while facial processing is performed locally on the smartphone. However, the phone must be positioned where the camera can clearly view the driver. Lighting, camera obstruction, device performance, and background-operation restrictions may also delay or interrupt warnings.

WingDriver is a smartphone-based driver-monitoring system that uses computer vision and artificial intelligence to detect drowsiness, fatigue, distraction, and inattention. It provides immediate audible or visual warnings to help the driver regain attention. The provider also offers related business solutions for fleets, ride-sharing services, insurers, and vehicle manufacturers. However, monitoring depends on the smartphone camera maintaining a usable view of the driver and on the mobile device having sufficient processing capability. Broader business deployment also requires coordination with the provider.

Fatigue Alert – Stay Awake is a deployed application that uses supported AirPods or Beats headphones to monitor the user's head posture. When downward head movement indicates possible drowsiness, the application provides warnings through sound, vibration, system notifications, or Apple Watch haptic feedback. It can continue operating in the background or while the device screen is locked. However, the system requires compatible Apple devices and supported headphones and focuses mainly on head-posture monitoring rather than facial indicators. It is also presented as a general stay-awake application rather than a system exclusively intended for driving.

Anti Sleep Pilot is a dashboard-mounted device and mobile application that continuously calculates a driver's fatigue level. Its calculation considers the driver's personal risk profile, fatigue condition before the trip, and driving information registered through built-in sensors. It also administers periodic reaction-time tests and provides audible and visual recommendations when the driver should take a break. However, the system requires initial profile configuration and periodic interaction from the driver. Its documentation also notes that low-speed traffic, GPS conditions, and some driving situations may affect trip-state interpretation or cause distraction.

The alertme Drowsy Driver Device is a commercially deployed wearable warning device developed by resqme, Inc. It is worn behind the ear and detects downward head nodding associated with possible dozing. When the user's head tilts forward by approximately 15 to 20 degrees, the

battery-powered device produces a 90-decibel audible alarm. The device does not require a smartphone, mobile application, or internet connection. However, it must be worn correctly, detects only head-nodding movement, and provides only a direct audible warning.

The reviewed systems demonstrate different approaches to driver-state monitoring and warning. Drowsy Driving Alert and WingDriver use smartphone cameras and artificial intelligence to monitor visible driver behaviors. Fatigue Alert – Stay Awake uses wearable motion information to monitor head posture. Anti Sleep Pilot estimates fatigue through driver profiles, sensor information, and reaction-time testing. Meanwhile, alertme uses a simple wearable mechanism to detect downward head movement and produce an immediate alarm.

The systems collectively show the progress and remaining integration gaps in driver-drowsiness and alertness monitoring. Smartphone-based applications provide accessible camera monitoring but depend on phone placement, facial visibility, and device performance. Wearable applications require compatible accessories, while fatigue-estimation systems require profile configuration and periodic driver interaction. Simple physical alarms can operate without internet access but generally monitor only one behavior and do not provide organized records.

Overall, DriveAlert combines selected driver-monitoring and record-management functions in an external-camera and Android-based prototype. Its configuration includes an external IoT camera with an infrared-capable sensor, Android-based local processing, offline IoT-to-mobile communication, monitoring of prolonged eye closure, yawning, and head nodding, and onboard speaker warning output. Confirmed warning events are stored locally for Driver review and descriptive summaries, while separate eligible Stage 3 transition records and persistence updates support approved Trusted Contact sharing. DriveAlert investigates whether these functions can operate together under controlled testing conditions.

Comparative Matrix

This section presents a comparative matrix developed to organize and assess relevant deployed systems in relation to DriveAlert: An IoT-Based Driver Drowsiness Monitoring and Alert System Using Computer Vision. It compares Drowsy Driving Alert, WingDriver, Fatigue Alert – Stay Awake, Anti Sleep Pilot, and the alertme Drowsy Driver Device according to their features, limitations, platform details, support mechanisms, and monetization schemes. The comparison identifies the different mobile, wearable, and embedded approaches used for driver-state monitoring and warning and supports the positioning of DriveAlert as an integrated driver-focused system.

Table 1.1

COMPARATIVE MATRIX

Related Studies/ Systems	**Features**	**Limitations**	**Platform Detail**	**Support**	**Monetization Scheme**
Name: "Drowsy Driving Alert – Drive Safe" URL: https://www.drowsydrivingalert.app/ Year: Current deployed application / 2026 listing Proponent(s): 1. Trung Huynh	1. Monitors the driver's face and eyes. 2. Detects extended eye closure. 3. Provides audio and visual alerts. 4. Performs processing on the smartphone.	1. Requires proper smartphone positioning. 2. Depends on clear facial visibility. 3. Lighting, obstruction, and device conditions may affect warnings.	Mobile application (Android and iOS)	Official website, FAQ, support email, and app-store support channels.	Free download with a seven-day trial, in-app purchases, and Pro subscription.
Name: "WingDriver" URL: https://wingdriver.com/ Year: Current deployed application / 2026 listing Proponent(s): 1. WingDriver, Inc.	1. Detects drowsiness, fatigue, and distraction. 2. Uses smartphone computer vision 3. Provides visual and audible warnings. 4. Offers business driver-safety solutions.	1. Relies on smartphone camera and processing capability. 2. Requires consistent visibility of the driver. 3. Business deployment requires provider coordination.	Mobile application (smartphone-based)	Support email, chat support, contact form, and business demonstration requests.	A free mobile application is available; business-service pricing is available through direct inquiry.

Table 1.2

COMPARATIVE MATRIX

Related Studies/ Systems	**Features**	**Limitations**	**Platform Detail**	**Support**	**Monetization Scheme**
Name: “Fatigue Alert – Stay Awake” URL: https://apps.apple.com/us/app/fatigue-alert-stay-awake/id6479893638 Year: Current deployed application / 2026 listing Proponent(s): 1. Xuhai Luo	1. Monitors head posture. 2. Provides sound and vibration warnings. 3. Supports system notifications and Apple Watch feedback. 4. Operates in the background or while the screen is locked.	1. Requires supported AirPods or Beats. headphones. 2. Limited to compatible Apple devices. 3. Focuses mainly on head posture	Mobile and wearable-supported application (iPhone, iPad, and Apple Watch)	Developer email, App. Store support channel, and developer response system	Free download with in-app purchases and an auto-renewing subscription.
Name: “Anti Sleep Pilot” URL: https://antisleeppilot.com/ Year: Current product documentation / 2026 Proponent(s): 1. ASP Technology ApS	1. Calculates the driver’s drowsiness level 2. Uses personal, trip, and sensor information. 3. Conducts reaction-time tests. 4. Provides audible and visual break recommendations	1. Requires a personal risk profile. 2. Requires periodic driver interaction. 3. GPS and low-speed conditions may affect trip-state interpretation.	Embedded dashboard device and mobile application (iOS)	FAQ, downloadable manual, product and app support pages, email support, and telephone assistance.	One-time hardware purchase and separately paid mobile application;

Table 1.3

COMPARATIVE MATRIX

Related Studies/ Systems	**Features**	**Limitations**	**Platform Detail**	**Support**	**Monetization Scheme**
Name: "alertme Drowsy Driver Device" URL: https://resqme.com/products/alertme-drowsy-driver-device Year: Current deployed product / 2026 listing Proponent(s): 1. resqme, Inc.	1. Detects downward head nodding. 2. Produces a 90-decibel alarm. 3. Uses lightweight battery-powered hardware. 4. Operates without a phone or internet connection.	1. Must be worn behind the ear. 2. Detects only head-nodding movement. 3. Provides only a local audible warning.	Wearable electronic device	Telephone and email support, FAQ, contact assistance, and return and refund support.	One-time hardware purchase.

The comparative matrix presents five driver-drowsiness, fatigue, and alertness monitoring and warning systems. Drowsy Driving Alert and WingDriver use smartphone-based computer vision to monitor visible driver behavior and provide immediate alerts. Fatigue Alert – Stay Awake uses compatible wearable accessories to monitor head posture. Anti Sleep Pilot evaluates fatigue through driver information, trip conditions, sensor data, and reaction-time testing. The alertme device uses a simpler wearable mechanism to recognize downward head movement and activate a loud alarm.

Clear distinctions exist among the systems. Drowsy Driving Alert focuses on facial and eye monitoring through the smartphone camera. WingDriver expands mobile monitoring to include drowsiness, fatigue, distraction, and business driver-safety applications. Fatigue Alert – Stay Awake depends on supported Apple accessories and monitors head posture. Anti Sleep Pilot does not primarily use computer vision but provides personalized fatigue calculation and

break recommendations. The alertme device operates independently of mobile and internet connectivity but detects only head-nodding movement.

DriveAlert combines external camera monitoring, Android-based detection, physical warning output, local record review, and controlled Trusted Contact sharing. It uses an external IoT camera with an infrared-capable sensor. The Android application receives the camera stream through a local Wi-Fi connection, evaluates prolonged eye closure, yawning, and downward head posture using the applicable measurement and temporal-confirmation conditions, and sends warning commands to the IoT device. Confirmed warning events are stored locally and summarized for Driver review. Separate eligible Stage 3 transition records and persistence updates are synchronized for access by an approved Trusted Contact when internet connectivity becomes available.

Table 2

COMPARATIVE MATRIX SUMMARY

Features	**DriveAlert**	**Drowsy Driving Alert**	**Wing Driver**	**Fatigue Alert**	**Anti Sleep Pilot**	**alertme**
Uses driver-state indicators for alert generation	✓	✓	✓	✓	✓	✓
Uses local computer-vision processing on a smartphone	✓	✓	✓	✗	✗	✗
Uses external monitoring hardware	✓	✗	✗	✓	✓	✓
Provides audible warning output	✓	✓	✓	✓	✓	✓
Stores drowsiness-alert records	✓	✗	✓	✗	✗	✗
Supports Trusted Contact synchronization	✓	✗	✗	✗	✗	✗
Uses a dedicated external monitoring camera	✓	✗	✗	✗	✗	✗

Table 2 presents a comparative evaluation of DriveAlert alongside existing driver-drowsiness, fatigue, and alertness monitoring and warning systems: Drowsy Driving Alert,

WingDriver, Fatigue Alert – Stay Awake, Anti Sleep Pilot, and the alertme Drowsy Driver Device. While each system addresses specific aspects of driver alertness, such as camera-based monitoring, head-posture detection, fatigue estimation, or audible warning, DriveAlert consolidates several of these functions into one driver-focused system. Specifically, DriveAlert combines an external infrared-supported IoT camera, Android-based local processing, offline IoT-to-mobile communication, monitoring of prolonged eye closure, yawning, and head nodding, onboard speaker warning output, drowsiness-alert records, basic drowsiness insights, and Trusted Contact synchronization. Drowsy Driving Alert uses smartphone-based facial and eye monitoring and provides immediate alerts, but its publicly documented functions primarily focus on mobile detection and warning. WingDriver provides mobile computer-vision monitoring for drowsiness, fatigue, and distraction and supports driving-related information, but its documented individual-user platform relies mainly on the smartphone camera. Fatigue Alert – Stay Awake monitors head posture through compatible wearable accessories and provides sound, vibration, and smartwatch warnings, but it focuses primarily on head movement. Anti Sleep Pilot estimates fatigue through personal information, trip conditions, sensor data, and reaction-time testing rather than camera-based facial monitoring. The alertme Drowsy Driver Device provides a simple offline audible warning when downward head nodding is detected, but its documented monitoring function is limited to that movement. In contrast, DriveAlert integrates external camera monitoring, offline mobile processing, multiple visible signs associated with drowsiness, physical warning output, drowsiness-alert record review, and approved Trusted Contact access within one cohesive system.

CHAPTER III

RESEARCH METHODOLOGY

This chapter outlines the software engineering methodology used in the development of "DriveAlert." It includes the planning and initiation phase, program workflow, validation board, and Gantt chart. It also covers the analysis and design phase, including the use case diagram, user interface design, storyboard, database design, and network design for the connection between the IoT camera device, mobile application, drowsiness-sign detection process, alert mechanism, basic drowsiness insights, and Trusted Contact synchronization. The chapter further discusses the development and construction phase by presenting the technology stack diagram, software and hardware specifications, program specifications, and list of modules used in developing the system.

Software Engineering Methodology

The development process for "DriveAlert: An IoT-Based Driver Drowsiness Monitoring and Alert System Using Computer Vision" will be based on the Agile methodology due to its iterative and flexible approach, which supports continuous improvement and frequent testing of system features. In the case of DriveAlert, where the system involves driver drowsiness monitoring, alert triggering, drowsiness alert history recording, basic drowsiness insights, and Trusted Contact synchronization, Agile supports a development cycle that allows the researchers to improve the system based on testing results and user feedback. Agile's emphasis on collaboration, feedback, and incremental progress provides a structured yet adaptable roadmap for the development process, helping the researchers manage system changes and ensure that the system requirements are addressed systematically.

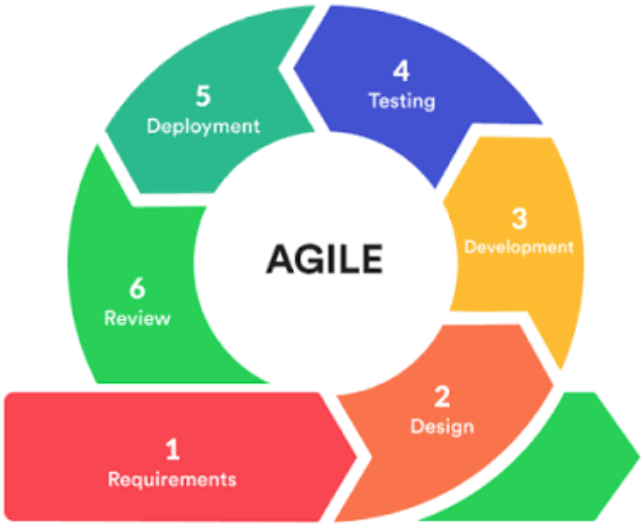


Figure 2: **Agile Methodology**

The team will use the iterative approach of Agile methodology to guide the development of DriveAlert and monitor the progress of activities throughout the project timeline. The project will be divided into ten two-week sprints, with each sprint following the Plan, Design, Develop, Test, Deploy, and Review activities. This approach is suitable because DriveAlert consists of several connected components, including the IoT device, mobile application, web portal, drowsiness-monitoring functions, warning system, drowsiness-alert record dashboard, and synchronization feature. Each sprint will produce a working system component that can be tested, reviewed, and improved before the next sprint begins.

Planning Phase. The team will identify and organize the requirements for each sprint using the survey results, validation board, related studies, project objectives, scope, and approved system design. The team will define the expected outputs, required resources, task assignments, and testing activities. Early sprints will focus on research, documentation, requirements, and system planning, while later sprints will focus on development, integration, testing, deployment, and evaluation.

Design Phase. The team will prepare and refine the designs needed for each system component. This phase includes the system architecture, diagrams, workflows, user interfaces, database design, network design, and testing procedures. The designs will show how the IoT device, mobile application, web portal, warning system, record-management functions, and synchronization feature will work together.

Development Phase. The team will convert the approved designs into functional system components. Development will include setting up the required development environments, configuring the database and account services, assembling and programming the IoT device, developing the mobile application and web portal, implementing drowsiness-monitoring and warning functions, and creating the drowsiness-alert record and synchronization features. Each component will be developed gradually according to the assigned sprint.

Testing Phase. Testing will be conducted during every sprint rather than only after the entire system has been completed. The team will test the functionality, accuracy, reliability, responsiveness, usability, and compatibility of each completed component. Testing will also cover device connection, video transmission, warning response, record storage, dashboard display, synchronization, low-light operation, and system performance. Identified errors, delays, incorrect detections, communication problems, and missing functions will be documented for correction.

Deployment Phase. At the end of each development sprint, the completed component will be placed in the DriveAlert prototype environment for integration and verification. Early deployments will include the project foundation, database, account functions, IoT connection, and

monitoring features. Later deployments will include the warning system, records, dashboard, synchronization, integrated application, web portal, and final IoT configuration.

Review Phase. After each sprint, the team will review the completed work based on testing results, documented issues, and feedback. The review will determine whether the system requirements were achieved and whether revisions are needed. Identified improvements will be added to the project backlog and addressed in the succeeding sprint. The final review will verify that DriveAlert satisfies the study objectives, scope, system requirements, and evaluation criteria.

Table 3.1

AGILE SPRINT PLAN

SDLC Phases	**Sprint 1:** Goal: Research and Manuscript Foundation Time Frame: 2 weeks	**Sprint 2:** Goal: Manuscript Completion and Development Transition Time Frame: 2 weeks	**Sprint 3:** Technical Foundation, IoT Sourcing, and Manuscript Compliance Time Frame: 2 weeks
Plan	Plan the research problem, objectives, scope, questionnaire, survey process, and literature requirements.	Plan the remaining methodology, diagrams, specifications, revisions, and development-transition activities.	Plan project setup, database foundations, IoT parts sourcing, manuscript revisions, and compliance requirements.
Design	Organize Chapters 1 and 2, the questionnaire format, and the comparative review structure.	Finalize workflows, UI designs, ERD, network design, technology stack, and system specifications.	Refine the database schema, proposed hardware layout, project structure, initial screens, and manuscript diagrams.
Develop	Conduct the survey, analyze responses, and complete the initial Chapters 1 and 2 drafts.	Complete Chapter 3, revise Chapters 1 to 3, organize appendices, and finalize the manuscript.	Configure projects and Firebase, establish local storage, source IoT parts, revise the manuscript, and prepare compliance documents.

Table 3.2

AGILE SPRINT PLAN

Test	Validate survey results, citations, objectives, research gap, scope, and section consistency.	Check consistency among requirements, diagrams, references, captions, modules, and manuscript sections.	Test project builds and initial database operations. Check parts compatibility, manuscript consistency, and document completeness.
Deploy	Compile the completed survey outputs and revised Chapters 1 and 2 for review.	Finalize the approved manuscript and prepare requirements, tools, assignments, and development resources.	Commit and push changes to GitHub. Submit manuscript revisions and compliance documents, and route the approval sheet.
Review	Review the research foundation, evidence quality, objectives, scope, and identified system need.	Confirm manuscript completeness, design alignment, technical feasibility, and readiness for development.	Review project readiness, parts availability, manuscript feedback, compliance status, and pending approvals.
SDLC Phases	**Sprint 4:** Goal: IoT Assembly, Communication, and Core Application Functions Time Frame: 2 weeks	**Sprint 5:** Goal: MediaPipe Integration and Live Facial Measurements Time Frame: 2 weeks	**Sprint 6:** Goal: Calibration, Drowsiness Detection, and Warning Control Time Frame: 2 weeks
Plan	Plan IoT assembly, Bluetooth setup, local Wi-Fi communication, video streaming, and account functions.	Plan MediaPipe integration, facial landmarks, EAR, MAR, head-pitch calculations, and lighting checks.	Plan guided calibration, detection thresholds, confirmation conditions, warning stages, and local alert records.

Table 3.3

AGILE SPRINT PLAN

Design	Finalize wiring, device setup flow, communication interfaces, account screens, and navigation.	Define measurement formulas, Driver-area selection, visibility feedback, and the live processing flow.	Define calibration steps, sign-confirmation rules, warning-stage transitions, and alert-record structures.
Develop	Assemble and program the IoT device. Implement Bluetooth provisioning, camera streaming, authentication, and initial mobile and web screens.	Integrate MediaPipe with the IoT camera stream. Implement facial measurements, Driver-area selection, and lighting checks.	Implement calibration, eye-closure, yawning, and head-down detection, staged speaker warnings, and local alert recording.
Test	Test power stability, camera capture, speaker output, device connections, video streaming, and account functions.	Test landmark stability, measurement consistency, processing delay, lighting conditions, and Android compatibility.	Test calibration, false alerts, missed detections, stage progression, speaker response, and offline record storage.
Deploy	Upload firmware to the IoT device. Commit and push tested firmware and application changes to GitHub.	Commit and push the tested camera-processing and facial-measurement changes to GitHub.	Commit and push tested calibration, detection, warning, and local-record changes to GitHub.
Review	Review hardware stability, connection reliability, account functions, and issues	Review processing performance, tracking reliability, lighting limitations, and	Review detection reliability, warning behavior, record accuracy, and readiness for Trusted Contact integration.

Table 3.4

AGILE SPRINT PLAN

	before MediaPipe integration.	readiness for calibration and detection.	
SDLC Phases	**Sprint 7:** Goal: Trusted Contact Sharing and System Integration Time Frame: 2 weeks	**Sprint 8:** Goal: Integrated Testing and System Refinement Time Frame: 2 weeks	**Sprint 9:** Goal: Controlled Evaluation and Release Preparation Time Frame: 2 weeks
Plan	Plan Trusted Contact connections, Stage 3 synchronization, dashboards, and system integration.	Prepare mobile, web, IoT, and controlled-condition test cases. Identify remaining manuscript revisions.	Schedule controlled evaluation, prioritize remaining fixes, and prepare manuscript completion tasks.
Design	Define approval rules, eligible shared records, synchronization flow, and mobile and web views.	Define expected results, failure scenarios, evaluation procedures, and acceptance criteria.	Finalize evaluation forms, demonstration scenarios, release checks, and manuscript formatting.
Develop	Implement connection requests, approved access, Stage 3 synchronization, and record summaries. Integrate mobile, web, and IoT functions.	Correct integration defects, refine interfaces and recovery behavior, and update manuscript descriptions and diagrams.	Resolve remaining defects, conduct controlled evaluation, analyze findings, and complete manuscript revisions.
Test	Test requests, approvals, revocation, offline queues, synchronized records,	Test detection, warnings, offline monitoring, synchronization, access control, usability, and device compatibility.	Run regression tests and verify detection, audible warnings, offline operation,

Table 3.5

AGILE SPRINT PLAN

	dashboards, and the integrated workflow.		synchronization, and approved access.
Deploy	Commit and push tested integration changes to GitHub. Deploy the web portal to a test environment.	Commit and push fixes to GitHub. Generate a test APK and update the web test deployment.	Commit and push verified changes. Generate the release-candidate APK and update the web deployment.
Review	Review sharing accuracy, access restrictions, integration stability, and unresolved issues before formal testing.	Review test evidence, unresolved defects, manuscript alignment, and readiness for controlled evaluation.	Review evaluation findings, release readiness, manuscript completeness, and requirements for manuscript final submission.

SDLC Phases	**Sprint 10:** Goal: Final Submission, Deployment, and Presentation Time Frame: 2 weeks
Plan	Plan final manuscript submission, system deployment, and presentation preparation.
Design	Finalize manuscript formatting, deployment settings, presentation materials, and demonstration flow.
Develop	Complete manuscript corrections and approval requirements. Resolve final system issues and prepare the demonstration.
Test	Verify manuscript completeness and test the final APK, web portal, IoT firmware, and demonstration workflow.
Deploy	Submit the manuscript. Commit and push final changes, generate the final APK, and deploy the web portal.
Review	Confirm manuscript submission, system readiness, presentation preparation, and project completion.

Planning-Initiation Phase

During this phase, the project's goals, scope, and requirements will be defined and clarified. This involves creating a Business Model Canvas to evaluate the business concept, a program workflow to illustrate user interactions with the DriveAlert application, and a validation board to identify the core problem and define a solution that addresses the needs of the intended users. Additionally, a Gantt chart will be used to outline tasks and their associated timeframes, and a functional decomposition diagram will break down complex processes into more minor, more manageable activities.

Business Model Canvas

The Business Model Canvas allows the researchers to visualize and adequately assess the business concept of the study. By diving into the Key Partners, Key Activities, Value Proposition, Customer Relationships, Customer Segments, Key Resources, Channels, Revenue Streams, and Cost Structure, the researchers will be able to gain comprehensive insights into the study's potential, uncovering the factors that aid to its success and areas that need improvement, making room for more informed decisions

Table 4

BUSINESS MODEL CANVAS

Key Partners	Key Activities	Value Proposition	Customer Relationships	Customer Segments
● Drivers of four-wheeled vehicles and larger vehicles ● Trusted Contact	● Development of the DriveAlert mobile and web applications ● Integration of MediaPipe Face Landmarker ● Development and calibration of drowsiness-sign detection thresholds ● Setup and testing of the IoT device ● Management of drowsiness-alert records and Trusted	● Real-time driver drowsiness monitoring ● Detection of prolonged eye closure, yawning, and head nodding ● Offline monitoring through local Wi-Fi connection ● Onboard speaker warning output using predefined audio warning alerts with adjustable volume and an	● Mobile app alerts and warning notifications ● Trusted Contact updates for Alert records ● Basic setup guide for the IoT device and mobile application ● User feedback and continuous improvement	● Drivers of four-wheeled vehicles and larger vehicles ● Trusted Contact

<table>
<tr><td></td><td>Contact synchronization
● System testing and continuous improvement
Key Resources
● MediaPipe Face Landmarker model
● Cloud infrastructure (Google Firebase & Google Cloud)
● Android and web development tools
● DriveAlert IoT device
● Alert records and Drowsiness insight summaries for Trusted Contact synchronization</td><td>enforced minimum level
● Alert history records and Drowsiness insight for driver review
● Trusted Contact synchronization for shared Alert record review</td><td>Channels
● Mobile application
● Website
● System demonstration
● Online promotion</td><td></td></tr>
<tr><td colspan="2">Cost Structure
● IoT device components and assembly
● Mobile and web application development
● Facial-analysis integration and system testing
● Cloud database and synchronization services
● Deployment, maintenance, and documentation</td><td colspan="3">Revenue Streams
● One-time purchase of the DriveAlert IoT device
● Subscription for additional user of DriveAlert device</td></tr>
</table>

Table 4 presents the Business Model Canvas for the DriveAlert system. The system serves drivers of four-wheeled vehicles and larger vehicles, and Trusted Contacts. Its key activities include developing the mobile and web applications, integrating the MediaPipe Face Landmarker model, calibrating drowsiness-sign detection thresholds, setting up and testing the IoT device, managing Alert records, and continuously improving the system. DriveAlert provides real-time drowsiness monitoring, offline local Wi-Fi operation, onboard speaker warnings, alert history and insight summaries, and Trusted Contact synchronization. Its key resources include the MediaPipe Face Landmarker model, cloud infrastructure, development tools, the DriveAlert IoT device, and

synchronized alert records. The system's possible revenue streams include the one-time purchase of the DriveAlert IoT device and a paid subscription for additional user of DriveAlert device.

Program Workflow

This section highlights the program workflow for the "DriveAlert's" app. It will present the structured sequence of steps or activities that outline the process flow within the application. This will serve as a visual representation or a detailed description of how tasks, operations, or processes are carried out from start to finish. It helps in understanding the logical progression of activities, identifying dependencies, and ensuring efficient execution of tasks which are essential for planning, managing, and optimizing processes in various domains such as software development, business operations, and project management.

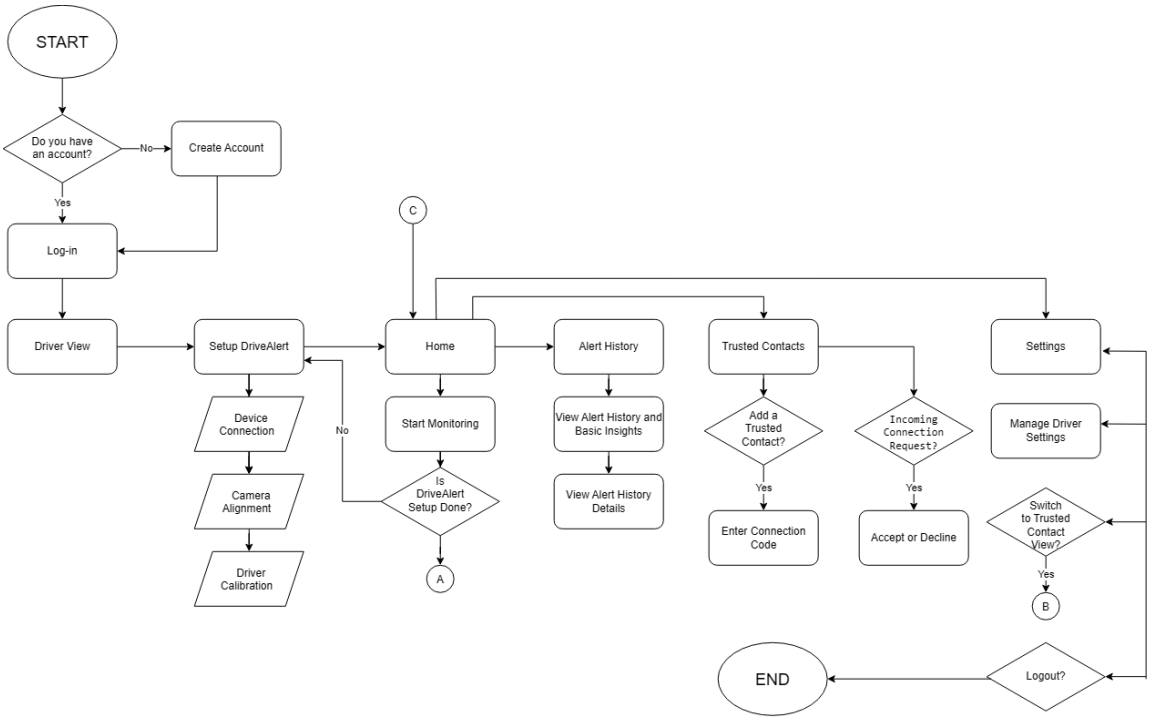


Figure 3: **Program Workflow (Drivers)**

Figure 3 presents the program workflow for the Driver view of the DriveAlert mobile application. The process begins with account creation or login, followed by access to the Driver view and DriveAlert setup, which includes device connection, camera alignment, and Driver calibration. From the Home screen, the Driver may start monitoring, review Alert History and basic insights, manage Trusted Contacts, or access Settings. When starting monitoring, the system checks whether the required setup is complete. If further setup is needed, the Driver returns to the setup

section; otherwise, the workflow continues through connector A, which leads to the separate monitoring workflow. In Alert History, the Driver may review recorded alerts and their details. The Trusted Contacts section allows the Driver to add a contact using a connection code and accept or decline incoming connection requests. Through Settings, the Driver may manage preferences, switch to the Trusted Contact view through connector B, or log out to end the workflow.

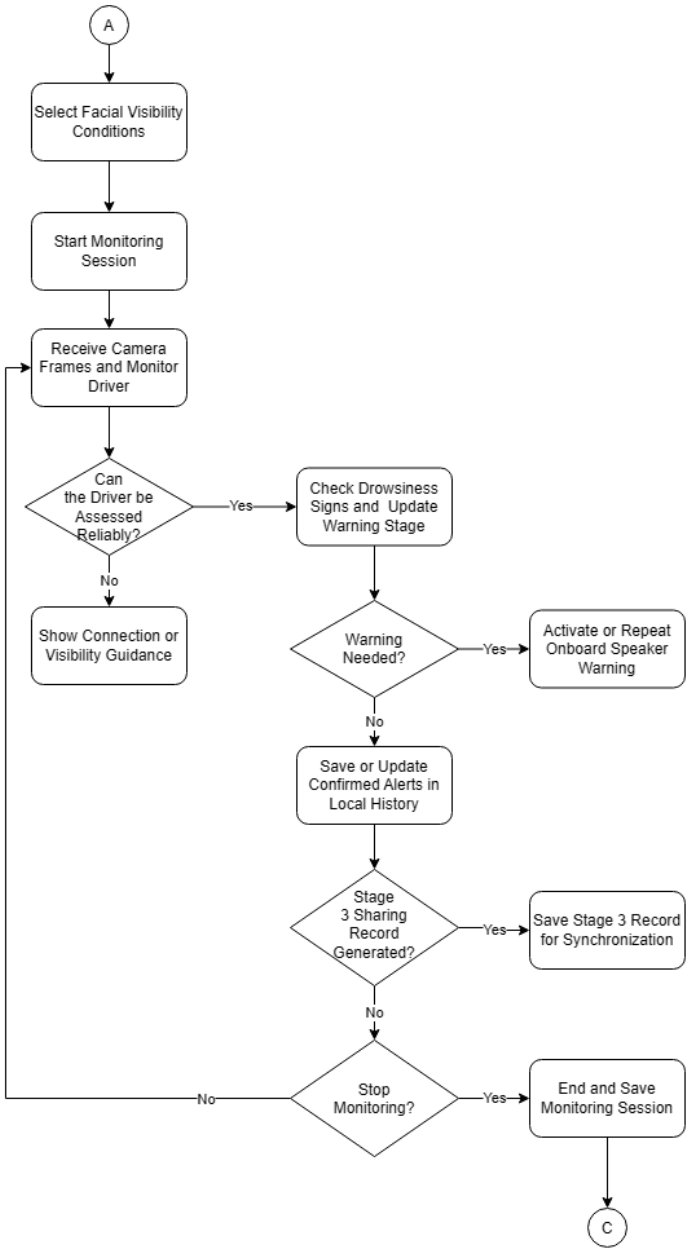


Figure 4: **Program Workflow (Monitoring Session)**

Figure 4 presents the continuation of the Driver monitoring workflow from connector A. The Driver selects the facial visibility condition and starts a monitoring session. The system receives camera frames and checks whether the Driver can be assessed reliably. If assessment is unavailable, it displays connection or visibility guidance. Otherwise, it checks the available drowsiness signs and updates the warning stage. When required, the system activates or repeats the onboard speaker warning. Confirmed Alerts are saved or updated in local history, while separate Stage 3 sharing records are saved for synchronization with eligible approved Trusted Contacts when internet access is available. Monitoring continues until the Driver chooses to stop, after which the session is ended and saved. Connector C returns the Driver to the Home screen.

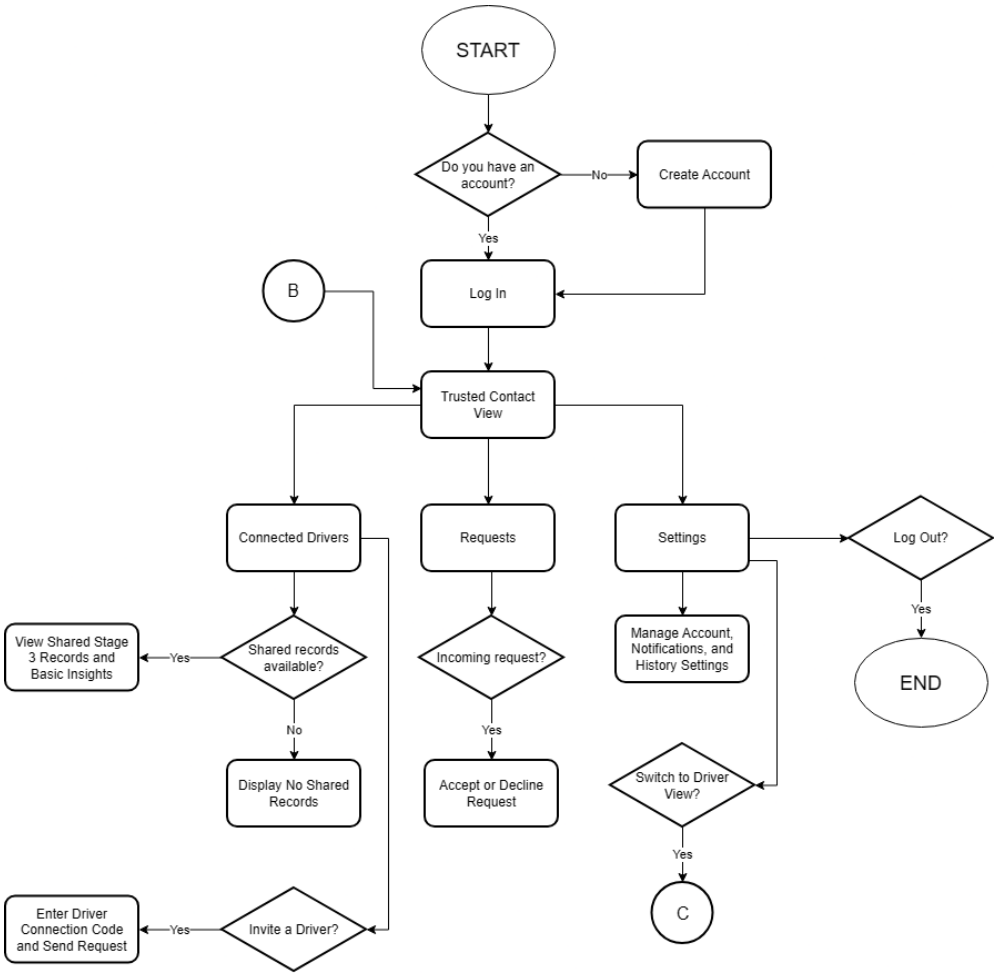


Figure 5: **Program Workflow (Trusted Contact)**

Figure 5 presents the program workflow for the Trusted Contact view of the DriveAlert mobile application. The user accesses this view after account creation and login or through connector B from the Driver workflow. The Trusted Contact may view Connected Drivers, manage Requests, or access Settings. Under Connected Drivers, the system checks whether shared records are available. If records exist, the Trusted Contact may review shared Stage 3 records and basic insights; otherwise, the system displays that no shared records are available. The Trusted Contact may also invite a Driver by entering the Driver's connection code and sending a request. In Requests, incoming connection requests may be accepted or declined. Settings allows the user to manage account information, notifications, and history settings, switch to the Driver view through connector C, or log out to end the workflow.

Validation Board (Stages 1 and 2)

This section presents the validation board which is a strategic tool used in the development process to systematically test and validate assumptions about a product or service. It helps teams identify the riskiest assumptions, design experiments to test these assumptions, and gather data to inform decision-making.

Table 5.1

VALIDATION BOARD

Experiments	**1**	**2**
User	Drivers of four-wheeled vehicles and larger vehicles	Potential Trusted Contacts of drivers
Problem	Drivers may experience drowsiness while driving and display observable signs such as heavy eyelids, frequent yawning, and head nodding. These signs may not always be recognized immediately, which may affect the driver's attentiveness while operating a vehicle.	Trusted Contacts may need a practical way to remain informed about drowsiness-alert information shared by a Driver, particularly when records cannot be synchronized immediately because internet connectivity is unavailable.
Solution	Develop DriveAlert, an IoT-based driver drowsiness monitoring and alert system that monitors prolonged eye closure, yawning, and head nodding, provides warning alerts, stores drowsiness-alert records, and	Provide a Trusted Contact feature in DriveAlert that allows an approved Trusted Contact to review synchronized drowsiness-alert records and related information after the Driver has authorized access.

Table 5.2

VALIDATION BOARD cont'd

	supports Trusted Contact synchronization.	
Riskiest Assumptions	Drivers experience drowsiness or sleepiness while driving, find warning alerts helpful, and are willing to consider using an affordable and easy-to-use driver-monitoring system.	Potential Trusted Contacts are willing to review Driver-approved alert records, consider the shared information useful, accept delayed synchronization, and are willing to use an easy-to-use Trusted Contact feature.
Method & Success Criteria	A survey was conducted among drivers to assess drowsiness or sleepiness while driving, perceived helpfulness of warning alerts, and willingness to use an affordable and easy-to-use monitoring system. Success requires at least 70% positive responses for drowsiness experience and willingness to use the system.	A survey was conducted among potential Trusted Contacts to assess willingness to review shared records, perceived usefulness, acceptance of Driver-approved access and delayed synchronization, and willingness to use the feature. Success requires at least 70% positive acceptance of the Trusted Contact feature.
Result & Destination	The survey showed that 95.2% of 42 respondents had experienced drowsiness or sleepiness while driving, while 81.0% were willing to use a system like DriveAlert if affordable and easy to use. Since both exceeded 70%, development will proceed.	The survey showed that 88.2% of 17 respondents were willing to review shared records, 94.1% found the information useful, and 82.4% accepted delayed synchronization and would probably use the feature. Since the results exceeded 70%, the feature will proceed.
Learnings	The findings showed that drowsiness while driving is relevant among the surveyed drivers and that respondents showed interest in an affordable and easy-to-use monitoring system. These results support further development of DriveAlert's monitoring and warning functions.	The findings showed that potential Trusted Contacts generally accepted Driver-approved access, delayed synchronization, and shared alert information. These results support retaining and refining DriveAlert's Trusted Contact and record-sharing functions.

Experiment 1. Drivers may experience drowsiness or sleepiness while driving and display observable signs such as heavy eyelids, frequent yawning, and head nodding. DriveAlert is proposed as an IoT-based driver-support system that uses computer vision to monitor prolonged eye closure, yawning, and head nodding, provide warning alerts, and store drowsiness-alert records. A survey was conducted among drivers to determine whether drowsiness or sleepiness occurs while driving and whether respondents are interested in an affordable and easy-to-use monitoring system. Success was defined as at least 70% of respondents reporting drowsiness or sleepiness while driving and at least 70% expressing willingness to use the proposed system. The results will guide the continued development and refinement of DriveAlert's monitoring, warning, usability, and alert-record functions.

Experiment 2. Potential Trusted Contacts may need a practical way to remain informed about drowsiness-alert information shared by an approved Driver. DriveAlert therefore includes a Trusted Contact feature that allows authorized users to review synchronized drowsiness-alert records and related information when internet connectivity becomes available. A survey was conducted among potential Trusted Contacts to assess their willingness to review shared records, perceived usefulness of the information, acceptance of Driver-approved access and delayed synchronization, and willingness to use the feature. Success was defined as at least 70% positive acceptance of the Trusted Contact feature. The results will guide the continued refinement of DriveAlert's Trusted Contact, access-control, record-sharing, and synchronization functions.

Gantt Chart/PERT Chart

This section presents the researchers' Gantt chart for the project. It will visually represent the project timeline, showing the start and finish dates of various tasks and activities. It consists of horizontal bars, with each bar representing a task, positioned along a timeline. Tasks are organized sequentially, with dependencies between them clearly marked, making it easy to see which tasks need to be completed before others can begin. The chart helps project managers track progress, allocate resources, and ensure timely completion of the project.

Table 6.1

GANTT CHART

Task ID	Task Name	Task Lead	Start Date	End Date	JUN 2026				JULY 2026				AUG 2026				SEPT 2026				OCT 2026				NOV 2026			
					W 1	W 2	W 3	W4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4
1	**INTRODUCTION**																											
1.1	Create an introduction	Basubas	June 25	June 30																								
1.2	Explain the problem, research gap and solution	Isagunde	June 25	June 30																								
1.3	Write the specific objectives	Basubas	June 25	June 30																								
1.4	Define what the system includes and excludes	Isagunde	June 25	June 30																								
1.5	Explain who benefits from the study	Catarig	June 25	June 30																								
1.6	Show the study process	Catarig	June 25	June 30																								
1.7	Define the study terms	Borces	June 25	June 30																								
1.8	Make the questionnaire	Basubas	June 25	June 30																								
1.9	Collect responses	Isagunde	July 1	July 7																								
1.10	Add citations	Basubas	July 1	July 7																								
1.11	Revise Chapter 1	All	July 1	July 7																								
2	**REVIEW OF RELATED LITERATURE AND STUDIES**																											
2.1	Identify and explain the theories	Catarig	June 25	June 30																								
2.2	Research journals	Borces	June 25	June 30																								
2.3	Gather and summarize literature	Isagunde	June 25	June 30																								
2.4	Review related studies	Basubas	June 25	June 30																								
2.5	Compare DriveAlert with existing systems and identify the research gap	Isagunde	July 1	July 7																								
2.6	Add citations, organize references, and revise Chapter 2	Basubas	July 1	July 7																								
3	**RESEARCH METHODOLOGY**																											
3.1	Select and explain the Agile methodology	Isagunde	July 9	July 14																								
3.2	Define the system requirements and development plan	Basubas	July 9	July 14																								
3.3	Prepare the Business Model Canvas	Basubas	July 9	July 14																								
3.4	Create the Driver and Trusted Contact program workflows	Borces	July 9	July 14																								
3.5	Complete the Validation Boards for Stages 1 and 2	Basubas	July 9	July 14																								
3.6	Prepare the Gantt charts	Isagunde	July 9	July 14																								
3.7	Create the Functional Decomposition Diagram	Catarig	July 9	July 14																								
3.8	Create the Use Case Diagram	Basubas	July 9	July 14																								
3.9	Design the mobile and web user interfaces	Borces	July 9	July 14																								
3.10	Create the storyboard	Basubas	July 9	July 14																								
3.11	Identify the required database entities, attributes, and relationships	Borces	July 9	July 14																								
3.12	Create the Entity Relationship Diagram	Basubas	July 15	July 21																								
3.13	Prepare the Data Dictionary	Catarig	July 15	July 21																								
3.14	Create the Network Design	Isagunde	July 15	July 21																								

Table 6.2

GANTT CHART

Task ID	Task Name	Task Lead	Start Date	End Date	JUN 2026				JULY 2026				AUG 2026				SEPT 2026				OCT 2026				NOV 2026			
					W 1	W 2	W 3	W4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4
3.15	Create the Network Model	Isagunde	July 15	July 21																								
3.16	Describe the system development	Basubas	July 9	July 14																								
3.17	Create the Technology Stack Diagram	Basubas	July 9	July 14																								
3.18	Define the software and hardware specifications	Borces	July 9	July 14																								
3.19	Prepare the Program Specification	Catarig	July 9	July 14																								
3.20	Prepare the Team Composition	Isagunde	July 9	July 14																								
3.21	Prepare the Appendices	Basubas	July 15	July 21																								
3.22	Prepare the List of Modules	Basubas	July 9	July 14																								
3.23	Revise the Gantt Chart and Agile Sprint Plan	Isagunde	July 15	July 21																								
3.24	Revise Chapter 3	All	July 15	July 21																								
3.25	Finalize the manuscript	All	July 15	July 21																								
4	**DEVELOPMENT TRANSITION PREPARATION**																											
4.1	Review the finalized manuscript requirements, diagrams, and module boundaries	All	July 15	July 21																								
4.2	Prepare development tools, accounts, hardware requirements, and project resources	Basubas and Borces	July 15	July 21																								
4.3	Assign module ownership, repository workflow, and coding standards	All	July 15	July 21																								
5	**PROJECT FOUNDATION, IOT SOURCING, AND MANUSCRIPT COMPLIANCE**																											
5.1	Create the Git repository and configure Android, web, and Firebase projects	Basubas, Isagunde, and Borces	July 23	July 29																								
5.2	Refine local and cloud schemas for accounts, connections, and device access	Borces	July 23	July 29																								
5.3	Identify suppliers and source the required IoT components	Basubas and All	July 23	July 29																								
5.4	Review component specifications, compatibility, and proposed wiring	Basubas and Borces	July 23	July 29																								
5.5	Establish local storage with separate Driver calibration and alert records	Borces	August 1	August 5																								
5.6	Prepare project structure, account layouts, and device invitation rules	Isagunde and Catarig	August 1	August 5																								
5.7	Revise the manuscript and diagrams based on review feedback	All	August 1	August 5																								
5.8	Prepare compliance documents and route the approval sheet	All	August 1	August 5																								
5.9	Check builds and documents, commit and push changes, and review readiness	All	August 1	August 5																								
6	**IOT ASSEMBLY, COMMUNICATION, AND CORE APPLICATION FUNCTIONS**																											
6.1	Assemble the camera, speaker, power connections, and mounting	Basubas and All	August 8	August 12																								
6.2	Configure camera firmware, capture settings, and device services	Basubas	August 8	August 12																								
6.3	Implement Bluetooth provisioning and local Wi-Fi connection	Basubas	August 8	August 12																								
6.4	Implement the IoT HTTP stream and Android video receiving	Basubas and Borces	August 8	August 12																								

Table 6.3

GANTT CHART

Task ID	Task Name	Task Lead	Start Date	End Date	JUN 2026				JULY 2026				AUG 2026				SEPT 2026				OCT 2026				NOV 2026			
					W 1	W 2	W 3	W4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4
6.5	Verify onboard speaker playback and initial warning commands	Basubas and Catarig	August 8	August 12										■														
6.6	Build accounts, device ownership, and simulated subscription invitations	Basubas and Isagunde	August 6	August 19									■	■	■													
6.7	Build initial web views and online Driver setup for offline device access	Isagunde and Catarig	August 6	August 19									■	■	■													
6.8	Test power, streaming, reconnection, and account functions	All	August 15	August 19											■													
6.9	Upload tested firmware, push changes to GitHub, and review connection readiness	Basubas and Borces	August 15	August 19											■													
7	**MEDIAPIPE INTEGRATION AND LIVE FACIAL MEASUREMENTS**																											
7.1	Integrate MediaPipe Face Landmarker into Android processing	Basubas	August 22	August 26												■												
7.2	Build live frame processing from the IoT camera stream	Basubas and Borces	August 22	August 26												■												
7.3	Identify eye and mouth landmarks and calculate EAR and MAR	Basubas and Borces	August 22	August 26												■												
7.4	Calculate head-pitch measurements from facial transformations	Basubas	August 22	August 26												■												
7.5	Implement Driver-area selection and tracking continuity	Basubas	August 22	August 26												■												
7.6	Implement facial-lighting and visibility checks	Basubas and Borces	August 27	August 31												■												
7.7	Develop camera-alignment and facial-visibility feedback	Isagunde	August 27	August 31												■												
7.8	Test landmark stability, lighting, delay, and Android compatibility	All	August 27	August 31												■												
7.9	Commit and push processing changes and review calibration readiness	Basubas and Borces	August 27	August 31												■												
8	**CALIBRATION, DROWSINESS DETECTION, AND WARNING CONTROL**																											
8.1	Implement guided Driver calibration and stored calibration values	Basubas	September 3	September 7													■											
8.2	Implement prolonged-eye-closure detection using calibrated EAR	Basubas	September 8	September 14														■										
8.3	Implement yawning detection using calibrated MAR	Catarig and Basubas	September 8	September 14														■										
8.4	Implement relative head-pitch and temporal head-down detection	Basubas	September 8	September 14														■										
8.5	Implement confirmation, episode grouping, and warning-stage rules	Basubas	September 3	September 7													■											
8.6	Connect staged warnings to the onboard speaker command	Basubas and Isagunde	September 8	September 14														■										
8.7	Implement local monitoring-session and alert-history records	Borces	September 3	September 7													■											
8.8	Test calibration, false alerts, stage progression, and offline records	All	September 8	September 14														■										
8.9	Commit and push tested changes and review detection reliability	All	September 8	September 14														■										
9	**TRUSTED CONTACT SHARING AND SYSTEM INTEGRATION**																											
9.1	Implement connection requests, approval, and Driver relationships	Borces and Catarig	September 17	September 21															■									

Table 6.4

GANTT CHART

Task ID	Task Name	Task Lead	Start Date	End Date	JUN 2026				JULY 2026				AUG 2026				SEPT 2026				OCT 2026				NOV 2026			
					W 1	W 2	W 3	W4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4
9.2	Implement approved access, revocation, and security rules	Borces	September 17	September 21																								
9.3	Create eligible Stage 3 transition and persistence records	Basubas and Borces	September 17	September 21																								
9.4	Implement synchronization retries and Android notifications using Worker, D1, and FCM	Basubas and Borces	September 17	September 21																								
9.5	Develop mobile shared-record views and Driver alert summaries	Isagunde and Catarig	September 17	September 30																								
9.6	Connect the web portal to approved Driver records and requests	Isagunde and Catarig	September 17	September 30																								
9.7	Integrate mobile, web, IoT warnings, records, and notifications	All	September 17	September 30																								
9.8	Test sharing eligibility, revocation, offline queues, and full workflow	All	September 24	September 30																								
9.9	Push tested changes, deploy the web test build, and review integration	Isagunde and Borces	September 24	September 30																								
10	**INTEGRATED TESTING AND SYSTEM REFINEMENT**																											
10.1	Test accounts, Driver invitations, simulated subscriptions, and offline setup	Borces and Catarig	October 1	October 7																								
10.2	Test camera tracking, lighting pauses, and connection recovery	Basubas and Isagunde	October 1	October 7																								
10.3	Test calibration, sign confirmation, warning stages, and speaker response	Basubas and Catarig	October 1	October 7																								
10.4	Test separate Driver calibration, alert histories, settings, and retention	Isagunde and Borces	October 1	October 7																								
10.5	Test approved sharing, revocation, synchronization, and retries	Borces and Catarig	October 1	October 7																								
10.6	Test web responsiveness, usability, and device compatibility	Isagunde and Catarig	October 1	October 7																								
10.7	Correct integrated defects and perform focused regression tests	All	October 8	October 14																								
10.8	Update manuscript descriptions and diagrams to match the system	All	October 8	October 14																								
10.9	Push fixes, generate a test APK, and update the web test deployment	Basubas and Isagunde	October 8	October 14																								
11	**CONTROLLED EVALUATION AND RELEASE PREPARATION**																											
11.1	Finalize controlled-evaluation procedures, forms, and device setup	All	October 15	October 21																								
11.2	Conduct controlled eye-closure, yawning, and head-down trials	All	October 15	October 28																								
11.3	Evaluate audible warnings, offline monitoring, and approved sharing	All	October 15	October 28																								
11.4	Record false alerts, missed detections, delays, and test results	Basubas and Borces	October 15	October 28																								
11.5	Resolve remaining mobile, web, IoT, and synchronization defects	All	October 15	October 28																								
11.6	Analyze evaluation findings against the study objectives	Basubas and Catarig	October 22	October 28																								
11.7	Complete manuscript findings, diagrams, and supporting evidence	Isagunde and Borces	October 22	October 28																								
11.8	Run complete regression tests and verify release readiness	All	October 22	October 28																								

Table 6.5

GANTT CHART

Task ID	Task Name	Task Lead	Start Date	End Date	JUN 2026				JULY 2026				AUG 2026				SEPT 2026				OCT 2026				NOV 2026			
					W 1	W 2	W 3	W4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4
11.9	Push verified changes, generate the release-candidate APK, and update the web	Basubas and Isagunde	October 22	October 28																								
12	**FINAL SUBMISSION, DEPLOYMENT, AND PRESENTATION**																											
12.1	Finalize manuscript formatting, references, and required corrections	All	November 1	November 5																								
12.2	Complete approval requirements, compliance documents, and appendices	All	November 1	November 5																								
12.3	Verify manuscript completeness and prepare the submission package	All	November 1	November 5																								
12.4	Submit the final manuscript	All	November 6	November 11																								
12.5	Resolve final system issues and verify corrected functions	All	November 6	November 11																								
12.6	Generate and install the final APK and upload verified IoT firmware	Basubas and Borces	November 6	November 11																								
12.7	Deploy and verify the Trusted Contact web portal and access rules	Isagunde and Borces	November 6	November 11																								
12.8	Prepare presentation materials and rehearse the demonstration	All	October 29	November 11																								
12.9	Push final changes and verify deployment, presentation, and project completion	All	November 6	November 11																								

LEGEND	COMPLETE
	ONGOING
	INCOMPLETE

The Gantt chart presents project tasks and their timeframes in a horizontal bar format, helping researchers organize activities, coordinate responsibilities, and track progress. It supports the Agile methodology by showing the work scheduled across sprints and helping the team meet project deadlines.

Functional Decomposition Diagram

This section presents the functional decomposition diagram, which illustrates the breakdown of various application processes into their respective components for a comprehensive understanding of the application

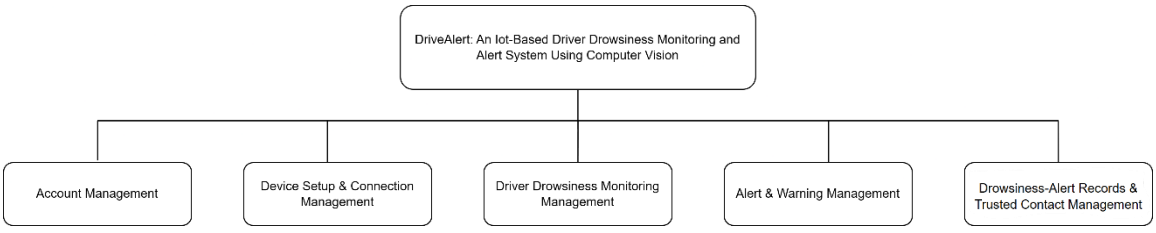


Figure 6.1: **Functional Decomposition Diagram**

Figure 6.1 illustrates that the “DriveAlert” system is composed of several key modules: Account Management, Device Setup & Connection Management, Driver Drowsiness Monitoring Management, Alert & Warning Management, and Drowsiness-Alert Records & Trusted Contact Management. These modules work together to provide a driver-focused drowsiness monitoring and alert system. Account Management handles user registration, login, and profile information. Device Setup & Connection Management supports the required app permissions and the local offline Wi-Fi connection between the IoT device and the driver’s smartphone. Driver Drowsiness Monitoring Management handles the live camera feed and detection of selected visible signs associated with drowsiness, such as prolonged eye closure, yawning, and head nodding. Alert & Warning Management manages threshold-based alerting, mobile notifications, warning-volume settings, and the activation of the onboard speaker using predefined warning sounds. The warning volume can be adjusted but cannot be muted or reduced below the enforced minimum level during active monitoring. Drowsiness-Alert Records & Trusted Contact Management stores drowsiness-alert history and supports the automatic synchronization of approved drowsiness-alert records to a Trusted Contact once an internet connection becomes available.

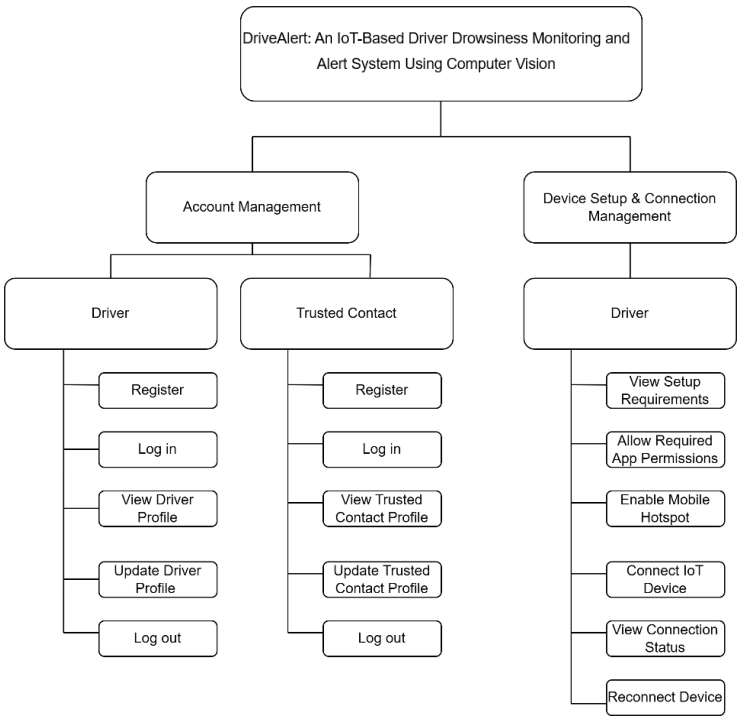


Figure 6.2: **Functional Decomposition Diagram cont’d**

Figure 6.2 presents two main modules of the “DriveAlert” system: Account Management and Device Setup & Connection Management. In the Account Management module, Drivers and

Trusted Contacts can register, log in, view their profiles, update their account information, and log out of the application. This module allows each user to access the system based on their role, where Drivers can use monitoring-related features while Trusted Contacts can access synchronized drowsiness-alert records after approval. The Device Setup & Connection Management module enables the Driver to view setup requirements, allow the required app permissions, enable the mobile hotspot, connect the IoT device, view the device connection status, and reconnect the device when needed. This supports the proper setup of the IoT camera device and the driver's smartphone before starting the drowsiness-monitoring process. Through these modules, DriveAlert ensures that users can securely access the application and that the device connection is prepared for real-time driver drowsiness monitoring.

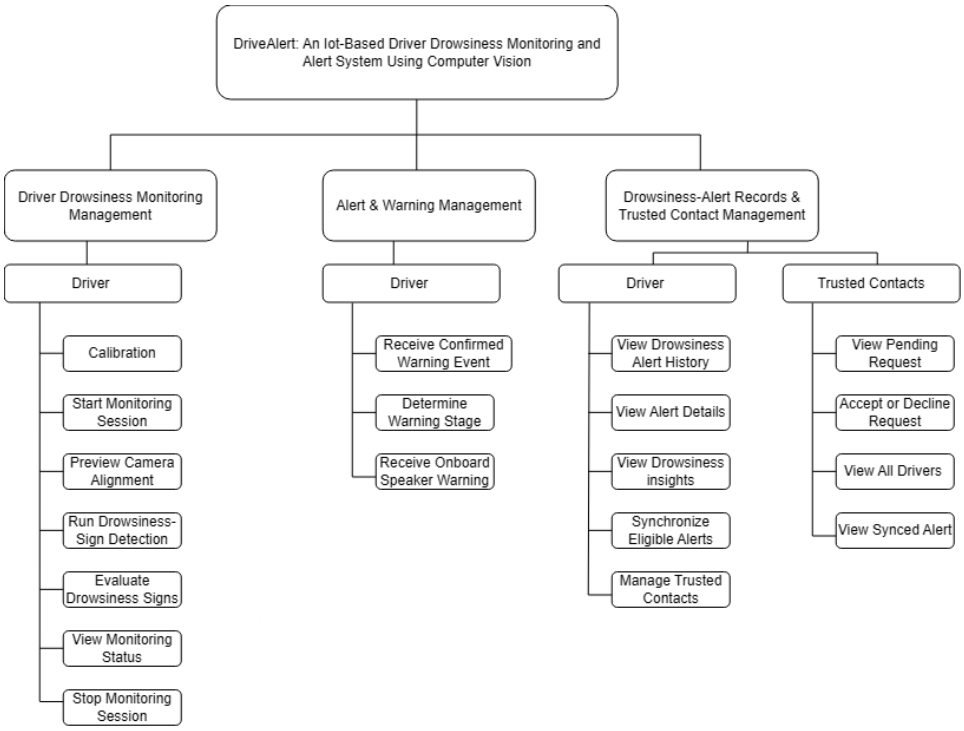


Figure 6.3: **Functional Decomposition Diagram cont'd**

Figure 6.3 presents the remaining main modules of the "DriveAlert" system: Driver Drowsiness Monitoring Management, Alert & Warning Management, and Drowsiness-Alert Records & Trusted Contact Management. The Driver Drowsiness Monitoring Management module allows the Driver to calibrate the system, start and stop a monitoring session, preview the camera alignment, run drowsiness-sign detection, evaluate the selected visible signs associated with drowsiness, and view

the current monitoring status. The system evaluates prolonged eye closure, yawning, and head nodding using Driver-specific thresholds together with the corresponding temporal or contextual confirmation conditions. The Alert & Warning Management module handles the warning process after a selected visible sign satisfies its complete confirmation condition. It allows the Driver to receive a confirmed warning event, determines the applicable Warning Stage, and provides the corresponding warning output through the onboard speaker. A threshold crossing alone does not immediately create a confirmed warning event. The Drowsiness-Alert Records & Trusted Contact Management module allows the Driver to view drowsiness-alert history, view alert details, view basic drowsiness insights, synchronize eligible alerts, and manage Trusted Contacts. The basic drowsiness insights include information derived from stored drowsiness-alert records, such as alert counts, detected-sign summaries, the highest Warning Stage reached, alert trends, and synchronization status. On the other hand, the Trusted Contact can view pending requests, accept or decline requests, view connected Drivers, and view synchronized drowsiness-alert records. Through these modules, DriveAlert supports Driver drowsiness monitoring, confirmed warning-event processing, Warning Stage management, warning output, drowsiness-alert record review, basic drowsiness insights, and synchronization of eligible drowsiness-alert records with approved Trusted Contacts when internet connectivity is available.

Use Case Diagram

This section presents the use case diagram, which provides a visual representation of the system's functional requirements and user interactions. For "DriveAlert," the diagram outlines the main scenarios where the Driver and Trusted Contact interact with the system to perform key functions related to account access, device setup, drowsiness monitoring, alert viewing, alert record management, and Trusted Contact sharing.

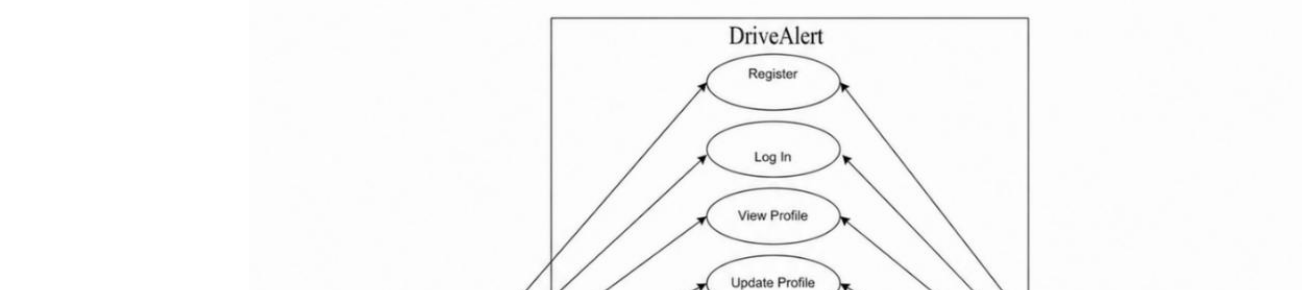


Figure 7.1: **Use Case Diagram**

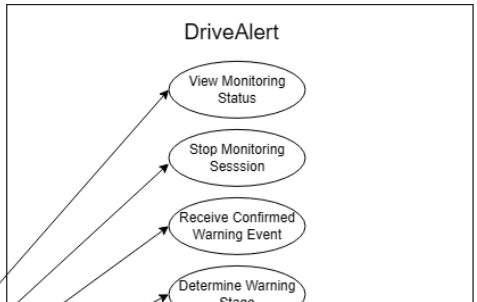


Figure 7.2: **Use Case Diagram cont'd**

Figures 7.1 and 7.2 presents the use case diagram of the "DriveAlert" system. The diagram shows two main actors: the Driver and the Trusted Contact. Both actors can register, log in, view their profile, update their profile, and log out of the application. The Driver can view setup requirements, allow the required app permissions, enable the mobile hotspot, connect the IoT device, view the device connection status, and reconnect the device when needed. The Driver can also start a monitoring session, preview camera alignment, run drowsiness-sign detection, detect selected visible signs associated with drowsiness, view monitoring status, and stop the monitoring session. For alert-related functions, the Driver can receive drowsiness alerts, receive onboard speaker warning output, view alert status, view drowsiness alert history, view alert details, and view basic drowsiness insights. The Driver can also add Trusted Contacts and send Trusted Contact requests. The Trusted Contact can view pending requests, accept or decline requests, view connected drivers, view shared drowsiness-alert records, view shared alert history, view shared alert details, and view shared basic drowsiness insights. These use cases show that the Driver

controls the device setup, monitoring process, drowsiness-sign detection, alert viewing, drowsiness alert summary review, and Trusted Contact request process. On the other hand, the Trusted Contact is limited to accepting or declining requests and reviewing shared drowsiness-alert records and related basic drowsiness insights from connected drivers. Unlike the Driver, the Trusted Contact does not control monitoring, device connection, drowsiness-sign detection, or warning output.

User Interface Design

The following figures present the implemented user interfaces of DriveAlert. They show the actual screens, available functions, and navigation of the Driver and Trusted Contact mobile application and the web portal.

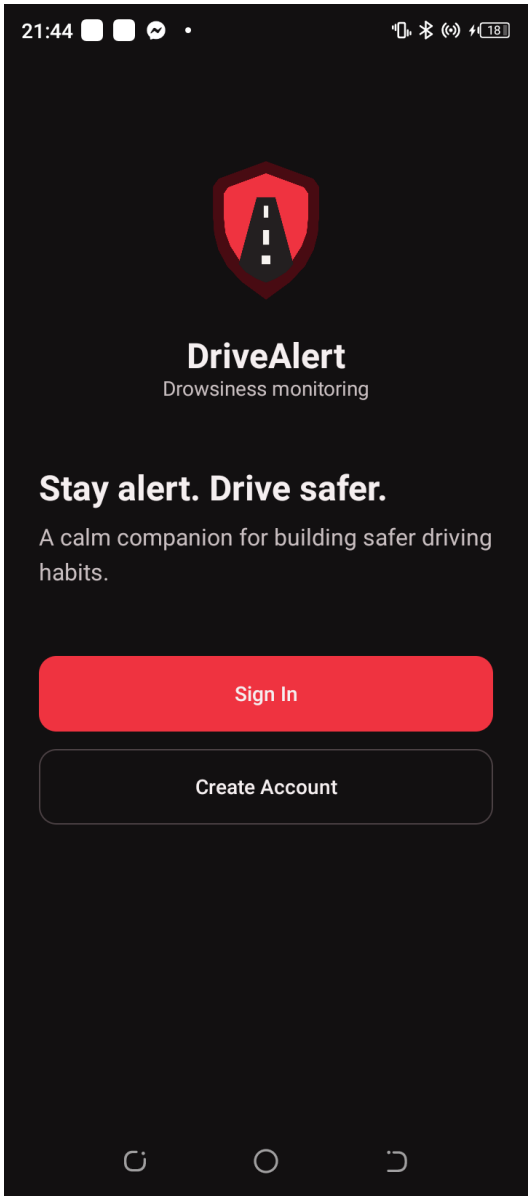


Figure 8: **Starter Screen (Driver and Trusted Contact)**

The Starter Screen is the first screen users see when they open DriveAlert. It introduces the application as a drowsiness-monitoring companion and provides two options: Sign In for existing users and Create Account for new users.

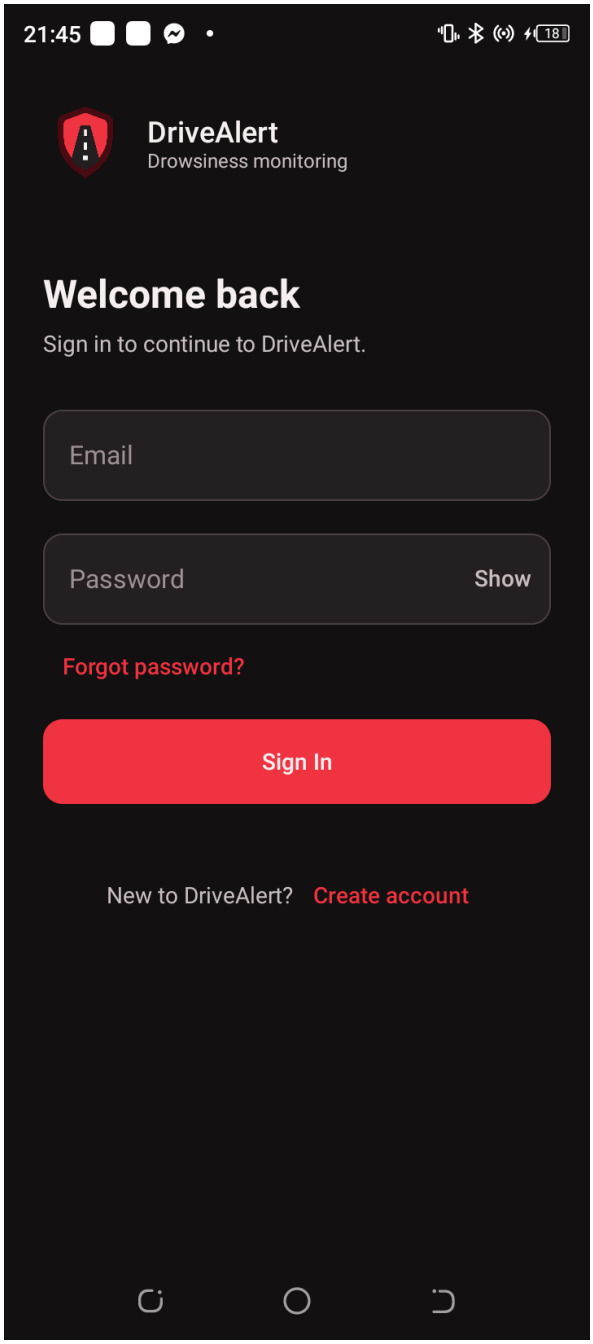


Figure 9: **Sign In Screen (Driver and Trusted Contact)**

The Sign In Screen allows existing users to access their DriveAlert accounts using their email address and password. Users may show or hide the password, recover a forgotten password, or proceed to the Create Account Screen to register.

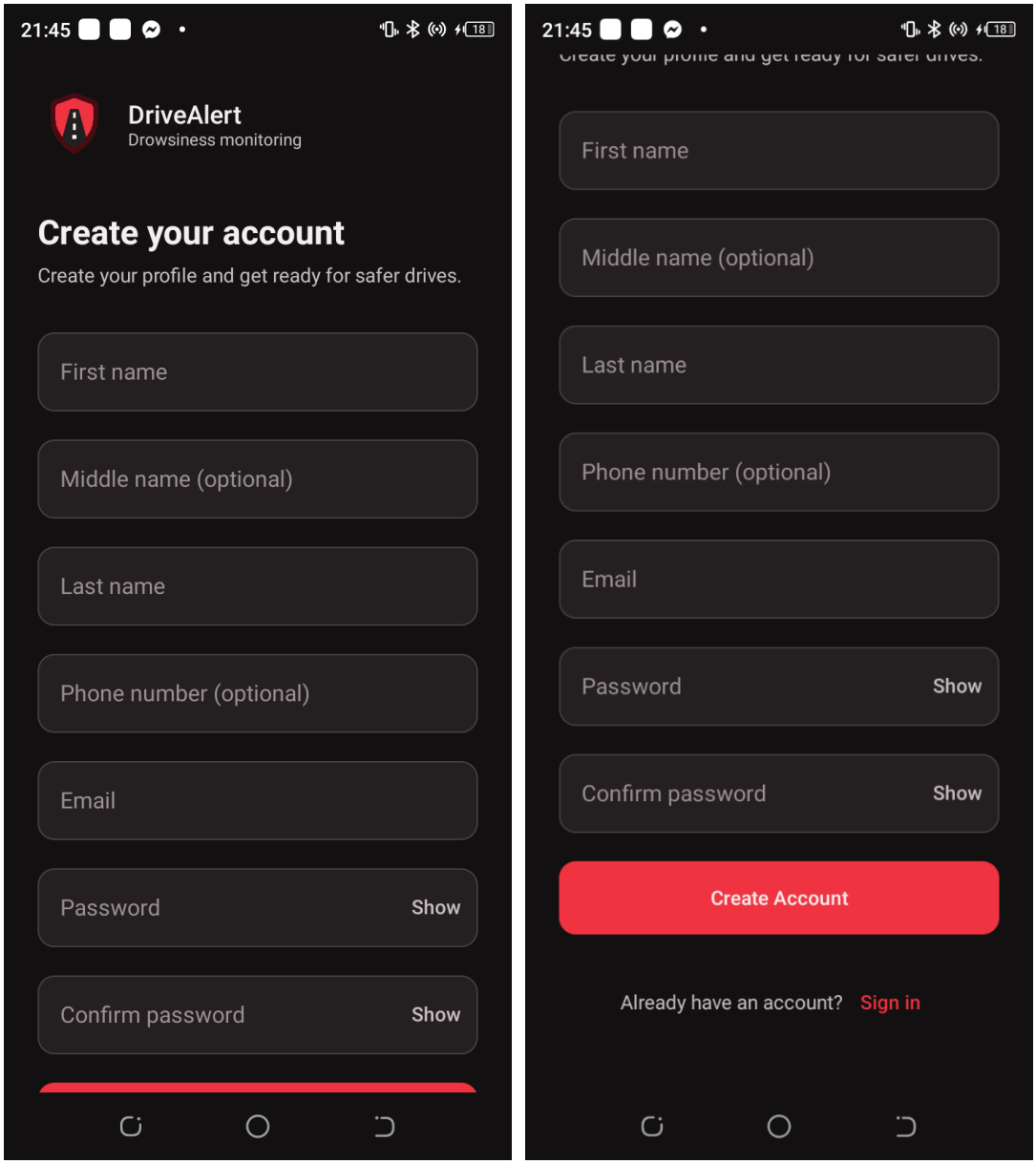


Figure 10: **Create Account Screen (Driver and Trusted Contact)**

The Create Account Screen allows new users to register for DriveAlert. It includes fields for first name, optional middle name, last name, optional phone number, email address, password, and password confirmation. Users can select Create Account to register or Sign in to return to the login screen.

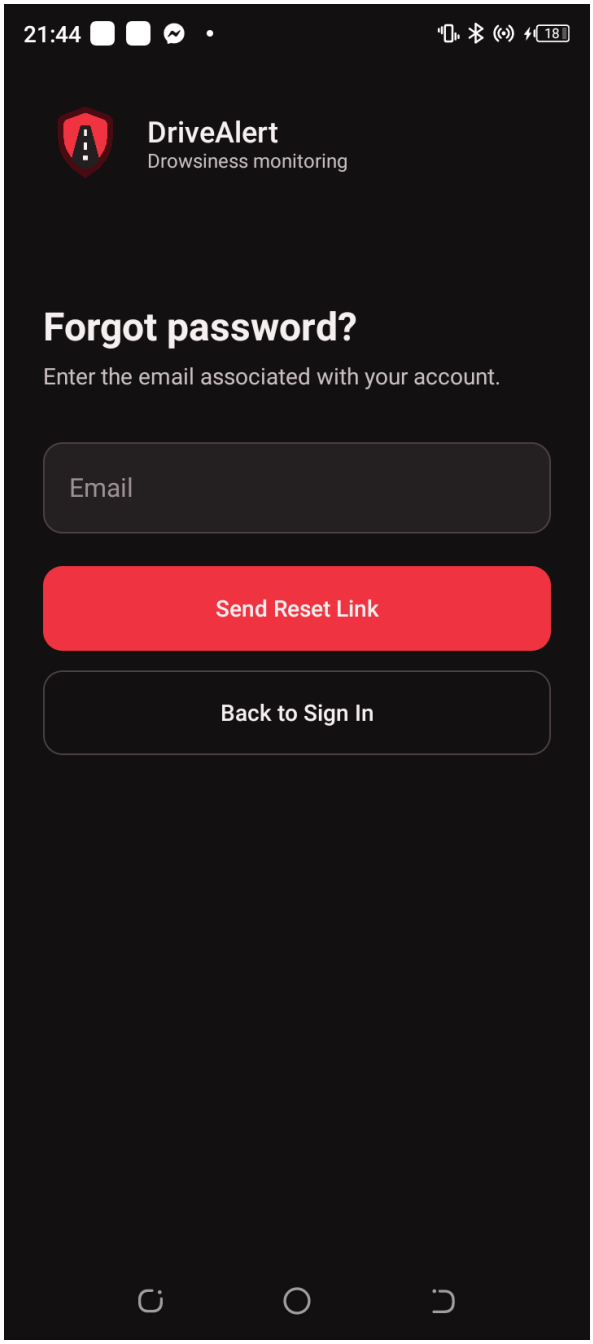


Figure 11: **Forgot Password Screen (Driver and Trusted Contact)**

The Forgot Password Screen helps users recover access to their DriveAlert accounts. Users enter the email address associated with their account and select Send Reset Link to request password-reset instructions. The Back to Sign In button returns users to the login screen.

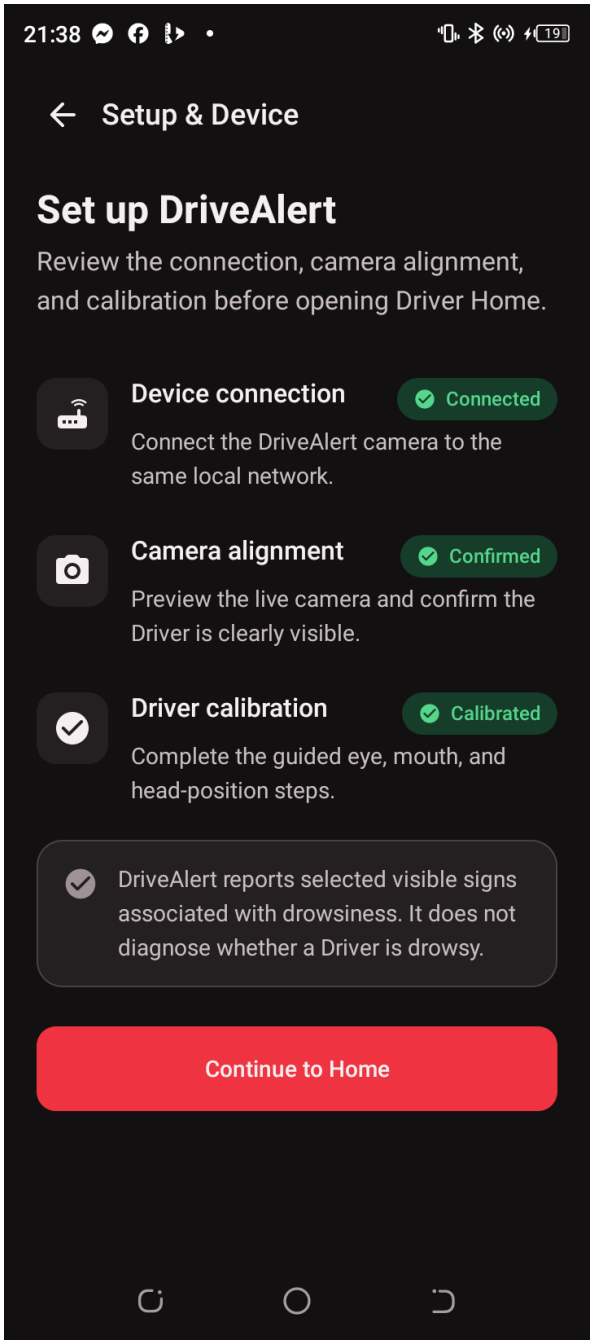


Figure 12: **Setup and Device Screen (Driver)**

The Setup and Device Screen guides the driver through the preparations needed before monitoring. It shows the status of the device connection, camera alignment, and driver calibration. The driver can open each setup item to complete the required checks or select Continue to Home to access the Driver Home Screen.

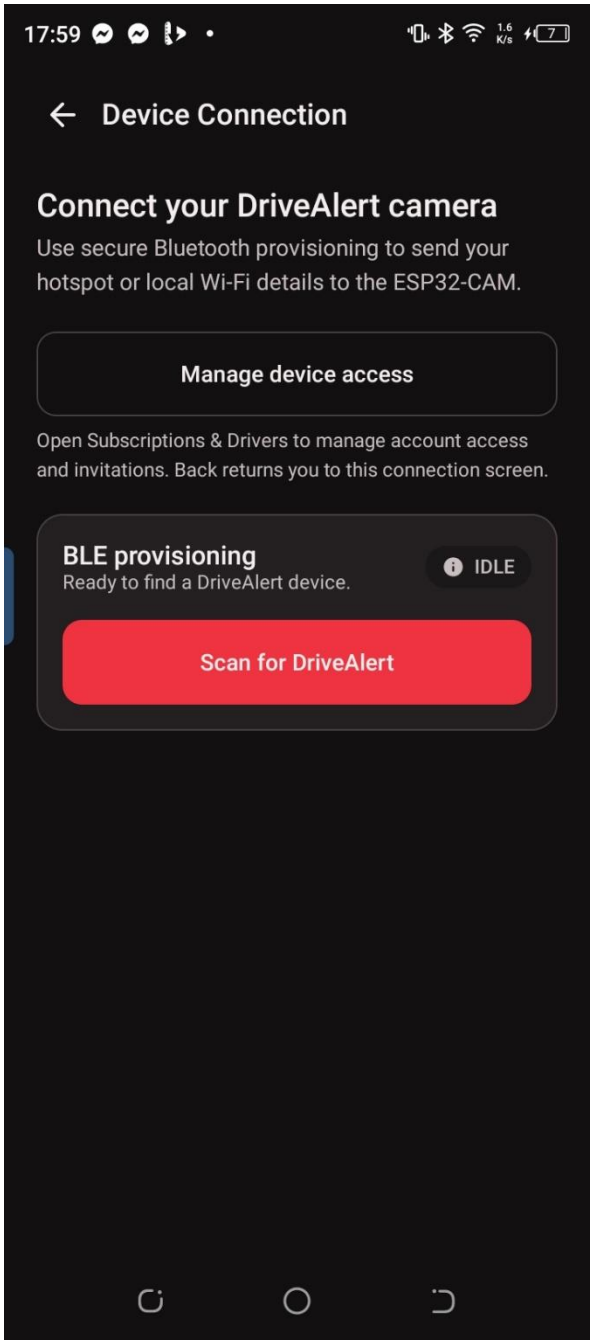


Figure 13: **Device Connection Screen (Driver)**

The Device Connection Screen allows the driver to find and connect the DriveAlert camera through secure Bluetooth provisioning. The driver can scan for a nearby device and provide hotspot or local Wi-Fi details during setup. Manage device access opens the Subscriptions and Drivers Screen for registration-related access, additional drivers, and invitations.

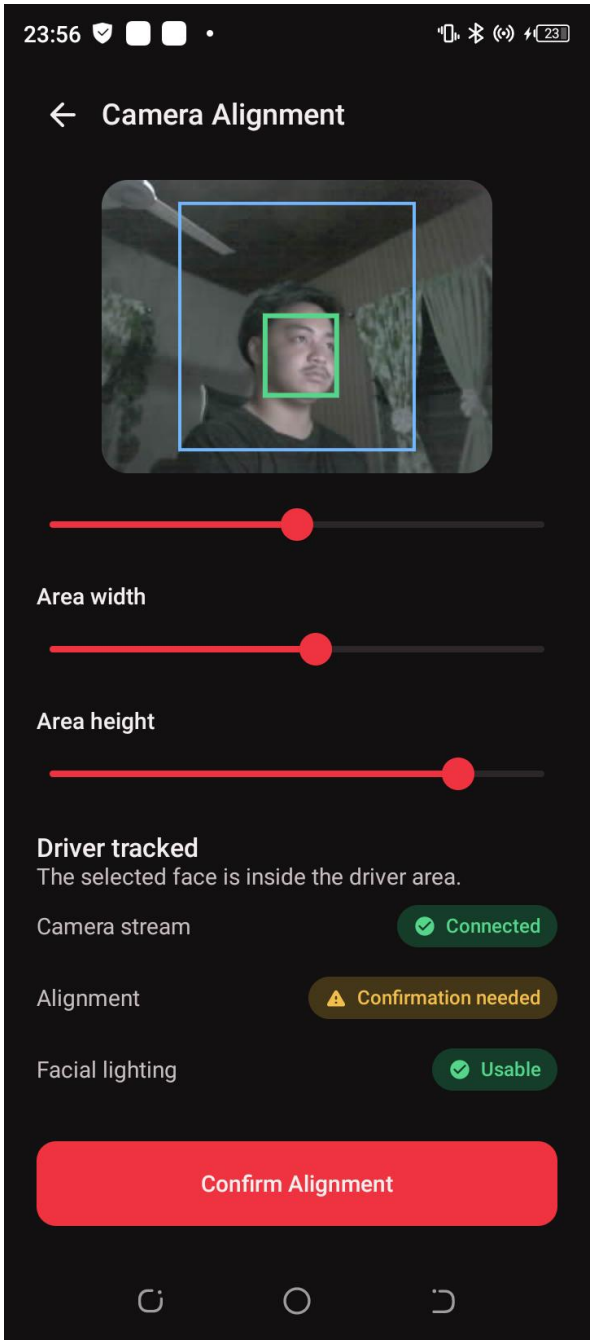


Figure 14: **Camera Alignment Screen (Driver)**

The Camera Alignment Screen helps the driver position the camera and define the driver area before monitoring. It displays the live camera preview with a blue selection box and controls for adjusting the area. The driver follows the positioning instructions and confirms the alignment while parked.

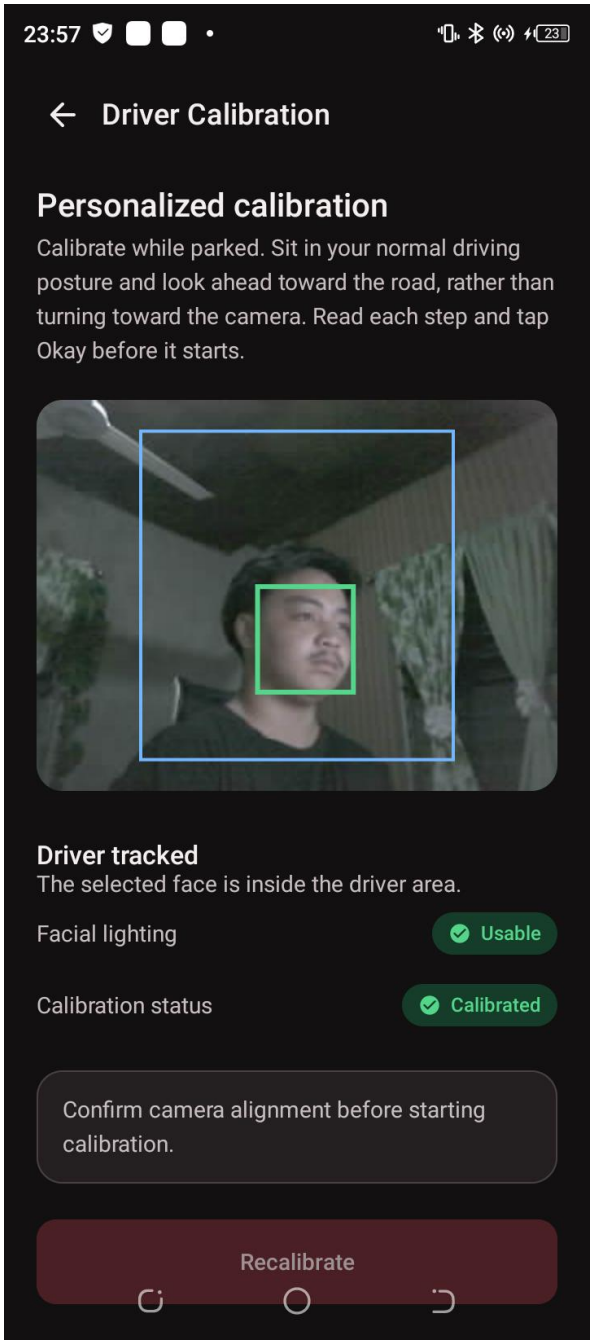


Figure 15: **Driver Calibration Screen (Driver)**

The Driver Calibration Screen guides the driver through personalized calibration while parked. It displays the live camera preview, driver visibility, facial lighting, and calibration status. The driver follows the guided eye, mouth, and head-position steps after confirming camera alignment.

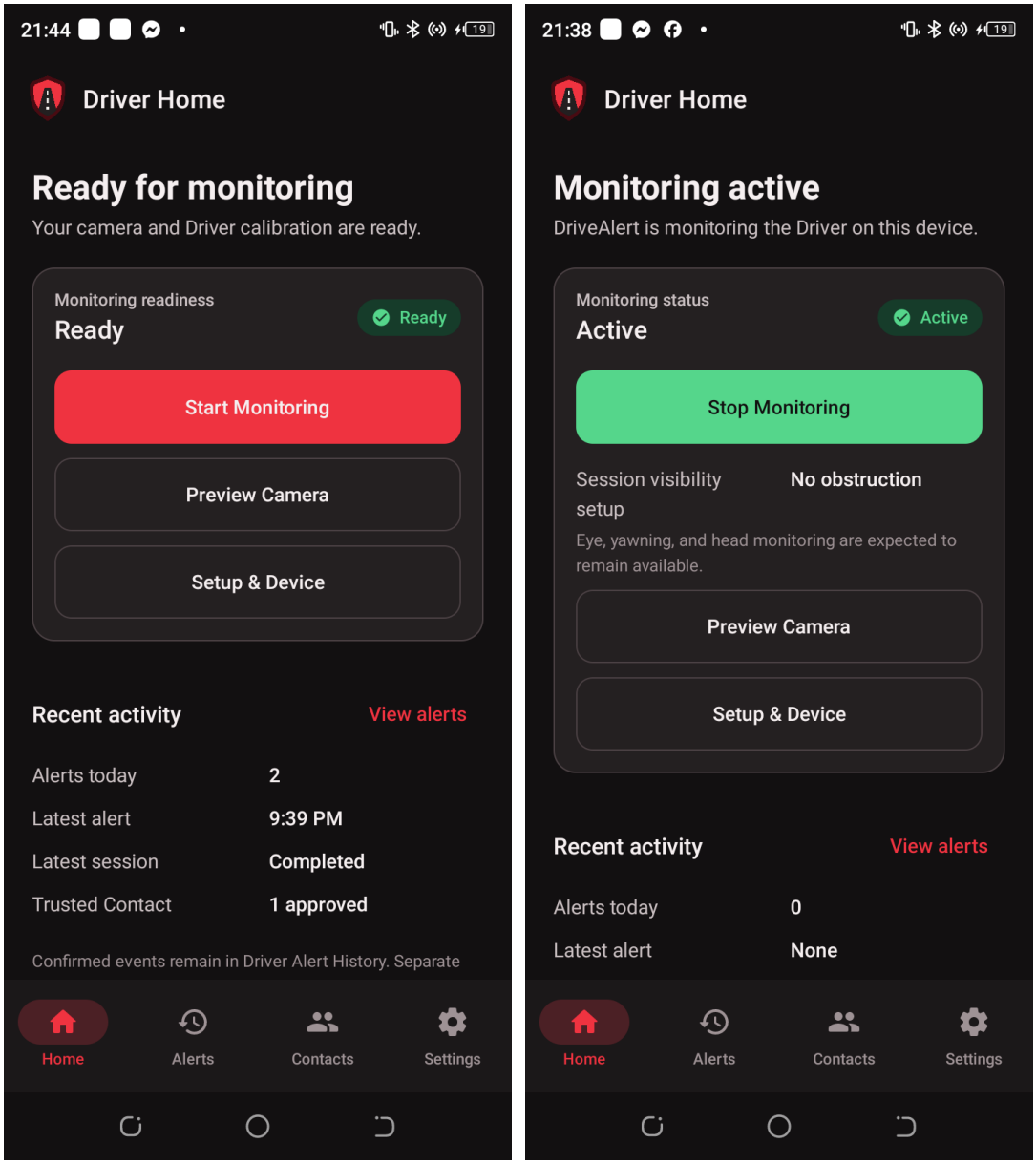


Figure 16: **Driver Home Screen (Driver)**

The Driver Home Screen summarizes monitoring readiness and recent activity. Before a session, it provides controls to start monitoring, preview the camera, and open Setup and Device. During a session, it shows the active monitoring status and Stop Monitoring control. Recent activity includes alert counts, the latest alert and session, and approved Trusted Contacts.

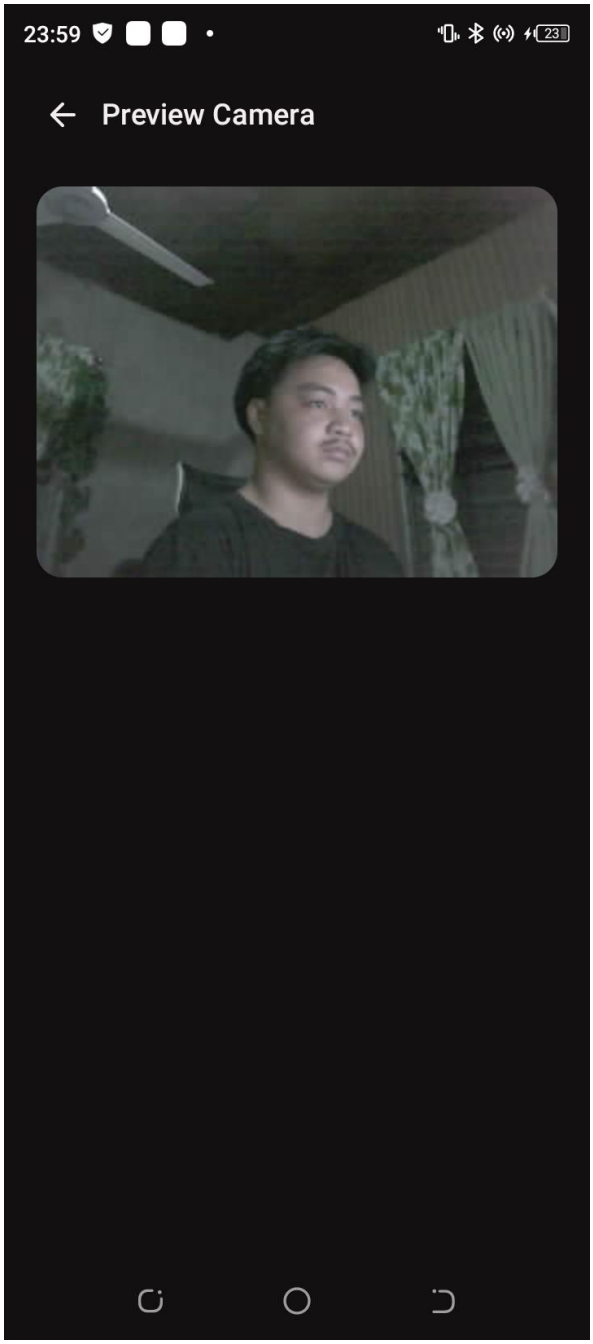


Figure 17: **Camera Preview Screen (Driver)**

The Camera Preview Screen displays the live video received from the connected DriveAlert camera. The driver can use the preview to check the camera view before returning to Driver Home. The back button returns the driver to the previous screen.

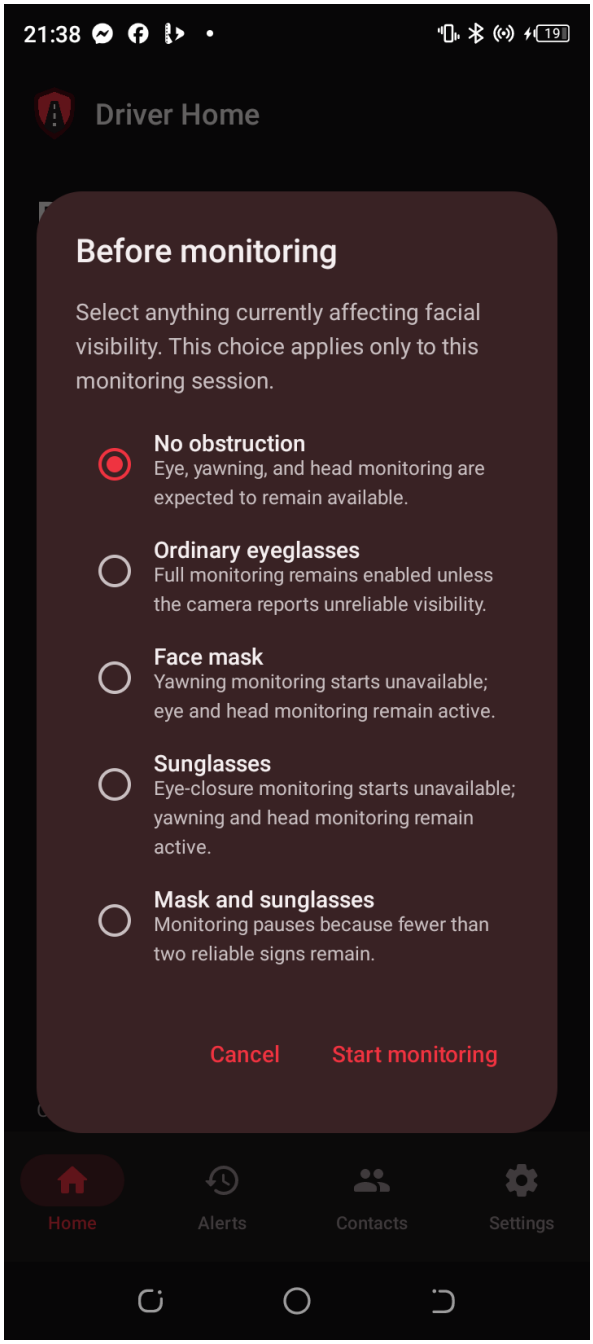


Figure 18: **Before Monitoring Dialog (Driver)**

The Before Monitoring dialog allows the driver to identify conditions that may affect facial visibility for the current session. The available options include no obstruction, ordinary eyeglasses, a face mask, sunglasses, or a mask with sunglasses. Each option explains its effect on monitoring availability. The driver can start monitoring or cancel.

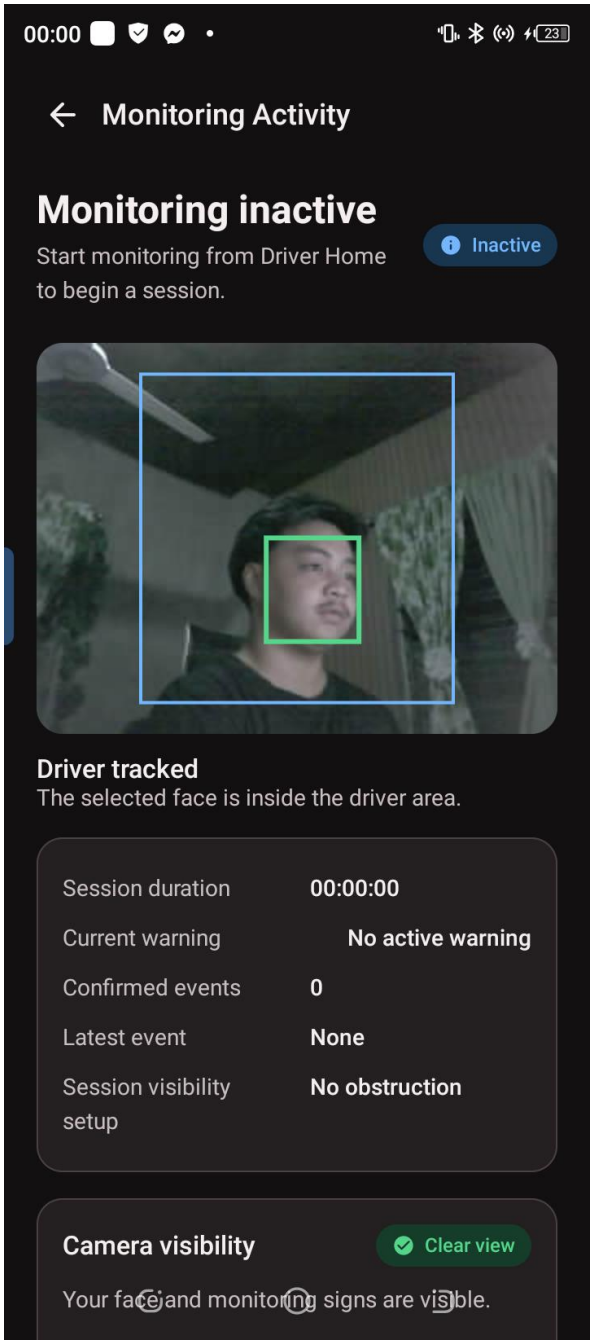


Figure 19: **Monitoring Activity Screen (Driver)**

The Monitoring Activity Screen displays the monitoring status and live camera preview. It shows driver tracking, session duration, the current warning, confirmed events, the latest event, and camera visibility. When monitoring is inactive, the screen directs the driver to start a session from Driver Home.

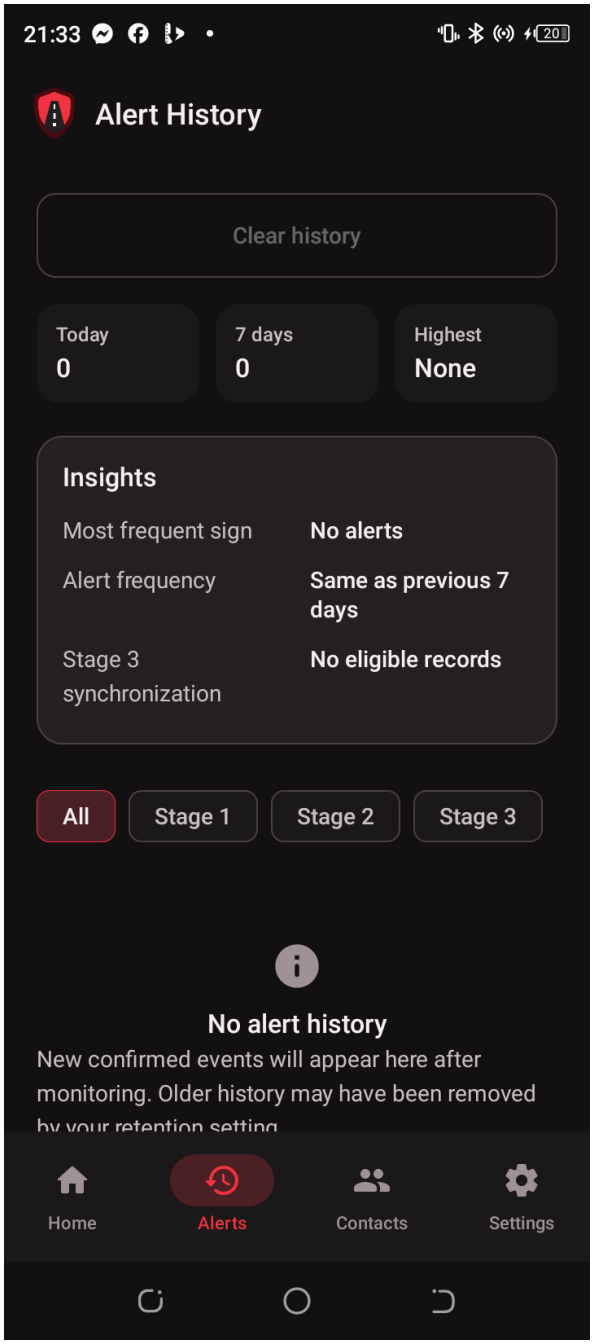


Figure 20: **Alert History Screen (Driver)**

The Alert History Screen presents saved confirmed warning events and a summary of recent alert activity. It shows alert counts for today and the past seven days, the highest recorded Warning Stage, and related insights. The driver can filter records by Warning Stage. When no records are available, the screen displays the No alert history message.

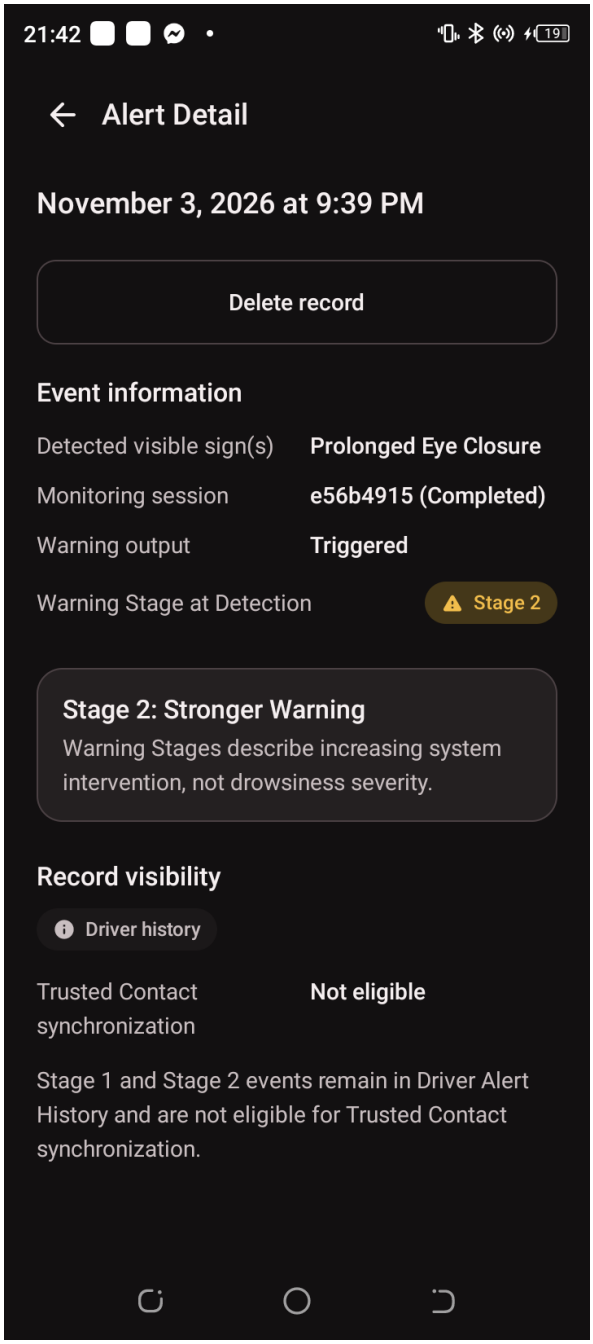


Figure 21: **Alert Detail Screen (Driver)**

The Alert Detail Screen displays a saved confirmed warning event. It shows the date and time, detected visible sign, monitoring session, warning output, and Warning Stage at detection. The screen also explains the record visibility and Trusted Contact synchronization eligibility. The driver can return to the alert list or delete the record.

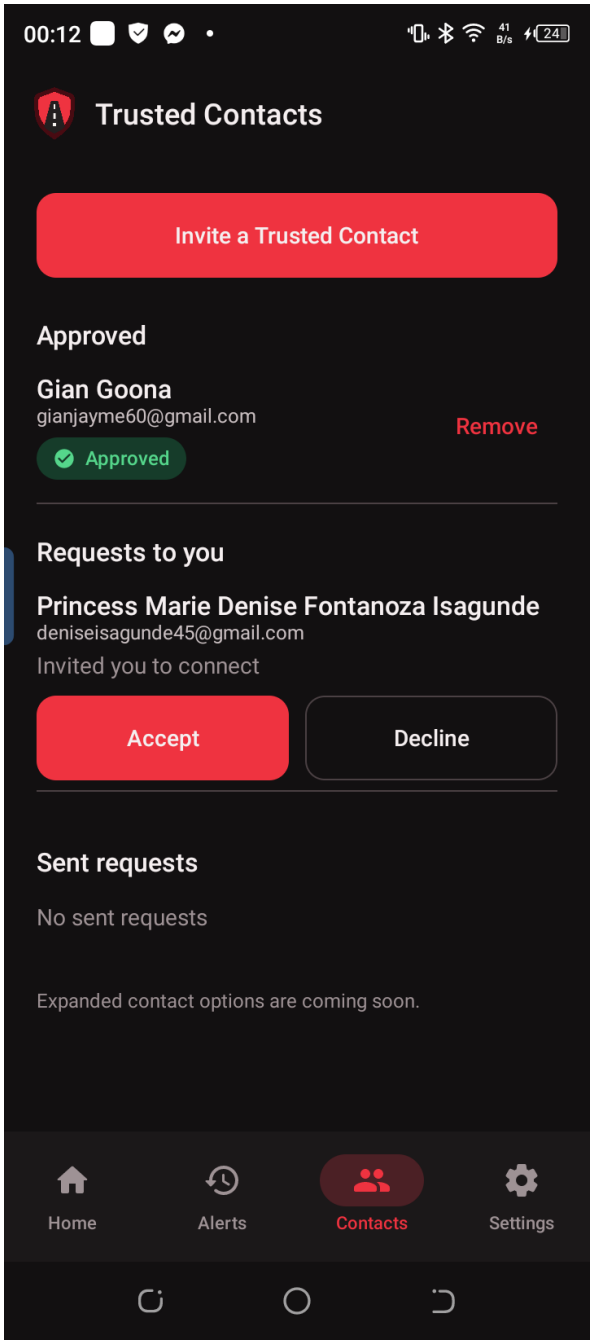


Figure 22: **Trusted Contacts Screen (Driver)**

The Trusted Contacts Screen allows the driver to review approved Trusted Contacts, incoming connection requests, and sent requests. Incoming requests display the account name and email address, together with Accept and Decline buttons. The driver can approve a connection, remove an approved contact, or select Invite a Trusted Contact to add another person.

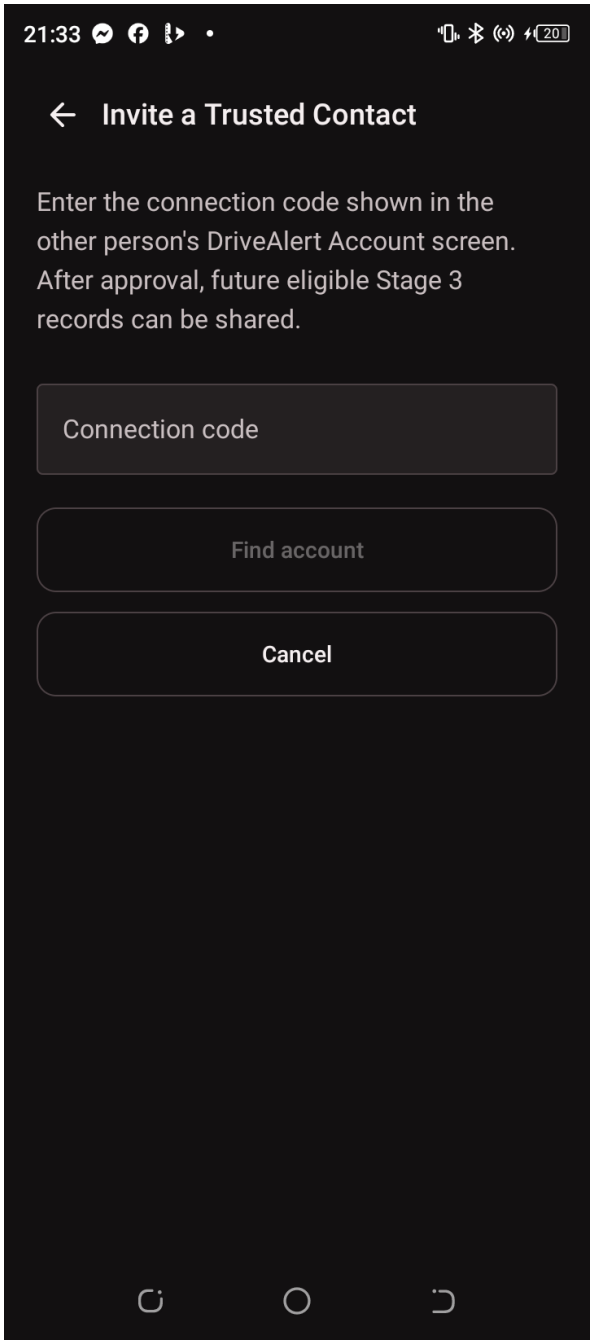


Figure 23: **Invite a Trusted Contact Screen (Driver)**

The Invite a Trusted Contact Screen allows the driver to find another DriveAlert account using its connection code. The driver enters the code displayed on the other person's Account Screen and selects Find account. After the connection is approved, future eligible Stage 3 records can be shared with that Trusted Contact.

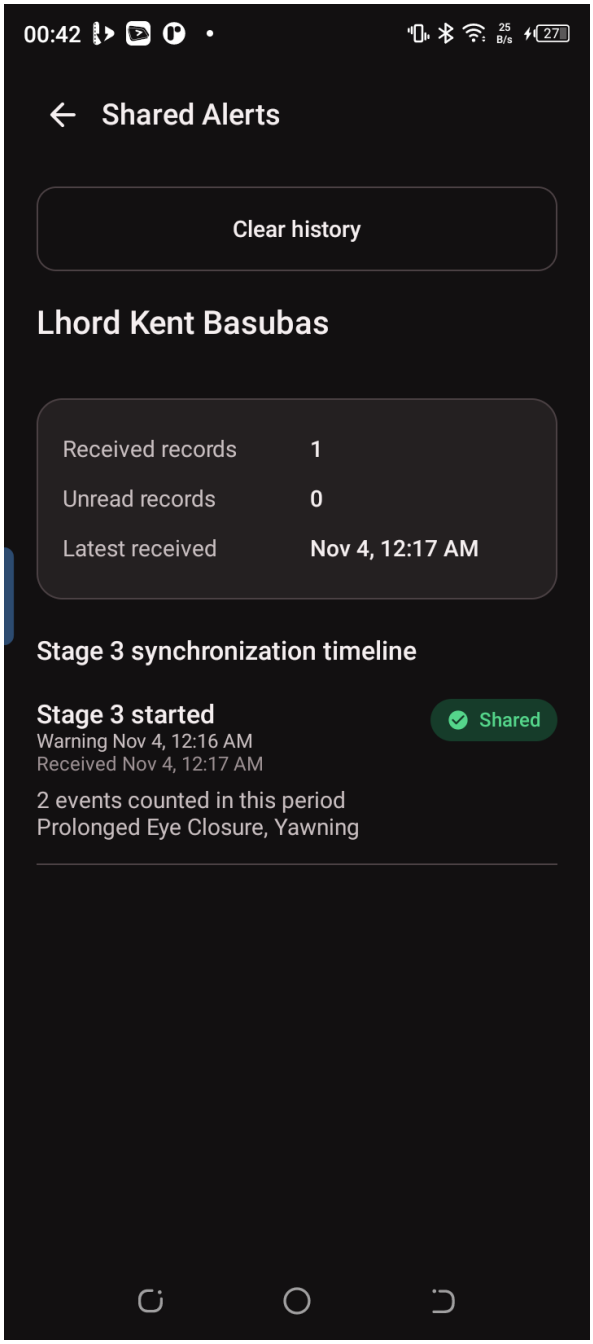


Figure 24: **Shared Alerts Screen (Trusted Contact)**

The Shared Alerts Screen allows a Trusted Contact to review eligible Stage 3 records shared by a connected driver. It displays the driver's name, received and unread record counts, and the latest received date and time. The Stage 3 synchronization timeline shows when a warning started, when the record was received, its sharing status, the number of events counted in the period, and the detected visible signs.

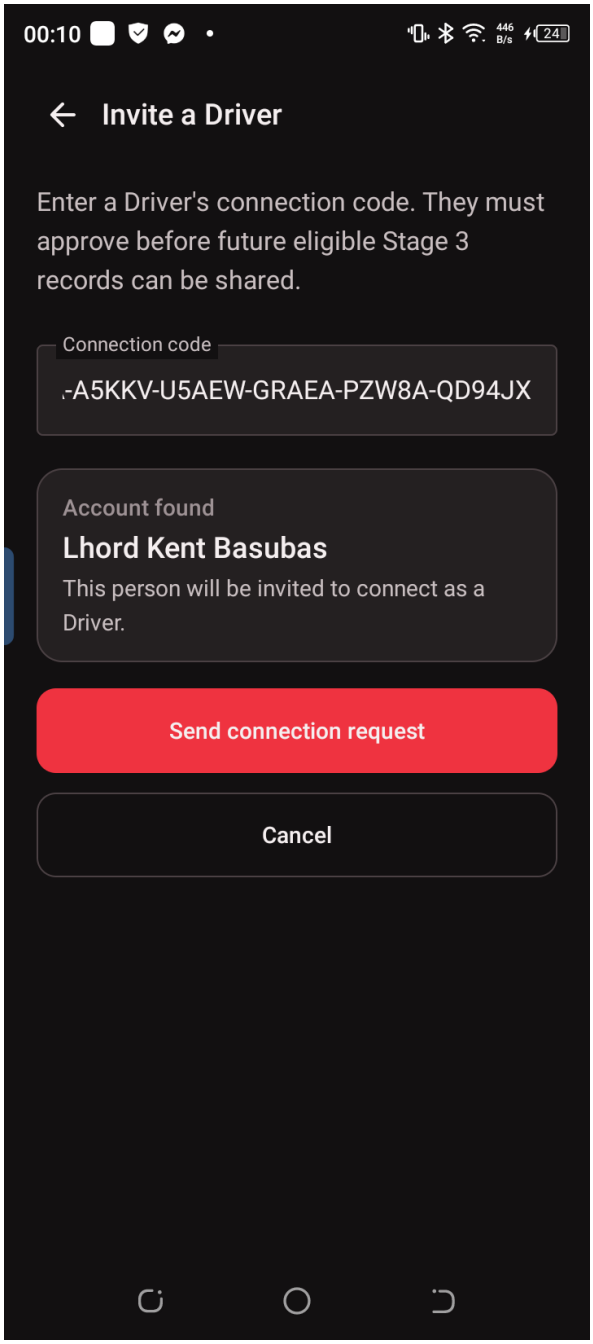


Figure 25: **Invite a Driver Screen (Trusted Contact)**

The Invite a Driver Screen allows a Trusted Contact to find a driver's account using its connection code. After an account is found, the screen displays the account name for verification. The user can select Send connection request to invite the driver or Cancel to return. The driver must approve the connection before future eligible Stage 3 records can be shared.

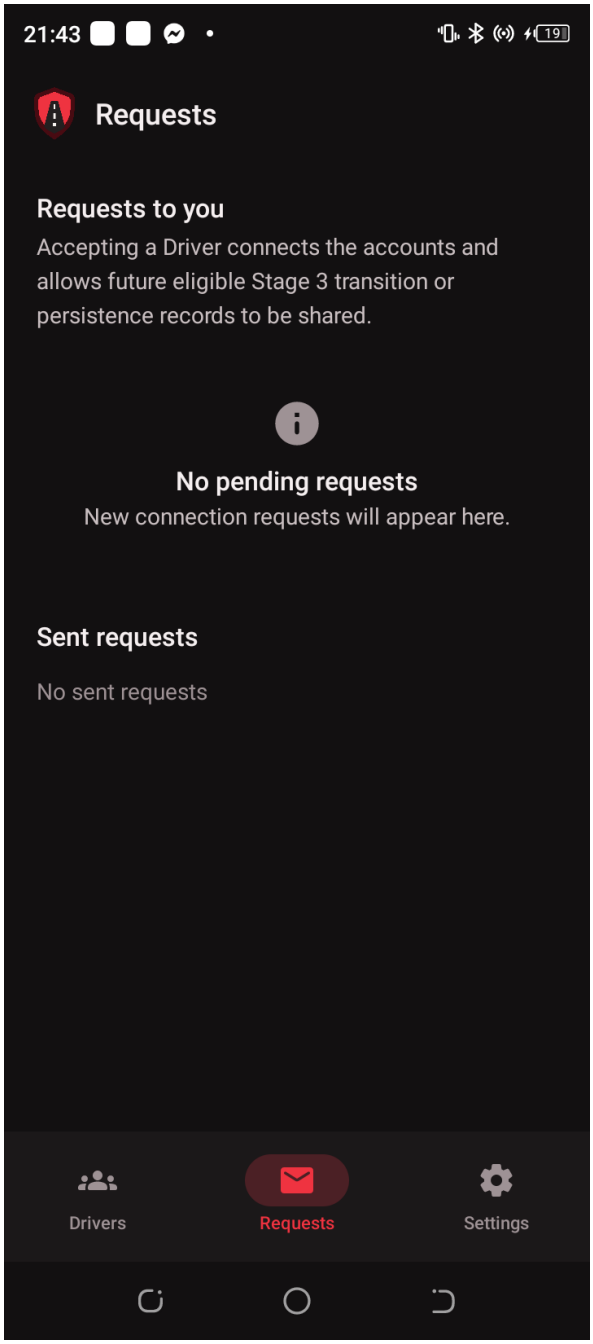


Figure 26: **Requests Screen (Trusted Contact)**

The Requests Screen allows a Trusted Contact to review incoming and sent connection requests. It explains that accepting a driver connects the accounts and allows future eligible Stage 3 transition or persistence records to be shared. When no requests are available, the screen displays separate empty states for incoming and sent requests.

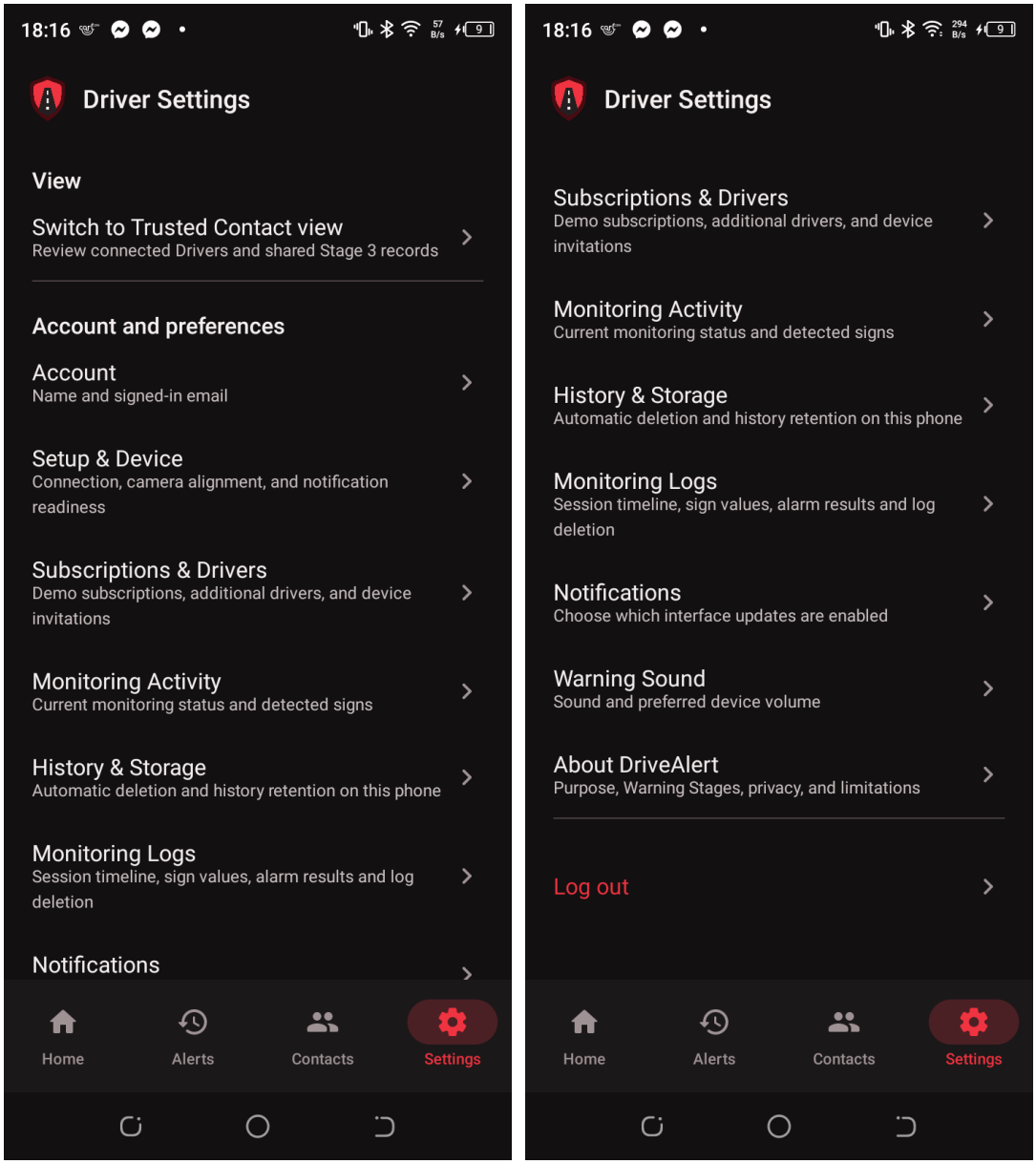


Figure 27: **Driver Settings Screen (Driver)**

The Driver Settings Screen provides access to the driver's account, device setup, monitoring activity, history, monitoring logs, notifications, and warning sound. Subscriptions and Drivers opens device-access management and invitations for additional drivers. The driver can also switch to Trusted Contact view, read information about DriveAlert, or log out.

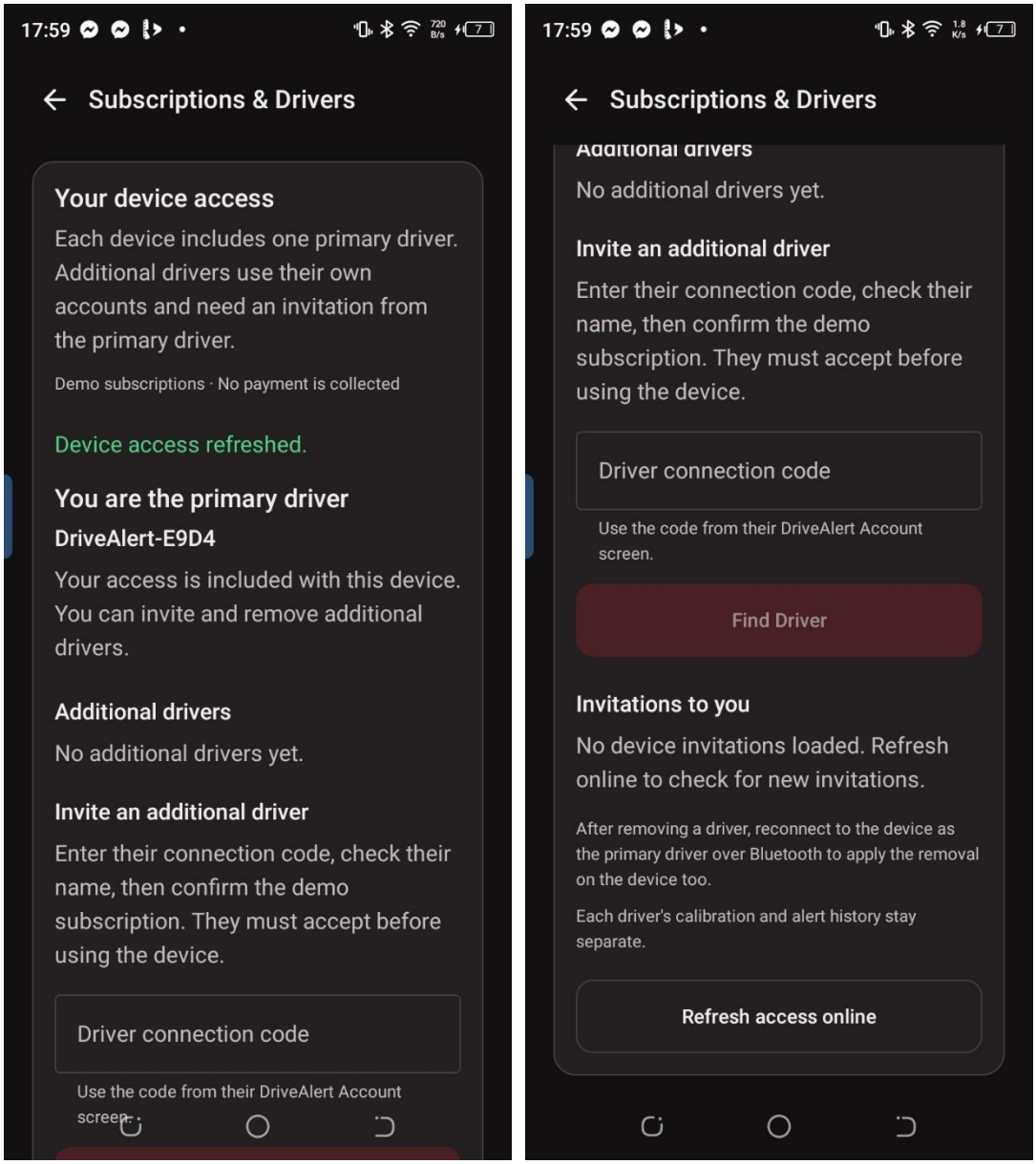


Figure 28: **Subscriptions and Drivers Screen (Driver)**

The Subscriptions and Drivers Screen shows the driver's device access and identifies the primary driver whose access is included with the device. The primary driver can invite additional drivers using their connection codes and refresh access online. The screen explains that subscriptions are simulated, no payment is collected, and each driver's calibration and alert history remain separate.

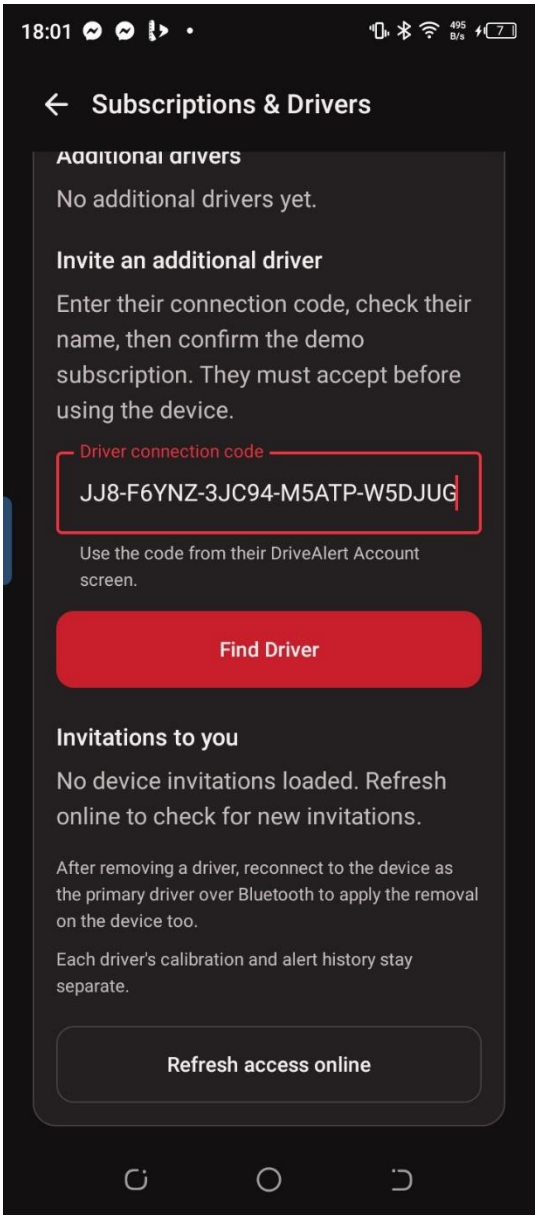


Figure 29: **Invite an Additional Driver Screen (Driver)**

The Invite an Additional Driver section allows the primary driver to enter another user's DriveAlert connection code. Selecting Find Driver looks up the intended account before the primary driver confirms the simulated subscription. The invited driver must accept the invitation before using the device.

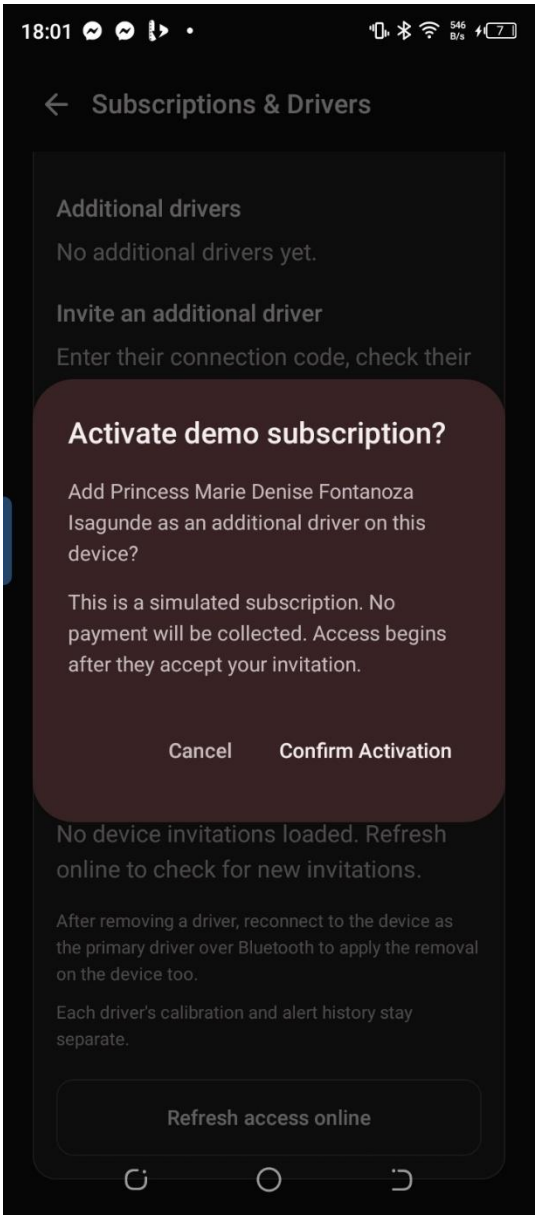


Figure 30: **Demo Subscription Activation Dialog (Driver)**

The Demo Subscription Activation dialog displays the selected account and asks the primary driver to confirm additional-driver access for the device. It clearly states that the subscription is simulated and no payment will be collected. Confirm Activation sends the device invitation, while Cancel returns without confirming it.

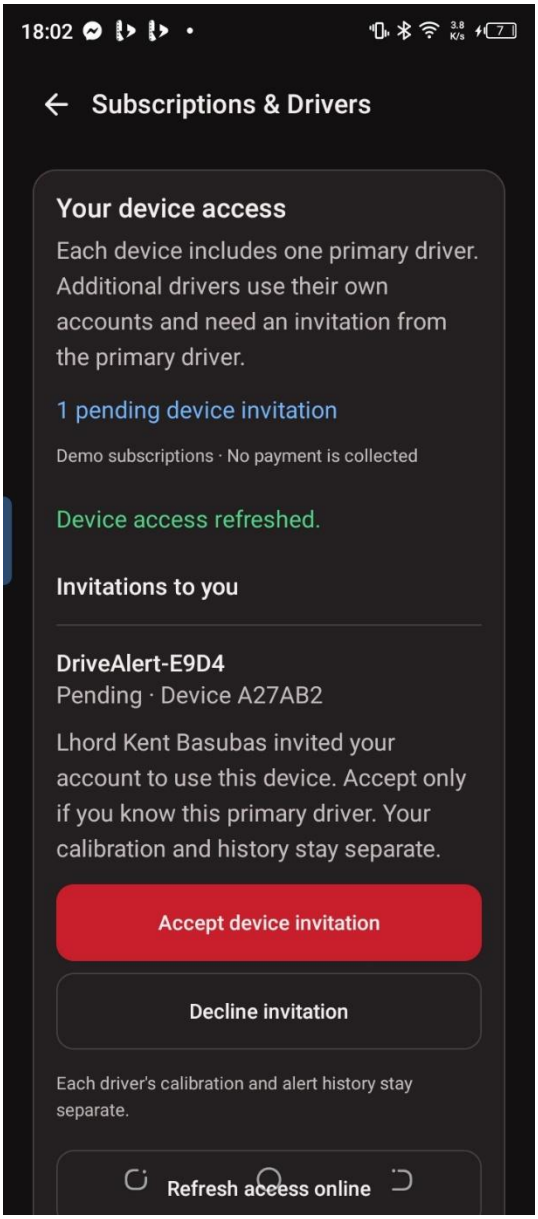


Figure 31: **Device Invitation Screen (Driver)**

The Device Invitation Screen shows a pending invitation received by an additional driver. It identifies the device and the primary driver who sent the invitation. The recipient may accept or decline while online. Accepted access can be saved for later offline device use, while calibration and alert history remain specific to the driver's account.

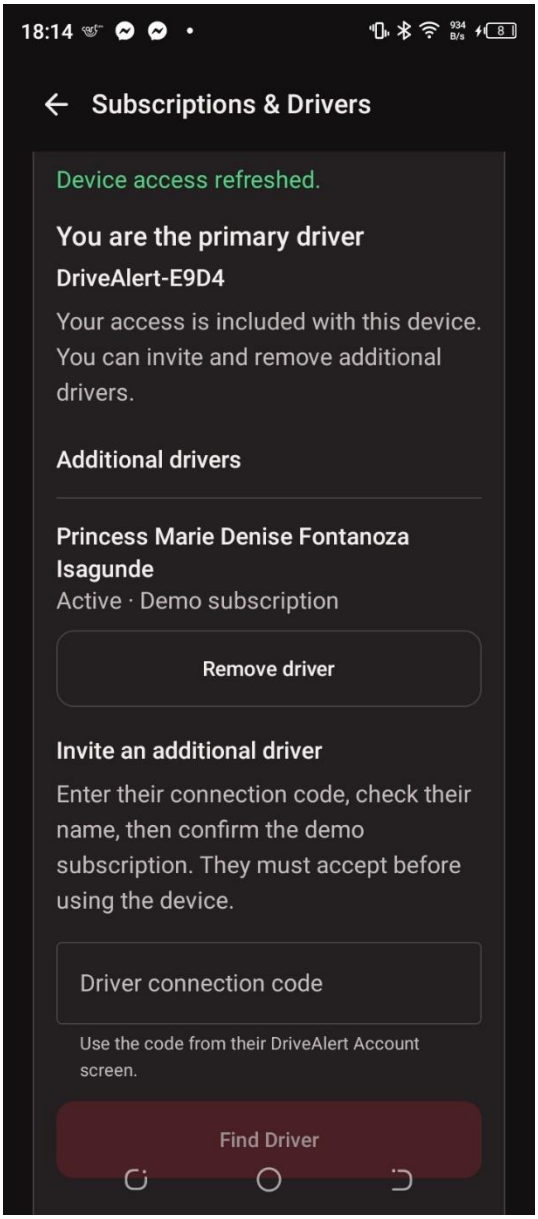


Figure 32: **Active Additional Driver Access Screen (Driver)**

The Active Additional Driver Access Screen shows an accepted account under the primary driver's additional-driver list with an Active demo subscription status. The primary driver can remove access or invite another driver. Removal preserves saved history; the primary driver must reconnect to the device through Bluetooth to synchronize the removal to the device.

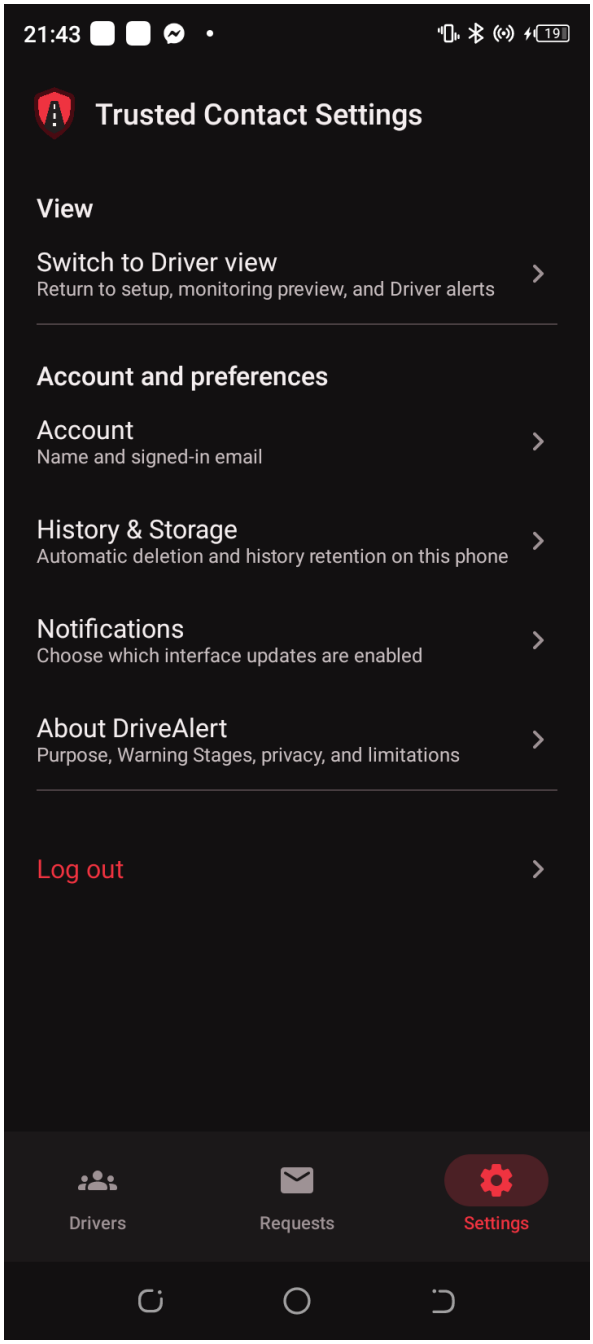


Figure 33: **Trusted Contact Settings Screen (Trusted Contact)**

The Trusted Contact Settings Screen provides access to account and application preferences for Trusted Contact users. It includes options to switch to Driver view, open the account profile, manage history and storage, adjust notifications, and view information about DriveAlert. The user can also log out from this screen.

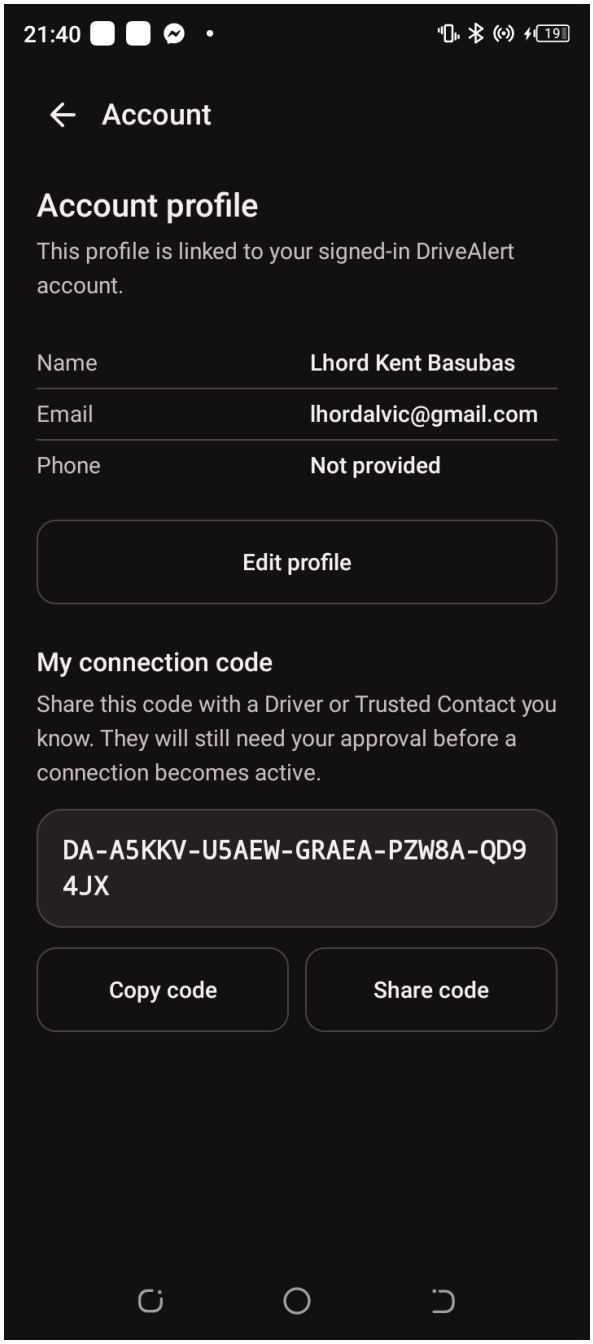


Figure 34: **Account Screen (Driver and Trusted Contact)**

The Account Screen displays the signed-in user's name, email address, and phone information. Users can select Edit profile to update their details. It also shows the account connection code and provides Copy code and Share code options. Connections still require approval before they become active.

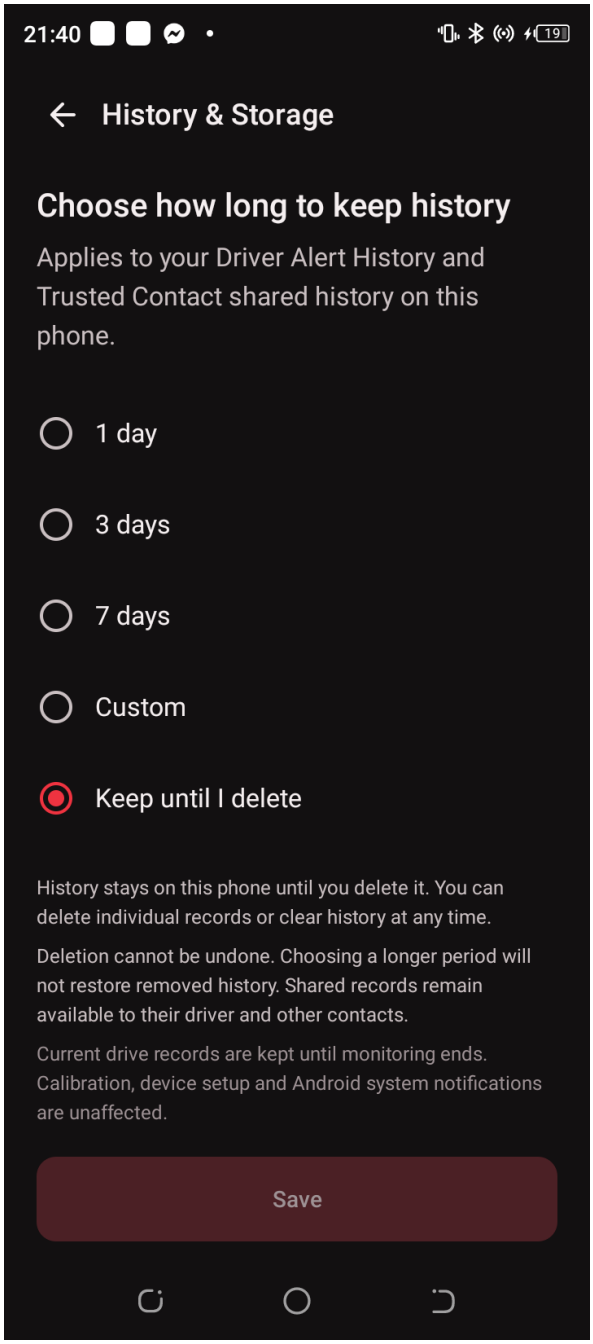


Figure 35: **History and Storage Screen (Driver and Trusted Contact)**

The History and Storage Screen allows users to choose how long Driver Alert History and Trusted Contact shared history remain on the phone. The available options include one day, three days, seven days, a custom period, or keeping history until it is deleted manually. The screen explains the effect of deletion and provides a Save button for applying changes.

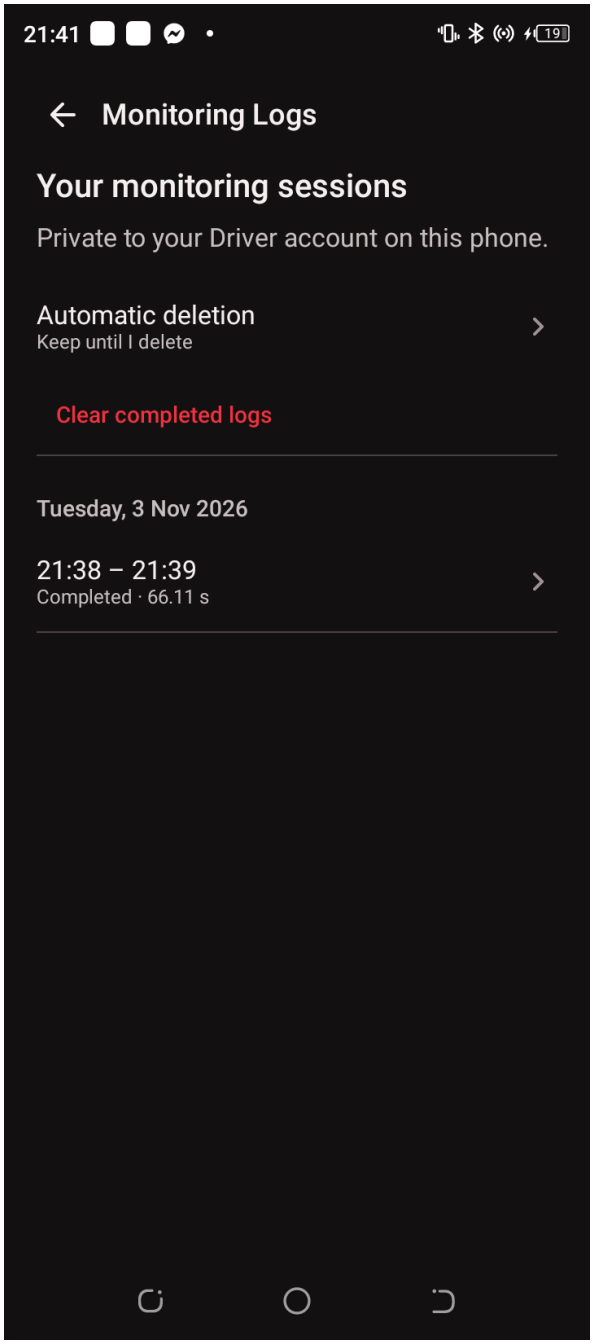


Figure 36: **Monitoring Logs Screen (Driver)**

The Monitoring Logs Screen lists monitoring sessions recorded for the signed-in Driver on the phone. Sessions are grouped by date and show their time and completion status. The driver can open a session to review its timeline, choose automatic deletion settings, or clear completed logs.

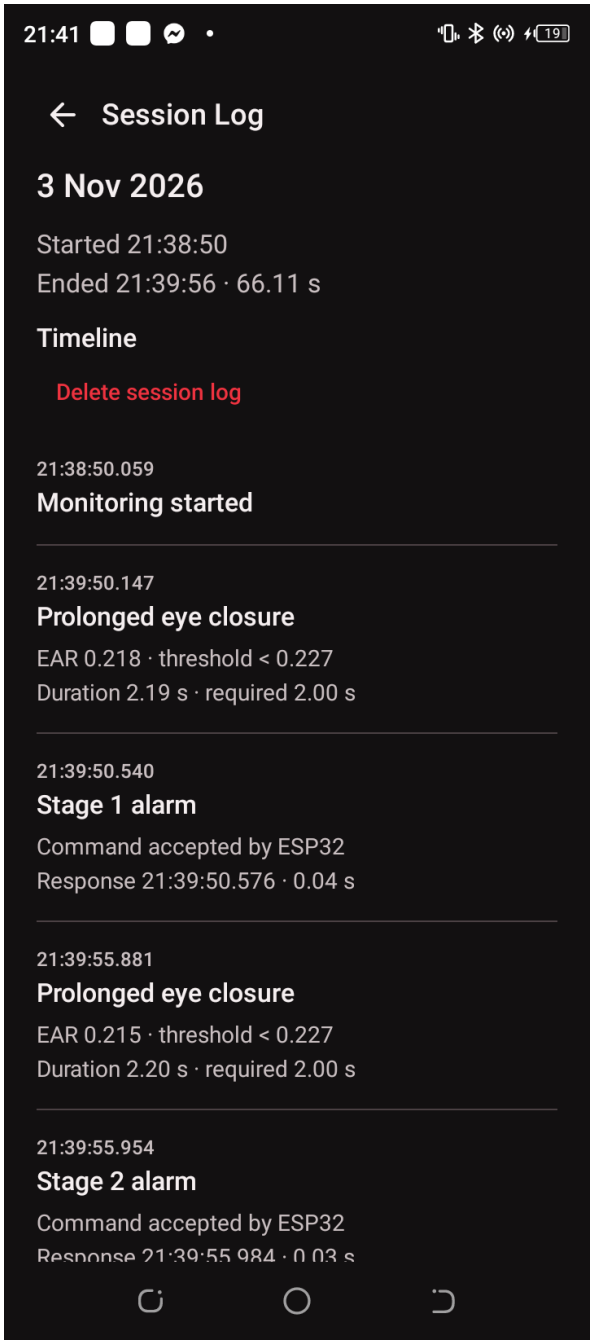


Figure 37: **Session Log Screen (Driver)**

The Session Log Screen displays the timeline of a selected monitoring session. It shows the session date, start and end times, detected signs, recorded measurements, and alarm requests. Each alarm entry indicates the response from the ESP32 device. The driver can review the recorded events or delete a completed session log.

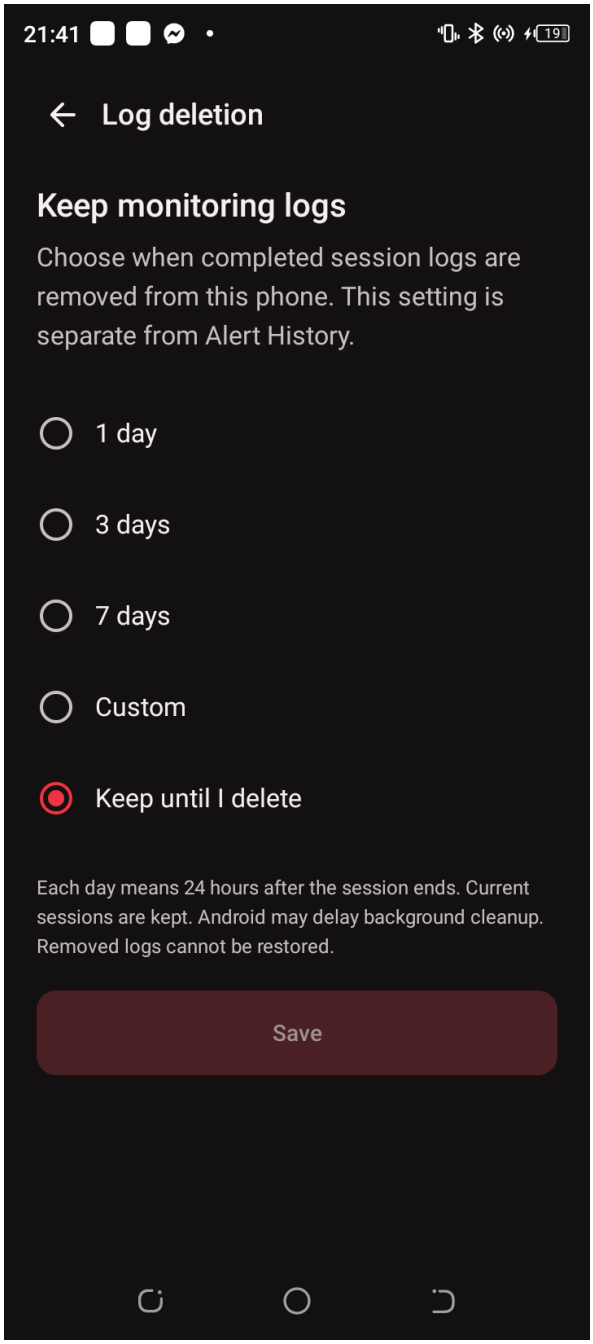


Figure 38: **Log Deletion Screen (Driver)**

The Log Deletion Screen controls how long completed monitoring-session logs remain on the phone. The driver can choose one day, three days, seven days, a custom period, or manual deletion. The screen explains that this setting is separate from Alert History and that current sessions are kept. Changes can be applied using Save.

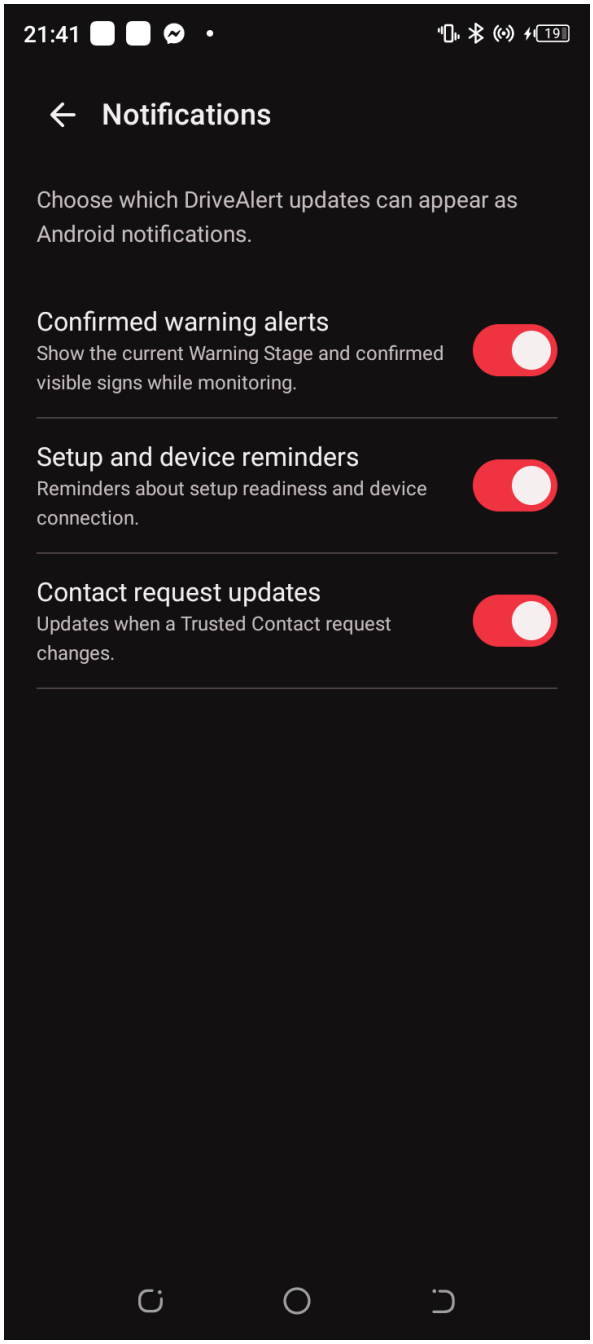


Figure 39: **Driver Notifications Screen (Driver)**

The Driver Notifications Screen allows the driver to choose which DriveAlert updates can appear as Android notifications. It provides separate switches for confirmed warning alerts, setup and device reminders, and contact request updates. Each setting includes a short explanation of the information it controls.

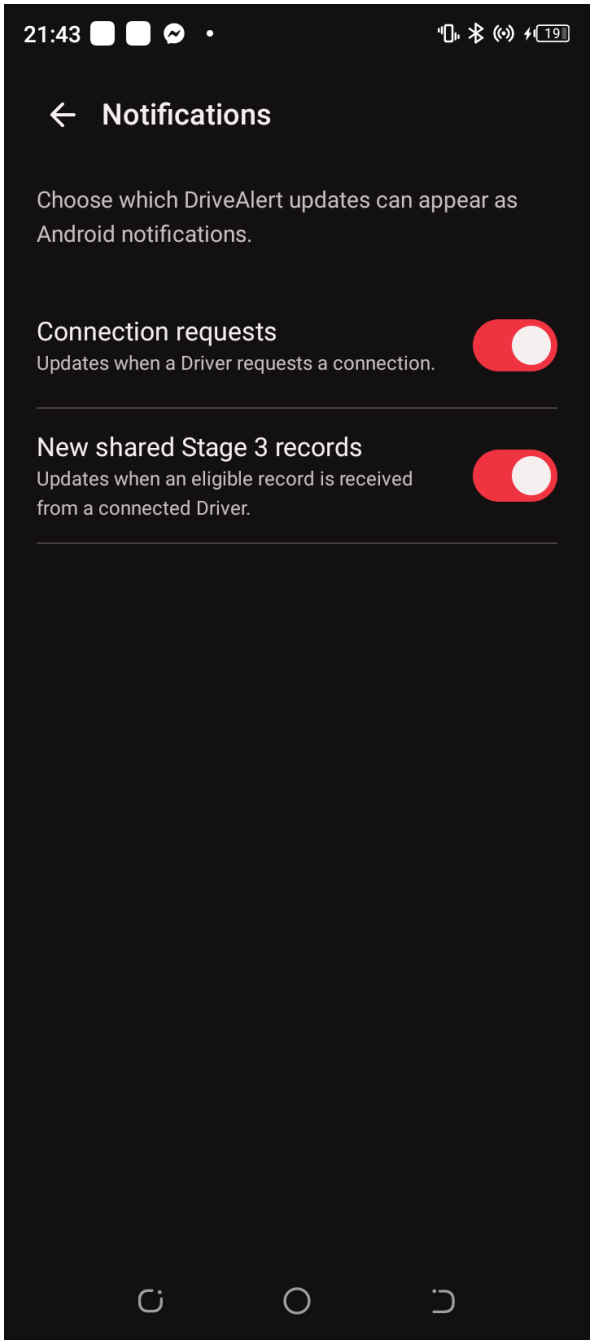


Figure 40: **Trusted Contact Notifications Screen (Trusted Contact)**

The Trusted Contact Notifications Screen allows the user to manage Android notifications related to shared records and connections. It provides switches for connection requests and new shared Stage 3 records. These options control updates about driver connection requests and eligible records received from connected drivers.

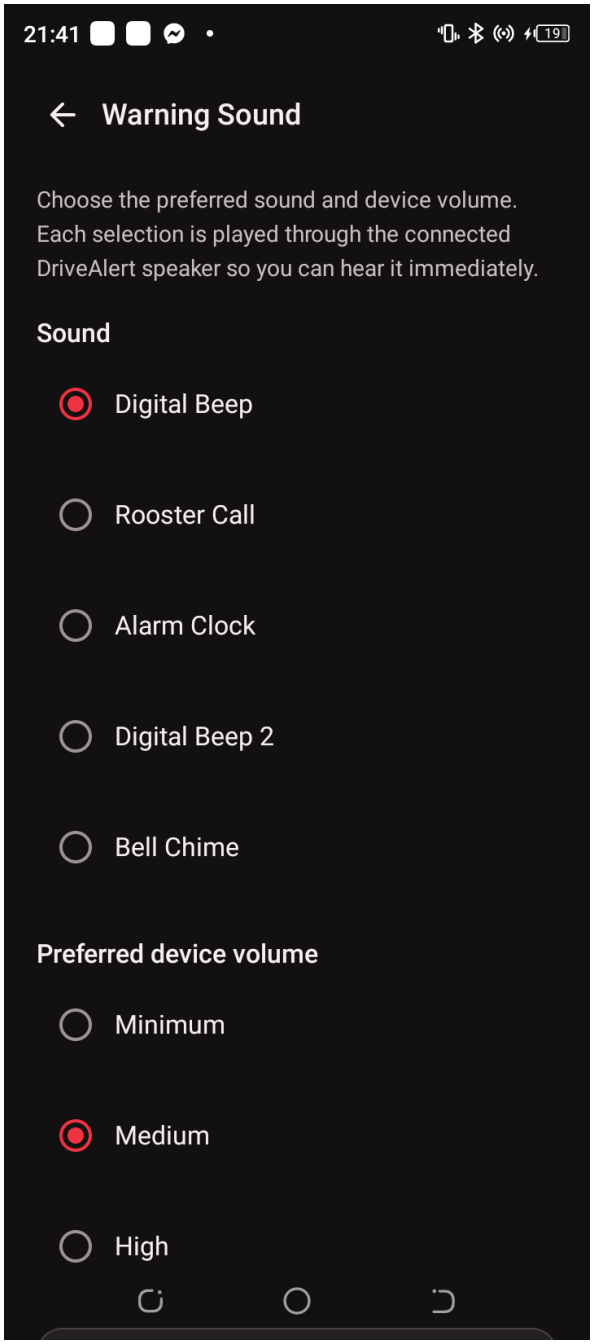


Figure 41: **Warning Sound Screen (Driver)**

The Warning Sound Screen allows the driver to choose a predefined warning sound and the preferred device volume. Available sounds include Digital Beep, Rooster Call, Alarm Clock, Digital Beep 2, and Bell Chime. The volume options are Minimum, Medium, and High. Each selection is played through the connected DriveAlert speaker for an immediate preview.

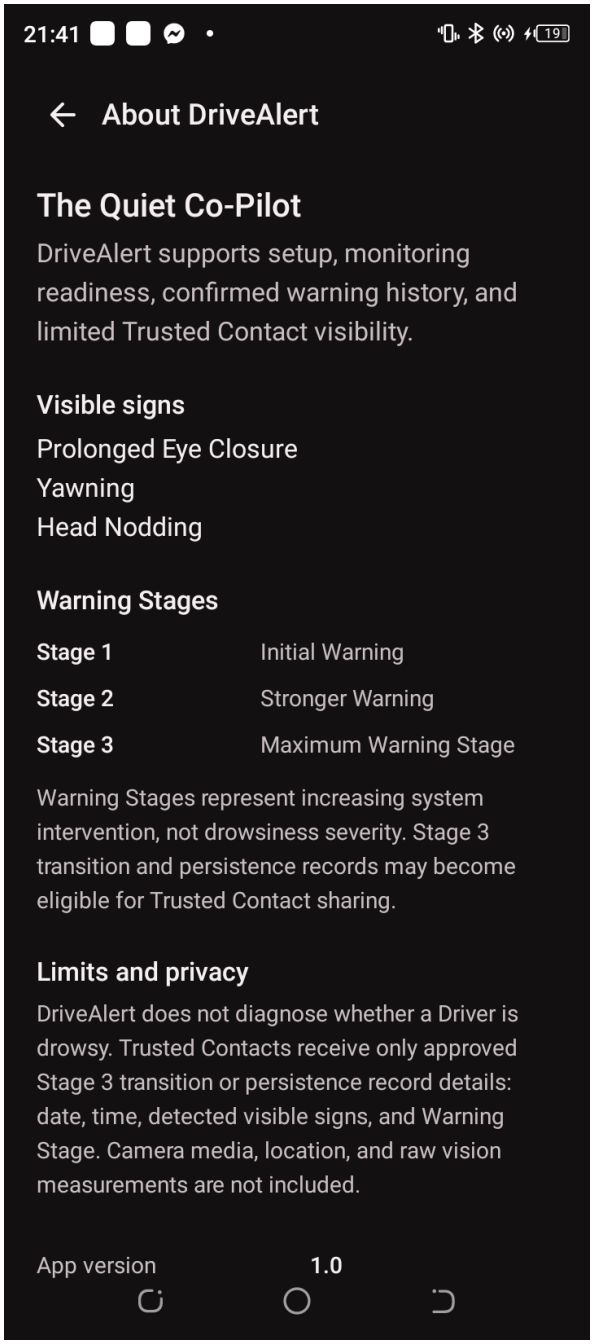


Figure 42: **About DriveAlert Screen (Driver and Trusted Contact)**

The About DriveAlert Screen explains the application's purpose, monitored visible signs, Warning Stages, privacy, and limitations. It identifies prolonged eye closure, yawning, and head nodding as the selected signs. The screen also explains that Warning Stages represent increasing system intervention and that approved Trusted Contacts receive only eligible Stage 3 record details.

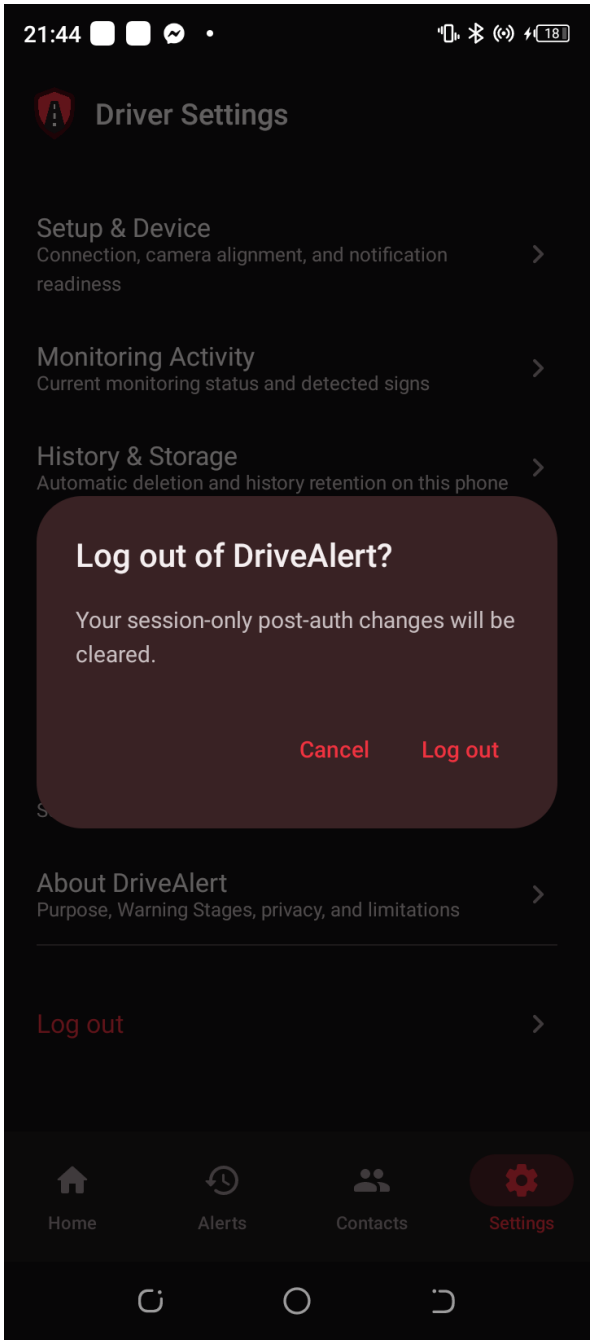


Figure 43: **Log Out Confirmation Dialog (Driver and Trusted Contact)**

The Log Out Confirmation dialog appears when a user selects Log out from Settings. It asks the user to confirm leaving the signed-in session and explains that session-only changes will be cleared. The user can select Log out to continue or Cancel to remain signed in.

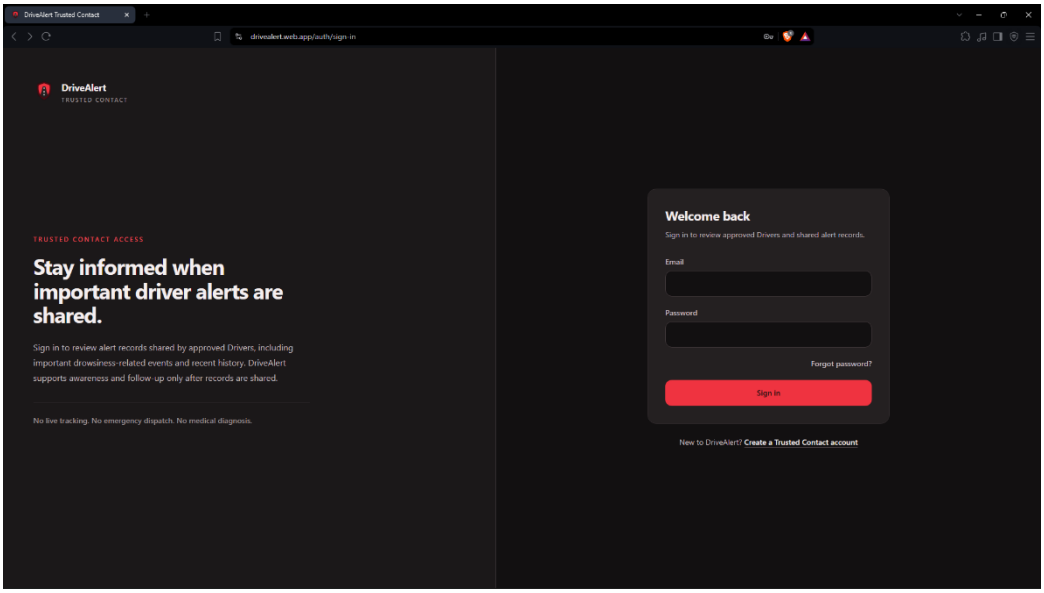


Figure 44: **Sign In Screen (Web)**

The Sign In Screen allows existing users to access the DriveAlert web portal using their email address and password. Users can select Forgot password to recover account access or create a Trusted Contact account through the registration link. After signing in, users can review approved drivers and shared alert records.

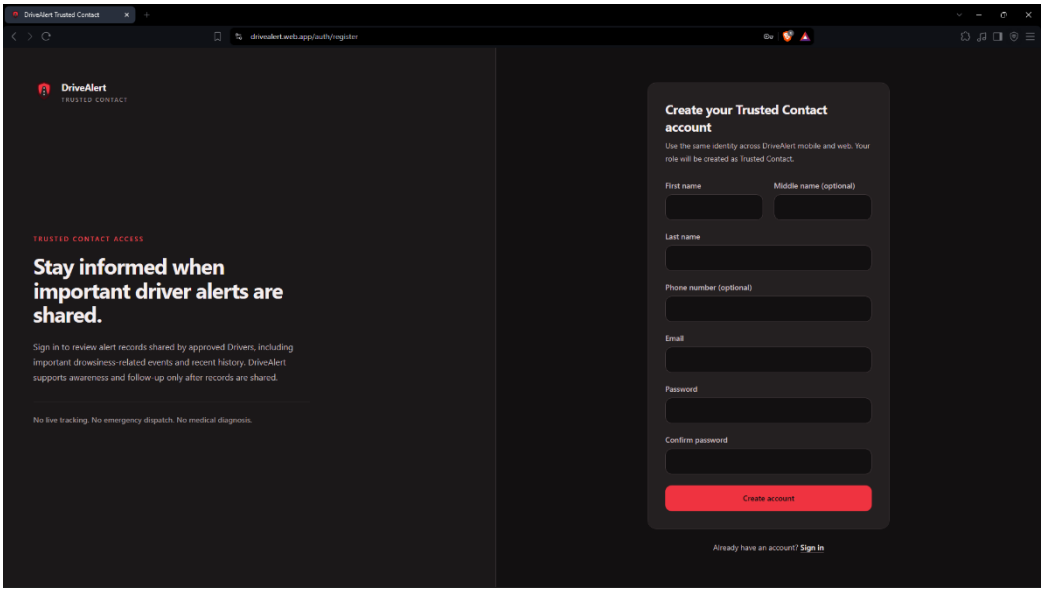


Figure 45: **Create Account Screen (Web)**

The Create Account Screen allows new users to register as Trusted Contacts. It includes fields for first name, optional middle name, last name, optional phone number, email address, password, and password confirmation. Users can select Create account to register or Sign in to access an existing account.

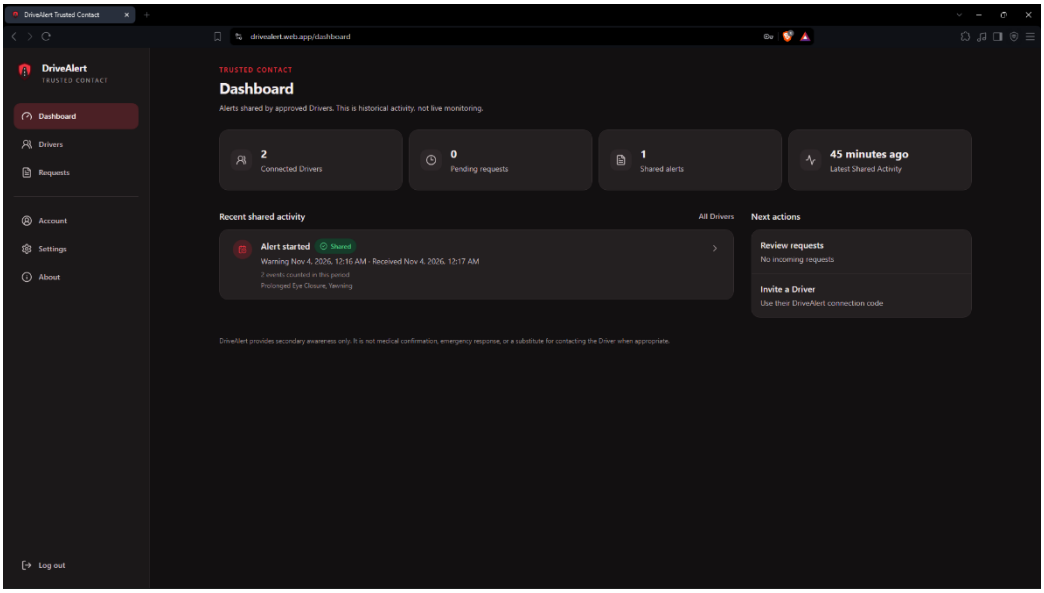


Figure 46: **Dashboard Screen (Web)**

The Dashboard Screen summarizes connected drivers, pending requests, shared alerts, and the latest shared activity. It lists recent shared records with their warning and received times, event counts, and detected signs. Users can review requests or invite a driver. The displayed activity represents shared historical records rather than live monitoring.

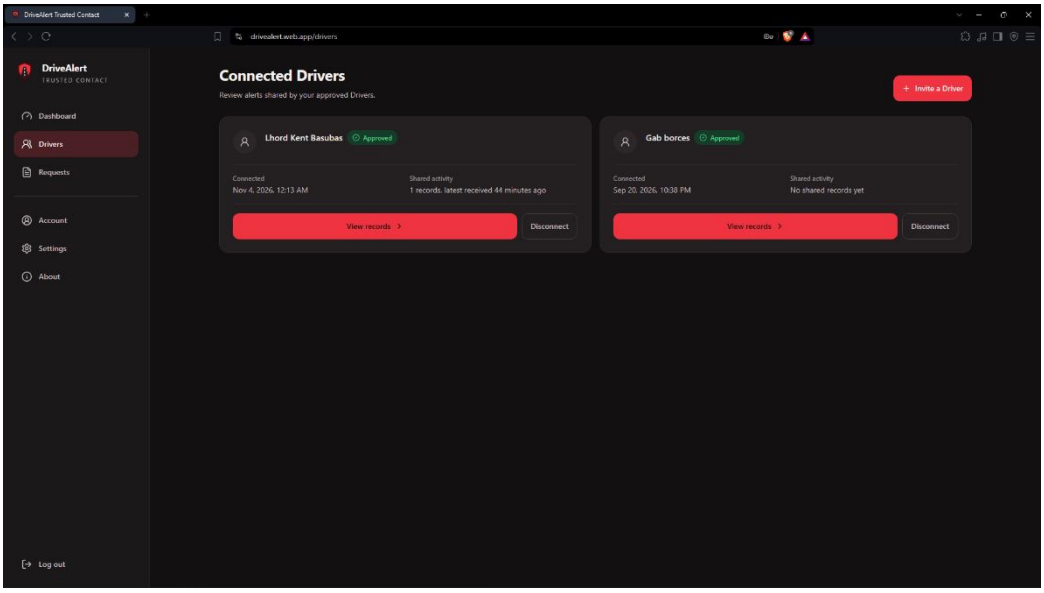


Figure 47: **Connected Drivers Screen (Web)**

The Connected Drivers Screen lists drivers whose connections have been approved. Each entry displays the driver's name, approval status, connection date, and shared-record activity. Users can select View records to open a driver's shared history, Disconnect to end a connection, or Invite a Driver to begin another connection.

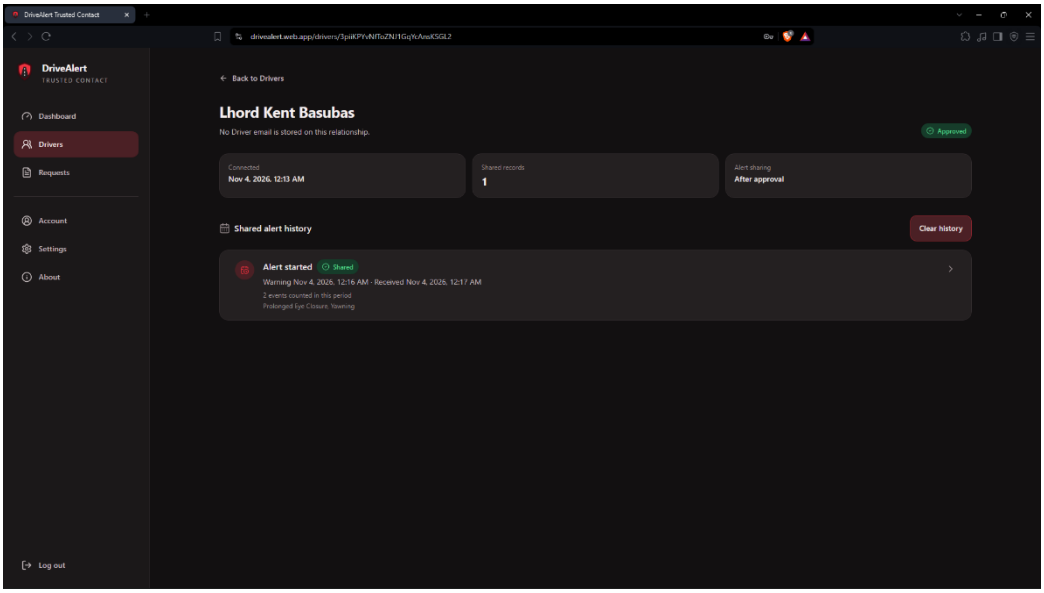


Figure 48: **Shared Alert History Screen (Web)**

The Shared Alert History Screen displays records received from a selected driver. It shows the driver's name, approval status, connection date, and shared-record count. Each history entry includes the warning time, received time, sharing status, event count, and detected signs. Users can open a record, clear the history, or return to Drivers.

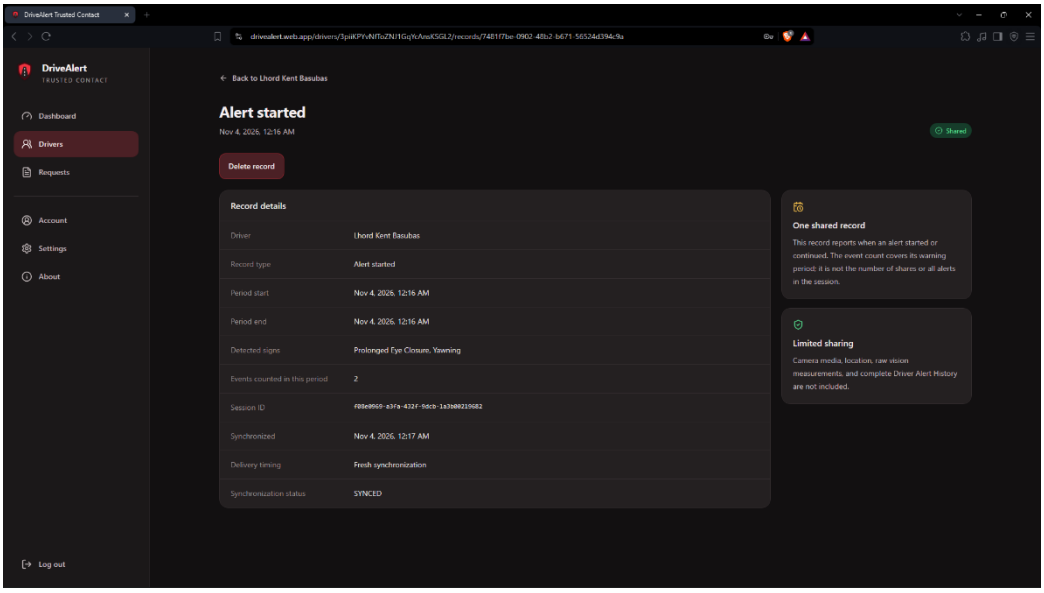


Figure 49: **Shared Alert Detail Screen (Web)**

The Shared Alert Detail Screen presents a selected shared record. It includes the driver, record type, period start and end, detected signs, event count, session identifier, and synchronization information. It explains that the event count covers the warning period. Users can delete the record or return to the driver's shared history.

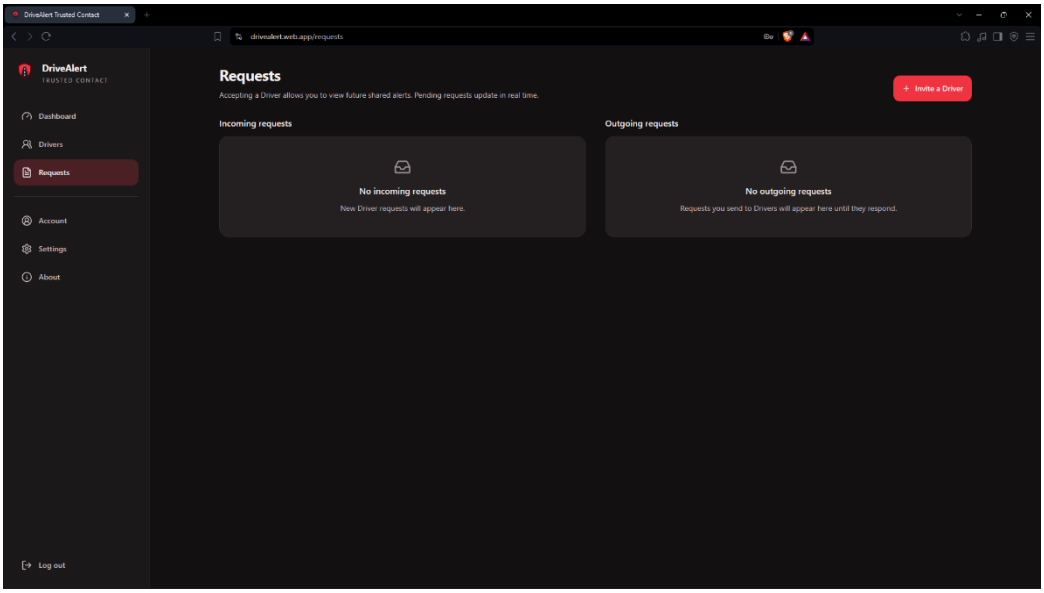


Figure 50: **Requests Screen (Web)**

The Requests Screen separates incoming and outgoing driver connection requests. It explains that accepting a driver allows future shared alerts to be viewed. Empty panels indicate when no requests are available. Users can select Invite a Driver to begin a new connection request.

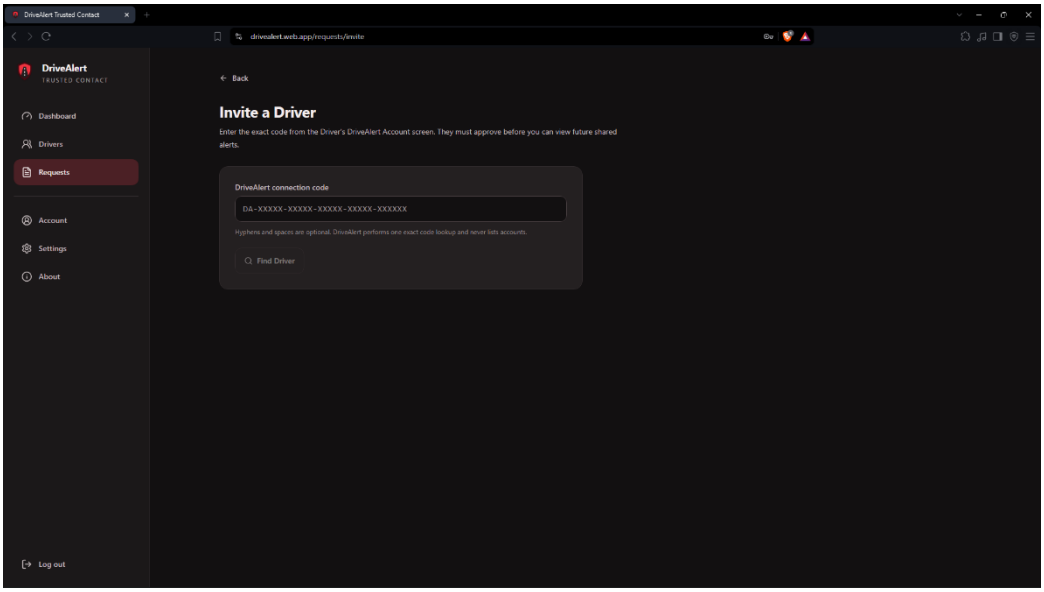


Figure 51: **Invite a Driver Screen (Web)**

The Invite a Driver Screen allows users to find a driver using the connection code displayed on the driver's Account Screen. Users enter the code and select Find Driver to look up the account. The driver must approve the connection before future shared alerts can be viewed. The Back link returns to the previous screen.

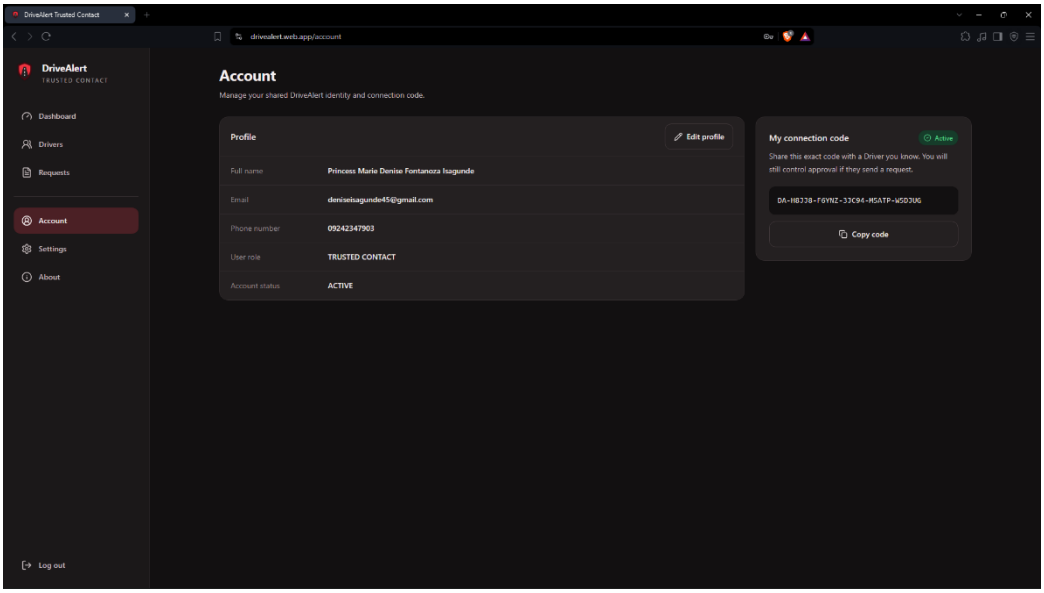


Figure 52: **Account Screen (Web)**

The Account Screen displays the user's full name, email address, phone number, role, and account status. Users can select Edit profile to update their information. The connection-code panel displays the active account code and a Copy code button for use when connecting with a driver.

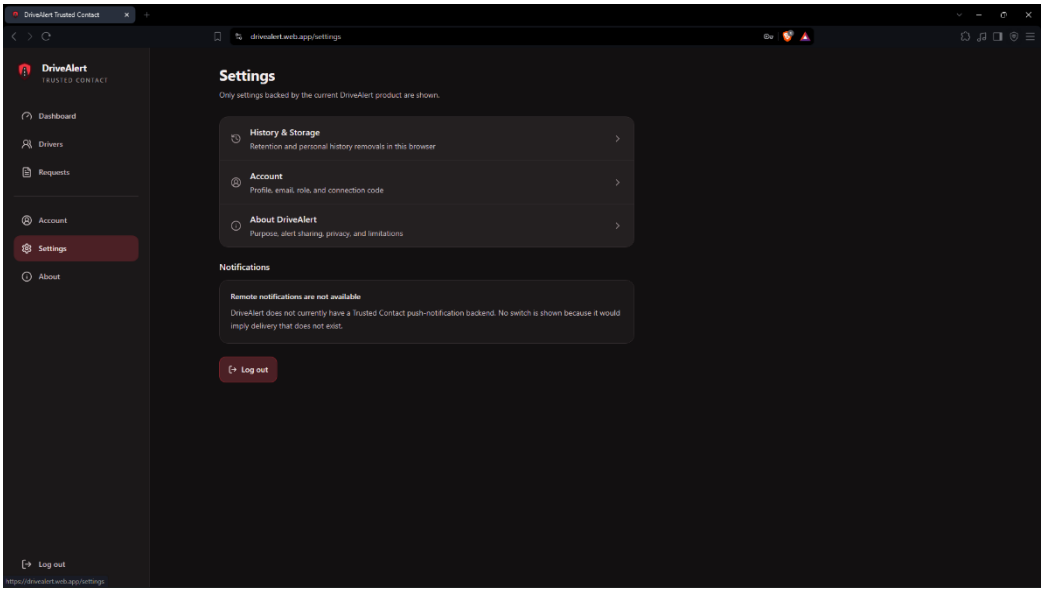


Figure 53: **Settings Screen (Web)**

The Settings Screen provides access to History and Storage, Account, and About DriveAlert. It also includes a Log out button. The notification area explains that remote notifications are currently unavailable, allowing users to distinguish the portal's supported settings from features that are not available.

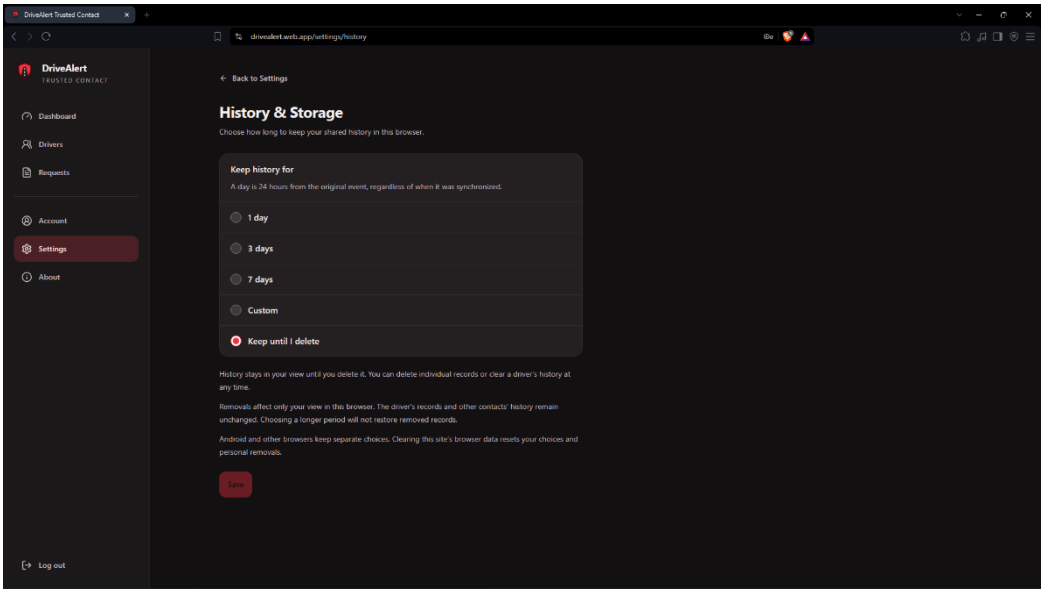


Figure 54: **History and Storage Screen (Web)**

The History and Storage Screen lets users choose how long shared history remains visible in the current browser. Options include one day, three days, seven days, a custom period, or manual deletion. Users can save their choice. The screen explains that removals affect only their browser view and do not restore deleted history.

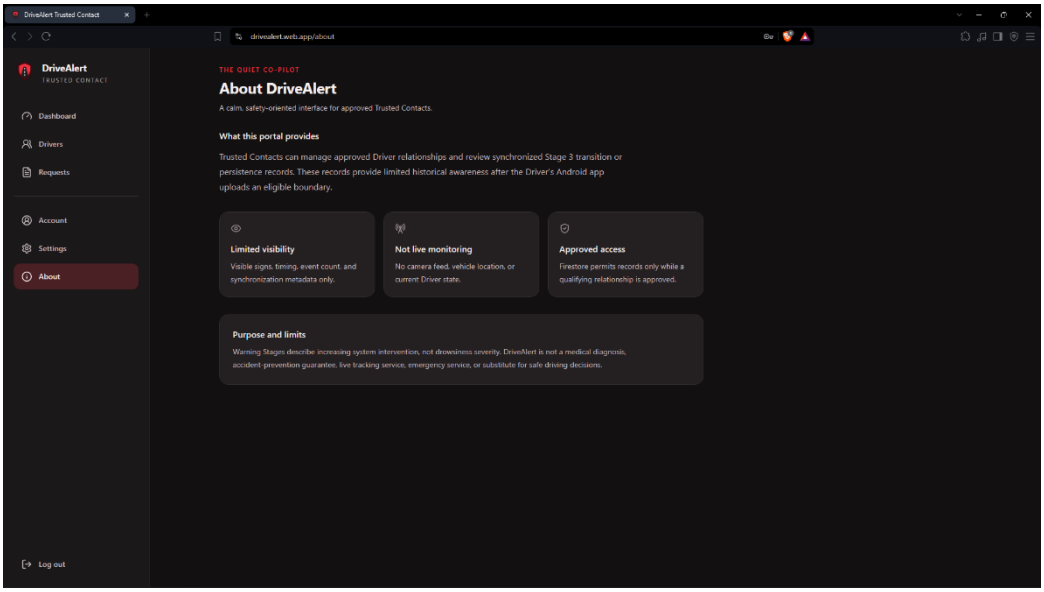


Figure 55: **About DriveAlert Screen (Web)**

The About DriveAlert Screen explains the web portal's purpose, approved access, privacy, and limitations. It describes the sharing of eligible Stage 3 transition or persistence records and the limited information available to Trusted Contacts. It also clarifies that the portal does not provide live camera monitoring, vehicle location, or medical diagnosis.



Figure 56: **Driver Storyboard**

Figure 56 illustrates the DriveAlert Driver workflow. Users sign in or register, then prepare the camera connection, alignment, and calibration before opening Driver Home. The Driver can preview the camera, select a facial-visibility condition, and start or stop monitoring. Alert History presents recorded warnings, while Alert Detail displays the selected event and sharing eligibility. Trusted Contacts allows the Driver to invite contacts and manage connection requests. Driver Settings provides access to Account, Monitoring Activity, History and Storage, Monitoring Logs, Notifications, Warning Sound, and About DriveAlert. Session Log displays recorded session events, while Log Deletion controls completed-log retention. Approved contacts may receive eligible Stage 3 records. The workflow ends with Log Out confirmation and a return to the Starter screen.



Figure 57: **Trusted Contact Storyboard**

Figure 57 illustrates the DriveAlert Trusted Contact mobile workflow. Users sign in or register and enter Trusted Contact view. Connected Drivers displays approved connections or an empty state when no Drivers are connected. Invite a Driver allows account lookup through a connection code, while Requests separates incoming and sent requests. Approval is required before future eligible Stage 3 records can be shared. Selecting an approved Driver opens Shared Alerts, which displays the received warning record, timing, event count, and detected signs. Trusted Contact Settings provides access to Account, History and Storage, Notifications, and About DriveAlert. Users may manage retention and notification preferences, review their connection code, or confirm Log Out to return to the Starter screen.

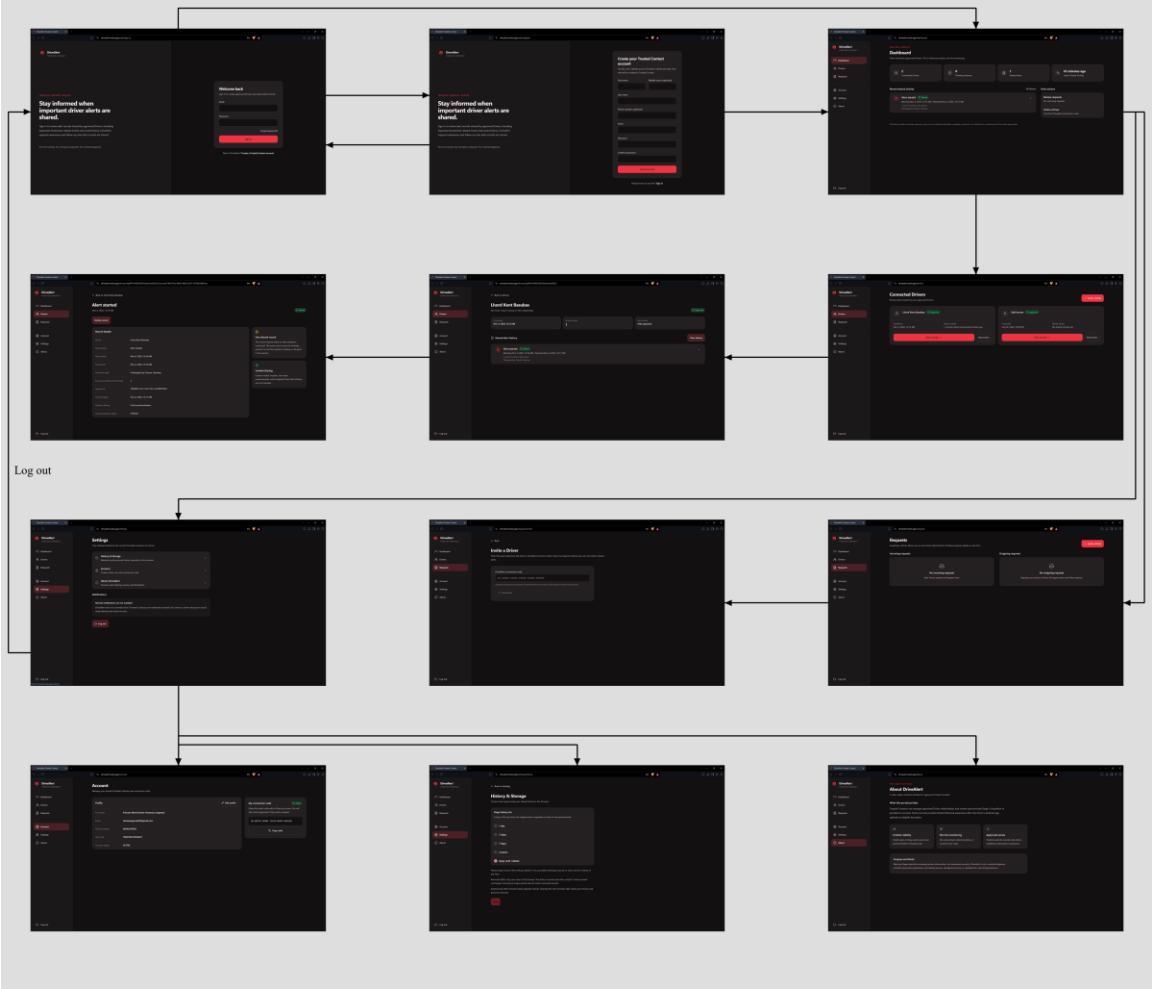


Figure 58: **Trusted Contact Storyboard (Web)**

Figure 58 illustrates the DriveAlert Trusted Contact web workflow. Users sign in or create an account, then open the Dashboard to review connection and shared-activity summaries. Connected Drivers lists approved Drivers, and View records opens a Driver's shared alert history. Selecting a record displays its warning period, detected signs, event count, and synchronization details. Requests and Invite a Driver support connection management using a Driver's connection code and approval. Settings provides access to Account, History and Storage, and About DriveAlert. Users can review their connection code, choose browser history retention, and read the portal's purpose and limitations. Shared records provide historical awareness. Selecting Log out ends the session and returns the user to Sign In.

Database Design

This section refers to the process of creating a detailed data structure for storing, managing, and organizing data in a database management system (DBMS). It involves designing the layout, relationships, and rules that govern how data is stored and accessed within the database.

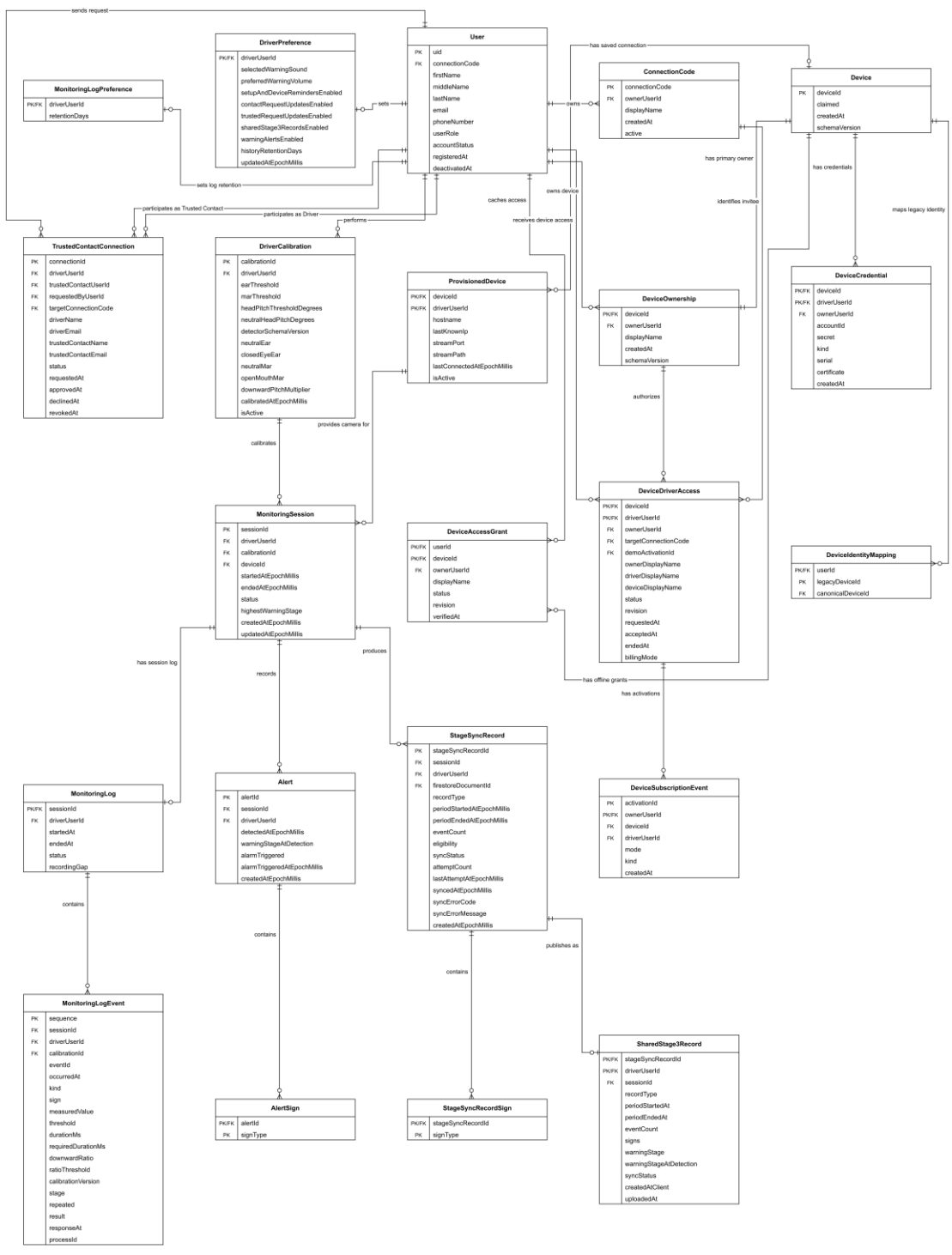


Figure 59: **Entity Relationship Diagram**

FK denotes logical references across Room and Firestore. Repeated account references remain in the attributes; only declared Room foreign keys are enforced by SQL.

Figure 59 presents the Entity Relationship Diagram (ERD) of DriveAlert, combining Room's local monitoring data with Cloud Firestore's account, sharing, and device-access records. Its 22 entities support personalized calibration, warning history, approved Trusted Contact sharing, and simulated additional-driver subscriptions. FK labels identify logical references; only relationships explicitly declared as Room foreign keys are enforced by SQL.

User stores account identity, contact details, role, status, registration date, and connection code. The role identifies a Driver, Trusted Contact, or both. ConnectionCode associates an account lookup code with its owner, display name, creation date, and active status. Users enter these codes to request connections or invite additional Drivers.

DriverPreference stores warning-sound selection, volume, notification preferences, history retention, and update time. MonitoringLogPreference separately stores the Driver's retention setting for completed diagnostic logs.

DriverCalibration stores personalized EAR, MAR, and head-pitch thresholds, calibration measurements, detector schema version, date, and active status. A Driver may retain multiple calibration records. MonitoringSession references the calibration used, preserving the relationship between each session and its detection thresholds.

Device identifies a registered camera by its canonical Bluetooth identifier and claim marker. DeviceOwnership assigns one primary Driver to each device and stores its display name and registration metadata. ProvisionedDevice separately stores each Driver's saved camera connection, including hostname, network address, stream settings, last connection time, and active status. Its composite key combines driverUserId and deviceId. MonitoringSession records the Driver, device, calibration, timing, status, and highest Warning Stage.

TrustedContactConnection records the Driver, Trusted Contact, requester, target connection code, status, and request, approval, decline, and revocation times. Drivers and Trusted Contacts may have multiple approved connections. This relationship permits future eligible Stage 3 sharing and is separate from additional-driver device access.

Alert records confirmed warning events, including detection time, Warning Stage, onboard alarm status, and alarm-trigger time. AlertSign associates each event with prolonged eye closure, yawning, or head nodding. Warning Stages describe increasing system intervention rather than medical drowsiness severity.

StageSyncRecord stores local Stage 3 transition and persistence records, including session references, warning-period boundaries, event counts, sharing eligibility, and synchronization metadata. StageSyncRecordSign identifies the detected signs. Eligible records are published as

SharedStage3Record documents in Cloud Firestore for approved Trusted Contacts. These records provide historical awareness rather than live camera monitoring.

MonitoringLog and MonitoringLogEvent store diagnostic sessions and their timeline entries, including timing, measured values, thresholds, signs, durations, calibration references, Warning Stages, and alarm-response results. DeviceAccessGrant caches server-verified account access locally in Room, while DeviceIdentityMapping resolves older local identifiers to canonical device identifiers.

DeviceDriverAccess stores invitations and additional-driver memberships, including their status, activation reference, and acceptance or ending times. DeviceSubscriptionEvent records simulated activations; no real payment is collected. DeviceCredential stores account-specific authorization credentials. An invited Driver accepts and completes setup online, after which cached access and credentials permit offline device use. Each Driver retains separate calibration and alert history. Together, these records support one included primary Driver per device and controlled access for additional Drivers.

Data Dictionary

The data dictionary describes the Room and Cloud Firestore entities used by DriveAlert for accounts, monitoring, alert sharing, and simulated additional-driver subscriptions. PK identifies primary keys and FK identifies logical references; only declared Room foreign keys are enforced by SQL. A dash in Field Size means no fixed length is declared. OPTIONAL fields may be absent, while NULL permits a null value. Subscription activation is simulated and no payment is collected.

Table 7

User

Column Name	**Data Type**	**Field Size**	**Null**	**Description**
uid (PK)	String	—	NOT NULL	Firebase Authentication UID and User document identifier.
connectionCode (FK)	String	28	OPTIONAL	Logical reference to the account's ConnectionCode document.
firstName	String	—	OPTIONAL	User's first name.
middleName	String	—	OPTIONAL	User's middle name.
lastName	String	—	OPTIONAL	User's last name.
email	String	—	OPTIONAL	User's email address.
phoneNumber	String	—	OPTIONAL	User's phone number.
userRole	String	—	OPTIONAL	Account role: DRIVER, TRUSTED_CONTACT, or BOTH.
accountStatus	String	—	OPTIONAL	Account status: ACTIVE or DEACTIVATED.
registeredAt	Timestamp	—	OPTIONAL	Account registration timestamp.
deactivatedAt	Timestamp	—	NULL	Account deactivation timestamp; may be absent or null.

Table 7 stores account profiles in Cloud Firestore. Each profile is identified by the Firebase Authentication UID and includes identity, contact, role, and account-status information. The connection code links the account to its lookup record.

Table 8

ConnectionCode

Column Name	Data Type	Field Size	Null	Description
connectionCode (PK)	String	28	NOT NULL	Primary document identifier; normalized 28-character lookup code.
ownerUserId (FK)	String	—	NOT NULL	Owner UID; logical reference to User.uid.
displayName	String	—	NOT NULL	Account name displayed during connection-code lookup.
createdAt	Timestamp	—	NOT NULL	Timestamp when the connection-code record was created.
active	Boolean	—	NOT NULL	Whether the connection code is active.

Table 8 stores account lookup codes in Cloud Firestore. The normalized code is the document identifier rather than a stored body field. Each code identifies its owner and provides the display name, creation timestamp, and active status used during connection requests.

Table 9

TrustedContactConnection

Column Name	Data Type	Field Size	Null	Description
connectionId (PK)	String	—	NOT NULL	Primary identifier matching the connection document ID.
driverUserId (FK)	String	—	NOT NULL	Driver UID; logical reference to User.uid.
trustedContactUserId (FK)	String	—	NOT NULL	Trusted Contact UID; logical reference to User.uid.
requestedByUserId (FK)	String	—	NOT NULL	Requester UID; logical reference to User.uid.

Table 9.1

TrustedContactConnection cont'd

Column Name	**Data Type**	**Field Size**	**Null**	**Description**
targetConnectionCode (FK)	String	28	NOT NULL	Logical reference to the target ConnectionCode document.
driverName	String	—	NOT NULL	Driver name saved with the connection.
driverEmail	String	—	NOT NULL	Driver email saved with the connection.
trustedContactName	String	—	NOT NULL	Trusted Contact name saved with the connection.
trustedContactEmail	String	—	NOT NULL	Trusted Contact email saved with the connection.
status	String	—	NOT NULL	Connection state: PENDING, APPROVED, DECLINED, or REVOKED.
requestedAt	Timestamp	—	NOT NULL	Timestamp when the connection was requested.
approvedAt	Timestamp	—	NULL	Timestamp when the request was approved.
declinedAt	Timestamp	—	NULL	Timestamp when the request was declined.
revokedAt	Timestamp	—	NULL	Timestamp when the connection was revoked.

Table 9 stores Driver and Trusted Contact relationships in Cloud Firestore. It identifies both participants, the requester, and the target connection code, and records the request status and approval, decline, or revocation timestamps. Approved relationships permit access to eligible shared records.

Table 10

DriverCalibration

Column Name	Data Type	Field Size	Null	Description
calibrationId (PK)	TEXT	—	NOT NULL	Primary identifier of the calibration record.
driverUserId (FK)	TEXT	—	NOT NULL	Driver UID; logical reference to User.uid.
earThreshold	REAL	—	NOT NULL	Personalized eye aspect ratio threshold for eye closure.
marThreshold	REAL	—	NOT NULL	Personalized mouth aspect ratio threshold for yawning.
headPitchThresholdDegrees	REAL	—	NOT NULL	Head-pitch threshold in degrees for head nodding.
neutralHeadPitchDegrees	REAL	—	NOT NULL	Neutral head-pitch baseline in degrees.
detectorSchemaVersion	INTEGER	—	NOT NULL	Version of the calibration detector schema.
neutralEar	REAL	—	NOT NULL	Eye aspect ratio measured with the eyes open.
closedEyeEar	REAL	—	NOT NULL	Eye aspect ratio measured with the eyes closed.
neutralMar	REAL	—	NOT NULL	Mouth aspect ratio measured with the mouth relaxed.
openMouthMar	REAL	—	NOT NULL	Mouth aspect ratio measured with the mouth open.
downwardPitchMultiplier	REAL	—	NOT NULL	Multiplier applied to the downward head-pitch baseline.
calibratedAtEpochMillis	INTEGER	—	NOT NULL	Calibration completion time in epoch milliseconds.
isActive	INTEGER	—	NOT NULL	Whether this record is active; stored as 0 or 1.

Table 10 stores personalized calibration records locally in Room. Each record contains eye, mouth, and head-pitch thresholds, baseline measurements, detector version, completion time, and active status. Monitoring sessions retain the identifier of the calibration used during detection.

Table 11

ProvisionedDevice

Column Name	**Data Type**	**Field Size**	**Null**	**Description**
deviceId (PK/FK)	TEXT	—	NOT NULL	Composite key component; logical Device reference for canonical IDs.
driverUserId (PK/FK)	TEXT	—	NOT NULL	Composite primary-key component; logically references User.uid.
hostname	TEXT	—	NULL	Optional device hostname.
lastKnownIp	TEXT	—	NOT NULL	Last known device IP address.
streamPort	INTEGER	—	NOT NULL	Network port used for the camera stream.
streamPath	TEXT	—	NOT NULL	Path used for the camera stream.
lastConnectedAtEpochMillis	INTEGER	—	NOT NULL	Last successful device connection time in epoch milliseconds.
isActive	INTEGER	—	NOT NULL	Whether this record is active; stored as 0 or 1.

Table 11 stores Driver-specific camera connection details in Room. The composite primary key combines driverUserId and deviceId. Canonical device identifiers logically reference the Device registry, while legacy identifiers can be resolved through DeviceIdentityMapping. Device ownership and additional-driver access are managed separately from these connection details.

Table 12

DriverPreference

Column Name	Data Type	Field Size	Null	Description
driverUserId (PK/FK)	TEXT	—	NOT NULL	Driver UID; logical reference to User.uid.
selectedWarningSound	TEXT	—	NOT NULL	Selected warning-sound enum value.
preferredWarningVolume	TEXT	—	NOT NULL	Preferred warning volume: MINIMUM, MEDIUM, or HIGH.
setupAndDeviceRemindersEnabled	INTEGER	—	NOT NULL	Enables setup and device reminders; 0 or 1.
contactRequestUpdatesEnabled	INTEGER	—	NOT NULL	Enables Driver contact-request updates; 0 or 1.
trustedRequestUpdatesEnabled	INTEGER	—	NOT NULL	Enables Trusted Contact request updates; 0 or 1.
sharedStage3RecordsEnabled	INTEGER	—	NOT NULL	Enables shared Stage 3 record notifications; 0 or 1.
warningAlertsEnabled	INTEGER	—	NOT NULL	Enables warning-alert notifications; 0 or 1.
historyRetentionDays	INTEGER	—	NULL	History retention in days; null keeps history until deleted.
updatedAtEpochMillis	INTEGER	—	NOT NULL	Last record update time in epoch milliseconds.

Table 12 stores account-specific warning, notification, and history preferences in Room. One record contains the selected sound, volume, notification choices, and retention period for a Driver account. Boolean choices are persisted as SQLite integer values.

Table 13

MonitoringLogPreference

Column Name	Data Type	Field Size	Null	Description
driverUserId (PK/FK)	TEXT	—	NOT NULL	Driver UID; logical reference to User.uid.
retentionDays	INTEGER	—	NULL	Monitoring-log retention in days; null keeps logs until deleted.

Table 13 stores the Driver's monitoring-log retention setting in Room. The Driver identifier is the primary key, and the nullable retention period determines how long local diagnostic logs remain available. This setting is separate from alert-history retention.

Table 14

MonitoringSession

Column Name	Data Type	Field Size	Null	Description
sessionId (PK)	TEXT	—	NOT NULL	Primary identifier of the monitoring session.
driverUserId (FK)	TEXT	—	NOT NULL	Driver UID; logical reference to User.uid.
calibrationId (FK)	TEXT	—	NOT NULL	Room foreign key to DriverCalibration.calibrationId.
deviceId (FK)	TEXT	—	NOT NULL	Logical device reference combined with driverUserId.
startedAtEpochMillis	INTEGER	—	NOT NULL	Session start time in epoch milliseconds.
endedAtEpochMillis	INTEGER	—	NULL	Session end time in epoch milliseconds.
status	TEXT	—	NOT NULL	Session state: ACTIVE, PAUSED, COMPLETED, or INTERRUPTED.
highestWarningStage	TEXT	—	NULL	Highest warning stage reached: STAGE_1, STAGE_2, or STAGE_3.

Table 14.1

MonitoringSession cont'd

Column Name	Data Type	Field Size	Null	Description
createdAtEpochMillis	INTEGER	—	NOT NULL	Record creation time in epoch milliseconds.
updatedAtEpochMillis	INTEGER	—	NOT NULL	Last record update time in epoch milliseconds.

Table 14 stores monitoring sessions in Room. Each session identifies the Driver, provisioned device, and calibration used, together with its timing, status, and highest warning stage. The calibration relationship is enforced locally; the account and device references are maintained by the application.

Table 15

Alert

Column Name	Data Type	Field Size	Null	Description
alertId (PK)	TEXT	—	NOT NULL	Primary identifier of the confirmed alert.
sessionId (FK)	TEXT	—	NOT NULL	Room foreign key to MonitoringSession.sessionId.
driverUserId (FK)	TEXT	—	NOT NULL	Driver UID; logical reference to User.uid.
detectedAtEpochMillis	INTEGER	—	NOT NULL	Alert detection time in epoch milliseconds.
warningStageAtDetection	TEXT	—	NOT NULL	Warning stage when the alert was detected.
alarmTriggered	INTEGER	—	NOT NULL	Whether the onboard alarm was triggered; 0 or 1.
alarmTriggeredAtEpochMillis	INTEGER	—	NULL	Onboard alarm activation time in epoch milliseconds.
createdAtEpochMillis	INTEGER	—	NOT NULL	Record creation time in epoch milliseconds.

Table 15 stores confirmed alert events in Room. Each alert belongs to a monitoring session and records the Driver, detection time, warning stage, and onboard alarm information. Warning stages describe system intervention and do not represent medical drowsiness severity.

Table 16

AlertSign

Column Name	**Data Type**	**Field Size**	**Null**	**Description**
alertId (PK/FK)	TEXT	—	NOT NULL	Composite key component and Room foreign key to Alert.alertId.
signType (PK)	TEXT	—	NOT NULL	Detected sign enum: eye closure, yawning, or head nodding.

Table 16 associates an alert with its detected visible signs in Room. The alert identifier and sign type form a composite primary key, preventing duplicate sign entries for the same alert. Deleting an alert also removes its associated sign rows.

Table 17

StageSyncRecord

Column Name	**Data Type**	**Field Size**	**Null**	**Description**
stageSyncRecordId (PK)	TEXT	—	NOT NULL	Primary identifier of the local synchronization record.
sessionId (FK)	TEXT	—	NOT NULL	Room foreign key to MonitoringSession.sessionId.
driverUserId (FK)	TEXT	—	NOT NULL	Driver UID; logical reference to User.uid.
firestoreDocumentId (FK)	TEXT	—	NOT NULL	Cloud record ID within the Driver's stageSyncRecords subcollection.
recordType	TEXT	—	NOT NULL	Stage 3 transition or persistence record type.
periodStartedAtEpochMillis	INTEGER	—	NOT NULL	Warning-period start time in epoch milliseconds.
periodEndedAtEpochMillis	INTEGER	—	NULL	Warning-period end time in epoch milliseconds.

Table 17.1

StageSyncRecord cont'd

Column Name	Data Type	Field Size	Null	Description
eventCount	INTEGER	—	NOT NULL	Number of confirmed events within the record's warning period.
eligibility	TEXT	—	NOT NULL	Sharing eligibility: pending evaluation, eligible, or no approved contact.
syncStatus	TEXT	—	NOT NULL	Synchronization state of the record.
attemptCount	INTEGER	—	NOT NULL	Number of synchronization attempts.
lastAttemptAtEpochMillis	INTEGER	—	NULL	Last synchronization attempt time in epoch milliseconds.
syncedAtEpochMillis	INTEGER	—	NULL	Successful synchronization time in epoch milliseconds.
syncErrorCode	TEXT	—	NULL	Error code from the last failed synchronization.
syncErrorMessage	TEXT	—	NULL	Error message from the last failed synchronization.
createdAtEpochMillis	INTEGER	—	NOT NULL	Record creation time in epoch milliseconds.

Table 17 stores local Stage 3 transition and persistence records in Room. Each record summarizes a warning period and maintains sharing eligibility, synchronization attempts, status, and errors. The Firestore document identifier connects the local record to its cloud representation.

Table 18

StageSyncRecordSign

Column Name	Data Type	Field Size	Null	Description
stageSyncRecordId (PK/FK)	TEXT	—	NOT NULL	Composite key component and Room foreign key to StageSyncRecord.
signType (PK)	TEXT	—	NOT NULL	Detected sign enum: eye closure, yawning, or head nodding.

Table 18 associates a local Stage 3 synchronization record with its detected visible signs in Room. The record identifier and sign type form a composite primary key. Deleting the synchronization record also deletes its associated sign rows.

Table 19

MonitoringLog

Column Name	Data Type	Field Size	Null	Description
sessionId (PK/FK)	TEXT	—	NOT NULL	Primary log identifier; logically matches MonitoringSession.sessionId.
driverUserId (FK)	TEXT	—	NOT NULL	Driver UID; logical reference to User.uid.
startedAt	INTEGER	—	NOT NULL	Log start time in epoch milliseconds.
endedAt	INTEGER	—	NULL	Log end time in epoch milliseconds.
status	TEXT	—	NOT NULL	Lifecycle status of the diagnostic log.
recordingGap	INTEGER	—	NOT NULL	Whether the log contains a recording gap; 0 or 1.

Table 19 stores local diagnostic monitoring logs in Room. Each log uses the monitoring session identifier and records its timing, status, and recording-gap flag. Logs have an independent retention lifecycle, so the session association is logical rather than a declared Room foreign key.

Table 20

MonitoringLogEvent

Column Name	**Data Type**	**Field Size**	**Null**	**Description**
sequence (PK)	INTEGER	—	NOT NULL	Auto-generated primary key ordering local log events.
sessionId (FK)	TEXT	—	NOT NULL	Room foreign key to MonitoringLog.sessionId.
driverUserId (FK)	TEXT	—	NOT NULL	Driver UID; logical reference to User.uid.
calibrationId (FK)	TEXT	—	NULL	Optional logical reference to DriverCalibration.calibrationId.
eventId	TEXT	—	NOT NULL	Unique event identifier; protected by a unique index.
occurredAt	INTEGER	—	NOT NULL	Event occurrence time in epoch milliseconds.
kind	TEXT	—	NOT NULL	Type of monitoring event recorded.
sign	TEXT	—	NULL	Visible sign associated with the event, when applicable.
measuredValue	REAL	—	NULL	Measured value associated with the event.
threshold	REAL	—	NULL	Detection threshold used for the event.
durationMs	INTEGER	—	NULL	Observed sign duration in milliseconds.
requiredDurationMs	INTEGER	—	NULL	Required sign duration in milliseconds.
downwardRatio	REAL	—	NULL	Measured downward head-pitch ratio.
ratioThreshold	REAL	—	NULL	Required downward head-pitch ratio.

Table 20.1

MonitoringLogEvent cont'd

Column Name	**Data Type**	**Field Size**	**Null**	**Description**
calibrationVersion	INTEGER	—	NULL	Calibration version recorded for the event.
stage	INTEGER	—	NULL	Warning stage recorded for the event.
repeated	INTEGER	—	NOT NULL	Whether the event is repeated; 0 or 1.
result	TEXT	—	NULL	Outcome associated with the event.
responseAt	INTEGER	—	NULL	Event response time in epoch milliseconds.
processId	TEXT	—	NULL	Identifier of the application process recording the event.

Table 20 stores individual diagnostic events in Room. Events belong to a monitoring log and may include detection measurements, thresholds, timing, warning stages, and outcomes. The generated sequence provides row identity, while the unique event identifier prevents duplicate events.

Table 21

SharedStage3Record

Column Name	Data Type	Field Size	Null	Description
stageSyncRecordId (PK/FK)	String	—	NOT NULL	Record document ID within the Driver path; references the local StageSyncRecord.
driverUserId (PK/FK)	String	—	NOT NULL	Parent Driver UID and part of the document-path key; references User.uid.
sessionId (FK)	String	—	NOT NULL	Logical reference to the local MonitoringSession.sessionId.
recordType	String	—	NOT NULL	Stage 3 transition or persistence record type.
periodStartedAt	Timestamp	—	NOT NULL	Timestamp at the start of the warning period.
periodEndedAt	Timestamp	—	NULL	Timestamp at the end of the warning period.
eventCount	Integer	—	NOT NULL	Number of confirmed events within the record's warning period.
signs	Array	—	NOT NULL	Embedded array of detected visible-sign names.
warningStage	String	—	NOT NULL	Shared warning stage; always STAGE_3.
warningStageAtDetection	Integer	—	NOT NULL	Shared warning-stage number; always 3.
syncStatus	String	—	NOT NULL	Cloud synchronization state; always SYNCED.
createdAtClient	Timestamp	—	NOT NULL	Client timestamp when the local record was created.
uploadedAt	Timestamp	—	NOT NULL	Server timestamp when the shared record was uploaded.

Table 21 stores synchronized Stage 3 records in the Driver's Cloud Firestore subcollection. The Driver identifier and record identifier identify the document path. Approved Trusted Contacts can review the shared warning period, visible signs, event count, and synchronization timestamps without receiving camera media.

Table 22

Device

Column Name	**Data Type**	**Field Size**	**Null**	**Description**
deviceId (PK)	String	12	NOT NULL	Canonical 12-character Bluetooth device identifier and document ID.
claimed	Boolean	—	NOT NULL	Claim marker; true for a registered device.
createdAt	Timestamp	—	NOT NULL	Server timestamp when the device was first registered.
schemaVersion	Integer	—	NOT NULL	Registry document schema version; currently 1.

Table 22 stores the public device-claim marker in Cloud Firestore. Each canonical Bluetooth identifier has one registration record. The marker contains no owner identity or credentials and is created atomically with the private ownership and owner-credential records.

Table 23

DeviceOwnership

Column Name	**Data Type**	**Field Size**	**Null**	**Description**
deviceId (PK/FK)	String	12	NOT NULL	Primary key and logical reference to Device.deviceId.
ownerUserId (FK)	String	—	NOT NULL	Primary Driver UID; logical reference to User.uid.
displayName	String	80	NOT NULL	Device label shown to its primary and invited drivers.
createdAt	Timestamp	—	NOT NULL	Server timestamp when primary ownership was created.
schemaVersion	Integer	—	NOT NULL	Ownership document schema version; currently 1.

Table 23 stores the device's single primary Driver in Cloud Firestore. Primary access is included with the device, and the primary Driver may invite additional accounts. Ownership remains separate from Driver-specific camera connection records and cannot be transferred through the current prototype.

Table 24

DeviceDriverAccess

Column Name	**Data Type**	**Field Size**	**Null**	**Description**
deviceId (PK/FK)	String	12	NOT NULL	Composite document-key component; references Device.deviceId.
driverUserId (PK/FK)	String	—	NOT NULL	Composite document-key component; references the invited User.uid.
ownerUserId (FK)	String	—	NOT NULL	Primary Driver UID; logical reference to User.uid.
targetConnectionCode (FK)	String	28	NOT NULL	Logical reference to the invited account's ConnectionCode.
demoActivationId (FK)	String	1–80	NOT NULL	Current activation ID within the primary Driver's DeviceSubscriptionEvent path.
ownerDisplayName	String	—	NOT NULL	Primary Driver name saved with the invitation.
driverDisplayName	String	—	NOT NULL	Invited Driver name saved with the invitation.
deviceDisplayName	String	—	NOT NULL	Device label saved with the invitation.
status	String	—	NOT NULL	PENDING, ACTIVE, DECLINED, CANCELLED, REMOVED, or LEFT.

Table 24.1

DeviceDriverAccess cont’d

Column Name	**Data Type**	**Field Size**	**Null**	**Description**
revision	Integer	—	NOT NULL	Positive revision increased on acceptance, ending access, or reinvitation.
requestedAt	Timestamp	—	NOT NULL	Server timestamp when this invitation was sent.
acceptedAt	Timestamp	—	NULL	Server timestamp of acceptance; null before acceptance.
endedAt	Timestamp	—	NULL	Server timestamp when access or the invitation ended; otherwise null.
billingMode	String	—	NOT NULL	Subscription mode; always SIMULATED.

Table 24 stores invitations and additional-driver memberships in Cloud Firestore. The device and invited Driver identify one membership. Status and revision track acceptance, cancellation, removal, leaving, and reinvitation. The current activation reference connects the membership to its simulated subscription event; no real payment is collected.

Table 25

DeviceSubscriptionEvent

Column Name	**Data Type**	**Field Size**	**Null**	**Description**
activationId (PK)	String	1–80	NOT NULL	Document-path identifier; composite key with ownerUserId, not a stored body field.
ownerUserId (PK/FK)	String	—	NOT NULL	Primary Driver UID and parent-path key; references User.uid.
deviceId (FK)	String	12	NOT NULL	Logical reference to the Device receiving additional-driver access.
driverUserId (FK)	String	—	NOT NULL	Invited account UID; logical reference to User.uid.
mode	String	—	NOT NULL	Activation mode; always SIMULATED.
kind	String	—	NOT NULL	Event type; always ACTIVATION.
createdAt	Timestamp	—	NOT NULL	Server timestamp of the immutable activation event.

Table 25 stores immutable simulated-activation events under the primary Driver's User document. Each event identifies the device and invited account. Stable activation identifiers prevent duplicate events when a confirmation is retried. The event is an access audit record rather than a payment receipt or recurring billing transaction.

Table 26

DeviceCredential

Column Name	**Data Type**	**Field Size**	**Null**	**Description**
deviceId (PK/FK)	String	12	NOT NULL	Composite document-key component; references Device.deviceId.
driverUserId (PK/FK)	String	—	NOT NULL	Composite document-key component; identifies the authorized User.uid.
ownerUserId (FK)	String	—	NOT NULL	Primary Driver UID; logical reference to User.uid.
accountId	String	32	NOT NULL	32-character account identifier derived from the Firebase UID.
secret	String	64	NOT NULL	Private 64-character hexadecimal device-authorization secret.
kind	String	—	NOT NULL	Credential role: OWNER or DRIVER.
serial	Integer	—	NOT NULL	Owner serial is 0; Driver serial identifies the invitation revision.
certificate	String	0 / 64	NOT NULL	Owner value is empty; Driver value is a 64-character authorization certificate.
createdAt	Timestamp	—	NOT NULL	Server timestamp when this credential was issued.

Table 26 stores private account-specific authorization credentials in Cloud Firestore. Device and account identifiers form the document key. The primary Driver issues delegated credentials, and an additional Driver can download their own credential only after acceptance. Android caches credentials in encrypted storage for later offline authorization.

Table 27

DeviceAccessGrant

Column Name	**Data Type**	**Field Size**	**Null**	**Description**
userId (PK/FK)	TEXT	—	NOT NULL	Composite primary-key component; logically references the signed-in User.uid.
deviceId (PK/FK)	TEXT	12	NOT NULL	Composite primary-key component; logically references Device.deviceId.
ownerUserId (FK)	TEXT	—	NOT NULL	Primary Driver UID; logical reference to User.uid.
displayName	TEXT	—	NOT NULL	Device label saved in the verified local access cache.
status	TEXT	—	NOT NULL	Cached OWNER or membership state; only OWNER and ACTIVE permit access.
revision	INTEGER	—	NOT NULL	Last server-confirmed access revision; prevents stale grants restoring access.
verifiedAt	INTEGER	—	NOT NULL	Time of successful server verification in epoch milliseconds.

Table 27 stores server-verified device access locally in Room for each account and device. Only OWNER and ACTIVE grant access, together with a saved credential. Revision values preserve newer removals against stale data. Local access allows later offline use without combining different Drivers' calibration or history.

Table 28

DeviceIdentityMapping

Column Name	Data Type	Field Size	Null	Description
userId (PK/FK)	TEXT	—	NOT NULL	Composite primary-key component; logically references User.uid.
legacyDeviceId (PK)	TEXT	—	NOT NULL	Composite key component identifying the previous local device ID.
canonicalDeviceId (FK)	TEXT	12	NOT NULL	Logical reference to the canonical Device.deviceId.

Table 28 stores account-specific mappings from older local camera identifiers to the canonical device ID in Room. It supports migration of existing device connections while retaining historical identifiers. Its references are maintained by the application and are not declared Room foreign-key constraints.

Network Design

This section includes the planning and implementation of "DriveAlert's" computer network infrastructure. It involves evaluating, analyzing, and sizing the network to ensure it meets the needs of the application and its users. A well-structured network diagram representing the overall network architecture serves as a roadmap for physically assembling the network.

Network Model

A network model is a conceptual framework that describes the architecture, components, and interactions within a computer network. It outlines how different devices and services communicate with each other to achieve seamless connectivity and efficient data transfer.

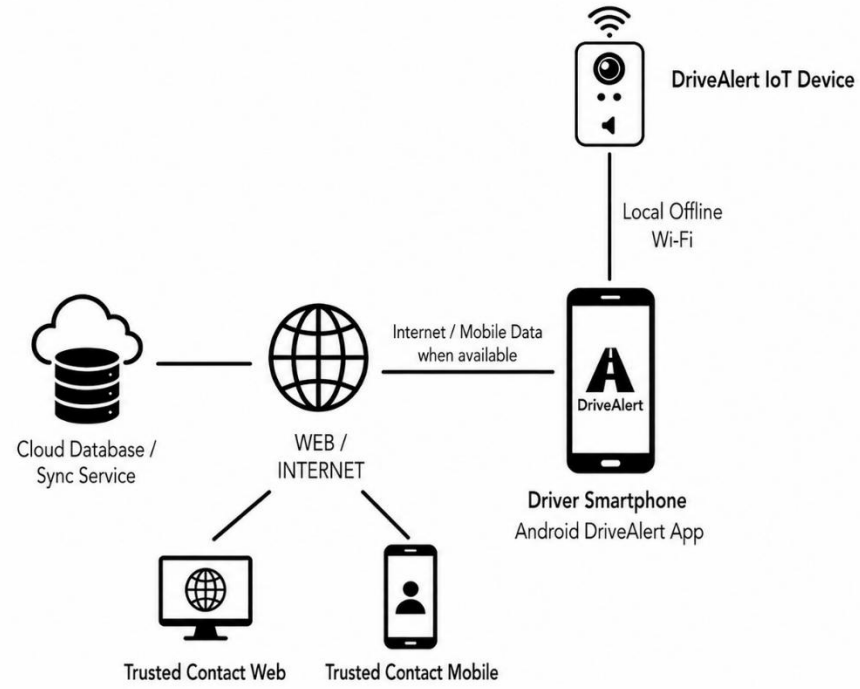


Figure 60: **DriveAlert Network Model**

The network model for "DriveAlert" consists of a DriveAlert IoT device, a driver smartphone, a cloud database or synchronization service, and Trusted Contact access through mobile and web platforms. The IoT device connects directly to the driver's Android application through local offline Wi-Fi, allowing real-time monitoring and alerting without requiring internet connectivity during active use. When internet or mobile data becomes available, drowsiness-alert records are automatically synchronized to the cloud database. These records can then be accessed by an approved Trusted Contact through the mobile application or web platform, ensuring efficient data flow for drowsiness-alert record synchronization and review.

Network Topology

This section presents a visual illustration of the network topology, which represents how devices and components are integrated into a computer network. The topology diagram shows the different links between network nodes and helps in understanding the data flow, addressing mechanisms, and protocols for efficient data transfer.

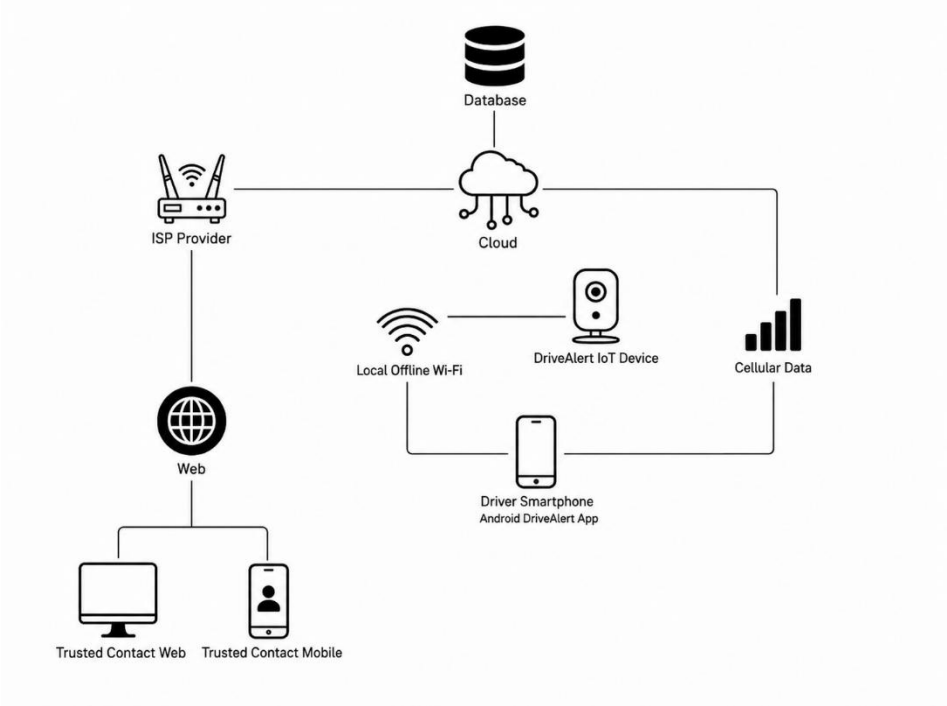


Figure 61: **DriveAlert Network Topology**

The network topology of "DriveAlert" centers around a cloud-based database and synchronization service, which supports the storage and synchronization of drowsiness-alert records across the system. The platform operates through two main communication paths: a local offline Wi-Fi connection between the DriveAlert IoT device and the driver's Android smartphone for real-time monitoring and alerting, and an internet or cellular data connection for synchronizing records to the cloud when connectivity is available. Approved Trusted Contacts may then access the synchronized drowsiness-alert records through either a web platform or a mobile device, allowing flexible monitoring and review from different locations. This topology provides both offline functionality for active monitoring and online accessibility for record synchronization, making the system suitable for DriveAlert's core functions such as drowsiness-sign detection, warning alerts, drowsiness-alert record storage, and Trusted Contact synchronization.

Development/Construction/Build Phase

During the development phase, everything required to implement the project is organized. a timeline is developed, materials and tools are ordered, and instructions are given

Technology Stack Diagram

The table shown below is the technology stack diagram, which is the needed requirements and technology for the proposed system and is categorized as six sections; mobile application, web development, server, AI model, Hardware and future development.

Table 29

TECHNOLOGY STACK

MOBILE DEVELOPMENT	WEB DEVELOPMENT	SERVER	AI MODEL	HARDWARE/IOT	FUTURE DEVELOPMENT
• Kotlin • Android SDK • Jetpack Compose • Material 3 • Room / SQLite • Work Manager	• TypeScript • React • HTML / CSS • Tailwind CSS • Vite	• Firebase Auth • Cloud Firestore • Firebase Hosting • Firebase Cloud Messaging • Cloudflare Workers • Cloudflare D1	• MediaPipe Face Landmarker • FaceAttrib Net	• ESP32-CAM • OV3660 Camera • DFPlayer Mini • Speaker • microSD Card • C/C++ Firmware • BLE / Wi-Fi • HTTP / MJPEG	• Android Studio • Arduino IDE / CLI • Arduino ESP32 Core • Gradle • Git

Table 29 presents the technology stack used to develop DriveAlert. The Android application uses Kotlin, the Android SDK, Jetpack Compose, and Material 3. Room provides local SQLite storage, while WorkManager handles background synchronization. The Trusted Contact web portal uses React, TypeScript, HTML, CSS, Tailwind CSS, and Vite, with Firebase Hosting for deployment.

Firebase Authentication manages account access, while Cloud Firestore stores cloud records, connections, and simulated additional-driver access. The notification module uses Firebase Cloud Messaging, Cloudflare Workers, and D1 when deployed and configured. MediaPipe Face Landmarker supports facial-landmark detection, while FaceAttribNet runs through TensorFlow Lite for facial-attribute assessment.

The IoT component uses an ESP32-CAM with an OV3660 camera, C/C++ firmware, and a DFPlayer Mini with a speaker and microSD card for warning audio. BLE supports setup and authorization, while Wi-Fi carries video and warning commands. Android Studio, Gradle, Arduino IDE/CLI, and Git support development, with iOS considered for future expansion.

Software Specification

This section presents the software that will be used, like the operating system, programming languages for both the front-end and backend of the application, platforms, and database requirements that will be used to develop the entire system of the DriveAlert. It is similar to creating a blueprint or a set of instructions that researchers must follow to develop the program

correctly. Before the construction process can commence, the researchers must create an accurate design that includes the layout of software specifications.

Table 30

SOFTWARE SPECIFICATION

Component	**Specification**
Operating System	Windows, Android OS Version 8 and above
Text Editor/IDE	Visual Studio Code & Android Studio
Programming Language	Kotlin, JavaScript, C/C++
Web Browser	Microsoft Edge, Opera, Google Chrome
Mobile	Android
Database	Cloud Firestore, SQLite
Framework	ReactJS, Android Material Design

Table 30 outlines the software specification for developing the “DriveAlert” system. The researchers chose Windows and Android OS Version 8 and above to ensure compatibility with the development environment and supported Android devices. Visual Studio Code and Android Studio are used as the main development tools, while Kotlin, JavaScript, and C/C++ serve as the programming languages for the mobile application, web component, IoT firmware, and AI-related processes. Supported web browsers include Microsoft Edge, Opera, and Google Chrome, allowing users to access the web-based features through commonly used browsers. The mobile platform specified is Android, which supports the driver-side monitoring application. Cloud Firestore serves as the cloud database for synchronized alert records and Trusted Contact access, while SQLite is used for local record storage on the driver’s smartphone. ReactJS is used for building the web interface, while Android Material Design supports a clean and user-friendly mobile interface. These specifications ensure that DriveAlert is developed using suitable tools and technologies to support its core features, including driver drowsiness monitoring, alert record storage, IoT communication, and Trusted Contact synchronization.

Hardware Specification

This section offers a comprehensive and detailed description of the hardware's objectives, specifying the exact requirements and constraints for each component essential to fulfill the specifications.

Table 31

HARDWARE SPECIFICATION

PC	
CPU	64-bit or 32-bit support 2.5GHz or faster processor Intel i5 10th Gen+ or AMD Ryzen 5 3000 series or higher
Memory	At least 8 GB RAM or above
Storage	10 GB Available space or more
GPU	Integrated or Dedicated Graphics Intel UHD Graphics or AMD Graphics or Nvidia Graphics
Mobile	
CPU	Octa Core or more 2.0 GHz or faster processor
Memory	At least 4 GB RAM or above
Storage	6 GB available space or more

Table 31 outlines the hardware specifications for “DriveAlert,” detailing the minimum requirements for both personal computers (PCs) and Android mobile devices to support system development and operation. For PC development, the specifications include a CPU with 64-bit or 32-bit support and a minimum clock speed of 2.5 GHz, such as Intel i5 10th generation or AMD Ryzen 5 3000 series and above. A minimum of 8 GB RAM and at least 10 GB of available storage are required, along with integrated or dedicated graphics such as Intel UHD Graphics, AMD Graphics, or Nvidia Graphics. For Android smartphones, the requirements include an octa-core processor with a minimum clock speed of 2.0 GHz, at least 4 GB of RAM, and 6 GB of available storage. These specifications are intended to ensure that DriveAlert can be developed, tested, and operated properly, particularly for mobile-based driver drowsiness monitoring, alert record handling, and communication with the IoT device.

IoT Architecture

This section presents how DriveAlert's hardware components and mobile application connect and work together for monitoring and alerting.

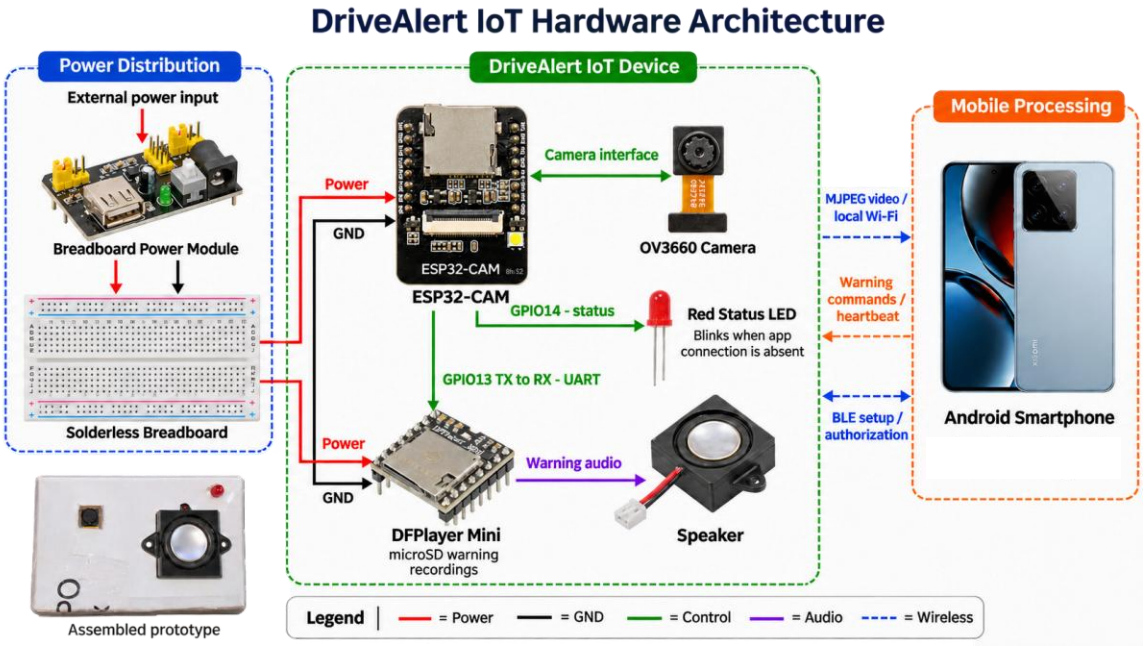


Figure 62: DriveAlert IoT Hardware Architecture

Figure 62 presents the updated DriveAlert IoT hardware architecture. The breadboard power module and solderless breadboard distribute operating power and provide a common ground for the ESP32-CAM and DFPlayer Mini. The assembled enclosure contains the camera, warning speaker, and red status LED.

The ESP32-CAM operates the OV3660 camera and sends its MJPEG video stream to the Driver's Android smartphone over local Wi-Fi. The smartphone performs facial analysis and monitoring, then sends warning commands and heartbeat messages back to the device. BLE supports setup and device authorization. The red status LED is controlled through GPIO14 and blinks when the application connection is absent.

For warning output, the ESP32-CAM sends UART commands from GPIO13 to the DFPlayer Mini, which plays recordings stored on its microSD card through the speaker. Once account and device setup are complete, camera streaming, monitoring, and warning output operate locally without internet. Online access supports account and device-access management and synchronization of eligible Stage 3 records for approved Trusted Contacts.

Program Specification

The program specification outlines the expected goals of the system and serves as a guideline for the development phase of the proposed system. It divides the application's functionalities into modules, and team members are assigned to work on these modules accordingly.

List of Modules

The modules and features required to run the "DriveAlert" application efficiently are outlined in this section. Different programmers are given the modules to build the different features.

Table 32.1

LIST OF MODULES

Programmer/ Process Owner	**Modules**	**Driver**	**Trusted Contact**
Isagunde	**Account Management**		
	Register	*	*
	Log in	*	*
	View Profile	*	*
	Update Profile	*	*
	Log out	*	*
No. of Points *(1 point per module per user)*		1	1
Basubas	**Device Setup & Connection Management**		
	View Setup Requirements	*	
	Allow Required App Permissions	*	
	Enable Mobile Hotspot	*	
	Connect IoT Device	*	
	View Connection Status	*	
	Reconnect Device	*	
No. of Points *(1 point per module per user)*		1	
Basubas	**Driver Drowsiness Monitoring Management**		
	Calibration	*	
	Start Monitoring Session	*	
	Preview Camera Alignment	*	
	Run Drowsiness Detection	*	
	Detect Drowsiness Signs	*	
	View Monitoring Status	*	

Table 32.2

LIST OF MODULES cont'd

	Stop Monitoring Session	*	
No. of Points *(1 point per module per user)*		1	
Basubas	**Alert & Warning Management**		
	Receive Confirmed Warning Event	*	
	Determine Warning Stage	*	
	Receive onboard speaker warning	*	
No. of Points *(1 point per module per user)*		1	
Catarig & Borces	**Drowsiness-alert Records & Trusted Contact Management**		
	View Drowsiness Alert History	*	
	View Alert Details	*	
	View Drowsiness Insights	*	
	Synchronize Eligible Alerts	*	
	Manage Trusted Contacts	*	
	Accept or Decline Request		*
	View All Drivers		*
	View Synced Alert		*
No. of Points *(1 point per module per user)*		1	1
Total number of modules per user		5	2
Total number of modules		7	

Testing/Quality Assurance Phase

The following section presents the application testing performed by the developers to evaluate the performance and functionality of both the mobile and web platforms. The testing process included Unit Testing, Integration Testing, and Alpha/Acceptance Testing to verify that all system components functioned as intended.

Unit Testing

Unit testing checks the individual functions of DriveAlert against their expected results. The following test cases cover the units listed under the system modules.

Table 33

UNIT TESTING

Module Name	**Unit Name**	**Date Tested**	**Test Case ID**	**Test Case Description**	**Expected Results**	**Actual Results**	**Remarks**
Account Management	Register		U1.1.1	Mobile: Register with valid details.	An account is created; the user selects a role.	Performed as expected	Passed
	Register		U1.1.2	Web: Register with valid details.	A Trusted Contact account is created.	Performed as expected	Passed
	Register		U1.1.3	Mobile and Web: Register with missing required fields.	Registration is blocked with field errors.	Performed as expected	Passed
	Register		U1.1.5	Mobile and Web: Register with an existing email.	The duplicate account is rejected.	Performed as expected	Passed
	Register		U1.1.6	Mobile and Web: Register with a short password.	At least eight characters are required.	Performed as expected	Passed
	Register		U1.1.7	Mobile and Web: Enter mismatched passwords.	Registration is blocked until they match.	Performed as expected	Passed
	Log in		U1.2.1	Mobile and Web: Log in with valid	The correct account opens.	Performed as expected	Passed

Table 33.1

UNIT TESTING cont'd

Module Name	Unit Name	Date Tested	Test Case ID	Test Case Description	Expected Results	Actual Results	Remarks
Account Management	Log in		U1.2.1	credentials.			
	Log in		U1.2.2	Mobile and Web: Log in with incorrect credentials.	Sign-in fails with an error.	Performed as expected	Passed
	Log in		U1.2.3	Mobile and Web: Log in with empty fields.	Required-field errors prevent sign-in.	Performed as expected	Passed
	View Profile		U1.3.1	Mobile and Web: Open the signed-in profile.	The correct account details appear.	Performed as expected	Passed
	Update Profile		U1.4.1	Mobile and Web: Save valid profile changes.	Updated details are saved and displayed.	Performed as expected	Passed
	Update Profile		U1.4.2	Mobile and Web: Save a blank required name.	The update is rejected.	Performed as expected	Passed
	Log out		U1.5.1	Mobile and Web: Log out from the account.	The session ends and sign-in appears.	Performed as expected	Passed

Table 33.2

UNIT TESTING cont'd

Module Name	Unit Name	Date Tested	Test Case ID	Test Case Description	Expected Results	Actual Results	Remarks
Device Setup & Connection Management	View Setup Requirements		U2.1.1	Mobile: Open device setup instructions.	Setup requirements are displayed.	Performed as expected	Passed
	Allow Required App Permissions		U2.2.1	Mobile: Grant required Bluetooth permissions.	BLE setup can proceed.	Performed as expected	Passed
	Allow Required App Permissions		U2.2.2	Mobile: Deny required Bluetooth permissions.	The affected setup step is blocked.	Performed as expected	Passed
	Enable Mobile Hotspot		U2.3.1	Mobile: Enable the phone hotspot manually.	The hotspot is available for setup.	Performed as expected	Passed
	Connect IoT Device		U2.4.1	Mobile: Connect an authorized camera device.	The device connects and streams frames.	Performed as expected	Passed
	Connect IoT Device		U2.4.2	Mobile: Submit Wi-Fi setup without an SSID.	Credential validation blocks submission.	Performed as expected	Passed
	Connect IoT Device		U2.4.4	Mobile: Confirm an additional-driver	A pending invitation is created without	Performed as expected	Passed

Table 33.3

UNIT TESTING cont'd

Module Name	Unit Name	Date Tested	Test Case ID	Test Case Description	Expected Results	Actual Results	Remarks
Device Setup & Connection Management	Connect IoT Device		U2.4.4	invitation.	real payment.		
	Connect IoT Device		U2.4.5	Mobile: Accept online, then complete device setup.	The invited Driver can use cached access offline.	Performed as expected	Passed
	View Connection Status		U2.5.1	Mobile: Disconnect a connected device.	The displayed connection status changes.	Performed as expected	Passed
	Reconnect Device		U2.6.1	Mobile: Reconnect a saved, reachable device.	The camera connection is restored.	Performed as expected	Passed
Driver Drowsiness Monitoring Management	Calibration		U3.1.1	Mobile: Complete all calibration phases.	Driver-specific calibration is saved.	Performed as expected	Passed
	Calibration		U3.1.2	Mobile: Leave calibration incomplete.	Incomplete calibration cannot be saved.	Performed as expected	Passed
	Start Monitoring Session		U3.2.1	Mobile: Start with all monitoring requirements met.	One monitoring session starts.	Performed as expected	Passed

Table 33.4

UNIT TESTING cont'd

Module Name	Unit Name	Date Tested	Test Case ID	Test Case Description	Expected Results	Actual Results	Remarks
Driver Drowsiness Monitoring Management	Start Monitoring Session		U3.2.2	Mobile: Start without saved calibration.	Monitoring is blocked.	Performed as expected	Passed
	Start Monitoring Session		U3.2.3	Mobile: Start without a connected camera stream.	Monitoring is blocked until connected.	Performed as expected	Passed
	Preview Camera Alignment		U3.3.1	Mobile: Open camera alignment preview.	Live IoT camera frames appear.	Performed as expected	Passed
	Run Drowsiness Detection		U3.4.1	Mobile: Lose the face during detection.	Assessment pauses; confirmed history remains.	Performed as expected	Passed
	Detect Drowsiness Signs		U3.5.1	Mobile: Detect prolonged eye closure.	Prolonged eye closure is confirmed.	Performed as expected	Passed
	Detect Drowsiness Signs		U3.5.2	Mobile: Detect yawning.	Yawning is confirmed.	Performed as expected	Passed
	Detect Drowsiness Signs		U3.5.3	Mobile: Detect head nodding.	Head nodding is confirmed.	Performed as expected	Passed
	View Monitoring Status		U3.6.1	Mobile: View an	The current	Performed as	Passed

Table 33.5

UNIT TESTING cont'd

Module Name	Unit Name	Date Tested	Test Case ID	Test Case Description	Expected Results	Actual Results	Remarks
Driver Drowsiness Monitoring Management	View Monitoring Status		U3.6.1	active monitoring session.	monitoring state appears.	expected	
	Stop Monitoring Session		U3.7.1	Mobile: Stop an active session.	Detection stops and the session end is saved.	Performed as expected	Passed
Alert & Warning Management	Receive Confirmed Warning Event		U4.1.1	Mobile: Receive a confirmed drowsiness event.	The alert is recorded and warning logic runs.	Performed as expected	Passed
	Determine Warning Stage		U4.2.1	Mobile: Receive the first confirmed event.	The warning advances to Stage 1.	Performed as expected	Passed
	Determine Warning Stage		U4.2.2	Mobile: Receive further distinct confirmed events.	The warning advances through Stages 2 and 3.	Performed as expected	Passed
	Determine Warning Stage		U4.2.3	Mobile: Complete quiet warning periods.	The stage decreases stepwise to normal.	Performed as expected	Passed
	Receive onboard speaker warning		U4.3.1	Mobile: Trigger a warning on a connected device.	The requested onboard warning plays.	Performed as expected	Passed

Table 33.6

UNIT TESTING cont'd

Module Name	Unit Name	Date Tested	Test Case ID	Test Case Description	Expected Results	Actual Results	Remarks
Drowsiness-alert Records & Trusted Contact Management	View Drowsiness Alert History		U5.1.1	Mobile: Open the Driver's alert history.	Only that Driver's records appear.	Performed as expected	Passed
	View Alert Details		U5.2.1	Mobile: Open a saved alert.	The selected alert and signs appear.	Performed as expected	Passed
	View Drowsiness Insights		U5.3.1	Mobile: View insights for saved alerts.	Summaries match the Driver's records.	Performed as expected	Passed
	Synchronize Eligible Alerts		U5.4.1	Mobile: Share an eligible Stage 3 record online.	Approved contacts can access the shared record.	Performed as expected	Passed
	Synchronize Eligible Alerts		U5.4.2	Mobile: Restore internet after an offline eligible alert.	The pending record can synchronize.	Performed as expected	Passed
	Manage Trusted Contacts		U5.5.1	Mobile and Web: Send a valid contact request.	The connection remains pending approval.	Performed as expected	Passed
	Manage Trusted Contacts		U5.5.2	Mobile and Web: Use an invalid or own	The request is blocked.	Performed as expected	Passed

Table 33.7

UNIT TESTING cont'd

Module Name	Unit Name	Date Tested	Test Case ID	Test Case Description	Expected Results	Actual Results	Remarks
Drowsiness-alert Records & Trusted Contact Management	Manage Trusted Contacts		U5.5.2	connection code.			
	Accept or Decline Request		U5.6.1	Mobile and Web: Accept a pending incoming request.	The connection becomes approved.	Performed as expected	Passed
	Accept or Decline Request		U5.6.2	Mobile and Web: Decline a pending incoming request.	The request closes without granting access.	Performed as expected	Passed
	View All Drivers		U5.7.1	Mobile and Web: Open the connected Driver list.	Approved connected Drivers are displayed.	Performed as expected	Passed
	View Synced Alert		U5.8.1	Mobile and Web: Open an approved Driver's shared alert.	Permitted record details appear without camera media.	Performed as expected	Passed
	View Synced Alert		U5.8.2	Mobile and Web: Open an alert after online connection revocation.	Shared-record access is refused.	Performed as expected	Passed

Integration Testing

Integration testing checks how DriveAlert's modules exchange data and complete workflows across the Driver application, camera device, cloud records, and Trusted Contact mobile and web interfaces.

Table 34

INTEGRATION TESTING

Test Case ID	Module 1	Integration Process	Module 2	Pre-condition	Result	Remarks
IT-1	Account Management	After registration, the Driver opens setup and continues to Driver Home without returning to view selection.	Device Setup & Connection Management	New Driver account; online registration completed.	Performed as expected	Passed
IT-2	Device Setup & Connection Management	An authorized camera connects; saved Driver calibration allows one monitoring session to start.	Driver Drowsiness Monitoring Management	Authorized camera available; alignment confirmed; Driver calibrated.	Performed as expected	
IT-3	Driver Drowsiness Monitoring Management	Confirmed detection events update the warning stage and trigger its onboard speaker warning.	Alert & Warning Management	Active monitoring; visible sign confirmed; speaker connected.	Performed as expected	
IT-4	Alert & Warning Management	A confirmed alert is saved under the current Driver and appears with matching details in Driver Alert History.	Drowsiness-alert Records & Trusted Contact Management	Signed-in Driver; active session; confirmed alert event.	Performed as expected	Passed
IT-5	Drowsiness-alert Records & Trusted Contact Management (View Drowsiness Alert History)	The Driver opens Drowsiness Insights; its summaries match the saved alerts in that Driver's history.	Drowsiness-alert Records & Trusted Contact Management (View Drowsiness Insights)	Known alert records saved for the current Driver.	Performed as expected	Passed

Table 34.1

INTEGRATION TESTING cont'd

Test Case ID	**Module 1**	**Integration Process**	**Module 2**	**Pre-condition**	**Result**	**Remarks**
IT-6	Drowsiness-alert Records & Trusted Contact Management (Driver)	A connection request is accepted; the approved relationship appears for the Driver and Trusted Contact on their respective screens.	Drowsiness-alert Records & Trusted Contact Management (Trusted Contact)	Driver and Trusted Contact online; pending request exist.	Performed as expected	Passed
IT-7	Alert & Warning Management	An eligible Stage 3 record synchronizes; the approved Trusted Contact opens the same shared details in the mobile app and web portal.	Drowsiness-alert Records & Trusted Contact Management	Eligible Stage 3 record; approved connection; internet available.	Performed as expected	Passed
IT-8	Alert & Warning Management	An eligible record remains pending while offline; when internet returns, synchronization makes it available to the approved Trusted Contact.	Drowsiness-alert Records & Trusted Contact Management	Eligible record queued; approved contact; internet unavailable.	Performed as expected	Passed
IT-9	Drowsiness-alert Records & Trusted Contact Management (Driver)	The Driver revokes the connection; shared-record access is refused on the Trusted Contact mobile app and web portal.	Drowsiness-alert Records & Trusted Contact Management (Trusted Contact)	Approved connection; shared record visible; both clients online.	Performed as expected	Passed
IT-10	Device Setup & Connection Management	The additional Driver accepts a device invitation and completes online setup; offline monitoring uses separate calibration and history.	Driver Drowsiness Monitoring Management	Simulated subscription invitation pending; internet for setup; camera available.	Performed as expected	

Alpha/Acceptance Testing

This section is conducted to evaluate how the system is judged by the people who use it, based on its performance and functionality from the user's perspective. It ensures that the system meets the needs of every role and is ready for deployment, with the ratings below filled in by the evaluators, who mark one rating per row.

Alpha Testing

Testing is conducted internally by the development team to find and fix problems before release and to confirm that the system works correctly for every type of user.

Table 35

ALPHA TESTING

Test Criteria	Poor	Fair	Good	Very Good
Graphical User Interface (GUI)				
Consistency (The user interface is of the same formatting style and icons throughout the system.)			X	
Reusability (The system contains reusable GUI components such as familiar buttons, text and check boxes, and other tools.)			X	
Forgiveness and Tolerance (The interface displays message or confirmation prompts that would allow the users to undo or redo critical actions.)			X	
Simplicity (The GUI design include simple GUI buttons, such as simple screens with clear, uncrowded messages.)			X	
Readability (The interface has appropriate colors, font sizes, and styles that is convenient to the target users.)			X	
Clarity (Displayed error, help, and warning messages are clear, concise, and as elementary as possible to assist user in operating the software.)			X	
Flexibility (The system includes user preferences setting to allow changes, for example, increasing the font size.)		X		
User-friendliness (The GUI design must be user-friendly, by providing helpful, courteous, and non-offending messages.)			X	
System Performance				
Conformance to the Requirements (The system effectively met all the identified features and/or requirements.)		X		
Conformance to the Objectives (All specific objectives of the system are met by the program.)		X		

Efficiency (The entire system functions efficiently. It doesn't have delay in any transaction.)		X		
Security (The system is secured. Login details are authenticated. Input parameters are ensured prior to the execution of the next transaction.)			X	
Integrity (The software allows the registered user to have control over its own private information.)			X	
Overall Impression (In general, the program or system is functional and useful.)			X	

Acceptance Testing

Acceptance testing allows Drivers and Trusted Contacts to assess DriveAlert's functions against their needs. Evaluators identify critical criteria, record Accept or Reject after trying the applicable features, and add comments before approval.

Table 36

ACCEPTANCE TESTING

PROJECT IDENTIFICATION	
Project Name: DriveAlert: An IoT-Based Driver Drowsiness Monitoring and Alert System Using Computer Vision	Date Created: June 2026
Project Sponsor/Owner: DriveAlert Team	Project Manager: Lhord Kent R. Basubas
Project Adviser: Roche A. Plando	Dean: Eric P. Ortega

ACCEPTANCE CRITERIA TEST MATRIX

		Critical		Test Results		
Number	**Acceptance Criterion Description**	**Yes**	**No**	**Accept**	**Reject**	**Comments**
1	I can register, sign in with valid credentials, and access the appropriate Driver or Trusted Contact view.	✓		✓		
2	I can view and update my profile, and log out of my account.	✓		✓		
3	As an authorized Driver, I can complete device setup, connect the camera, and view its connection status.	✓		✓		

4	As a Driver, I can complete calibration and use my own saved calibration during monitoring.	✓		✓		
5	As a Driver, I can start and stop monitoring with the authorized camera.	✓		✓		
6	Monitoring detects prolonged eye closure, yawning, and head nodding under supported conditions.	✓		✓		
7	Confirmed events update the warning stage and produce the corresponding onboard speaker warning.	✓		✓		
8	As a Driver, I can review my own alert history, details, and record-based insights.	✓		✓		
9	I can accept or decline connection requests and revoke approved connections to end shared-record access.	✓		✓		
10	Stage 3 records queued offline synchronize when internet returns and appear for approved contacts on mobile and web.	✓		✓		
11	An additional Driver can accept a simulated subscription invitation without payment, complete setup online, and monitor offline with separate calibration and history.	✓		✓		
12	As a Trusted Contact, I see only approved shared records, excluding camera media, location, and complete Driver history.	✓		✓		

OTHER CONSIDERATIONS		
Business Objectives – Did the software application/system meet the business objectives?		
Yes	No	State whether the system met each applicable requirement and describe any unmet requirement:
Does the software application/system require any changes prior to installation? If so, please describe.		
Yes	No	Identify any changes or requirements needed before installation:

APPROVAL			
Project Sponsor/Owner		Signature	Date
Yes	No		June 2026

Implementation/Deployment Phase

This section covers the installation and configuration of DriveAlert for use by Drivers and Trusted Contacts. It includes installing the Android application, uploading the ESP32-CAM firmware, preparing the camera and onboard warning components, configuring Firebase services, and deploying the Trusted Contact web portal through Firebase Hosting. Device registration and initial authorization require an internet connection. Once setup is complete, authorized Drivers can monitor offline while maintaining separate calibration and alert histories. This phase also includes user orientation and verification of camera connectivity, calibration, monitoring, onboard warnings, and eligible Stage 3 record sharing before deployment.

Deployment Diagram

This section illustrates how DriveAlert's components are distributed across devices and network environments. It shows the connections among the ESP32-CAM and its warning components, the Android application that processes camera frames and stores local records, Firebase services, and the Trusted Contact web portal. It also distinguishes local communication used for setup, monitoring, and warnings from internet connections used for account management, device authorization, and shared-record synchronization.

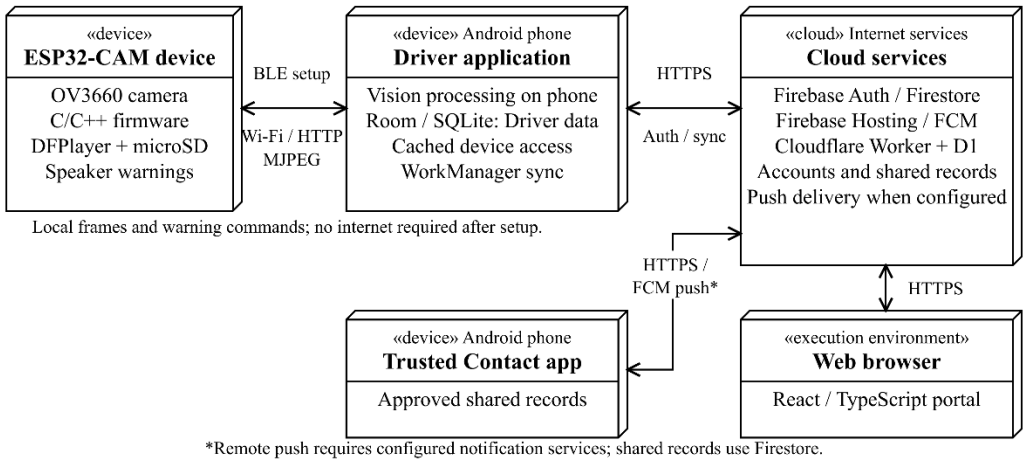


Figure 63: **DriveAlert Deployment Diagram**

The deployment diagram presents DriveAlert's architecture, showing the connections among the ESP32-CAM device, the Driver's Android application, cloud services, and the Trusted Contact mobile application and web portal. The ESP32-CAM uses an OV3660 camera to capture frames and transmit them to the Driver's phone through a local Wi-Fi HTTP/MJPEG connection. BLE supports device setup and authorization. The device also receives warning commands from the phone and plays the selected recordings through the DFPlayer Mini, microSD card, and speaker. After initial online setup, authorized Drivers can use the device offline.

The Android application, developed using Kotlin and Jetpack Compose, performs vision processing on the phone. MediaPipe Face Landmarker supplies facial landmarks, while personalized calibration and timing rules support the detection of prolonged eye closure, yawning, and head nodding. TensorFlow Lite supports facial-attribute assessment through FaceAttribNet. Room manages local SQLite records, keeping each Driver's calibration, monitoring sessions, and alert history separate. WorkManager schedules eligible Stage 3 record synchronization and retries when internet connectivity becomes available.

Firebase Authentication manages sign-in, while Cloud Firestore stores profiles, approved connections, shared records, device ownership, and simulated additional-driver access. Firebase Hosting serves the React and TypeScript Trusted Contact web portal. Approved Trusted Contacts can review shared historical records through the Android application or a web browser. When configured, a Cloudflare Worker and D1 database support notification registration and delivery tracking, while Firebase Cloud Messaging delivers Android push notifications. These internet-based services supplement local monitoring and onboard warnings; camera frames are processed on the Driver's phone rather than uploaded for cloud analysis.

Implementation Budget/Cost Specification

This section outlines the estimated costs required to complete DriveAlert. It includes expenses for hardware, software, and other resources necessary for the development and implementation of the system.

Table 37

COST SPECIFICATION

Items	Unit	Amount
Hardware:		
IoT parts (complete device set)	1	₱ 1,500
Android smartphone	1	₱ 5,000

Software and Services:		
AI subscription	1	₱ 4,000
Android Studio / Arduino IDE	1	₱ 0
Firebase services	1	₱ 0
Cloudflare Worker / D1	1	₱ 0
MediaPipe / TensorFlow Lite	1	₱ 0
React / TypeScript / Vite / Git	1	₱ 0
Total		**₱ 10,500**

Table 37 details DriveAlert's estimated development and implementation cost of ₱10,500. The hardware allocation includes ₱1,500 for the IoT device parts and ₱5,000 for an Android smartphone. The IoT components include the ESP32-CAM with an OV3660 camera, DFPlayer Mini, speaker, microSD card, and supporting power and wiring components. The camera streams frames to the smartphone through a local Wi-Fi connection, while the DFPlayer Mini and speaker provide audible warnings. The smartphone performs vision processing, applies the Driver's calibration, and stores monitoring sessions and alert history locally.

The software and services allocation include ₱4,000 for an AI subscription used during development. Android Studio, Arduino IDE, React, TypeScript, Vite, Git, MediaPipe, and TensorFlow Lite are recorded at no additional cost. Firebase supports authentication, cloud records, and web hosting, while Cloudflare Worker and D1 support the configured Android notification service. These cloud services are also allocated ₱0 for the project's usage. The simulated additional-driver subscription does not collect real payments and is separate from the AI development subscription.

Software verification covered account access, Driver-specific history, insights, approved connections, and shared-record synchronization across Android and the web portal. Tests also checked that eligible records remain queued while offline and synchronize when connectivity returns. Physical camera streaming, complete monitoring, and onboard speaker playback still require verification using the assembled device.

Software Specification

DriveAlert's software specifications support its Android application, Trusted Contact web portal, cloud services, and camera-device firmware. The Android application requires Android 8.0 or higher, as defined by its minimum SDK configuration. The web portal is accessed through a browser rather than installed as a desktop application.

The Android application is developed using Kotlin and Android Studio, with Jetpack Compose and Material 3 providing the interface. Room manages local SQLite records, while WorkManager schedules eligible Stage 3 synchronization and retries. MediaPipe Face Landmarker extracts facial landmarks for calibrated detection rules, while FaceAttribNet runs through TensorFlow Lite for supporting facial-attribute assessment. The Trusted Contact portal uses React, TypeScript, Vite, and Tailwind CSS. Arduino IDE or Arduino CLI supports development of the ESP32-CAM's C/C++ firmware.

Firebase Authentication manages sign-in, Cloud Firestore stores cloud records and access relationships, and Firebase Hosting serves the web portal. When configured, Cloudflare Worker, D1, and Firebase Cloud Messaging support Android push notifications. Initial device registration and authorization require internet access, while authorized monitoring can operate offline after setup.

Hardware Specification

DriveAlert's hardware specifications cover the development computer, Android smartphones, and assembled IoT device. The computer supports application development, firmware uploading, and testing. Vision processing runs on the Driver's smartphone, so a dedicated computer server or desktop graphics card is not required during monitoring.

For Android deployment, Table 31 specifies an octa-core processor running at approximately 2.0 GHz, at least 4 GB RAM, and 6 GB available storage. These figures represent a proposed testing configuration rather than verified minimum requirements. The phone also needs Wi-Fi and Bluetooth Low Energy for device communication and setup. Internet access supports account services, authorization, and synchronization.

The IoT device consists of an ESP32-CAM with an OV3660 camera, a DFPlayer Mini, speaker, microSD card containing warning recordings, and supporting power and wiring components. The camera supplies frames to the phone over local Wi-Fi, while the audio components play warning commands received from the application. Each authorized Driver maintains separate calibration and alert history when using the shared device.

Human Resource Specification

This section describes the two types of users in DriveAlert: Drivers and Trusted Contacts. Each role has specific responsibilities and access to the system's features. Drivers manage device

setup, monitoring, and personal records, while approved Trusted Contacts review shared historical alerts through the mobile application or web portal.

Table 38

HUMAN SPECIFICATION

Users	Description
Driver	The Driver uses the Android application to connect an authorized DriveAlert device, align the camera, and complete the guided personal calibration while parked. The Driver starts and stops monitoring, previews the camera, receives onboard warnings, and reviews personal alert history, insights, and monitoring logs. The Driver can change warning preferences and manage Trusted Contact requests and approved connections. A primary Driver may also invite additional Drivers through a simulated subscription that collects no real payment. Device registration, invitation acceptance, and initial authorization require internet access. After setup, authorized Drivers can use the device offline while keeping their calibration and alert history separate.
Trusted Contact	The Trusted Contact uses the Android application or web portal to request and manage approved connections with Drivers. The Trusted Contact reviews eligible shared Stage 3 transition and persistence records, including their warning periods, detected signs, event counts, and synchronization details. The Trusted Contact can accept or decline requests, disconnect from a Driver, and manage shared-history retention and personal record removal. Shared records become available after an eligible upload and provide historical awareness. Trusted Contact access does not include live camera video, vehicle location, complete Driver alert history, or control of the Driver's monitoring session.

User Guide

This section provides instructions for navigating DriveAlert and using its features. It explains the system configuration, user access levels, and the main mobile and web procedures.

1. Introduction

DriveAlert is an IoT-based Driver drowsiness monitoring and alert system using computer vision. Its Android application processes frames from an ESP32-CAM to recognize prolonged eye closure, yawning, and head nodding using the Driver's calibration. Confirmed events produce warning stages and are saved in the Driver's alert history. Approved Trusted Contacts can review eligible Stage 3 records through the mobile application or web portal.

The Android application uses Kotlin, Jetpack Compose, and Room for local records. Firebase supports accounts, connections, device access, and shared records, while the React and TypeScript web portal serves Trusted Contacts. After initial online setup, authorized Drivers can monitor offline. Eligible shared records synchronize when internet access becomes available.

2. System Summary

2.1 System Configuration

The mobile application runs on Android 8.0 or higher. Drivers use Wi-Fi for camera streaming and warning commands, and Bluetooth Low Energy for device setup and authorization. Monitoring requires a connected camera, confirmed alignment, and an active Driver calibration. The phone performs vision processing and stores local records.

Trusted Contacts can also open https://drivealert.web.app in a web browser. Internet access is required for registration, email verification, connection requests, device invitations, and loading cloud records. Saved local history can be reviewed without reconnecting the camera. Android and browser history preferences are managed separately.

2.2 User Access Levels

DriveAlert provides access according to the selected user view and approved device or Driver connections.

Driver: Uses the Android application to complete setup, start and stop monitoring, review personal alerts and insights, manage warning preferences, and approve or remove Trusted Contacts.

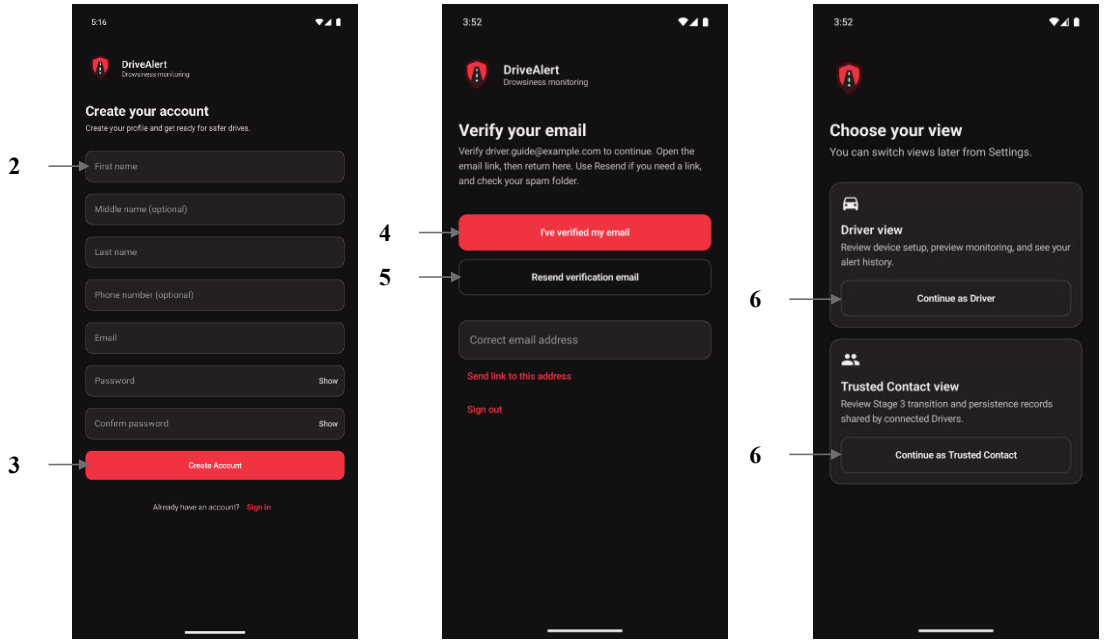
Trusted Contact: Uses the Android application or web portal to manage Driver connections and review approved shared historical records. This view does not provide live camera video, vehicle location, or complete Driver history.

Device access: One primary Driver is included per device. The primary Driver may invite additional Drivers through a simulated subscription without real payment. Additional Drivers must accept online and complete device setup; each Driver keeps separate calibration and alert history.

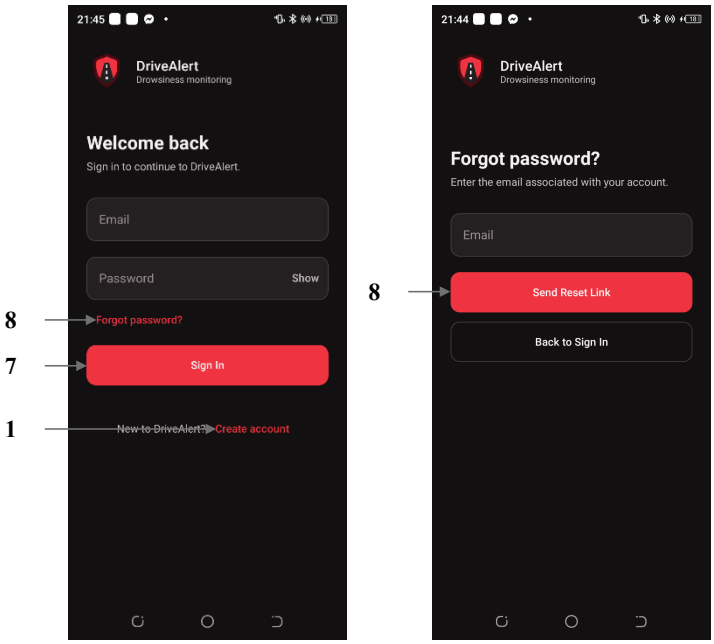
2.3 User Manual

Mobile

1. On the starter screen, tap Create Account to register.
2. Enter your first name, last name, email, password, and password confirmation. Middle name and phone number are optional.
3. Tap Create Account. Correct any highlighted fields before submitting again.
4. Open the verification email and follow its link. Return to DriveAlert and tap I've verified my email.
5. Tap Resend verification email if needed, and check your spam folder.
6. On Choose your view, tap Continue as Driver or Continue as Trusted Contact.

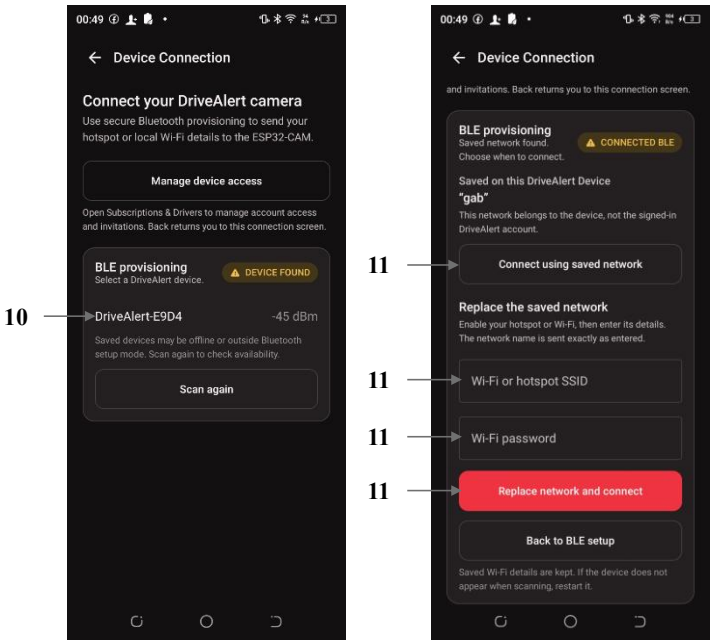


7. On Sign In, enter your registered email and password, then tap Sign In.
8. For a forgotten password, tap Forgot password?, enter your email, and tap Send Reset Link. Follow the email instructions.

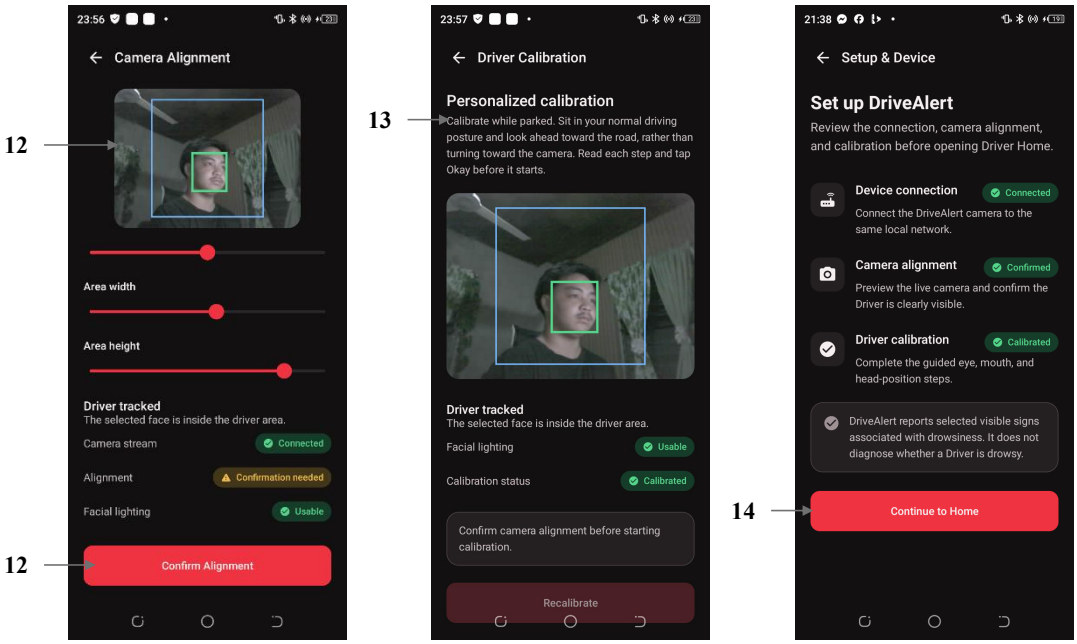


9. In Driver view, open Setup & Device, then Device connection.
10. Enable the requested Bluetooth permissions. Select your DriveAlert device from the discovered list; use Scan again if needed.

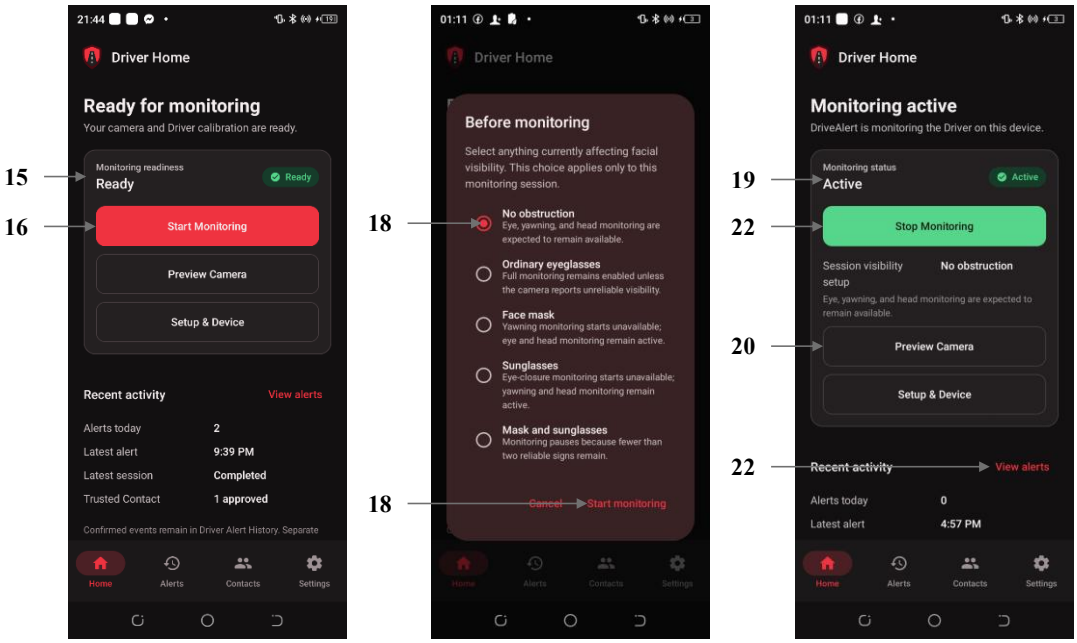
11. Tap Connect using saved network when the saved network is available. To use another network, enable your hotspot or Wi-Fi, enter its SSID and password, and tap Replace network and connect. Initial device registration and authorization require internet access.



12. In Setup & Device, open Camera alignment and adjust the camera until the Driver is clearly visible. Confirm the Driver area when the screen permits it.
13. Open Driver calibration while parked. Follow the on-screen instructions and complete the guided steps.
14. Return to Setup & Device, review readiness, and tap Continue to Home.

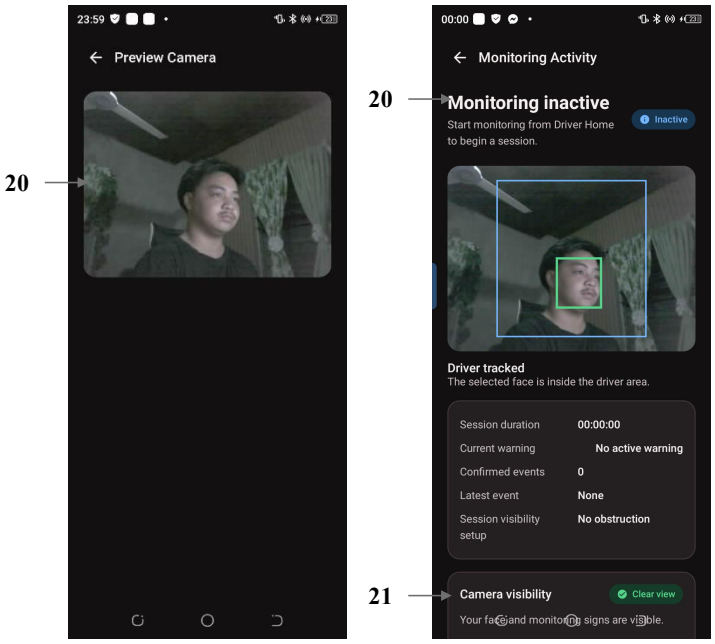


15. In Driver view, open Home. The readiness card shows whether monitoring is ready or setup is needed.
16. Tap Start Monitoring. If setup is incomplete, the app opens the required connection, alignment, or calibration screen.
17. Once the camera is connected, alignment is confirmed, and calibration is active, tap Start Monitoring again.
18. In Before monitoring, select the option describing accessories affecting facial visibility, then tap Start monitoring.
19. The Home status changes to Monitoring active when the session starts.



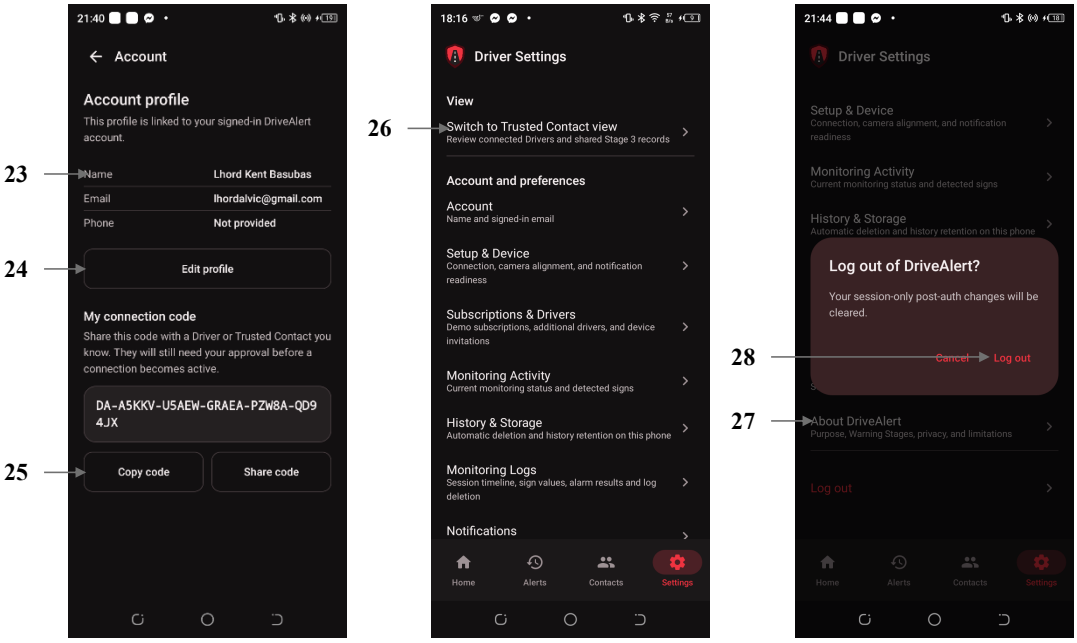
20. Tap Preview Camera to view the camera image. Open Settings, then Monitoring Activity, to review the session's status and detected signs.
21. If monitoring reports paused or unavailable visibility, follow its on-screen guidance before continuing.

22. Return to Home and tap Stop Monitoring to end the session. Tap View alerts to review saved events.



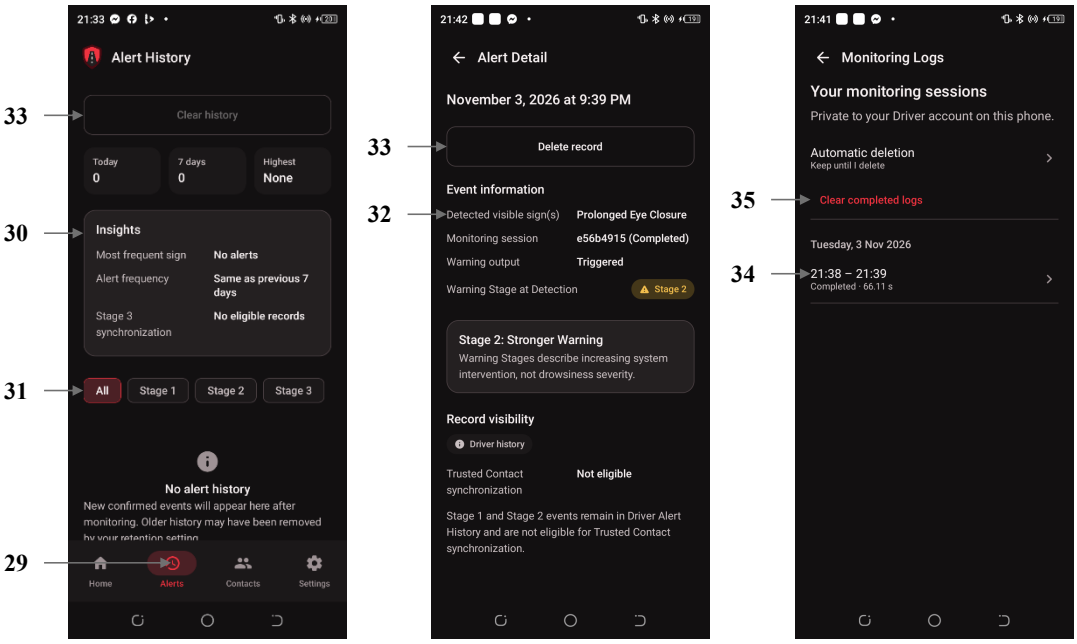
23. Open Settings, then Account, to view your name, email, phone number, and connection code.
24. Tap Edit profile, update the available name or phone fields, and save the changes.
25. Under My connection code, tap Copy code to share it with the person you want to connect with.
26. In Settings, use Switch to Trusted Contact view or Switch to Driver view to change your working view.
27. Open About DriveAlert to read the system's purpose, warning stages, privacy information, and limitations.

28. Tap Log out in Settings and confirm the prompt to sign out.



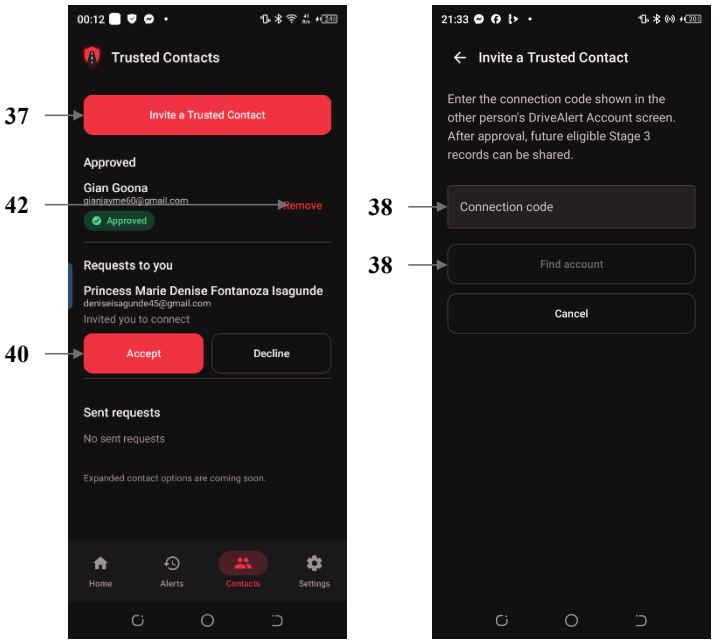
29. Open Alerts to view Alert History for your signed-in Driver account.
30. The summary shows today's alerts, the seven-day count, and the highest warning stage. Insights summarize the saved records.
31. Use All or the warning-stage filters to change the displayed list.
32. Tap an alert to open Alert Detail and review its time, signs, warning stage, and available sharing information.
33. Tap Delete record to remove one saved alert, or Clear history on the list to remove available history. Review the confirmation first.
34. Open Settings, then Monitoring Logs, and tap an existing session to review its timeline.
35. Use Delete session log or Clear completed logs to remove completed logs. Log deletion leaves Alert History unchanged.

36. An empty history or log list means no matching records are available. Creating a new monitoring session requires the camera.



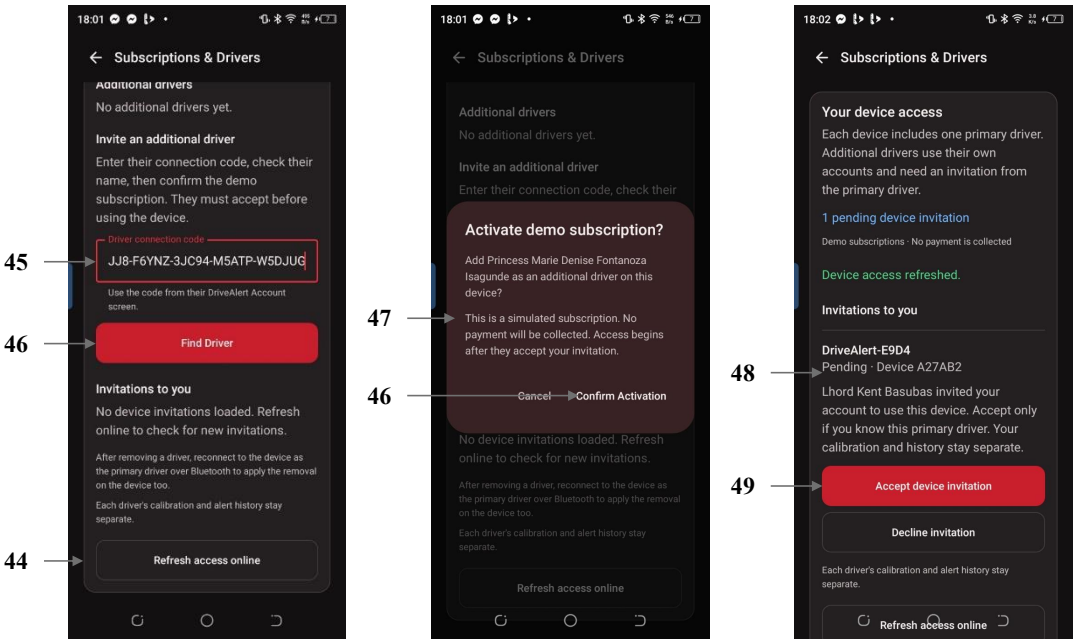
37. In Driver view, open Contacts and tap Invite a Trusted Contact.
38. Enter the code from the other person's Account screen and tap Find account.
39. Check the returned name before sending the request.
40. Incoming requests appear under Requests to you. Tap Accept or Decline and confirm approval when prompted.
41. Outgoing requests stay pending until accepted. Use Cancel request when needed.
42. To end an approved connection, tap Remove beside the contact and confirm Remove Trusted Contact?.

43. Approval permits future eligible Stage 3 sharing. It does not expose your complete local alert history.



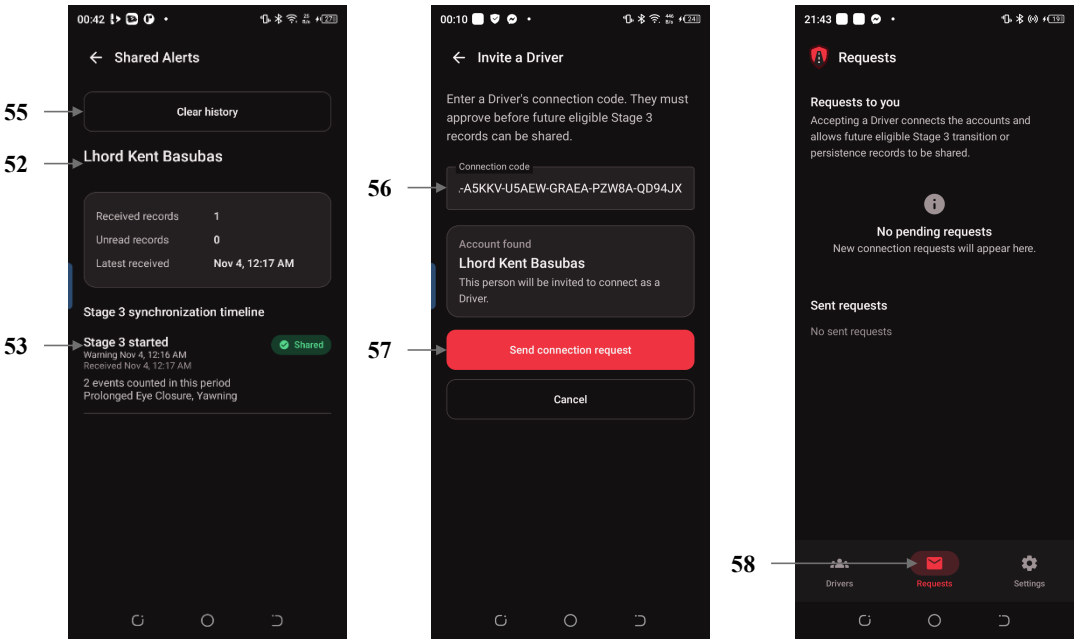
44. Open Settings, then Subscriptions & Drivers. Tap Refresh access online to load device access and invitations.
45. As the primary Driver of an already registered device, enter another Driver's connection code under Invite an additional driver.
46. Tap Find Driver, check the account name, then tap Confirm Activation to send the invitation.
47. The subscription is simulated. No payment is collected.
48. As the invited Driver, find Invitations to you and check the device and primary Driver's name.
49. Tap Accept device invitation or Decline invitation while online. After acceptance, complete pairing and device setup before monitoring.
50. The primary Driver may select Cancel invitation or Remove driver. An additional Driver may select Leave device.

51. Read the removal prompt. The primary Driver must reconnect over Bluetooth to apply removal to the device too. Saved Driver histories remain separate.



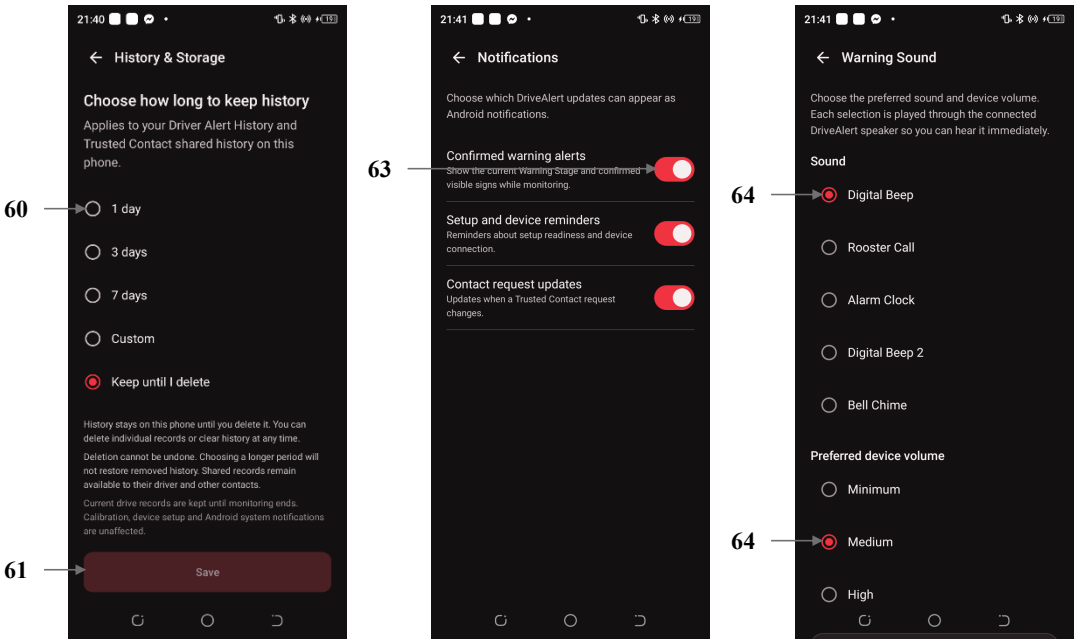
52. In Trusted Contact view, open Drivers and select an approved Driver to display Shared Alerts.
53. Tap a shared record to review its warning period, detected signs, event count, and synchronization information.
54. The event count covers the displayed warning period, not every alert in the monitoring session.
55. Tap Clear history to remove that Driver's shared history from your view on this phone. Other contacts' and the Driver's records are kept.
56. To request a connection, tap Invite a Driver, enter the Driver's code, and tap Find account.
57. Check the name and send the request. The Driver must approve before future eligible records can be shared.
58. Open Requests to accept or decline incoming requests, or cancel an outgoing request.

59. On the Drivers list, tap Disconnect and confirm Disconnect Driver? to end an approved connection.



60. Open Settings, then History & Storage. Choose 1 day, 3 days, 7 days, Custom, or Keep until I delete.
61. For Custom, enter 1 to 3,650 whole days, then tap Save. Review any deletion preview before confirming Save and apply.
62. A day means 24 hours from the event. Deleted history cannot be restored by choosing a longer retention period.
63. Open Notifications to change the available interface-update preferences for your current view.
64. In Driver view, open Warning Sound to choose a recording and preferred device volume.

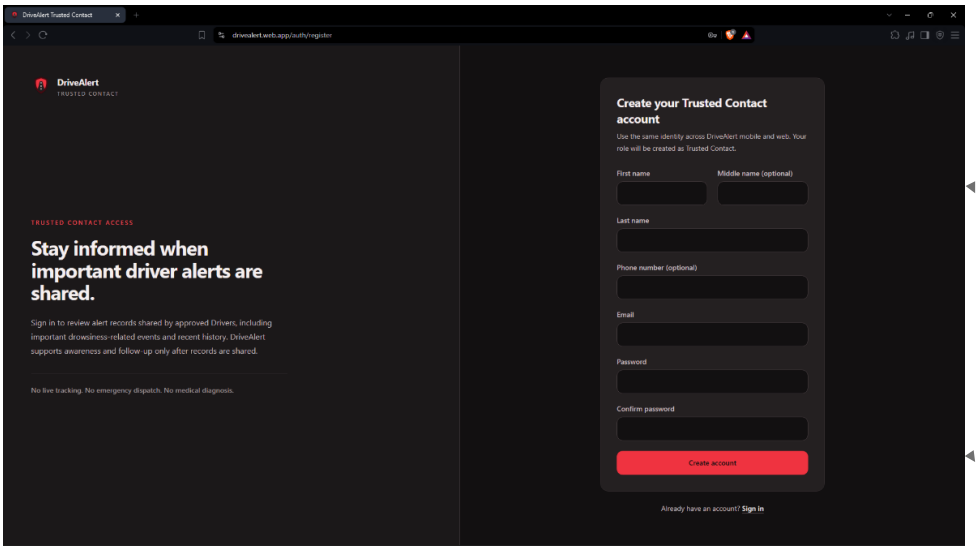
65. Preference selection does not require a camera stream. Onboard playback requires the device; remote push also depends on configured services and Android permissions.

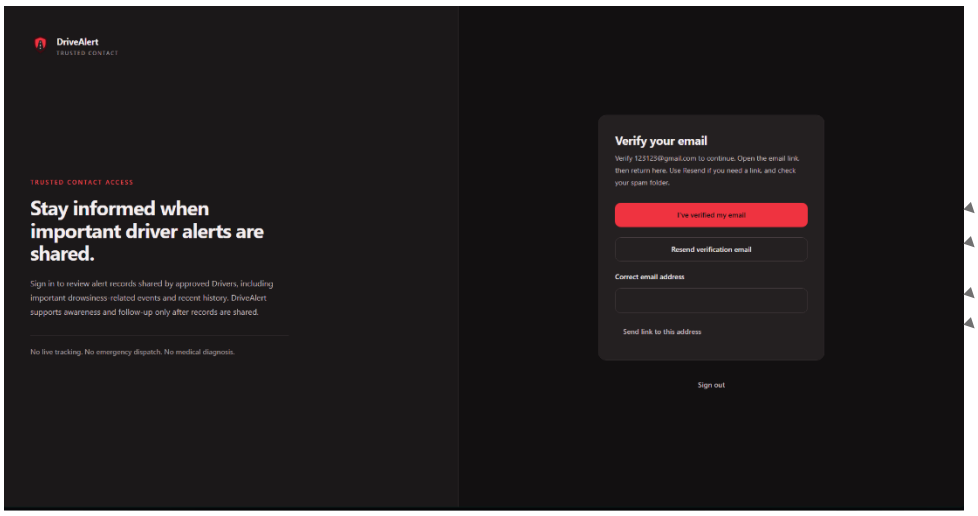


Web

1. Open https://drivealert.web.app. Click Create account to register a Trusted Contact account.
2. Enter your first name, last name, email, password, and confirmation. Middle name and phone number are optional.
3. Click Create account and complete the emailed verification link.
4. Return to the portal and click I've verified my email. Click Resend verification email if needed.

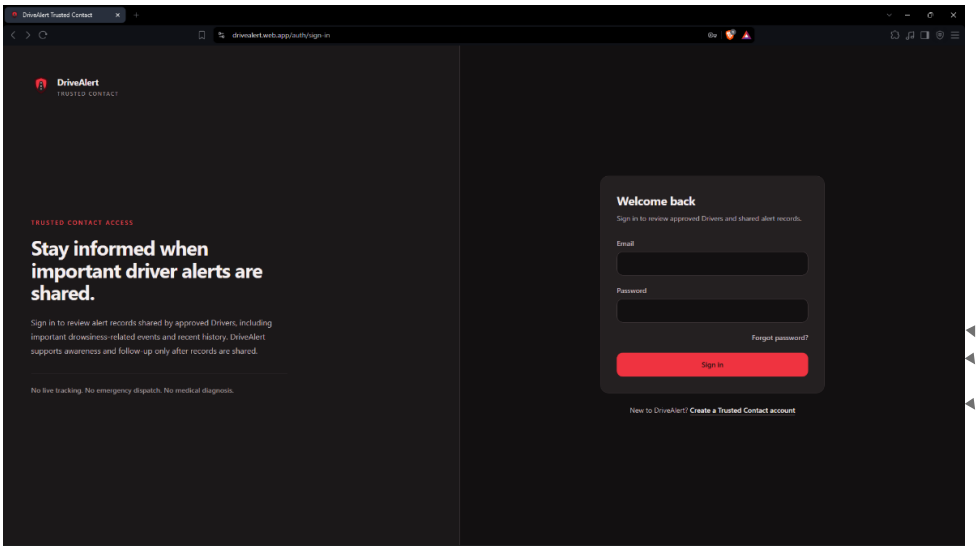
5. On the verification screen, use Correct email address and Send link to this address if an address correction is necessary.



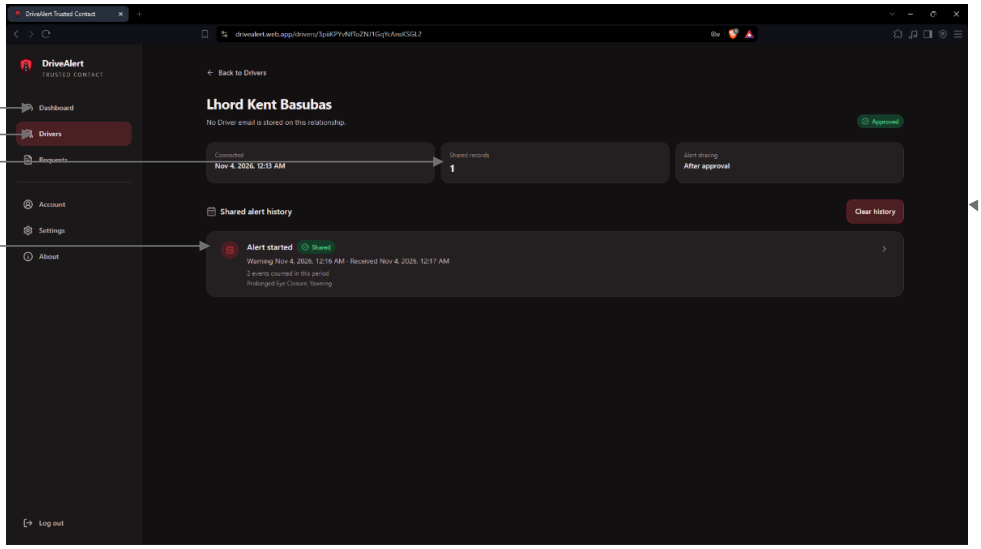


6. On Sign In, enter your registered email and password, then click Sign in.
7. Complete email verification if requested. Your account needs Trusted Contact access to open the portal.
8. For a forgotten password, click Forgot password?, enter your email, and click Send reset link.

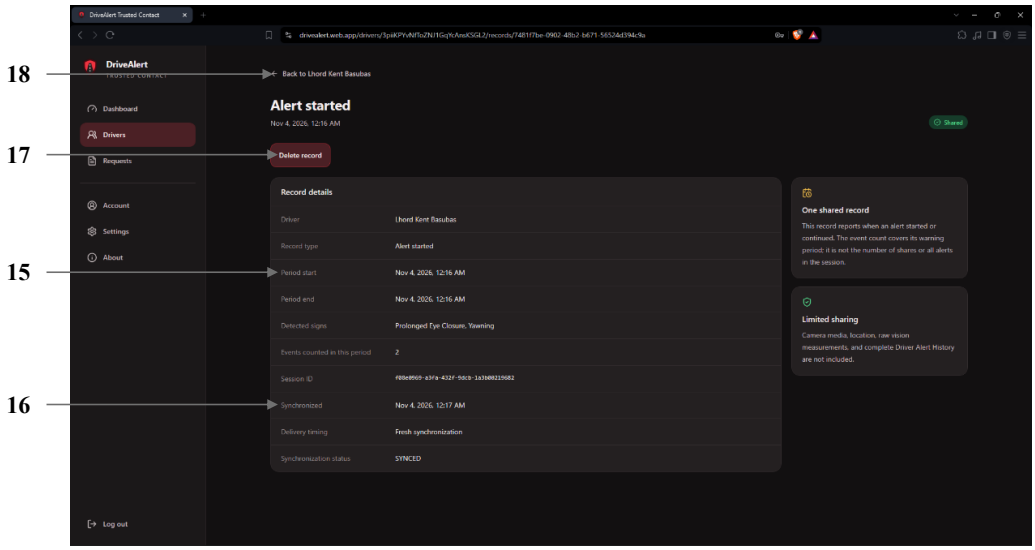
9. Follow the emailed link, then sign in with the new password. Check your spam folder if the message is missing.



10. Open Dashboard for an account overview. Use the sidebar to switch between Drivers, Requests, Account, Settings, and About.
11. Click Drivers to view approved connections, then select a Driver.
12. The Driver page shows approval information, the Shared records count, and Shared alert history.
13. Click a record to open its details. If no records appear, wait for an eligible upload after approval and check retention.
14. Click Clear history to remove that Driver's loaded records from your view in this browser. Review the confirmation before proceeding.

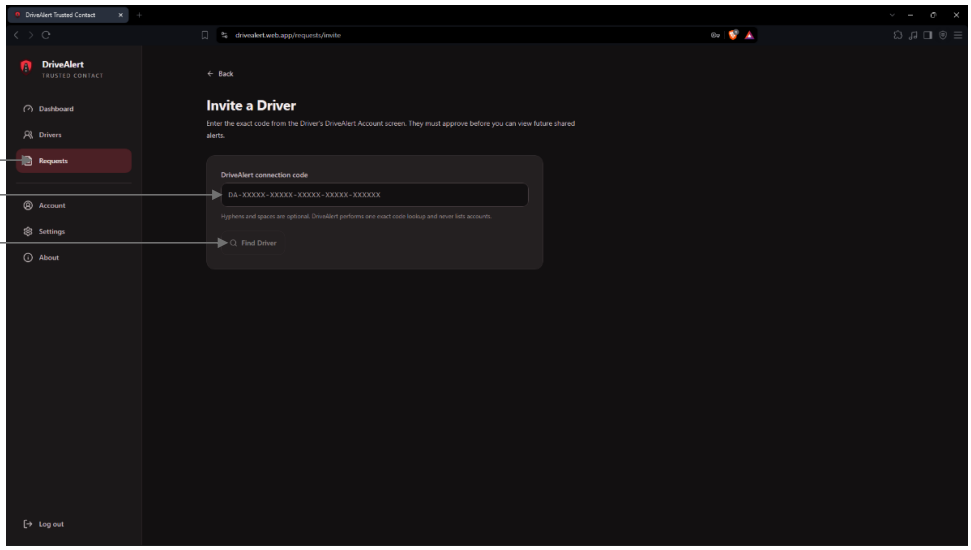


15. On the record page, review Record type, Period start, Period end, Detected signs, and Events counted in this period.
16. Review synchronization information to distinguish the warning period from the upload time.
17. Click Delete record to remove one item from this browser's history view. This does not delete the Driver's or other contacts' records.
18. Click Back to the Driver's name to return to the shared history.

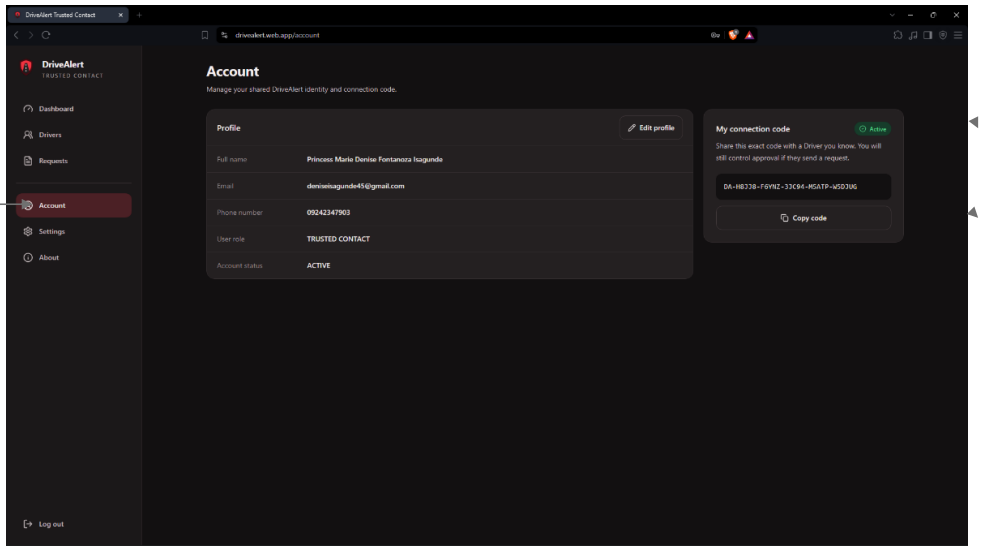


19. Open Requests and click Invite a Driver.
20. Enter the Driver's connection code and click Find Driver. Hyphens and spaces are optional.
21. Check the returned name before sending the request. The portal does not list public accounts.
22. Return to Requests to accept or decline incoming requests and cancel outgoing requests.

23. To end an approved connection, open Drivers, click Disconnect on the Driver's card, and confirm Disconnect Driver?.

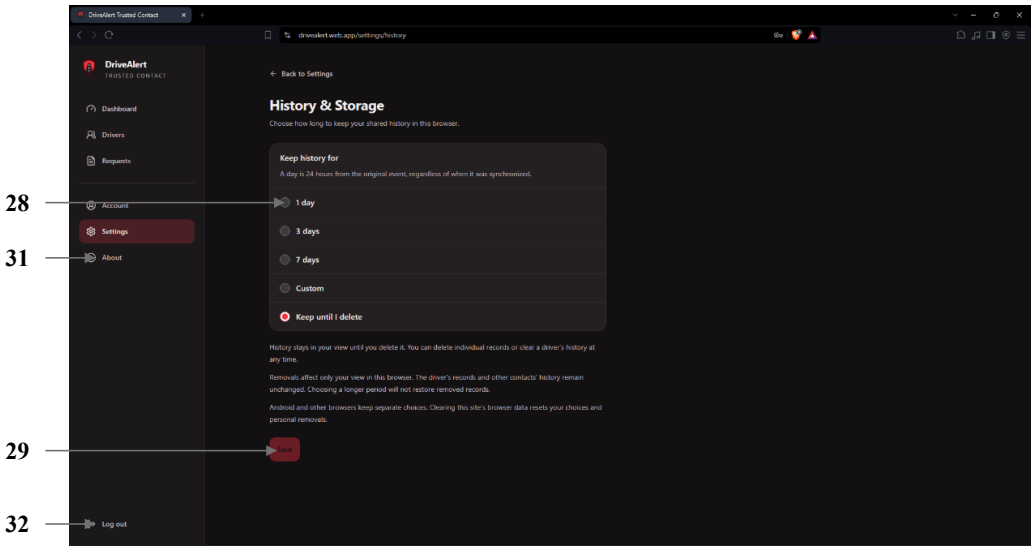


24. Open Account to view your profile, user role, account status, and connection code.
25. Click Edit profile, update the available name or phone fields, and save your changes.
26. Click Copy code to share your code with a Driver you know. Review the request before accepting it.
27. Use the Android application for Driver setup and monitoring; the web portal provides Trusted Contact functions.



28. Open Settings, then History & Storage. Choose 1 day, 3 days, 7 days, Custom, or Keep until I delete.

29. For Custom, enter 1 to 3,650 whole days and click Save. Review removal counts before confirming Save and apply.
30. These settings apply to this browser. Android and other browsers keep separate choices. Cleanup runs while the portal is open or when you return.
31. Open About for the system's purpose and limitations. Settings explains that browser remote push notifications are unavailable.
32. Click Log out in the sidebar or Settings to sign out. Clearing site data resets this browser's retention and personal removal settings.



Installation Guide

This guide assists Drivers and Trusted Contacts in installing and setting up the DriveAlert Android application.

1. Check Device Requirements
    a. The smartphone must run Android 8.0 (Oreo) or higher.
    b. Connect to the internet for registration, email verification, and initial device authorization. Driver setup also requires Wi-Fi, Bluetooth, and the DriveAlert camera device.
2. Install the Application
    a. Download DriveAlert.apk from the following GitHub release: https://github.com/LhordKent/DriveAlert/releases/tag/v1.0.0
    b. Open the downloaded APK. If prompted, allow Install unknown apps for the browser or file manager used to open it.
    c. Tap Install, then Open. If Android offers Update for an existing installation, tap Update.
3. Sign In
    a. Open DriveAlert. Create an account, or sign in with your registered email and password.
    b. Follow the verification link sent to your email, then return and tap I've verified my email when requested.
    c. On Choose your view, select Continue as Driver or Continue as Trusted Contact.
4. Complete Initial Setup
    a. For Drivers, open Setup & Device and connect the camera. Allow the requested Bluetooth or Nearby devices permissions; older Android versions may require Location permission for scanning.
    b. While parked, confirm camera alignment and follow the calibration instructions. Return to Home and tap Start Monitoring. After initial online setup, authorized Drivers can monitor offline.
    c. For Trusted Contacts, connect with a Driver and obtain approval to review eligible shared records. The web portal is also available at https://drivealert.web.app.

Project Road Map

This section presents DriveAlert's planned timeline and major milestones. It follows the Gantt chart and Agile sprint plan, showing the sequence of research, design, development, testing, and deployment activities from June to November 2026.

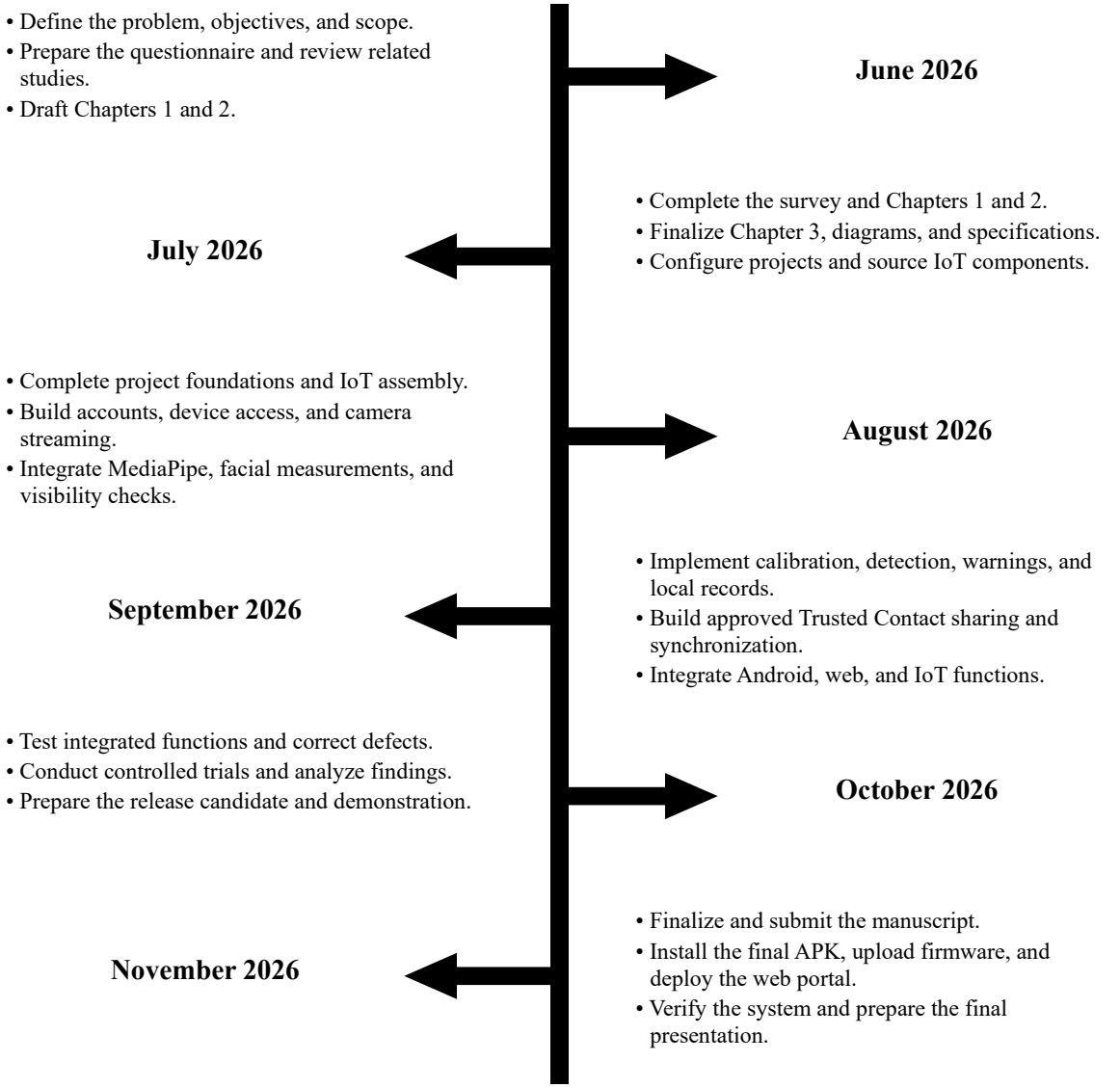


Figure 64: **Project Road Map**

Figure 64 presents the DriveAlert project road map, summarizing the activities scheduled from June 25 to November 11, 2026. The road map follows ten two-week sprints and the task dates in the Gantt chart, progressing from the research foundation to final manuscript submission, system deployment, and presentation.

In June 2026, the team will begin Sprint 1 by defining the research problem, objectives, scope, and study terms. The researchers will prepare the questionnaire, review related literature and studies, and develop the initial drafts of Chapters 1 and 2. These activities establish the basis for the proposed system.

In July 2026, the team will complete the survey, analyze the research foundation, and revise Chapters 1 and 2. Sprint 2 will complete Chapter 3, workflows, interface designs, database diagrams, and system specifications. Sprint 3 will begin project configuration, schema refinement, IoT component sourcing, and manuscript compliance work.

In August 2026, the team will finish Sprint 3 and complete the local-storage foundation and remaining compliance activities. Sprint 4 will cover IoT assembly, Bluetooth provisioning, camera streaming, account functions, and simulated additional-Driver access. Sprint 5 will integrate MediaPipe, facial measurements, Driver-area tracking, and lighting and visibility checks.

In September 2026, Sprint 6 will implement guided calibration, prolonged-eye-closure, yawning, and head-down detection, warning-stage rules, onboard audio, and local records. Sprint 7 will establish approved Trusted Contact connections, eligible Stage 3 records, synchronization retries, and shared mobile and web views. The team will integrate these functions and test the complete workflow.

In October 2026, Sprint 8 will focus on integrated testing, defect correction, regression checks, and manuscript alignment. Sprint 9 will conduct controlled trials, examine false alerts, missed detections, delays, and warning responses, and analyze the findings. The team will prepare the release-candidate APK and begin Sprint 10 presentation work on October 29.

In November 2026, the team will complete Sprint 10 by finalizing manuscript corrections, approval requirements, and the submission package. The final manuscript submission, APK installation, firmware upload, and web deployment are scheduled for November 6 to 11. Final verification, presentation preparation, and demonstration rehearsal will conclude the project by November 11.

CONCLUSION

Based on the data gathered, the proponents concluded that DriveAlert: An IoT-Based Driver Drowsiness Monitoring and Alert System Using Computer Vision is a beneficial innovation for supporting driver awareness and monitoring visible signs associated with drowsiness. Drivers may experience prolonged eye closure, yawning, and head nodding without immediately recognizing the need to stop and rest. Relying on personal awareness alone may allow these signs to go unnoticed, while the absence of organized alert records limits opportunities for reviewing recurring events and keeping Trusted Contacts informed. These concerns highlight the need for an integrated system that can assist Drivers through monitoring, warning alerts, and record sharing. With DriveAlert, Drivers are provided with a platform that uses an IoT camera and an Android application to monitor selected visible signs through computer vision and personalized calibration. Through local Wi-Fi communication, the application processes camera frames without requiring internet access during monitoring after initial setup and sends warning commands to an onboard speaker. Its dashboard allows Drivers to review alert history and basic insights, while eligible Stage 3 records are synchronized with approved Trusted Contacts through the mobile application or web portal when internet connectivity becomes available. Ultimately, DriveAlert offers a practical approach to combining local monitoring, audible warnings, and approved record sharing, helping Drivers recognize warning events and enabling Trusted Contacts to remain informed while supporting safer driving decisions.

RECOMMENDATION

The proponents identified several areas for future improvement and enhancement of the DriveAlert system, such as:

1. Improving the reliability of drowsiness-sign detection by refining Driver calibration, detection thresholds, and temporal-confirmation rules using more diverse Drivers and recorded scenarios, including normal facial movements, different head positions, and facial obstructions, to reduce false alerts and missed detections.
2. Enhancing the IoT hardware by using a more stable regulated power supply, improved camera mounting and enclosure, and suitable infrared illumination to address the prototype's lighting limitations and improve operation under low-light conditions.
3. Optimizing video processing, memory use, battery consumption, and temperature management on the Android application, together with broader device-compatibility testing, to improve monitoring stability and warning responsiveness across smartphones with different processing capabilities.
4. Developing an iOS version of the mobile application to support more users while retaining the Driver and Trusted Contact functions, local monitoring, and approved record-sharing features available in the Android system.
5. Expanding system evaluation to include more Drivers, compatible smartphones, and representative vehicle environments, beginning with controlled or simulated conditions, to assess detection reliability, warning response, usability, and continued operation beyond the configurations covered in the study.
6. Strengthening data security by improving the protection of user accounts, device authorization, calibration data, and alert records through stronger access controls, secure local storage, and regular review of cloud security rules to maintain privacy and prevent unauthorized access.

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APPENDICES

APPENDIX A

Transmittal Letter - Respondents

June 25, 2026

Dear Respondents,

Good day!

We, the student of the Bachelor of Science in Information Technology program at the University of Cebu – Banilad Campus, are currently developing a class project entitled "DriveAlert: An IoT Based Driver Fatigue Risk Monitoring and Alert System Using Computer Vision" as part of our requirements for Capstone Project 1. This course focuses on human-centered design, problem identification, and the development of innovative technology solutions based on real user needs.

The proposed project aims to develop an offline capable safety system that helps drivers prevent road accidents. The system utilizes a dashboard mounted camera and a mobile application to detect real time physical signs of driver fatigue, such as **Prolonged eye closure, Yawning, and Head slumping**, and immediately triggers customizable audio alerts. Before proceeding to the design and development stages, it is essential to validate the actual challenges drivers face regarding exhaustion, their current methods for staying awake during long trips, and their readiness to adopt a digital safety device.

In line with this, we respectfully request your permission to conduct surveys and brief interviews with selected drivers within Mandaue City, during [June 24–28, 2026]. We intend to survey approximately 10 individuals who have active experience in long-haul driving, daily driving, or public transport operations. The survey will focus on understanding their current methods for staying awake, the impact of physical exhaustion on their driving performance, and their readiness to utilize a digital safety tool for fatigue assessment.

We would be honored to conduct our research study within Cebu City and assure you that we will conduct ourselves professionally and respectfully at all times. Rest assured that our activities will not disrupt the daily driving routines of the respondents or their normal operations. Thank you very much for your consideration. We look forward to your positive response and the opportunity to work with you.

Respectfully yours,

LHORD KENT R. BASUBAS
Lead Researcher

Noted by:

Mr. ROCHÉ A. PLANDO
Capstone Adviser

Mr. ERIC P. ORTEGA
Chairperson, College of Computer Studies

APPENDIX B – Research Instrument

Survey Questionnaire

1. What is your primary role as a driver? (Check all that apply)

☐ Private vehicle owner / Everyday driver

☐ Taxi driver

☐ Grab / TNVS driver

☐ Other

2. On average, how many hours do you drive per day?

○ Less than 2 hours

○ 2-4 hours

○ 5-8 hours

○ More than 8 hours

3. When do you typically drive? (Check all that apply)

☐ Daytime (6:00 AM – 6:00 PM)

☐ Evening (6:00 PM – 12:00 AM)

☐ Late night / Early morning (12:00 AM – 6:00 AM)

4. Have you ever felt drowsy or sleepy while driving?

○ Yes

○ No

5. Have you experienced any of the following while driving? (Check all that apply)

☐ Heavy eyelids or difficulty keeping eyes open

☐ Frequent yawning

☐ Head nodding or drooping forward

☐ Missing a turn or road sign due to inattention

☐ None of the above

6. Have you ever almost had an accident because you felt drowsy while driving?

○ Yes

○ No

7. I often drive for long periods without taking a proper rest break.

○ 1 (Strongly Disagree)

○ 2

○ 3

○ 4

○ 5 (Strongly Agree)

8. I can manage my own drowsiness while driving without needing any external device or alert.

○ 1 (Strongly Disagree)

○ 2

○ 3

○ 4

○ 5 (Strongly Agree)

9. Are you currently using any device or method to help prevent drowsy driving?

○ Yes

○ No

10. If NO to Question 9, why not? (Check all that apply)

☐ I am not aware of any such device

☐ They are too expensive for me to afford

☐ I rely on coffee or energy drinks instead

☐ I do not think I need one

11. Most driver fatigue monitoring devices available today are too expensive for an ordinary driver to afford.

○ 1 (Strongly Disagree)

○ 2

○ 3

○ 4

○ 5 (Strongly Agree)

12. I have experienced poor or no mobile data signal in areas where I frequently drive.

○ 1 (Strongly Disagree)

○ 2

○ 3

○ 4

○ 5 (Strongly Agree)

13. I would be willing to add a Trusted Contact in DriveAlert who can receive my fatigue history records

○ 1 (Strongly Disagree)

○ 2

○ 3

○ 4

○ 5 (Strongly Agree)

14. Sharing fatigue-related records with a Trusted Contact would be helpful during long trips or late-night driving.

○ 1 (Strongly Disagree)

○ 2

○ 3

○ 4

○ 5 (Strongly Agree)

15. What fatigue-related information would you be comfortable sharing with your Trusted Contact? Check all that apply.

☐ Date and time of fatigue alert

☐ Number of fatigue alerts

☐ Trip/session fatigue summary

☐ I am not comfortable sharing any fatigue record

☐ Other: _____

16. Based on my driving experiences, I would benefit from receiving alerts when I begin to feel drowsy while driving.

○ 1 (Strongly Disagree)

○ 2

○ 3

○ 4

○ 5 (Strongly Agree)

17. I would feel more confident driving long distances if a fatigue monitoring device were available in my vehicle.

- 1 (Strongly Disagree)
- 2
- 3
- 4
- 5 (Strongly Agree)

18. Receiving alerts when I show signs of fatigue while driving would be helpful to me.

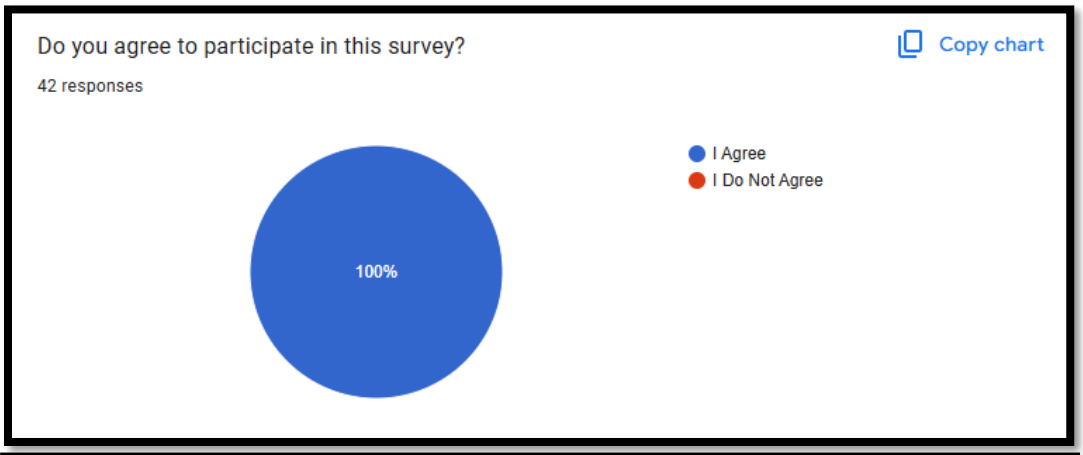
- 1 (Strongly Disagree)
- 2
- 3
- 4
- 5 (Strongly Agree)

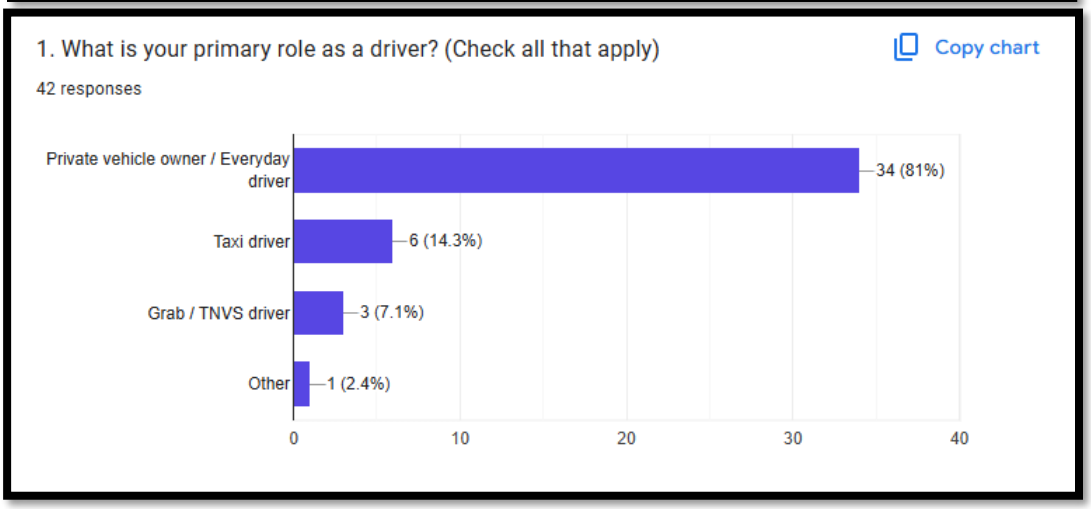
19. I would consider using a driver fatigue monitoring device if it were affordable and easy to use.

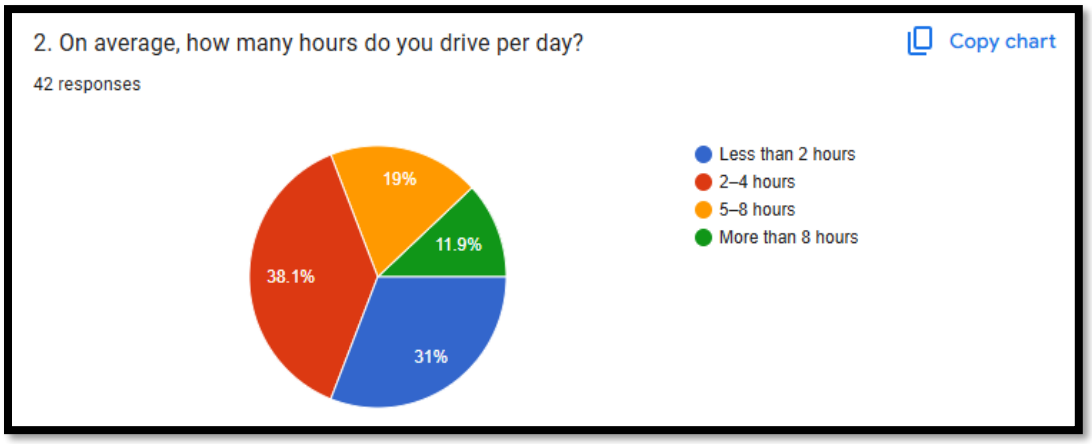
- 1 (Strongly Disagree)
- 2
- 3
- 4
- 5 (Strongly Agree)

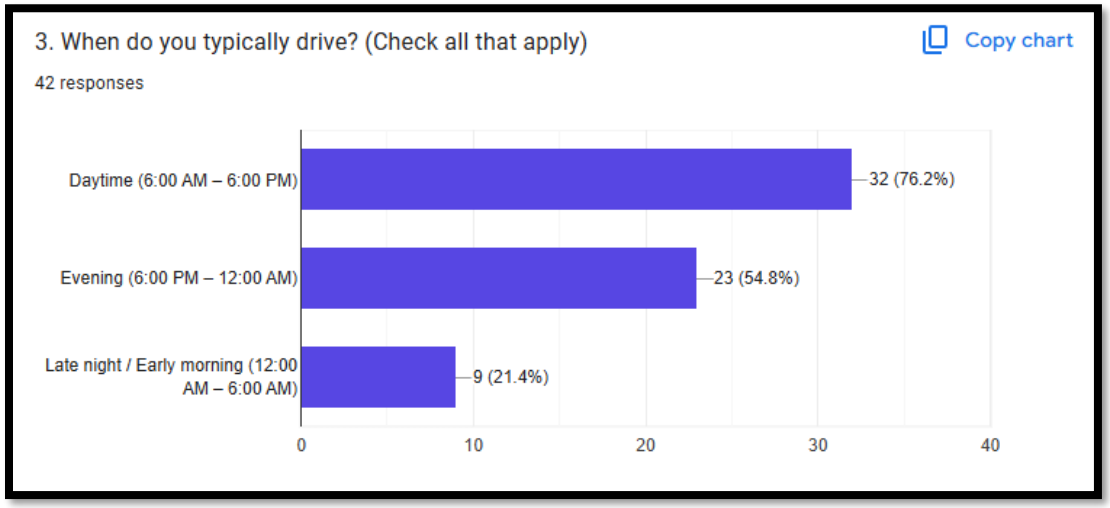
APPENDIX C

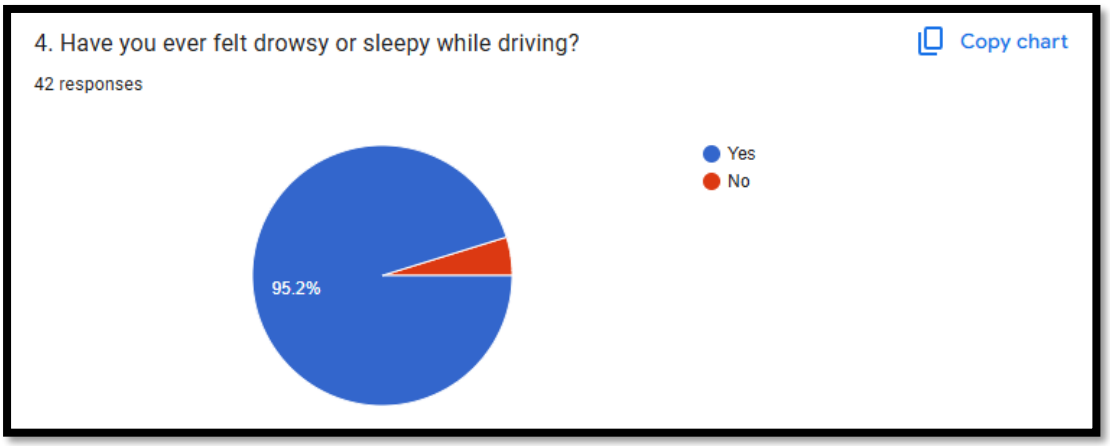
Results & Findings - Driver

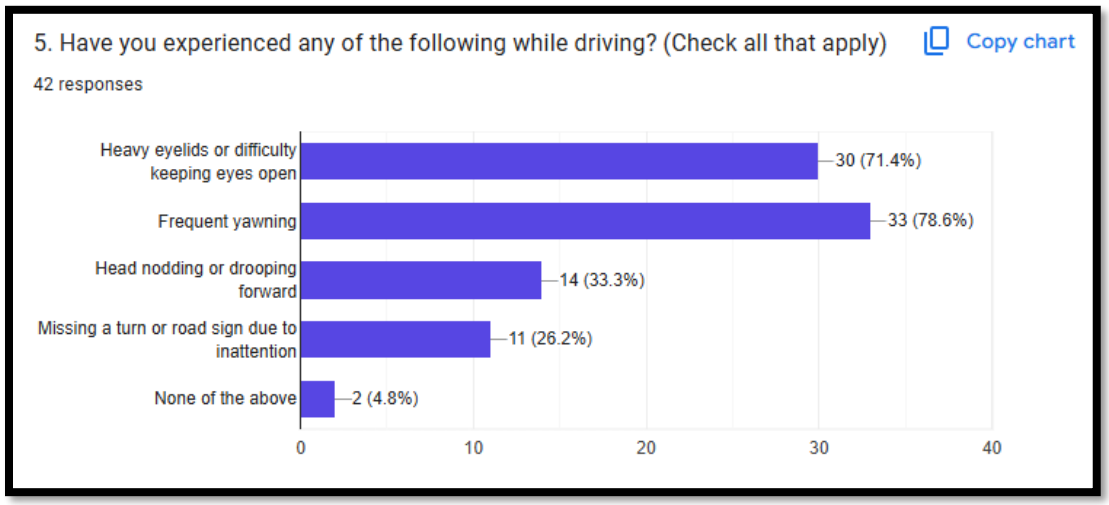


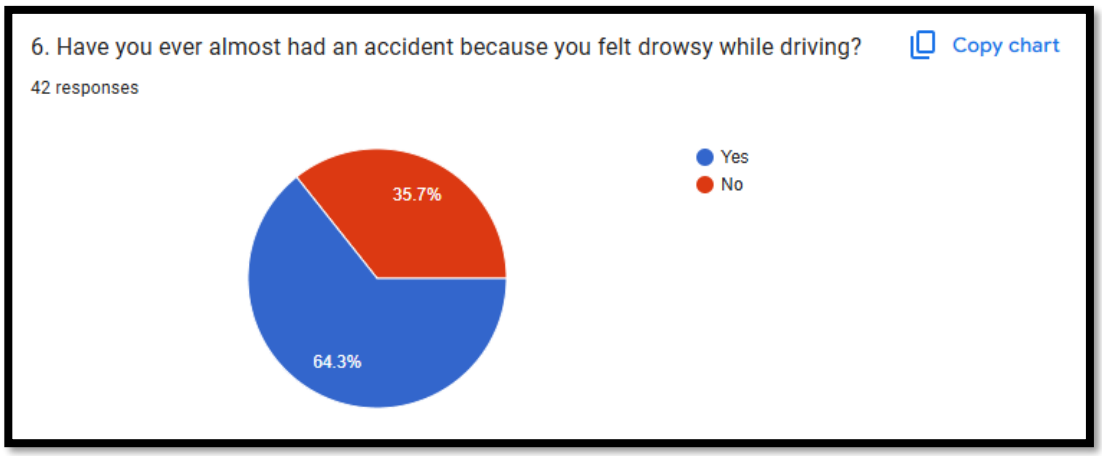


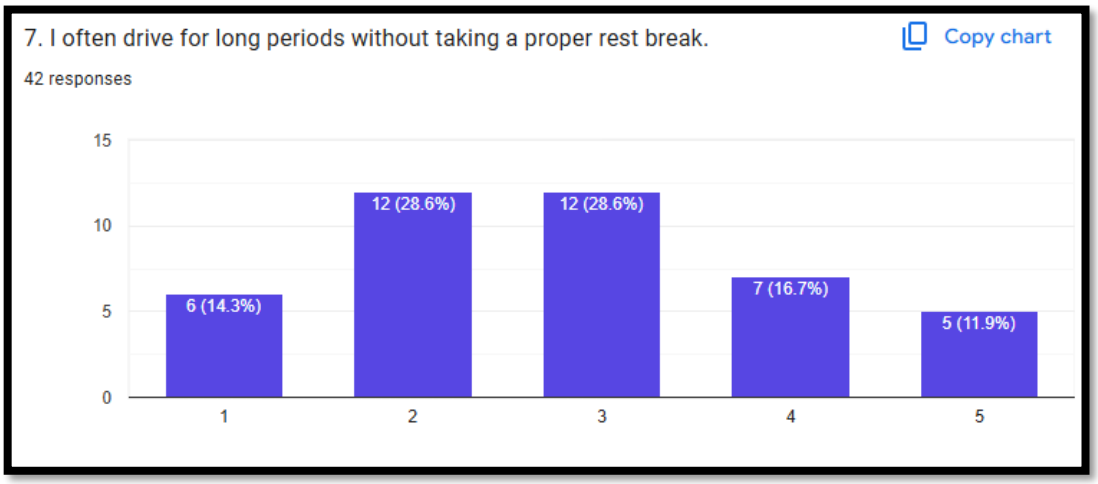


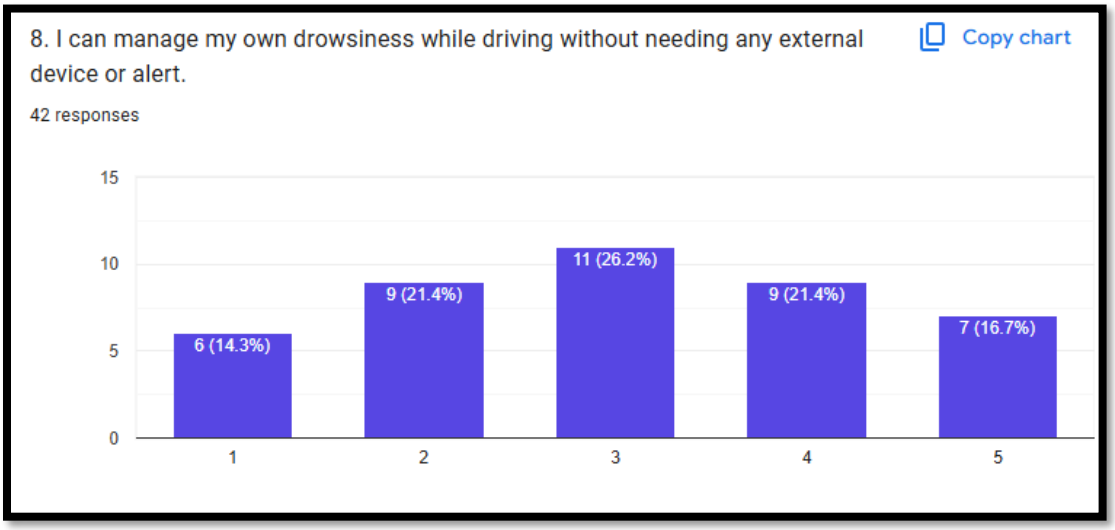

3. When do you typically drive? (Check all that apply)
Copy chart
42 responses
Daytime (6:00 AM – 6:00 PM)
32 (76.2%)
Evening (6:00 PM – 12:00 AM)
23 (54.8%)
Late night / Early morning (12:00 AM – 6:00 AM)
9 (21.4%)
0
10
20
30
40


4. Have you ever felt drowsy or sleepy while driving?
Copy chart
42 responses
Yes
No
95.2%


5. Have you experienced any of the following while driving? (Check all that apply)
Copy chart
42 responses
Heavy eyelids or difficulty keeping eyes open
30 (71.4%)
Frequent yawning
33 (78.6%)
Head nodding or drooping forward
14 (33.3%)
Missing a turn or road sign due to inattention
11 (26.2%)
None of the above
2 (4.8%)
0
10
20
30
40


6. Have you ever almost had an accident because you felt drowsy while driving?
Copy chart
42 responses
Yes
No
35.7%
64.3%

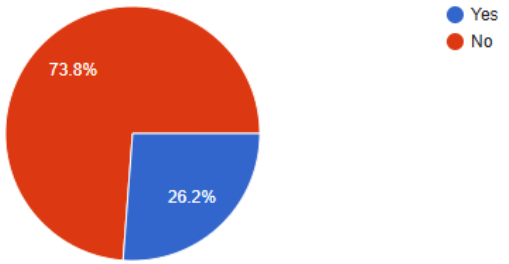

7. I often drive for long periods without taking a proper rest break.
Copy chart
42 responses
15
10
5
0
6 (14.3%)
12 (28.6%)
12 (28.6%)
7 (16.7%)
5 (11.9%)
1
2
3
4
5


8. I can manage my own drowsiness while driving without needing any external device or alert.
Copy chart
42 responses
15
10
5
0
6 (14.3%)
9 (21.4%)
11 (26.2%)
9 (21.4%)
7 (16.7%)
1
2
3
4
5

9. Are you currently using any device or method to help prevent drowsy driving?

Copy chart

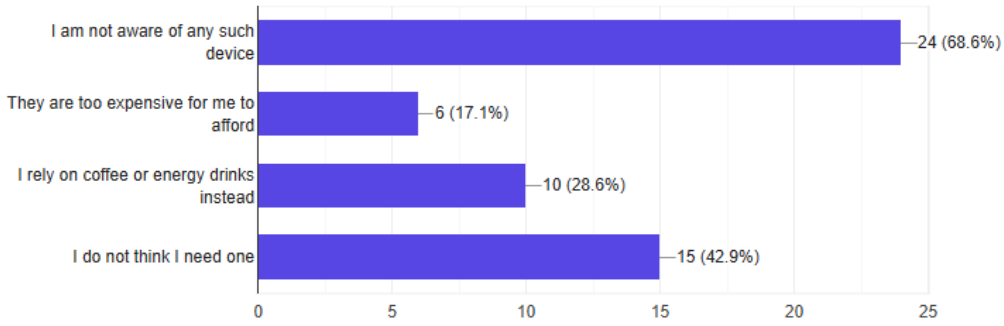
42 responses



10. If NO to Question 9, why not? (Check all that apply)

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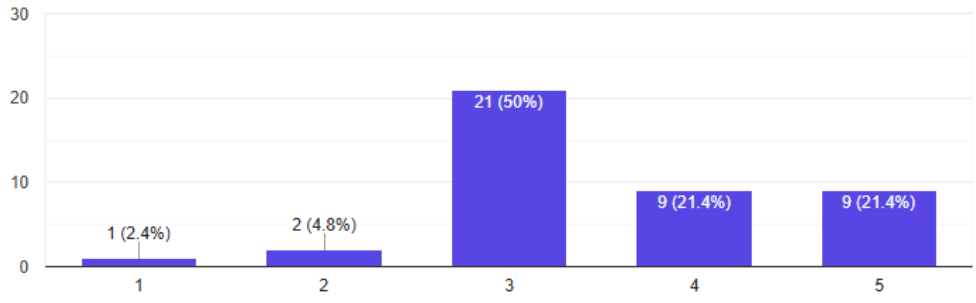
35 responses



11. Most driver fatigue monitoring devices available today are too expensive for an ordinary driver to afford.

Copy chart

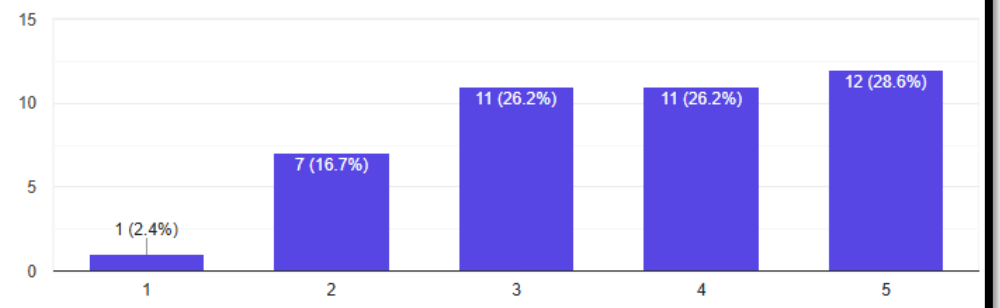
42 responses



12. I have experienced poor or no mobile data signal in areas where I frequently drive.

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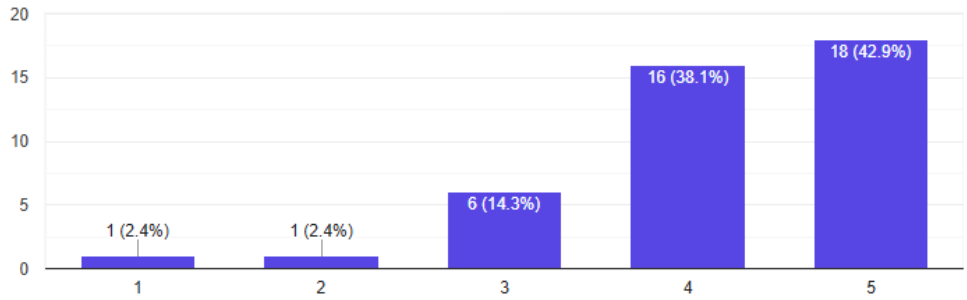
42 responses



13. I would be willing to add a Trusted Contact in DriveAlert who can receive my fatigue history records

Copy chart

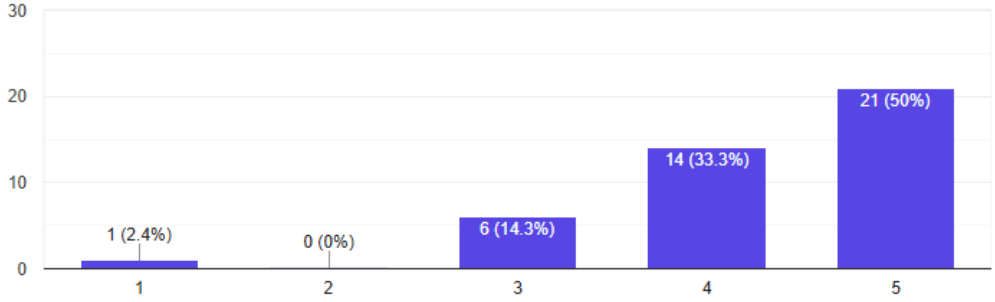
42 responses



14. Sharing fatigue-related records with a Trusted Contact would be helpful during long trips or late-night driving.

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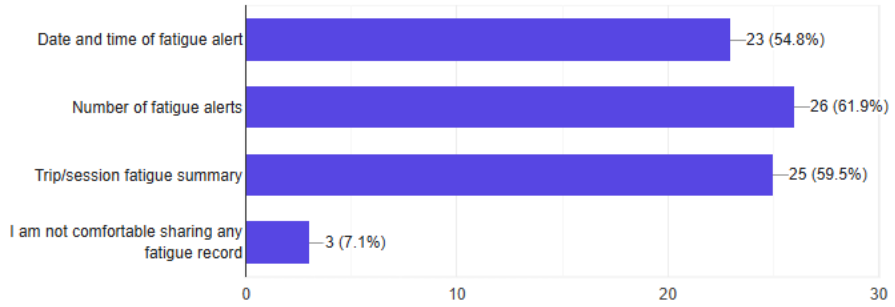
42 responses



15. What fatigue-related information would you be comfortable sharing with your Trusted Contact?
Check all that apply.

Copy chart

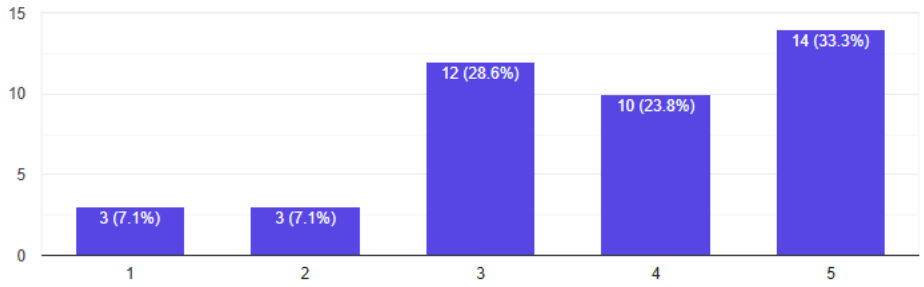
42 responses



16. Based on my driving experience, I would benefit from receiving alerts when I begin to feel drowsy while driving.

Copy chart

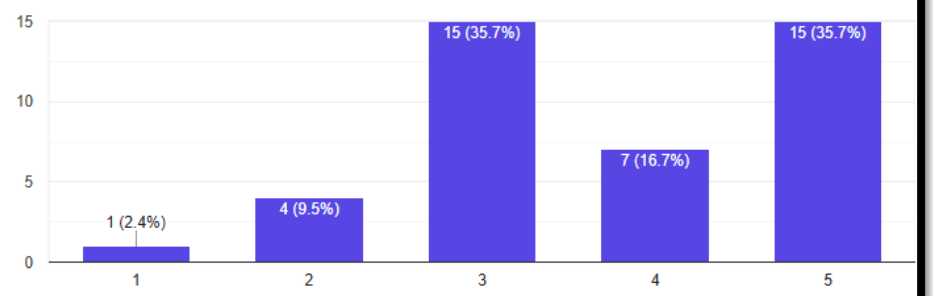
42 responses



17. I would feel more confident driving long distances if a fatigue monitoring device were available in my vehicle.

Copy chart

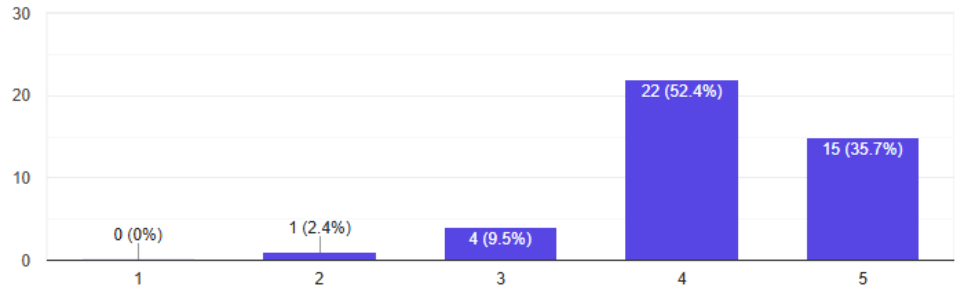
42 responses



18. Receiving alerts when I show signs of fatigue while driving would be helpful to me.

Copy chart

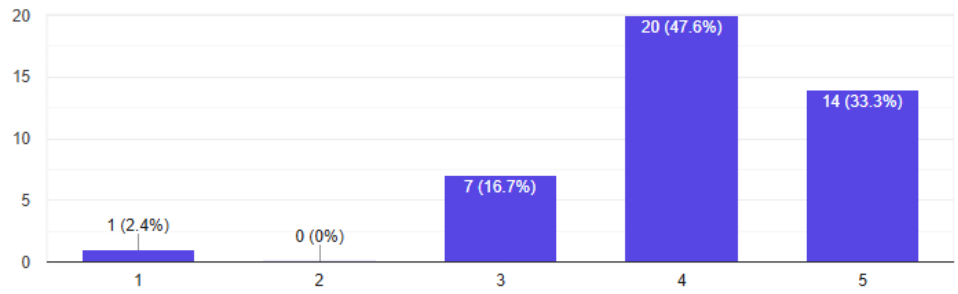
42 responses



19. I would consider using a driver fatigue monitoring device if it were affordable and easy to use.

Copy chart

42 responses



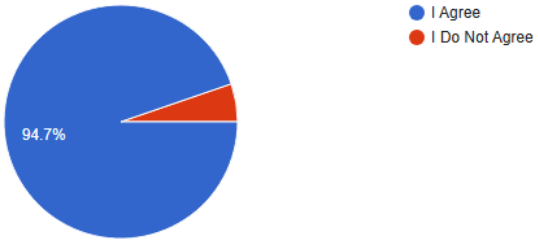
APPENDIX D

Results & Findings – Potential Trusted Contacts

1. Do you agree to participate in this survey?

Copy chart

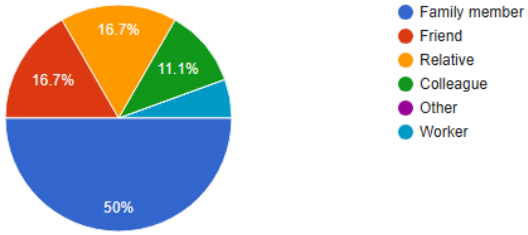
19 responses



3. What is your relationship to a driver whom you may potentially support as a Trusted Contact?

Copy chart

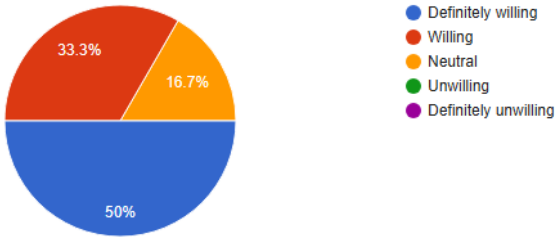
18 responses



4. If a system allowed a driver to select you as a Trusted Contact, how willing would you be to review drowsiness-alert records that the driver chooses to share with you?

Copy chart

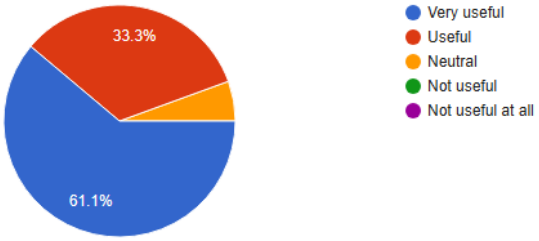
18 responses



5. How useful would it be for you to view information such as the date and time of alerts, detected drowsiness signs, alert counts, and recently synchronized records?

Copy chart

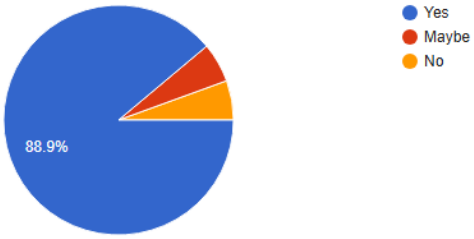
18 responses



6. Would you be comfortable viewing a driver's drowsiness-alert records only after the driver has approved you as a Trusted Contact?

Copy chart

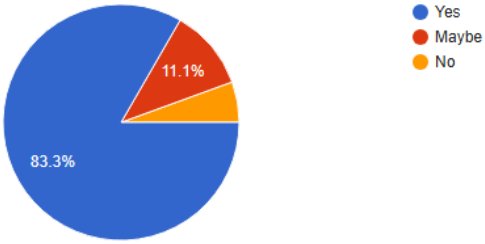
18 responses



7. If the system stored records while offline and shared them only when internet connectivity became available, would this be acceptable to you?

Copy chart

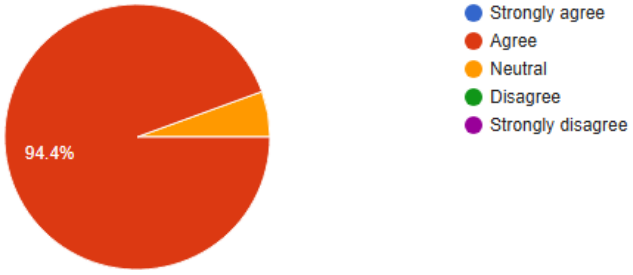
18 responses



8. Do you think access to shared drowsiness-alert records could help a Trusted Contact stay informed about alerts recorded during a driver's trips?

Copy chart

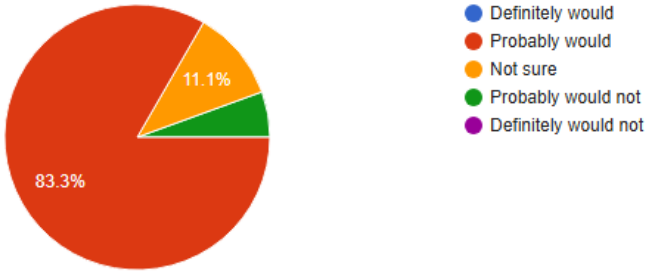
18 responses



9. If a system with this Trusted Contact feature were easy to use, would you consider using the feature when requested by a driver?

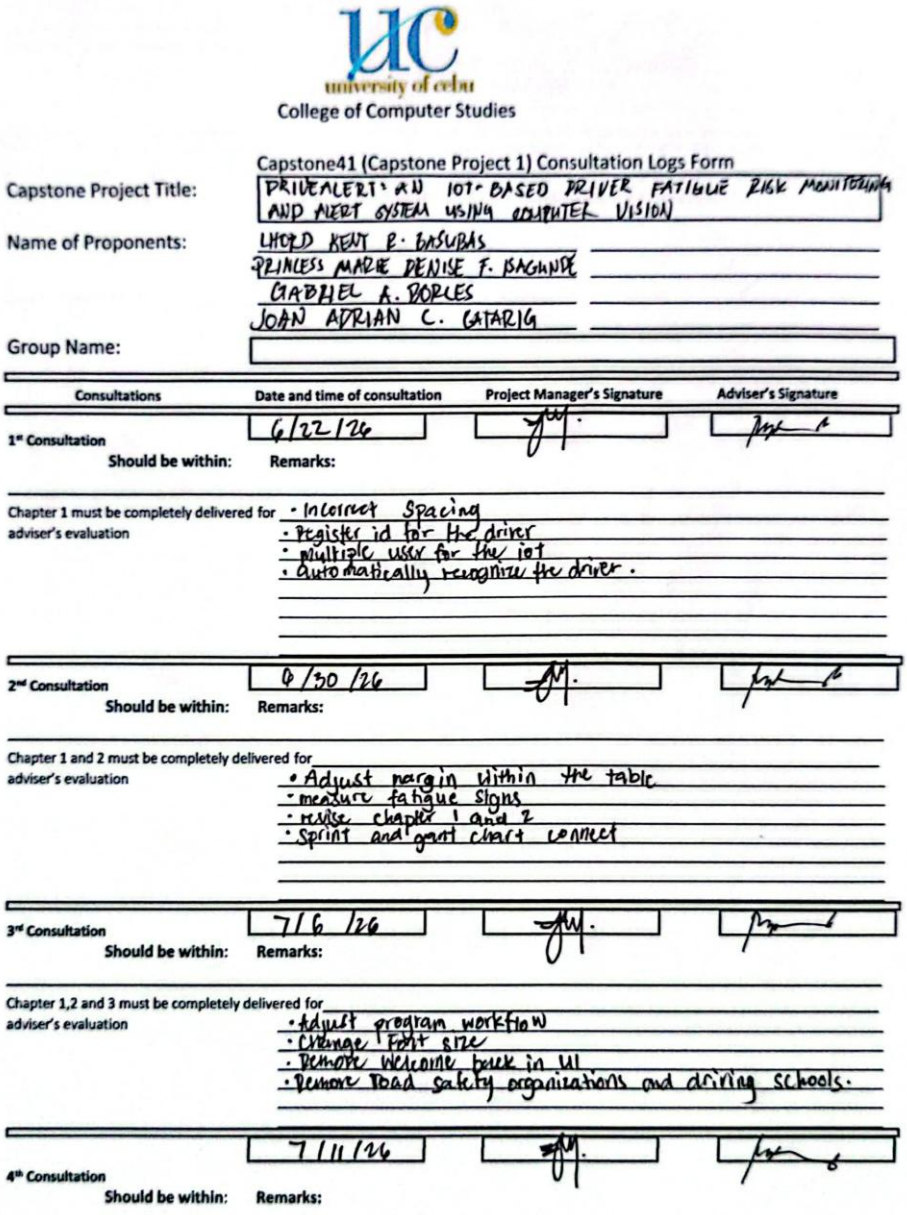
Copy chart

18 responses



APPENDIX E

Consultation logs - Adviser



university of cebu
College of Computer Studies

Capstone41 (Capstone Project 1) Consultation Logs Form

Capstone Project Title: DRIVEALERT: AN IOT-BASED DRIVER FATIGUE RISK MONITORING AND ALERT SYSTEM USING COMPUTER VISION

Name of Proponents: LHORD KENT B. BASUBAS
PRINCESS MARIE DENISE F. BAGUNDE
GABRIEL A. BORLES
JOAN ADRIAN C. CATARIG

Group Name:

Consultations	Date and time of consultation	Project Manager's Signature	Adviser's Signature
1st Consultation	6/22/26		

Should be within: Remarks:

Chapter 1 must be completely delivered for adviser's evaluation
- Incorrect Spacing
- Register id for the driver
- multiple users for the iot
- Automatically recognize the driver.

2nd Consultation	6/30/26		

Should be within: Remarks:

Chapter 1 and 2 must be completely delivered for adviser's evaluation
- Adjust margin within the table
- measure fatigue signs
- revise chapter 1 and 2
- Sprint and gant chart correct

3rd Consultation	7/6/26		

Should be within: Remarks:

Chapter 1,2 and 3 must be completely delivered for adviser's evaluation
- Adjust program workflow
- Change Font size
- Remove Welcome back in UI
- Remove Road safety organizations and driving schools.

4th Consultation	7/11/26		

Should be within: Remarks:

Chapter 1,2,3 including initial and final pages must be completely delivered for adviser's evaluation must be completely delivered for

- schematic diagram
- upgrade flow of the study (past tense)
- user role
- automatic synchronization (must reword
- alphabetical names (group members).

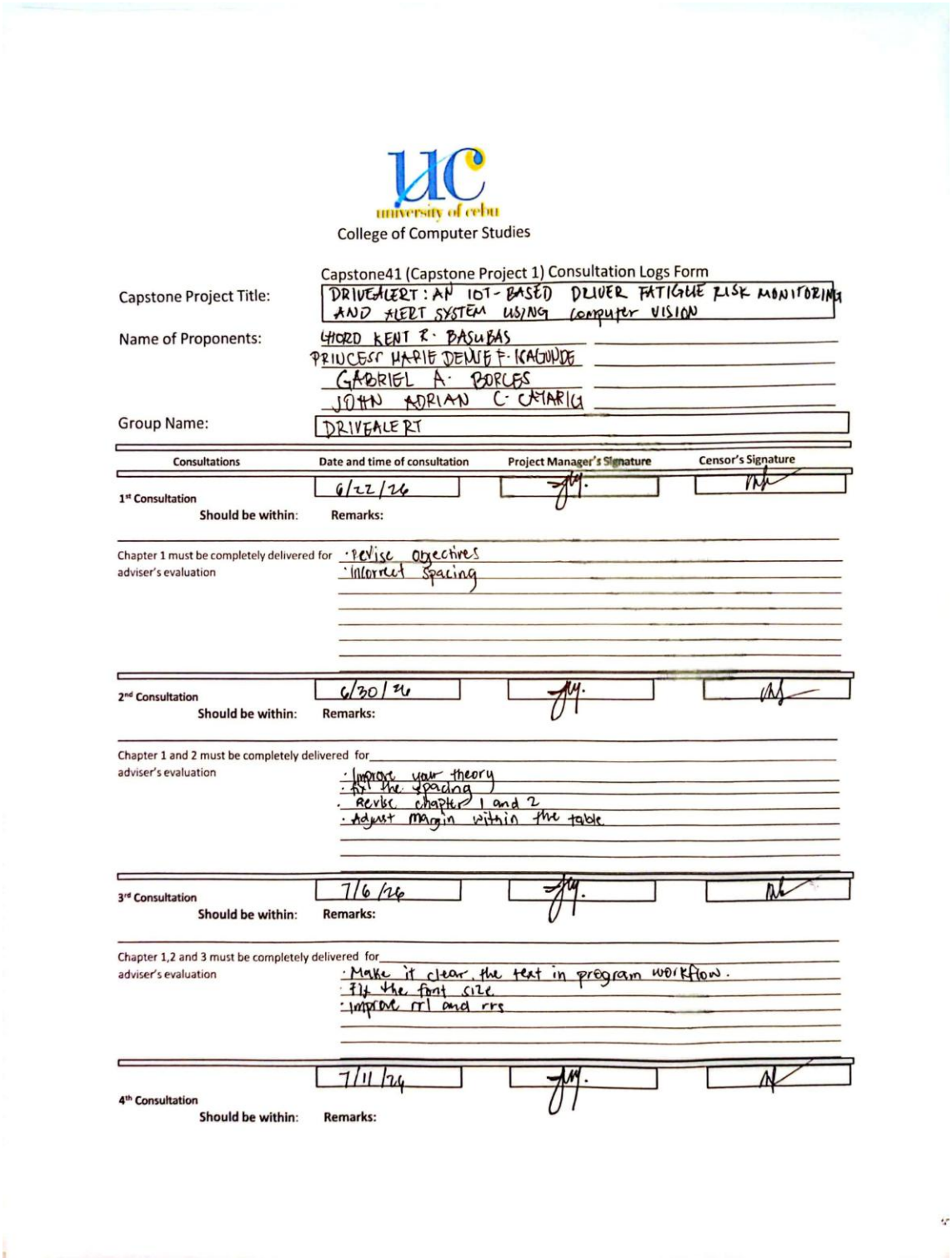
This is to certify that I have been regularly consulted by my advisees; have reviewed their system prototype as well as the required manuscript of the above-stated study. As their adviser, I therefore submit them ready for **Proposal Hearing** *as the Chapters I through 3 and the pertinent parts of their manuscript are complete.*

NOTE: Defense week:_____

Signed: _____
(Signature of Adviser over printed name)

APPENDIX F

Consultation Logs - Censor



university of cebu
College of Computer Studies

Capstone41 (Capstone Project 1) Consultation Logs Form

Capstone Project Title: DRIVEALERT: AN IOT-BASED DRIVER FATIGUE RISK MONITORING AND ALERT SYSTEM USING COMPUTER VISION

Name of Proponents: GHORD KENT R. BASUBAS
PRINCESS MARIE DENISE F. ICAGUNDE
GABRIEL A. BORCES
JOHN ADRIAN C. CAMARIG

Group Name: DRIVEALERT

Consultations	Date and time of consultation	Project Manager's Signature	Censor's Signature
1st Consultation	6/22/26		

Should be within: Remarks:

Chapter 1 must be completely delivered for adviser's evaluation

- Revise objectives
- Incorrect spacing

2nd Consultation	6/30/26		

Should be within: Remarks:

Chapter 1 and 2 must be completely delivered for adviser's evaluation

- Improve your theory
- Fix the spacing
- Revise chapter 1 and 2
- Adjust margin within the table

3rd Consultation	7/6/26		

Should be within: Remarks:

Chapter 1,2 and 3 must be completely delivered for adviser's evaluation

- Make it clear the text in program workflow.
- Fix the font size
- Improve rrl and rrs

4th Consultation	7/11/26		

Should be within: Remarks:

Chapter 1,2,3 including initial and final pages must be completely delivered for adviser's evaluation must be completely delivered for · Finalize chapter 1, 2 and 3

This is to certify that I have been regularly consulted by my advisees; have reviewed their system prototype as well as the required manuscript of the above-stated study. As their adviser, I therefore submit them ready for **Proposal Hearing** *as the Chapters I through 3 and the pertinent parts of their manuscript are complete.*

NOTE: Defense week:_____

Signed: ONEGA A. DEGFA
(Signature of Censor over printed name)

APPENDIX G

Censor's Certificate



University of Cebu

College of Computer Studies

Date: _____

CENSOR'S CERTIFICATE

This is to certify that the undersigned has reviewed and went through all the pages of the proposed projects study/research manuscript titled:

DRIVEALERT AN IOT-BASED DRIVER FATIGUE RISK MONITORING AND ALERT SYSTEM USING COMPUTER VISION

as against the set of structural rules that govern the composition of *sentences, phrases, and words* in the English language as well as the technical terms, syntax (format, etc.) and semantics appropriate for the information Technology and Computing fields.

Signed:

OMEGGA A. DELFIN
Grammarian

Conforme:

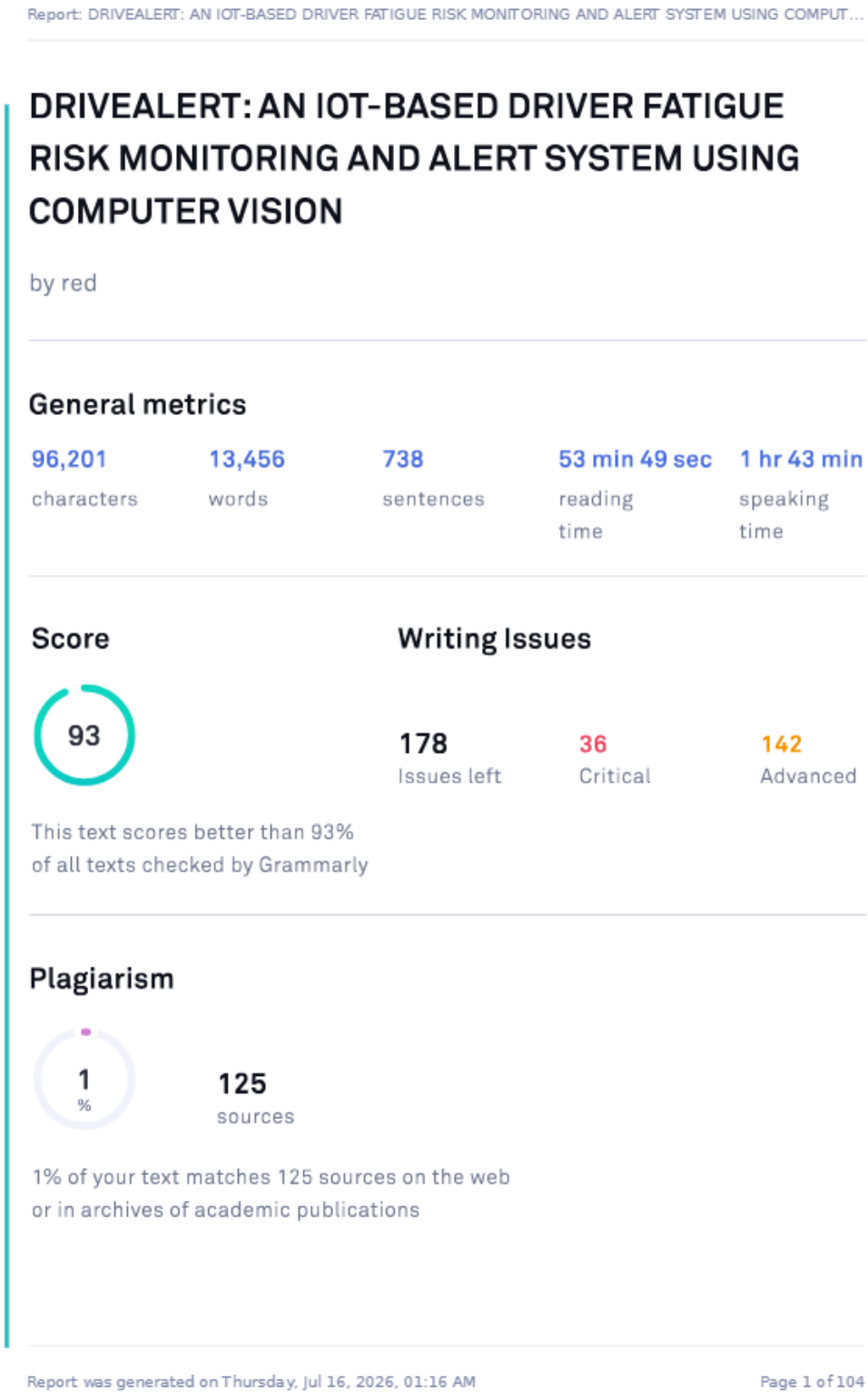
LHORD KENT R. BASUBAS
Project Manager

Noted:

RACHEL A. PLANADO
Adviser

APPENDIX H

Grammarly Results



TEAM COMPOSITION



Personal Information

Name	**:**	Lhord Kent R. Basubas
Address	**:**	Cabancalan Mandaue, Cebu
Contact Number	**:**	09566304870
Email Address	**:**	lhordalvic@gmail.com
Date of Birth	**:**	October 24, 2005
Place of Birth	**:**	Cebu City, Cebu
Civil Status	**:**	Single
Religion	**:**	Roman Catholic

Educational Background

2022 - Present	College Education University of Cebu - Banilad Campus Bachelor of Science in Information Technology Gov. M. Cuenco Ave, Cebu City, 6000 Cebu
2016 - 2022	Secondary Education Cabancalan National High School, Cebu City, 6000 Cebu
2010 - 2016	Elementary Education Cabancalan, Elementary School

Personal Information



Name	**:**	Princess Marie Denise F. Isagunde
Address	**:**	Pagsabungan Mandaue, Cebu
Contact Number	**:**	09915700829
Email Address	**:**	deniseisagunde45@gmail.com
Date of Birth	**:**	August 5, 2005
Place of Birth	**:**	Cebu City, Cebu
Civil Status	**:**	Single
Religion	**:**	Roman Catholic

Educational Background

2022 - Present	College Education University of Cebu - Banilad Campus Bachelor of Science in Information Technology Gov. M. Cuenco Ave, Cebu City, 6000 Cebu
2016 - 2022	Secondary Education SPM Academy INC, Cebu City, 6014 Cebu
2010 - 2016	Elementary Education Banilad Elementary School, Mandaue Cebu

Personal Information



Name	**:**	Gabriel A. Borces
Address	**:**	Lower Bacayan, Cebu City
Contact Number	**:**	09942664614
Email Address	**:**	gabrielborces69@gmail.com
Date of Birth	**:**	May 16, 2005
Place of Birth	**:**	Cebu City, Cebu
Civil Status	**:**	Single
Religion	**:**	Roman Catholic

Educational Background

2022 - Present	College Education University of Cebu - Banilad Campus Bachelor of Science in Information Technology Gov. M. Cuenco Ave, Cebu City, 6000 Cebu
2016 - 2022	Secondary Education Talamban National High School, Cebu City, 6000 Cebu
2010 - 2016	Elementary Education Golden Gate Childrens House, Cebu

Personal Information



Name	:	John Adrian C. Catarig
Address	:	Centro Mandaue, Cebu
Contact Number	:	09912080909
Email Address	:	Johnadriancatarig@gmail.com
Date of Birth	:	February 9, 2004
Place of Birth	:	Cebu City, Cebu
Civil Status	:	Single
Religion	:	Roman Catholic

Educational Background

2022 - Present	College Education University of Cebu - Banilad Campus Bachelor of Science in Information Technology Gov. M. Cuenco Ave, Cebu City, 6000 Cebu
2016 - 2022	Secondary Education University of Cebu, Lapu-Lapu Mandaue Campus, Cebu City, 6000 Cebu
2010 - 2016	Elementary Education St. Joseph's Academy